

All reasonable care has been taken to ensure that the information contained in this publication is accurate as at the date of printing. It should be noted however that the information above may be affected by changes occurring subsequent to the date of printing in the blend formulation or methods of application of any of the products referred to or in the requirements of any specification approval relating to any such products.

1 of 2



TYPICAL PHYSICAL CHARACTERISTICS

TEST	AWS 10	AWS 15	AWS 22	AWS 32	AWS 46	AWS 68	AWS 100	AWS 150
Kinematic Viscosity @ 40°C, cSt	10	15	22	32	46	68	100	150
@ 100°, cSt	2.55	3.2	4.25	4.25	6.65	8.6	11.1	14.5
Viscosity Index	75	75	102	102	102	102	97	97
Relative Density @ 20°C	0.875	0.880	0.865	0.870	0.875	0.880	0.880	0.885
Acidity, mgKOH/g	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Air Release Value @ 50°C, cSt	4	4	4	4	8	8	12	18
Pour Point, °C	-39	-39	-30	-30	-21	-21	-21	-21
Closed Flash Point, °C	160	160	170	200	220	220	220	220
Foam sequence 1, mins	30/0	30/0	30/0	30/0	30/0	30/0	30/0	-
Demulsification (D1401) mins	10	10	10	10	15	15	20	-
4-Ball 1hr Wear @ 30kg, 1640 rpm, mm			0.4	0.4	0.4	0.4	0.4	0.4
FZG (A/8.3/90) Fail Stage	11	11	11	11	12	12	12	12
Vickers HP Vane Pump Test				25	25	25		

OTHER INFORMATION

Castrol Hyspin AWS conforms to ISO Grade HM, DIN.51524 Part 2, Class HLP(+) and Hogglands Denison HF-O bench performance test.

Seal Compatibility: Suitable for use with Nitrile, Buna-N, Viton and Silicone seal materials.

Health and Safety information sheets are available for all Castrol products from the address below.

Castrol International, Pipers Way, Swindon, Wiltshire SN3 1RE, England., Telephone: Enquiries +44 (0)1793 512712, Technical Enquiries +44 (0)118 984 3311, Fax +44 (0)1793 453218

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Chevron SRI Grease

NLGI 2

Customer Benefits

Chevron SRI Grease delivers value through

Wide application range — Suitable for high rpm operation, operating temperatures ranging from -29°C to 177°C (-20°F to 350°F) and offers excellent rust protection in the presence of both salt and fresh water.

Excellent oxidation stability — Provides exceptional bearing life at operating temperatures in the range of 93°C to 177°C (199°F to 350°F).

Excellent rust protection — Provides superior rust protection as defined by ASTM D 1743.

Features

Chevron SRI Grease is a high temperature ball and roller bearing grease.

It is formulated with ISOSYN™ base stocks, a synthetic polyurea ashless organic thickener, and high performance rust and oxidation inhibitors (the latter to provide superior rust protection in severe applications). Its texture is smooth and buttery and its color is dark green.

Chevron SRI Grease passes ASTM D 1743-73 rust test with 5% synthetic sea water. It gives longer bearing life under high speed and high temperature operation than most other widely used antifriction bearing greases. ASTM D 3336 tests show that the life of a 204-K bearing lubricated with Chevron SRI Grease and operating at 150°C (302°F) and 10,000 rpm is about 3000 hours. This is nearly 10 times the life possible when using conventional lithium greases.

Applications

Chevron SRI Grease is recommended

- for use in a wide range of automotive and industrial applications
- for use in antifriction bearings operating at high speeds (10,000 rpm and greater)
- where the operating temperatures are on the order of 150°C (302°F) and higher
- where there is a likelihood that water or salt water will get into the bearings

It performs satisfactorily in bearings at temperatures as low as -30°C (-22°F).

Applications where Chevron SRI Grease will outperform most other greases include:

- As a "life-pack" lubricant by manufacturers of automotive generators, alternators, and starters to protect against the effects of moisture and road-splash
- Bearings on air-conditioning units in homes and other buildings
- Unsealed electric motor bearings operating under moist conditions
- Applications where silent operations are required

Original Equipment Manufacturers that specifically recommend Chevron SRI Grease are:

Bearing manufacturers: NSK, NTN, FAG, SKF, NMB, GMN Torrington, and American Koyo.

Electric motor manufacturers: Reliance Electric Company, U.S. Motors Division of Emerson Electric Company, Toshiba International, Magnetek National Coil Company, and Lincoln Electric.

Auto manufacturers: Chrysler, General Motors, and Ford.

Chevron SRI Grease is authorized by USDA for use in federally inspected meat and poultry plants as an H2 lubricant with no food contact.



Typical Test Data

NLGI Grade	2
<i>CPS Number</i>	254504
<i>MSDS Number</i>	6979
Operating Temperature, °C(°F)	
Minimum	-30(-22)
Maximum	177(350)
Penetration, at 25°C(77°F)	
Unworked	255
Worked	280
Dropping Point, °C(°F)	243(470)
High Temperature Life, hours at 177°C(350°F), ASTM D 3336	600
Lincoln Ventmeter, psig at 30 s, at	
75°F	225
30°F	425
0°F	750
-22°F	◆
Thickener, %	8.0
Type	Polyurea
ISO Viscosity Grade, Base Oil Equivalent	100
Viscosity, Kinematic*	
cSt at 40°C	116
cSt at 100°C	12.3
Viscosity, Saybolt*	
SUS at 100°F	606
SUS at 210°F	69.0
Viscosity Index*	97
Flash Point, °C(°F)*	260(500)
Pour Point, °C(°F)*	-15(+5)
Texture	Smooth, Buttery
Color	Dark Green

Typical test data are average values only. Minor variations which do not affect product performance are to be expected in normal manufacturing.

① Not tested at this temperature

* Determined on mineral oil extracted by vacuum filtration.



Material Safety Data Sheet

Chevron SRI Grease

MSDS: 6979

Revision #: 2 **Revision Date:** 02/05/02

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

CHEVRON SRI Grease

PRODUCT NUMBER(S): CPS254504 CPS254510

SYNONYM: CHEVRON SRI Grease NLGI 2
CHEVRON SRI Grease OEM NLGI 2

COMPANY IDENTIFICATION

Chevron Products Company
Lubricants and Specialty Products
6001 Bollinger Canyon Rd., T3325/B10
San Ramon, CA 94583
www.chevron-lubricants.com

EMERGENCY TELEPHONE NUMBERS

HEALTH (24 hr): (800)231-0623 or
(510)231-0623 (International)
TRANSPORTATION (24 hr): CHEMTREC
(800)424-9300 or (703)527-3887
Emergency Information Centers
are located in U.S.A.
Int'l collect calls accepted

PRODUCT INFORMATION: MSDS Request:(800)414-6737 email: lubemsds@chevron.com
Environmental, Safety, & Health Info: (925) 842-5535
Product Information: (800) 582-3835

2. COMPOSITION/INFORMATION ON INGREDIENTS

100.0 % CHEVRON SRI Grease

CONTAINING

COMPONENTS	AMOUNT	LIMIT/QTY	AGENCY/TYPE
HYDROTREATED DIST., HVY PARA			
Chemical Name: DISTILLATES, HYDROTREATED HEAVY PARAFFINIC			
CAS64742547	> 80.00%	5 mg/m3 (mist) 10 mg/m3 (mist) 5 mg/m3 (mist)	ACGIH TWA ACGIH STEL OSHA PEL
POLYUREA THICKENERS			

ADDITIVES < 10.00%
< 10.00%

COMPOSITION COMMENT:

All the components of this material are on the Toxic Substances Control Act Chemical Substances Inventory.

This product fits the ACGIH definition for mineral oil mist. The ACGIH TLV is 5 mg/m³, the OSHA PEL is 5 mg/m³.

3. HAZARDS IDENTIFICATION

IMMEDIATE HEALTH EFFECTS

EYE:

Not expected to cause prolonged or significant eye irritation.

SKIN:

Contact with the skin is not expected to cause prolonged or significant irritation. May cause skin discoloration. Not expected to be harmful to internal organs if absorbed through the skin. High-Pressure Equipment Information: Accidental high-velocity injection under the skin of materials of this type may result in serious injury. Seek medical attention at once should an accident like this occur. The initial wound at the injection site may not appear to be serious at first; but, if left untreated, could result in disfigurement or amputation of the affected part.

INGESTION:

Not expected to be harmful if swallowed.

INHALATION:

Contains a petroleum-based mineral oil. May cause respiratory irritation or other pulmonary effects following prolonged or repeated inhalation of oil mist at airborne levels above the recommended mineral oil mist exposure limit.

4. FIRST AID MEASURES

EYE:

No specific first aid measures are required because this material is not expected to cause eye irritation. As a precaution remove contact lenses, if worn, and flush eyes with water.

SKIN:

No specific first aid measures are required because this material is not expected to be harmful if it contacts the skin. As a precaution, remove clothing and shoes if contaminated. Use a waterless hand cleaner, mineral oil, or petroleum jelly to remove the material. Then wash skin with soap and water. Wash or clean contaminated clothing and shoes before reuse.

INGESTION:

No specific first aid measures are required because this material is not expected to be harmful if

swallowed. Do not induce vomiting. As a precaution, give the person a glass of water or milk to drink and get medical advice. Never give anything by mouth to an unconscious person.

INHALATION:

If exposed to excessive levels of material in the air, move the exposed person to fresh air. Get medical attention if coughing or respiratory discomfort occurs.

NOTE TO PHYSICIANS:

In an accident involving high-pressure equipment, this product may be injected under the skin. Such an accident may result in a small, sometimes bloodless, puncture wound. However, because of its driving force, material injected into a fingertip can be deposited into the palm of the hand. Within 24 hours, there is usually a great deal of swelling, discoloration, and intense throbbing pain. Immediate treatment at a surgical emergency center is recommended.

5. FIRE FIGHTING MEASURES

FIRE CLASSIFICATION:

Classification (29 CFR 1910.1200): Not classified by OSHA as flammable or combustible.

FLAMMABLE PROPERTIES:

FLASH POINT: >500F (>260C)

AUTOIGNITION: NDA

FLAMMABILITY LIMITS (% by volume in air): Lower: NA Upper: NA

EXTINGUISHING MEDIA:

CO₂, Dry Chemical, Foam, Water Fog

NFPA RATINGS: Health 1; Flammability 1; Reactivity 0.

FIRE FIGHTING INSTRUCTIONS:

This material will burn although it is not easily ignited. For fires involving this material, do not enter any enclosed or confined fire space without proper protective equipment, including self-contained breathing apparatus.

COMBUSTION PRODUCTS:

Normal combustion forms carbon dioxide and water vapor and may produce oxides of Mg, N, and Zn. Incomplete combustion can produce carbon monoxide.

6. ACCIDENTAL RELEASE MEASURES

CHEMTREC EMERGENCY NUMBER (24 hr): (800)424-9300 or (703)527-3887

International Collect Calls Accepted

ACCIDENTAL RELEASE MEASURES:

Clean up spills immediately, observing precautions in Exposure Controls/Personal Protection section.

7. HANDLING AND STORAGE

Container is not designed to contain pressure. Do not use pressure to empty container or it may rupture with explosive force. Empty containers retain product residue (solid, liquid, and/or vapor) and can be dangerous. Do not pressurize, cut, weld, braze, solder, drill, grind, or expose such containers to heat, flame, sparks, static electricity, or other sources of ignition. They may explode and cause injury or death. Empty containers should be completely drained, properly closed, and promptly returned to a drum reconditioner, or properly disposed of. Avoid contaminating soil or releasing this material into sewage and drainage systems and bodies of water.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

GENERAL CONSIDERATIONS:

Consider the potential hazards of this material (see Section 3), applicable exposure limits, job activities, and other substances in the work place when designing engineering controls and selecting personal protective equipment. If engineering controls or work practices are not adequate to prevent exposure to harmful levels of this material, the personal protective equipment listed below is recommended. The user should read and understand all instructions and limitations supplied with the equipment since protection is usually provided for a limited time or under certain circumstances.

ENGINEERING CONTROLS

Use in a well-ventilated area. If user operations generate an oil mist, use process enclosures, local exhaust ventilation, or other engineering controls to control airborne levels below the recommended mineral oil mist exposure limits.

PERSONAL PROTECTIVE EQUIPMENT

EYE/FACE PROTECTION:

No special eye protection is normally required.

SKIN PROTECTION:

No special protective clothing is normally required. Where splashing is possible, select protective clothing depending on operations conducted, physical requirements and other substances. Suggested materials for protective gloves include: <Nitrile> <Silver Shield> <Viton>

RESPIRATORY PROTECTION:

No respiratory protection is normally required. If user operations generate an oil mist, determine if airborne concentrations are below the recommended mineral oil mist exposure limits. If not wear a NIOSH approved respirator that provides adequate protection from measured concentrations of this material. Use the following elements for air-purifying respirators: particulate.

9. PHYSICAL AND CHEMICAL PROPERTIES

PHYSICAL DESCRIPTION:

Green grease.

pH: NA

VAPOR PRESSURE: <1 mm Hg @ 40C

VAPOR DENSITY
(AIR=1): Heavier than air.
BOILING POINT: >700F (>371C)
MELTING POINT: 215C (Min) Dropping Point
SOLUBILITY: Soluble in hydrocarbon solvents; insoluble in water.
SPECIFIC GRAVITY: 0.92 @ 20/20C
VOLATILE ORGANIC
COMPOUNDS (VOC): < 1.0 (wt. %), 9.2 g/l approx.
VISCOSITY: 11.8 cSt @ 100C Extracted Oil

10. STABILITY AND REACTIVITY

HAZARDOUS DECOMPOSITION PRODUCTS:

None known.

CHEMICAL STABILITY:

Stable.

CONDITIONS TO AVOID:

No data available.

INCOMPATIBILITY WITH OTHER MATERIALS:

May react with strong oxidizing agents, such as chlorates, nitrates, peroxides, etc.

HAZARDOUS POLYMERIZATION:

Polymerization will not occur.

11. TOXICOLOGICAL INFORMATION

EYE EFFECTS:

The mean 24-hour Draize eye irritation score in rabbits is 1.67/110.

SKIN EFFECTS:

For a 4-hour exposure, the Primary Irritation Index (PII) in rabbits is:

1.3. The acute dermal LD50 in rats is >5.0 g/kg. This material did not cause sensitization reactions in a Modified Buehler guinea pig test.

ACUTE ORAL EFFECTS:

The acute oral LD50 in rats is >5.0 g/kg.

ACUTE INHALATION EFFECTS:

The acute respiratory toxicity is based on data for a similar material.

ADDITIONAL TOXICOLOGY INFORMATION:

This product contains petroleum base oils which may be refined by various processes including severe solvent extraction, severe hydrocracking, or severe hydrotreating. None of the oils requires a cancer warning under the OSHA Hazard Communication Standard (29 CFR 1910.1200). These oils have not been listed in the National Toxicology Program (NTP) Annual Report nor have they been classified by the International Agency for Research on Cancer (IARC) as; carcinogenic to humans (Group 1), probably carcinogenic to humans (Group 2A), or possibly carcinogenic to humans (Group 2B).

12. ECOLOGICAL INFORMATION

ECOTOXICITY:

The toxicity of this material to aquatic organisms has not been evaluated. Consequently, this material should be kept out of sewage and drainage systems and all bodies of water.

ENVIRONMENTAL FATE:

This material is not expected to be readily biodegradable.

13. DISPOSAL CONSIDERATIONS

Oil collection services are available for used oil recycling or disposal. Place contaminated materials in containers and dispose of in a manner consistent with applicable regulations. Contact your sales representative or local environmental or health authorities for approved disposal or recycling methods.

14. TRANSPORT INFORMATION

The description shown may not apply to all shipping situations. Consult 49CFR, or appropriate Dangerous Goods Regulations, for additional description requirements (e.g., technical name) and mode-specific or quantity-specific shipping requirements.

DOT SHIPPING NAME: NONE

DOT HAZARD CLASS: NONE

DOT IDENTIFICATION NUMBER: NONE

DOT PACKING GROUP: N/A

ADDITIONAL INFO: Petroleum Lubricating Grease - Not hazardous by US DOT.
ADR/RID Hazard Class - Not applicable

15. REGULATORY INFORMATION

SARA 311 CATEGORIES:	1. Immediate (Acute) Health Effects:	NO
	2. Delayed (Chronic) Health Effects:	NO
	3. Fire Hazard:	NO
	4. Sudden Release of Pressure Hazard:	NO
	5. Reactivity Hazard:	NO

REGULATORY LISTS SEARCHED:

01=SARA 313	11=NJ RTK	22=TSCA Sect 5(a)(2)
02=MASS RTK	12=CERCLA 302.4	23=TSCA Sect 6
03=NTP Carcinogen	13=MN RTK	24=TSCA Sect 12(b)
04=CA Prop 65-Carcin	14=ACGIH TWA	25=TSCA Sect 8(a)

05=CA Prop 65-Repro Tox	15=ACGIH STEL	26=TSCA Sect 8(d)
06=IARC Group 1	16=ACGIH Calc TLV	27=TSCA Sect 4(a)
07=IARC Group 2A	17=OSHA PEL	28=Canadian WHMIS
08=IARC Group 2B	18=DOT Marine Pollutant	29=OSHA CEILING
09=SARA 302/304	19=Chevron TWA	30=Chevron STEL
10=PA RTK	20=EPA Carcinogen	31=OSHA STEL

The following components of this material are found on the regulatory lists indicated.

DISTILLATES, HYDROTREATED HEAVY PARAFFINIC
is found on lists: 14,15,17,

NEW JERSEY RTK CLASSIFICATION:

Under the New Jersey Right-to-Know Act L. 1983 Chapter 315 N.J.S.A. 34:5A-1 et. seq., the product is to be identified as follows:

Petroleum Oil (Grease)

WHMIS CLASSIFICATION:

This product is not considered a controlled product according to the criteria of the Canadian Controlled Products Regulations.

16. OTHER INFORMATION

NFPA RATINGS: Health 1; Flammability 1; Reactivity 0;

HMIS RATINGS: Health 1; Flammability 1; Reactivity 0;

(0-Least, 1-Slight, 2-Moderate, 3-High, 4-Extreme, PPE:- Personal Protection Equipment Index recommendation, *- Chronic Effect Indicator). These values are obtained using the guidelines or published evaluations prepared by the National Fire Protection Association (NFPA) or the National Paint and Coating Association (for HMIS ratings).

REVISION STATEMENT:

This revision updates Sections 1, 5, 7, 8, 9, 10, 12, and 15.

ABBREVIATIONS THAT MAY HAVE BEEN USED IN THIS DOCUMENT:

TLV - Threshold Limit Value	TWA - Time Weighted Average
STEL - Short-term Exposure Limit	TPQ - Threshold Planning Quantity
RQ - Reportable Quantity	PEL - Permissible Exposure Limit
C - Ceiling Limit	CAS - Chemical Abstract Service Number
A1-5 - Appendix A Categories	() - Change Has Been Proposed
NDA - No Data Available	NA - Not Applicable

Prepared according to the OSHA Hazard Communication Standard (29 CFR 1910.1200) and the ANSI MSDS Standard (Z400.1) by the Toxicology and Health Risk Assessment Unit, CRTC, P.O. Box 1627, Richmond, CA 94804

The above information is based on the data of which we are aware and is believed to be correct as of the date hereof. Since this information may be applied under conditions beyond our control and with which we may be unfamiliar and since data made available subsequent to the date hereof may suggest modification of the information, we do not assume any responsibility for the results of its use. This information is furnished upon condition that the person receiving it shall make his own determination of the suitability of the material for his particular purpose

THIS IS THE LAST PAGE OF THIS MSDS

Information About

Dow Corning® 702 **Diffusion Pump** **Fluid**

Dow Corning® 704 **Diffusion Pump** **Fluid**

Dow Corning® 705 **Diffusion Pump** **Fluid**

Type

Silicone (702 Fluid); single-component silicone material (704, 705 Fluid)

Physical Form

Fluid

Special Properties

Thermal stability; resists oxidation; chemically inert

Primary Uses

Fast pumping of large volumes of gas or vapor in production operations (702 Fluid); producing vacuums of 10-5 to 10-8 torr, untrapped, and 10-10 to 10-11 torr, trapped (704 Fluid); producing ultrahigh and ultraclean vacuum (705 Fluid)

DESCRIPTION

Dow Corning® 702, 704 and 705 Diffusion Pump Fluids are designed for high vacuum and for fast pumping of large volumes of gas or vapor in production operations.

Dow Corning 702 Diffusion Pump Fluid

is a general-purpose fluid designed for fast pumping of large volumes of gas. It is used to produce vacuums in the range of 10-5 to 10-7 torr. It is also used in vapor ejector pumps that attain vacuums of 10-4 to 10-5 torr.

Dow Corning 704 Diffusion Pump Fluid

is a single-component fluid for high vacuums of 10-6 to 10-8 torr (untrapped) and 10-10 to 10-11 torr (trapped). It performs well in tough, rugged applications and offers quick pumpdown, even after exposure to air at operating temperatures.

Dow Corning 705 Diffusion Pump Fluid

is a colorless to straw-colored, single-component fluid designed for ultrahigh vacuum applications in the range of 10-9 to 10-10 torr (untrapped) and 10-11 torr (trapped). The vapor pressure and backstreaming rate of *Dow Corning 705 Diffusion Pump Fluid* are so low that use of traps or refrigeration is unnecessary for some ultrahigh and ultraclean vacuum applications. It has the highest phenyl content of all silicone diffusion pump fluids and the best resistance to radiation. High spontaneous ignition temperature makes this fluid ideal for use in human-rated space simulation chambers.

BENEFITS

Dow Corning Diffusion Pump Fluids offer the following benefits:

- Shorter conditioning runs: Single-component silicone fluids reach maximum potential in far less time than that required for multi-component organic fluids.
- Faster pumping: Diffusion pumps using silicone fluids can operate against 20 to 300 percent higher forepressures than those using organic fluids. For higher gas through-put at the high-pressure end of the pump's operating range,

heater input can be increased 20 to 30 percent.

- Minimal backstreaming: The vapor pressure of single-component silicone fluids from Dow Corning is so low that the use of traps – or refrigeration of existing traps – is unnecessary for many applications.
- Longer service life: Thermal and chemical stability of silicone fluids allows exceptionally long runs without deterioration or contamination.
- Cleaner systems, less maintenance required: Low vapor pressure of silicone fluids at baffle temperatures results in low migration rates. Jets and boiler surfaces stay clean; silicone fluids exhibit virtually no breakdown or decomposition under operational conditions.
- Faster cycling, reduced downtime, less frequent fluid replacement: The recovery rate of silicone fluids after exposure to air at operating temperatures is many times faster than that of organic fluids. Time is saved between cycles because the outstanding resistance of silicone fluids from Dow Corning to oxidation and hydrolysis allows release of the vacuum without cooling the pump.

USES

Dow Corning Diffusion Pump Fluids can be used in a variety of applications including:

Aerospace

- Space simulation chambers

Electronics

- Evacuating television and cathode-ray tubes
- Evacuating power and microwave tubes
- Vapor deposition of thin films by sputtering or evaporation in devices and microcircuits

Metallurgy

- Electron-beam operations; high-vacuum furnaces
- Melting, degassing and sintering refractory metals; thick-film deposition

TYPICAL PROPERTIES

These values are not intended for use in preparing specifications.

	<i>Dow Corning 702 Diffusion Pump Fluid</i>	<i>Dow Corning 704 Diffusion Pump Fluid</i>	<i>Dow Corning 705 Diffusion Pump Fluid</i>
Ultimate Vacuum, torr,			
untrapped	10-6	10-7 to 10-8	10-9 to 10-10
trapped	—	to 10-11	10-11
Extrapolated Vapor Pressure, torr,			
25°C (77°F)	1 x 10-6	2 x 10-8	3 x 10-10
Specific Gravity at 25°C (77°F)	1.07	1.07	1.09
Viscosity at 25°C (77°F), cSt	45	39	175
Flash Point, open cup, °C (°F)	193 (380)	221 (430)	243 (469)
Boiling Point, at 0.5 torr, °C (°F)	180 (356)	215 (419)	245 (473)
Typical Boiler Temperature, °C (°F)	190 (374)	220 (428)	250 to 270 (482 to 518)
Surface Tension, dynes/cm	30	37.3	36.5
Heat of Vaporization, kcal/g mole	21.7/190°C (374°F)	25.5/200°C (392°F)	28.2/250°C (482°F)
Molecular Description	Mixed phenylmethyl- dimethyl cyclosiloxane	Tetramethyltetra- phenyltrisiloxane	Penta phenyltri- methyltrisiloxane
Molecular Weight	—	484	546
Vapor Pressure Equation,	A = 10.3	A = 11.025	A = 12.31
log10 P = A - B/T	B = 4820	B = 5570	B = 6490

(P = vapor pressure, torr; T = absolute temperature, °K)

Specification Writers: Please obtain copies of the *Dow Corning Sales Specification* for these products and use them as a basis for your specifications. They may be obtained from any *Dow Corning Sales Office*, or from *Dow Corning Customer Service* in Midland, MI. Call (517) 496-6000.

Vacuum Coatings

- Aluminizing television tubes
- Optical coatings
- Decorative coatings for novelties and automobile parts
- Coating architectural glass, plastic and metal sheet

Atomic Energy

- Processing metals and fuels
- Accelerators

Research

- Ultrahigh vacuum and ultraclean vacuum for thin-film, surface-emissivity, lubrication and metal-fatigue studies
- Thermonuclear, plasma physics

LIMITATIONS

Because *Dow Corning 704* Diffusion Pump Fluid has been known to crystallize under rare and undefined conditions below 21°C (70°F), *Dow Corning* considers *Dow Corning 702* Diffusion

Pump Fluid the preferred choice for sealed transducers that will encounter low temperature.

This product is neither tested nor represented as suitable for medical or pharmaceutical uses.

PERFORMANCE DATA

Radiation Resistance

Dow Corning Diffusion Pump Fluids exhibit good resistance to gamma radiation because of their high phenyl content. *Dow Corning 704* Diffusion Pump Fluid, for example, requires approximately 1800 mega-rads of irradiation to double the original viscosity. The phenyl content of *Dow Corning 702* Diffusion Pump Fluid is slightly lower. And *Dow Corning 705* Diffusion Pump Fluid – with the highest phenyl content of all silicone diffusion pump fluids – has the best resistance to radiation. (See Table I.)

Seal Compatibility

Immersion Tests at 70°C (158°F) have shown that butyl and Viton®1 rubbers, as well as others, are suitable for sealing pumps using *Dow Corning 705* Diffusion Pump Fluid.

Oxidation Resistance

Silicone fluids are not oxidized by air at operating temperatures and are not subject to hydrolysis by water vapor. Their chemical resistance gives them long life and eliminates frequent replacement. Also, the silicone fluids, being inert, do not react with metal parts, elastomer seals and gasses such as hydrogen and carbon monoxide.

Rapid Cycling

The results of cycling tests show that the capability of silicone fluids was practically unchanged after 1000 cycles. In comparison, the vacuum obtainable with an organic ester deteriorated by a factor of about 100 after 400 cycles. In the test, a single-stage glass diffusion pump and a Knudsen vacuum gauge were used.

1Viton is a registered trademark of E.I. du Pont de Nemours and Company.

During each cycle, the pump was operated for 12 minutes and then exposed to atmosphere for three minutes. The heater was off during exposure, but the fluid remained close to the operating temperature. Test results were:

Dow Corning 702 Fluid

<i>Number of Cycles</i>	<i>Pressure, torr</i>
0	3.3 x 10-6
492	3.8 x 10-6
1016	3.8 x 10-6

Dow Corning 704 Fluid

<i>Number of Cycles</i>	<i>Pressure, torr</i>
0	2.3 x 10-6
994	3.0 x 10-6
2072	3.3 x 10-6

Dow Corning 705 Fluid

<i>Number of Cycles</i>	<i>Pressure, torr</i>
0	2.4 x 10-6
101	2.4 x 10-6
1082	3.0 x 10-6

Tests under the same conditions with the organic ester fluid, di (2-ethylexyl) phthalate, gave the following results:

<i>Number of Cycles</i>	<i>Pressure, torr</i>
0	2.0 x 10-6
107	3.5 x 10-5
220	6.0 x 10-5
412	1.6 x 10-4

Explosion Resistance

Silicone diffusion pump fluids, in the absence of less stable contaminants or high-temperature ignition sources, pump air and oxygen with little danger of fire or explosion. This conclusion, drawn from Dow Corning test data, is supported by service experience both in continuous pumping of pure oxygen and in repeated venting of hot pumps to atmospheric pressure.

Dow Corning 704 Diffusion Pump Fluid has excellent resistance to explosion, as shown by tests conducted in a pure oxygen atmosphere. (See Table II.)

Table I: Radiation Resistance After Exposure to Cobalt 60 Source¹

Fluid	Dosage, megarads	Pressure, torr	
		Before Exposure	After Exposure
Organic	100	2 x 10-6	1.1 x 10-5
<i>Dow Corning 704</i> Diffusion Pump Fluid	280	2 x 10-6	3 x 10-6

¹Samples tested in single-stage glass diffusion pump. Pressure measured with Knudsen gauge.

Table II: Explosion Resistance in Pure Oxygen Atmosphere¹

Fluid	Cell Pressure Range, ton	Cell Temp. Range, °C (°F)	Explosions at:		
			Cell Pressure, torr	Cell Temp., °C (°F)	Reservoir Temp., °C (°F)
<i>Dow Corning 704</i> Diffusion Pump Fluid	40 to 734	304 to 483 (580 to 901)	300	478 (893)	269 (516)
			455	430 (806)	267 (513)
			633	429 (804)	267 (513)

¹*Journal of Vacuum Science and Technology*, Vol. II, Number 5, Sept.-Oct. 1965.

Table III: Ignition Temperatures of Dow Corning 705 Diffusion Pump Fluid vs. Hydrocarbon Diffusion Pump Fluid

Fluid	Partial Fluid Pressure, torr	Volume Fluid Vapor, percent	Spontaneous Ignition Temperature, °C (°F)
<i>Dow Corning 705</i> Diffusion Pump Fluid	23	3	475 (890)
	37	4.9	464 (870)
	72	9.5	440 (824)
Hydrocarbon diffusion pump fluid	8.1	1.1	300 (572)
	21	2.7	325 (617)
	30	4.0	340 (644)

Oxygen was mixed with vapors of the pump fluids at known pressures and temperatures to determine the temperature at which various mixtures would explode. The total pressure of the mixture was one atmosphere. Ignition temperatures are shown in Table III.

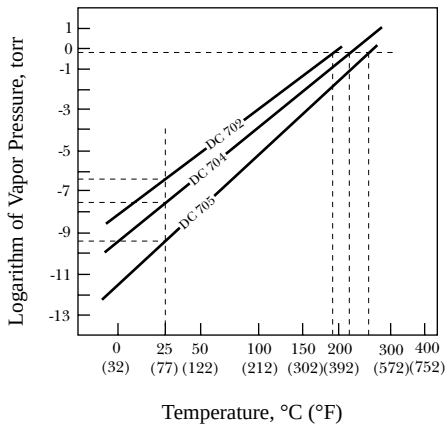
Spontaneous ignition temperatures of *Dow Corning 705* Diffusion Pump Fluid were found in this test to be approximately 200°C (392°F) above the normal

operating temperature in a diffusion pump.

Vapor Pressure

As Figure 1 shows, *Dow Corning 705* Diffusion Pump Fluid has extremely low vapor pressure. The broken line at 25°C (77°F) marks the vapor pressures with a baffle at that temperature. Broken lines at right indicate operating temperatures in modern commercial pumps producing 0.8 torr boiler pressure.

Figure 1: Vapor Pressure



Backstreaming

Contamination rates at surfaces in a vacuum chamber are controlled by the vapor pressure of the condensed fluid film on the baffle. Because of its low vapor pressure, contamination rates with *Dow Corning* 705 Diffusion Pump Fluid in water-baffled systems can be up to 100 times lower than with *Dow Corning* 704 Diffusion Pump Fluid. The difference is even greater in comparison with most organic fluids.

Contamination can be stopped by trapping; however, the rate for *Dow Corning* 705 Diffusion Pump Fluid is so low that ultrahigh vacuum can be obtained with a water-cooled baffle only. *Dow Corning* 705 Diffusion Pump Fluid

also makes ultraclean operation possible without trapping for moderate periods of time.

SHIPPING LIMITATIONS

None.

STORAGE AND SHELF LIFE

When stored in the original, sealed container, at or below 25°C (77°F), *Dow Corning* Diffusion Pump Fluids have a shelf life of 60 months from date of manufacture, although no inherent limitations on the useful life of this product are known to exist.

PACKAGING

Dow Corning Diffusion Pump Fluids are available in 16.9-fl oz (500-mL) bottles, 8.5-lb (3.9-kg), 42.5-lb (19.3-kg) pails and 425-lb (192.8-kg) drums. All weights net.

SAFE HANDLING INFORMATION
PRODUCT SAFETY INFORMATION REQUIRED FOR SAFE USE IS NOT INCLUDED. BEFORE HANDLING, READ PRODUCT AND MATERIAL SAFETY DATA SHEETS AND CONTAINER LABELS FOR SAFE USE, PHYSICAL AND HEALTH HAZARD INFORMATION. THE MATERIAL SAFETY DATA SHEET IS AVAILABLE FROM YOUR *DOW CORNING* REPRESENTATIVE, OR DISTRIBUTOR, OR BY WRITING TO *DOW CORNING* CUSTOMER SERVICE, OR BY CALLING (517) 496-6000.

LIMITED WARRANTY – PLEASE READ CAREFULLY

Dow Corning believes that the information in this publication is an accurate description of the typical characteristics and/or uses of the product or products, but it is your responsibility to thoroughly test the product in your specific application to determine its performance, efficacy and safety. Suggestions of uses should not be taken as inducements to infringe any particular patent.

Unless *Dow Corning* provides you with a specific written warranty of fitness for a particular use, *Dow Corning's* sole warranty is that the product or products, as supplied, will meet *Dow Corning's* then current sales specifications. **DOW CORNING SPECIFICALLY DISCLAIMS ANY OTHER EXPRESS OR IMPLIED WARRANTY, INCLUDING THE WARRANTIES OF MERCHANTABILITY AND OF FITNESS FOR USE.** Your exclusive remedy and *Dow Corning's* sole liability for breach of warranty is limited to refund of the purchase price or replacement of any product shown to be other than as warranted, and *Dow Corning* expressly disclaims any liability for incidental or consequential damages.

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AGP3952

Form No. 10-838-98

**DOW CORNING CORPORATION
Material Safety Data Sheet**

Page: 1 of 7

DOW CORNING(R) HIGH VACUUM GREASE**1. IDENTIFICATION OF THE SUBSTANCE AND OF THE COMPANY**

Dow Corning Corporation
South Saginaw Road
Midland, Michigan 48686

24 Hour Emergency Telephone: (989) 496-5900
Customer Service: (989) 496-6000
Product Disposal Information: (989) 496-6315
CHEMTREC: (800) 424-9300

MSDS No.: 01018817

Revision Date: 2002/01/18

Generic Description: Silicone compound
Physical Form: Viscous Liquid
Color: Translucent white
Odor: Odorless

NFPA Profile: Health 0 Flammability 1 Instability/Reactivity 0

Note: NFPA = National Fire Protection Association

2. OSHA HAZARDOUS COMPONENTS

None present. This is not a hazardous material as defined in the OSHA Hazard Communication Standard.

3. EFFECTS OF OVEREXPOSUREAcute Effects

Eye: Direct contact may cause temporary redness and discomfort.
Skin: No significant irritation expected from a single short-term exposure.
Inhalation: No significant effects expected from a single short-term exposure.
Oral: Low ingestion hazard in normal use.

Prolonged/Repeated Exposure Effects

Skin: No known applicable information.
Inhalation: No known applicable information.
Oral: No known applicable information.

Signs and Symptoms of Overexposure

No known applicable information.

Medical Conditions Aggravated by Exposure

No known applicable information.

The above listed potential effects of overexposure are based on actual data, results of studies performed upon similar compositions, component data and/or expert review of the product. Please refer to Section 11 for the detailed toxicology information.

DOW CORNING(R) HIGH VACUUM GREASE**4. FIRST AID MEASURES**

Eye: Immediately flush with water.

Skin: No first aid should be needed.

Inhalation: No first aid should be needed.

Oral: No first aid should be needed.

Comments: Treat symptomatically.

5. FIRE FIGHTING MEASURES

Flash Point: 212 °F / 100 °C (Closed Cup)

Autoignition Temperature: Not determined.

Flammability Limits in Air: Not determined.

Extinguishing Media: Carbon dioxide (CO2). Water spray. Dry chemical. Foam. Water can be used to cool fire exposed containers.

Fire Fighting Measures: Self-contained breathing apparatus and protective clothing should be worn in fighting large fires involving chemicals. Determine the need to evacuate or isolate the area according to your local emergency plan.

Unusual Fire Hazards: None.

Hazardous Decomposition Products

Thermal breakdown of this product during fire or very high heat conditions may evolve the following hazardous decomposition products: Carbon oxides and traces of incompletely burned carbon compounds. Silicon dioxide. Formaldehyde.

6. ACCIDENTAL RELEASE MEASURES

Containment/Clean up: Sections 13 and 15 of this MSDS provide information regarding certain federal and state requirements. Clean up remaining materials from spill with suitable absorbant. For large spills, provide diking or other appropriate containment to keep material from spreading. If diked material can be pumped, store recovered material in appropriate container. Clean area as appropriate since some silicone materials, even in small quantities, may present a slip hazard. Final cleaning may require use of steam, solvents or detergents. Observe all personal protection equipment recommendations described in Sections 5 and 8. Local, state and federal laws and regulations may apply to releases and disposal of this material, as well as those materials and items employed in the cleanup of releases. You will need to determine which federal, state and local laws and regulations are applicable.

Note: See section 8 for Personal Protective Equipment for Spills. Call Dow Corning Corporation, (989) 496-5900, if additional information is required.

DOW CORNING(R) HIGH VACUUM GREASE**7. HANDLING AND STORAGE**

Use with adequate ventilation. None. Avoid eye contact.

Use reasonable care and store away from oxidizing materials.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION**Component Exposure Limits**

There are no components with workplace exposure limits.

Engineering Controls

Local Ventilation: None should be needed.

General Ventilation: Recommended.

Personal Protective Equipment for Routine Handling

Eyes: Use proper protection - safety glasses as a minimum.

Skin: Washing at mealtime and end of shift is adequate.

Suitable Gloves: No special protection needed.

Inhalation: No respiratory protection should be needed.

Suitable Respirator: None should be needed.

Personal Protective Equipment for Spills

Eyes: Use proper protection - safety glasses as a minimum.

Skin: Washing at mealtime and end of shift is adequate.

Inhalation/Suitable
Respirator: No respiratory protection should be needed.

Precautionary Measures: Avoid eye contact. Use reasonable care.

Comments: None.

Note: These precautions are for room temperature handling. Use at elevated temperature or aerosol/spray applications may require added precautions.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical Form: Viscous Liquid

Color: Translucent white

Odor: Odorless

Specific Gravity @ 25°C: 1.1

DOW CORNING(R) HIGH VACUUM GREASE

Viscosity: 2000000 cSt
Freezing/Melting Point: Not determined.
Boiling Point: > 35C/95F
Vapor Pressure @ 25°C: Not determined.
Vapor Density: Not determined.
Solubility in Water: Not determined.
pH: Not determined.
Volatile Content: Not determined.

Note: The above information is not intended for use in preparing product specifications. Contact Dow Corning before writing specifications.

10. STABILITY AND REACTIVITY

Chemical Stability: Stable.

Hazardous Polymerization: Hazardous polymerization will not occur.

Conditions to Avoid: None.

Materials to Avoid: Oxidizing material can cause a reaction.

11. TOXICOLOGICAL INFORMATION**Component Toxicology Information**

No known applicable information.

Special Hazard Information on Components

No known applicable information.

12. ECOLOGICAL INFORMATION**Environmental Fate and Distribution**

Complete information is not yet available.

Environmental Effects

Complete information is not yet available.

Fate and Effects in Waste Water Treatment Plants

Complete information is not yet available.

Ecotoxicity Classification Criteria

Hazard Parameters (LC50 or EC50)	High	Medium	Low
Acute Aquatic Toxicity (mg/L)	<=1	>1 and <=100	>100
Acute Terrestrial Toxicity	<=100	>100 and <= 2000	>2000

This table is adapted from "Environmental Toxicology and Risk Assessment", ASTM STP 1179, p.34, 1993.

DOW CORNING(R) HIGH VACUUM GREASE

This table can be used to classify the ecotoxicity of this product when ecotoxicity data is listed above. Please read the other information presented in the section concerning the overall ecological safety of this material.

13. DISPOSAL CONSIDERATIONS**RCRA Hazard Class (40 CFR 261)**

When a decision is made to discard this material, as received, is it classified as a hazardous waste? No

State or local laws may impose additional regulatory requirements regarding disposal.

Call Dow Corning Corporate Environmental Management, (989) 496-6315, if additional information is required.

14. TRANSPORT INFORMATION**DOT Road Shipment Information (49 CFR 172.101)****Ocean Shipment (IMDG)**

Not subject to IMDG code.

Air Shipment (IATA)

Not subject to IATA regulations.

Call Dow Corning Transportation, (989) 496-8577, if additional information is required.

15. REGULATORY INFORMATION

Contents of this MSDS comply with the OSHA Hazard Communication Standard 29 CFR 1910.1200.

TSCA Status: All chemical substances in this material are included on or exempted from listing on the TSCA Inventory of Chemical Substances.

EPA SARA Title III Chemical Listings**Section 302 Extremely Hazardous Substances:**

None.

Section 304 CERCLA Hazardous Substances:

None.

Section 312 Hazard Class:

Acute: No
Chronic: No
Fire: No
Pressure: No
Reactive: No

DOW CORNING(R) HIGH VACUUM GREASE**Section 313 Toxic Chemicals:**

None present or none present in regulated quantities.

Supplemental State Compliance Information**California**

Warning: This product contains the following chemical(s) listed by the State of California under the Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65) as being known to cause cancer, birth defects or other reproductive harm.

None known.

Massachusetts

<u>CAS Number</u>	<u>Wt %</u>	<u>Component Name</u>
7631-86-9	7.0 - 13.0	Silica, amorphous

New Jersey

<u>CAS Number</u>	<u>Wt %</u>	<u>Component Name</u>
63148-62-9	> 60.0	Polydimethylsiloxane
7631-86-9	7.0 - 13.0	Silica, amorphous
70131-67-8	7.0 - 13.0	Dimethyl siloxane, hydroxy-terminated

Pennsylvania

<u>CAS Number</u>	<u>Wt %</u>	<u>Component Name</u>
63148-62-9	> 60.0	Polydimethylsiloxane
7631-86-9	7.0 - 13.0	Silica, amorphous
70131-67-8	7.0 - 13.0	Dimethyl siloxane, hydroxy-terminated

16. OTHER INFORMATION

DOW CORNING(R) HIGH VACUUM GREASE

Prepared by: Dow Corning Corporation

These data are offered in good faith as typical values and not as product specifications. No warranty, either expressed or implied, is hereby made. The recommended industrial hygiene and safe handling procedures are believed to be generally applicable. However, each user should review these recommendations in the specific context of the intended use and determine whether they are appropriate.

(R) indicates Registered Trademark

Product Information

Specialty Lubricants

DOW CORNING

Dow Corning® 55 O-Ring Lubricant

FEATURES

- Slightly swells rubber for better sealing
- Oxidation resistant
- Serviceable over a wide temperature range

COMPOSITION

- Soft silicone-based grease

Silicone-based grease provides dynamic lubrication and sealing between rubber and metal

USES

Dow Corning 55 O-Ring Lubricant is primarily used for dynamic lubrication and sealing between rubber and metal parts in pneumatic valves. Typical applications are found within the aircraft, automotive and other industries.

Dow Corning 55 O-Ring Lubricant has also been used as a temporary remedy for gear box seal leaks.

TYPICAL PROPERTIES

These values are not intended for use in preparing specifications.

Method	Test	Unit	Result
ASTM D 217	Color		Light beige
	Specific Gravity		0.98
	Penetration, worked 60 strokes		290
	Consistency, NLGI		No. 2
ASTM D 566	Dropping Point	°F (°C)	428 (220)
Fed Std 791 ¹	Bleed, 24 hours at 302°F (150°C)	percent	3.0
	Evaporation, 24 hours at 302°F (150°C)	percent	1.4
DIN 51802	Corrosion Testing, Emcore		0, no corrosion
ASTM D 1264	Water Washout Resistance	percent	4.2

¹Federal Standard 791, Method 321.1.

Specification Writers: Please obtain a copy of the Dow Corning Sales Specification for this product and use it as a basis for your specifications. It may be obtained from any Dow Corning Sales Office, or from Dow Corning Customer Service in Midland, MI. Call (517) 496-6000.

DESCRIPTION

Dow Corning 55 O-Ring Lubricant is a silicone-based material that helps ensure positive lubrication and sealing by slightly swelling rubber O-rings and seals. This product is heat stable and oxidation resistant, and is serviceable from approximately -85 to 351°F (-65 to 177°C).

COMPATIBILITY

Dow Corning 55 O-Ring Lubricant has been shown to swell natural rubber. However, compatibility of the lubricant may vary with the plasticizer content of specific

materials (especially elastomers). Small-scale compatibility testing should be conducted prior to the use of this product in any application.

Dow Corning 55 O-Ring Lubricant should not be used with the following materials unless thoroughly tested for your specific application:

- Silicone rubber (SR)
- Polycarbonate (PC)
- Acrylonitrile-butadiene-styrene (ABS)
- Liquid oxygen (or other strong oxidizers)

USE LIMITATIONS

This material is neither tested nor represented as suitable for medical or pharmaceutical uses.

SHIPPING LIMITATIONS

None.

STORAGE AND SHELF LIFE

When stored at or below 90°F (32°C), *Dow Corning* 55 O-Ring Lubricant has a shelf life of 60 months from date of manufacture. Refer to product packaging for "Use By" date.

PACKAGING

Dow Corning 55 O-Ring Lubricant is supplied in 5.3-oz (150-g) tubes, 8-lb (3.6-kg) cans, 40-lb (18.1-kg) pails and 400-lb (181.4-kg) drums. All weights, net.

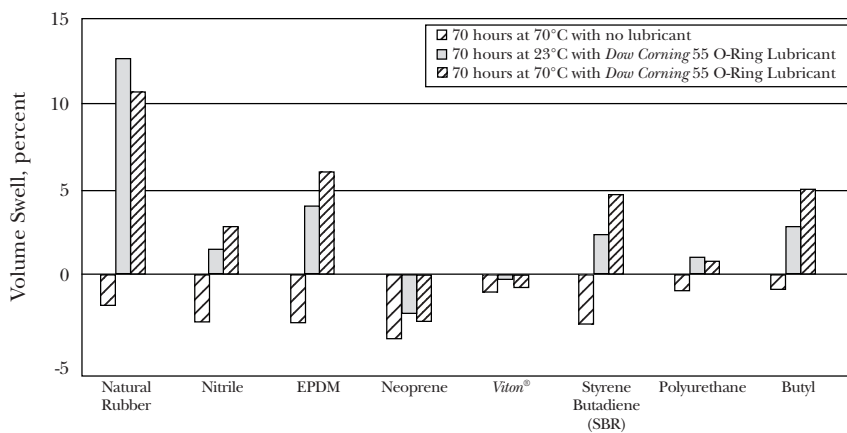
SAFE HANDLING INFORMATION

PRODUCT SAFETY INFORMATION REQUIRED FOR SAFE USE IS NOT INCLUDED. BEFORE HANDLING, READ PRODUCT AND MATERIAL SAFETY DATA SHEETS AND CONTAINER LABELS FOR SAFE USE, PHYSICAL AND HEALTH HAZARD INFORMATION. THE MATERIAL SAFETY DATA SHEET IS AVAILABLE FROM YOUR DOW CORNING REPRESENTATIVE, OR DISTRIBUTOR, OR BY WRITING TO DOW CORNING CUSTOMER SERVICE, OR BY CALLING (517) 496-6000.

WARRANTY INFORMATION – PLEASE READ CAREFULLY

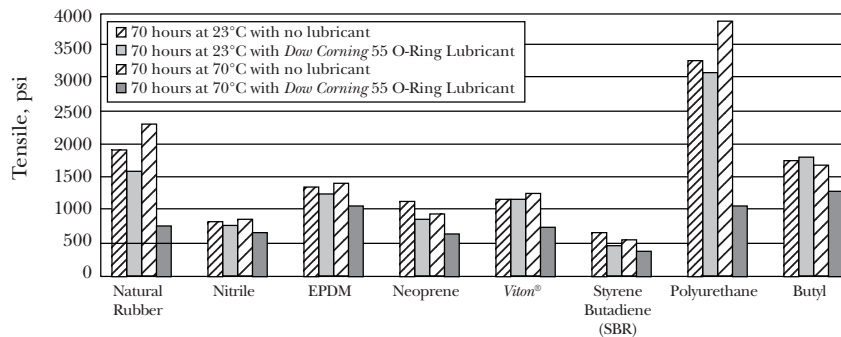
The information contained herein is offered in good faith and is

Figure 1. Percent Volume Swell of Various Elastomers (ASTM D 4289)



®Registered trademark of E.I. du Pont de Nemours Co.

Figure 2. Effect on Tensile Strength of Various Elastomers (ASTM D 412)



believed to be accurate. However, because conditions and methods of use of our products are beyond our control, this information should not be used in substitution for customer's tests to ensure that *Dow Corning*'s products are safe, effective, and fully satisfactory for the intended end use.

Dow Corning's sole warranty is that the product will meet the *Dow Corning* sales specifications in effect at the time of shipment. Your exclusive remedy for breach of such

warranty is limited to refund of purchase price or replacement of any product shown to be other than as warranted. *Dow Corning* specifically disclaims any other express or implied warranty of fitness for a particular purpose or merchantability. Unless *Dow Corning* provides you with a specific, duly signed endorsement of fitness for use, *Dow Corning* disclaims liability for any incidental or consequential damages. Suggestions of use shall not be taken as inducements to infringe any patent.

DOW CORNING CORPORATION
Material Safety Data Sheet**DOW CORNING(R) 55 O-RING LUBRICANT****1. IDENTIFICATION OF THE SUBSTANCE AND OF THE COMPANY**

Dow Corning Corporation
South Saginaw Road
Midland, Michigan 48686

24 Hour Emergency Telephone: (989) 496-5900
Customer Service: (989) 496-6000
Product Disposal Information: (989) 496-6315
CHEMTREC: (800) 424-9300

MSDS No.: 01889826

Revision Date: 2001/10/17

Generic Description: Silicone grease.
Physical Form: Grease
Color: Pink
Odor: Slight odor

NFPA Profile: Health 1 Flammability 1 Instability/Reactivity 0

Note: NFPA = National Fire Protection Association

2. OSHA HAZARDOUS COMPONENTS

<u>CAS Number</u>	<u>Wt %</u>	<u>Component Name</u>
10448-09-6	<0.1	Phenylheptamethyl cyclotetrasiloxane
33204-76-1	<0.1	2,6-cis-Diphenylhexamethyl cyclotetrasiloxane

The above components are hazardous as defined in 29 CFR 1910.1200.

3. EFFECTS OF OVEREXPOSUREAcute Effects

Eye: Direct contact may cause mild irritation.

Skin: No significant irritation expected from a single short-term exposure.

Inhalation: No significant effects expected from a single short-term exposure.

Oral: Low ingestion hazard in normal use.

Prolonged/Repeated Exposure Effects

Skin: Repeated or prolonged exposure may cause irritation.

Inhalation: No known applicable information.

Oral: Repeated ingestion or swallowing large amounts may injure internally.

Signs and Symptoms of Overexposure

No known applicable information.

Medical Conditions Aggravated by Exposure

DOW CORNING(R) 55 O-RING LUBRICANT

No known applicable information.

The above listed potential effects of overexposure are based on actual data, results of studies performed upon similar compositions, component data and/or expert review of the product. Please refer to Section 11 for the detailed toxicology information.

4. FIRST AID MEASURES

Eye: Immediately flush with water for 15 minutes.

Skin: No first aid should be needed.

Inhalation: No first aid should be needed.

Oral: Get medical attention.

Comments: Treat according to person's condition and specifics of exposure.

5. FIRE FIGHTING MEASURES

Flash Point: > 214 °F / > 101.1 °C (Closed Cup)

Autoignition Temperature: Not determined.

Flammability Limits in Air: Not determined.

Extinguishing Media: Water can be used to cool fire exposed containers. Water spray. Carbon dioxide (CO₂). Dry chemical. Foam.

Fire Fighting Measures: Self-contained breathing apparatus and protective clothing should be worn in fighting large fires involving chemicals.

Unusual Fire Hazards: None.

Hazardous Decomposition Products

Thermal breakdown of this product during fire or very high heat conditions may evolve the following hazardous decomposition products: Carbon oxides and traces of incompletely burned carbon compounds. Silicon dioxide. Formaldehyde. Nitrogen oxides. Metal oxides.

6. ACCIDENTAL RELEASE MEASURES

DOW CORNING CORPORATION
Material Safety Data Sheet**DOW CORNING(R) 55 O-RING LUBRICANT**

Containment/Clean up: Sections 13 and 15 of this MSDS provide information regarding certain federal and state requirements. Clean up remaining materials from spill with suitable absorbant. Clean area as appropriate since some silicone materials, even in small quantities, may present a slip hazard. Final cleaning may require use of steam, solvents or detergents. Observe all personal protection equipment recommendations described in Sections 5 and 8. Local, state and federal laws and regulations may apply to releases and disposal of this material, as well as those materials and items employed in the cleanup of releases. You will need to determine which federal, state and local laws and regulations are applicable.

Note: See section 8 for Personal Protective Equipment for Spills. Call Dow Corning Corporation, (989) 496-5900, if additional information is required.

7. HANDLING AND STORAGE

Use with adequate ventilation. Traces of benzene (carcinogen) may form if heated in air above 300 F (149 C). Provide ventilation to control vapor exposure within inhalation guidelines when handling at elevated temperatures. Review the OSHA benzene regulation for detailed information on safe handling requirements. Use at high temperatures may evolve or generate phenylheptamethylcyclotetrasiloxane, 2,6-cis-diphenylhexamethylcyclotetrasiloxane. Provide adequate ventilation to control within exposure guidelines. Personnel should wear organic vapor respirators until workplace exposure levels have been determined. Avoid eye contact. Do not take internally.

Use reasonable care and store away from oxidizing materials.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION**Component Exposure Limits**

<u>CAS Number</u>	<u>Component Name</u>	<u>Exposure Limits</u>
10448-09-6	Phenylheptamethyl cyclotetrasiloxane	Dow Corning guide: TWA-skin: 1.8 ppb.
33204-76-1	2,6-cis-Diphenylhexamethyl cyclotetrasiloxane	Dow Corning guide: TWA-skin: 0.4 ppb as vapor, 0.7 microgms/m3 as aerosol.

Engineering Controls

Local Ventilation: None should be needed.
General Ventilation: Recommended.

Personal Protective Equipment for Routine Handling

Eyes: Use proper protection - safety glasses as a minimum.
Skin: Washing at mealtime and end of shift is adequate.
Suitable Gloves: No special protection needed.
Inhalation: No respiratory protection should be needed.
Suitable Respirator: None should be needed.

DOW CORNING(R) 55 O-RING LUBRICANT**Personal Protective Equipment for Spills**

Eyes: Use proper protection - safety glasses as a minimum.

Skin: Washing at mealtime and end of shift is adequate.

Inhalation/Suitable Respirator: No respiratory protection should be needed.

Precautionary Measures: Avoid eye contact. Do not take internally. Use reasonable care.

Comments: Traces of benzene (carcinogen) may form if heated in air above 300 F (149 C). Provide ventilation to control vapor exposure within inhalation guidelines when handling at elevated temperatures. Review the OSHA benzene regulation for detailed information on safe handling requirements. Use at high temperatures may evolve or generate phenylheptamethylcyclotetrasiloxane, 2,6-cis-diphenylhexamethylcyclotetrasiloxane. Provide adequate ventilation to control within exposure guidelines. Personnel should wear organic vapor respirators until workplace exposure levels have been determined.

Note: These precautions are for room temperature handling. Use at elevated temperature or aerosol/spray applications may require added precautions.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical Form: Grease
Color: Pink
Odor: Slight odor
Specific Gravity @ 25°C: 1.1
Viscosity: Not determined.
Freezing/Melting Point: Not determined.
Boiling Point: Not determined.
Vapor Pressure @ 25°C: Not determined.
Vapor Density: Not determined.
Solubility in Water: Not determined.
pH: Not determined.
Volatile Content: Not determined.

Note: The above information is not intended for use in preparing product specifications. Contact Dow Corning before writing specifications.

10. STABILITY AND REACTIVITY

Chemical Stability: Stable.

Hazardous Polymerization: Hazardous polymerization will not occur.

Conditions to Avoid: None.

Materials to Avoid: Oxidizing material can cause a reaction.

11. TOXICOLOGICAL INFORMATION

DOW CORNING(R) 55 O-RING LUBRICANT**Component Toxicology Information**

This compound may contain low levels of phenylheptamethylcyclotetrasiloxane and 2,6-cis-diphenylhexamethylcyclotetrasiloxane which are estrogen-like chemicals. Estrogens are hormone substances which affect the reproductive system.

Special Hazard Information on Components

No known applicable information.

12. ECOLOGICAL INFORMATION**Environmental Fate and Distribution**

Complete information is not yet available.

Environmental Effects

Complete information is not yet available.

Fate and Effects in Waste Water Treatment Plants

Complete information is not yet available.

Ecotoxicity Classification Criteria

Hazard Parameters (LC50 or EC50)	High	Medium	Low
Acute Aquatic Toxicity (mg/L)	<=1	>1 and <=100	>100
Acute Terrestrial Toxicity	<=100	>100 and <= 2000	>2000

This table is adapted from "Environmental Toxicology and Risk Assessment", ASTM STP 1179, p.34, 1993.

This table can be used to classify the ecotoxicity of this product when ecotoxicity data is listed above. Please read the other information presented in the section concerning the overall ecological safety of this material.

13. DISPOSAL CONSIDERATIONS**RCRA Hazard Class (40 CFR 261)**

When a decision is made to discard this material, as received, is it classified as a hazardous waste? No

State or local laws may impose additional regulatory requirements regarding disposal.

Call Dow Corning Corporate Environmental Management, (989) 496-6315, if additional information is required.

14. TRANSPORT INFORMATION**DOT Road Shipment Information (49 CFR 172.101)**

Not subject to DOT.

DOW CORNING(R) 55 O-RING LUBRICANT**Ocean Shipment (IMDG)**

Not subject to IMDG code.

Air Shipment (IATA)

Not subject to IATA regulations.

Call Dow Corning Transportation, (989) 496-8577, if additional information is required.

15. REGULATORY INFORMATION

Contents of this MSDS comply with the OSHA Hazard Communication Standard 29 CFR 1910.1200.

TSCA Status: All chemical substances in this material are included on or exempted from listing on the TSCA Inventory of Chemical Substances.

EPA SARA Title III Chemical Listings**Section 302 Extremely Hazardous Substances:**

None.

Section 304 CERCLA Hazardous Substances:

None.

Section 312 Hazard Class:Acute: No
Chronic: Yes
Fire: No
Pressure: No
Reactive: No**Section 313 Toxic Chemicals:**

None present or none present in regulated quantities.

Supplemental State Compliance Information**California**

Warning: This product contains the following chemical(s) listed by the State of California under the Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65) as being known to cause cancer, birth defects or other reproductive harm.

None known.

Massachusetts

<u>CAS Number</u>	<u>Wt %</u>	<u>Component Name</u>
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DOW CORNING(R) 55 O-RING LUBRICANT

4485-12-5 15.0 - 40.0 Lithium stearate

New Jersey

<u>CAS Number</u>	<u>Wt %</u>	<u>Component Name</u>
63148-52-7	40.0 - 70.0	Dimethyl, phenylmethyl siloxane, trimethyl-terminated
4485-12-5	15.0 - 40.0	Lithium stearate
122-62-3	15.0 - 40.0	Di(2-ethylhexyl) sebacate

Pennsylvania

<u>CAS Number</u>	<u>Wt %</u>	<u>Component Name</u>
63148-52-7	40.0 - 70.0	Dimethyl, phenylmethyl siloxane, trimethyl-terminated
4485-12-5	15.0 - 40.0	Lithium stearate
122-62-3	15.0 - 40.0	Di(2-ethylhexyl) sebacate

16. OTHER INFORMATION

Prepared by: Dow Corning Corporation

These data are offered in good faith as typical values and not as product specifications. No warranty, either expressed or implied, is hereby made. The recommended industrial hygiene and safe handling procedures are believed to be generally applicable. However, each user should review these recommendations in the specific context of the intended use and determine whether they are appropriate.

(R) indicates Registered Trademark

Mobilgrease[®] 28

Synthetic Lubricating Grease

Description

Mobilgrease 28 is manufactured from a synthesized hydrocarbon fluid and a nonsoap thickener. Its consistency is between an NLGI No. 1 and No. 2 grease. It offers outstanding performance over wide temperature ranges, with the excellent retention and resistance to high-temperature degradation. In addition, it resists water washing, provides superior load-carrying ability, reduces frictional drag, and prevents excessive wear.

Mobilgrease 28 exceeds the requirements of U.S. Military Specification MIL-G-81322E Grease (Grade A), General-Purpose, Aircraft, and meets the requirements of U.S. Military Specification DOD-G-24508A (Navy) for shipboard auxiliary machinery. It can be designated U.S. Military Symbol WTR and NATO Symbol G-395.

Other tests show that Mobilgrease 28 prevents friction oxidation (fretting) and lubricates rolling element bearings under conditions of high speeds and temperatures. It has also shown superior ability to lubricate heavily loaded sliding mechanisms, such as wing flap screwjacks.

Advantages

- Improved friction reduction
- Low wear rates
- Low lubricant drag
- Wide temperature range
- High thermal stability
- Compatibility with mineral-oil-base greases
- Extreme-pressure characteristics
- High resistance to water washing

Application

Mobilgrease 28 is designed for the lubrication of plain and rolling bearings at low to high speeds, and splines, screws, worm gears, and other mechanisms where high friction reduction, low wear, and low lubricant friction losses are required. It provides minimum resistance to starting at extreme low temperatures (down to -54° C/-65° F), as well as low running torque.

Mobilgrease 28 is recommended for use in landing wheel assemblies, control systems and actuators, screwjacks, servo devices, sealed-bearing motors, oscillating bearings, and helicopter rotor bearings on military and civil aircraft and on naval shipboard auxiliary machinery. It also can be used where superseded specifications MIL-G-81322 (WP), MIL-G-7711A, MIL-G-3545B, and MIL-G-25760A may be called for. Mobilgrease 28 also is recommended for industrial lubrication, including sealed or repackable ball and roller bearings wherever extreme temperature conditions, high speeds, or water washing resistance are factors.

Typical applications include conveyor bearings, small alternator bearings operating at temperatures near 177° C (350° F), high-speed miniature ball bearings, and bearing situations where oscillatory motion, vibration, and fretting create problems.



Typical Properties

Physical characteristics are listed in the table.

Values not shown as maximum or minimum are typical and may vary slightly.

	Mobilgrease 28	MIL-G-81322E Requirements
Thickener Type	Non soap	--
Fluid Type	Synthetic Hydrocarbon	--
Color, Visual	Dark Red	--
Structure	Smooth, Buttery	--
Penetration, 25° C (77° F), 60 Strokes	305	265-320
100,000 Strokes	310	350 max
Dropping Point, °C (°F), IP 132	232 (450)	232 (450)
Corrosion Prevention,	Pass	Pass
Load Wear Index, kg	30	30 min
Copper Corrosion, 24 hr at 100° C (212° F)	None	None
Water Washout, 38° C (100° F)	1	20 max
Four-Ball Wear Test, 40 kg at 75° C (167° F), 1200 rpm, 1 hr, Scar Diam, mm 2266, max	1.30	1.30
Evaporation Loss, 22 hr at 177° C (350° F), wt %	10	12 max
Oil Separation, 30 hr at 177° C (350° F), vol %	5	10 max
Low Temperature Torque, -54° C (-65° F), after 100,000 Strokes, Starting/Running, g-cm	10,000/1,000	10,000/1,000
Rubber Swell, FTM 3603, L Type Synthetic 1 week at 70° C (158° F), vol %	6	10 max

Precautions

Mobilgrease 28 is manufactured from high quality petroleum base stocks , carefully blended with selected soaps and additives. As with all petroleum products, good personal hygiene and careful handling should always be practiced. Avoid prolonged contact to skin, splashing into the eyes, ingestion or vapour inhalation. High-pressure injection of any grease under the skin can cause serious delayed soft tissue damage and should be treated immediately by a physician. To avoid injection injuries, inspect greasing equipment regularly for worn hoses and fittings. Keep finger away from the nozzle and ensure the nozzle is firmly in place before discharging the grease. Please refer to our Imperial Oil Material Safety Data Sheet for further information.

Note: This product is not controlled under Canadian WHMIS legislation.

Due to continual product research and development, the information contained herein is subject to changes without notification. Additional important health and safety information on this product can be found in the Material Safety Data Sheet, which is available online at www.imperialoil.ca/mobil or by contacting Lubricants and Petroleum Specialities, Technical Services, Technical Help Desk: 1-800-268-3183

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Oct-01

Product Data Sheet

Mobilgrease 28 Synthetic Aviation Grease

Product Description

Mobilgrease 28 is a supreme performance, wide-temperature, antiwear grease designed to combine the unique features of a polyalphaolefin (PAO) synthetic base fluid with an organo-clay (non-soap) thickener. The wax-free nature of the synthetic base fluid, together with its low coefficient of traction compared with mineral oils, provide excellent low temperature pumpability, very low starting and running torque, and can reduce operating temperatures in the load zone of rolling element bearings. The clay thickener gives Mobilgrease 28 a high dropping point value of around 300 °C, which provides excellent stability at high temperatures. Mobilgrease 28 resists water washing, provides superior load-carrying ability, reduces frictional drag, and prevents excessive wear. Tests show that Mobilgrease 28 lubricates effectively rolling element bearings under conditions of high speeds and temperatures. Mobilgrease 28 has also shown superior ability to lubricate heavily loaded sliding mechanisms, such as wing flap screw jacks. Its consistency is between an NLGI No. 1 and No. 2 grease.

Mobilgrease 28 is of quality level U.S. Military Specification MIL-G-81322E, Grease (Grade A), General-Purpose, Aircraft, and is qualified against U.S. Military Specification DOD-G-24508A (Navy) for shipboard auxiliary machinery. It is designated U.S. Military Symbol WTR and NATO Symbol G-395.

For more than 30 years, Mobilgrease 28 has been the multi-purpose grease of choice for military and related aviation applications, worldwide.

Features & Benefits

A key factor in the development of Mobilgrease 28 is the close contacts between ExxonMobil product engineers and key OEMs to ensure that product offerings will provide exceptional performance in the continually evolving aviation equipment designs. This work has helped confirm the results from ExxonMobil's own laboratory tests showing the exceptional performance of Mobilgrease 28.

A particular requirement of aviation greases is the need to resist high temperature stresses, while providing excellent starting and low torque at low-temperature. To meet this combination of needs ExxonMobil product formulation scientists chose proprietary synthetic base oils for Mobilgrease 28 because of their exceptional thermal/oxidative resistance potential, and superb low-temperature capability. Formulators chose specific thickener chemistry and a proprietary additive combination which would maximize the benefits of the synthetic base oils.

Mobilgrease 28 meets the requirements of key military and commercial aviation specifications and has built up a superb reputation for performance and reliability among users around the world. Mobilgrease 28 provides the following advantages and potential benefits:

Features	Advantages and Potential Benefits
High viscosity index (VI) base stock with no wax content	Allows wide operating temperature range - outstanding high and low temperature performance. Provides thicker fluid films protecting against wear of equipment parts operating at high temperature. Causes low resistance during start-up at very low temperatures.
Low traction base oil	Results in low sliding friction, reduced heat build-up, potential energy savings.
Excellent protection against wear and corrosion	Superb bearing protection and extended bearing life and reduced bearing replacement costs.
Extreme-pressure protection characteristics	Avoids excessive wear, even under shock load.
High thermal/oxidative stability	Long relubrication intervals.
High resistance to water washout	Maintains excellent grease performance in adverse weather and other water-exposure conditions.

Applications

Mobilgrease 28 is designed for the lubrication of plain and rolling bearings at low to high speeds, and splines, screws, worm gears, and other mechanisms where high friction reduction, low wear, and low lubricant friction losses are required. The recommended operating temperature range is -54 °C to 177 °C with appropriate relubrication intervals.

Mobilgrease 28 is recommended for use in landing wheel assemblies, control systems and actuators, screw jacks, servo devices, sealed-bearing motors, oscillating bearings, and helicopter rotor bearings on military and civil aircraft. Subject to equipment manufacturer approvals, it can also be used on naval shipboard auxiliary machinery and where superseded specifications MIL-G-81322 (WP), MIL-G-7711A, MIL-G-3545B, and MIL-G-25760A may be called for.

Mobilgrease 28 also is recommended for industrial lubrication, including sealed or repackable ball and roller bearings wherever extreme temperature conditions, high speeds, or water washing resistance are factors. Typical industrial applications include conveyor bearings, small alternator bearings operating at temperatures near 177 °C (350 °F), high-speed miniature ball bearings, and bearing applications where oscillatory motion, and vibration create problems.

Specifications & Approvals

Mobilgrease 28 has the following Military qualifications:

MIL-G-81322E quality level
DOD-G-24508A approved

Typical Properties

	Test Method	Mobilgrease 28	MIL-G-81322E
NLGI Grade		1 1/2	
Thickener Type		Clay (non-soap)	
Colour	Visual	Dark Red	
Structure	Visual	Smooth, buttery	
Dropping Point, °C (°F)	ASTM D 2265	305 (581)	232 (450) min
Viscosity of Base Oil, cSt, at 40°C	ASTM D 445	29.3	
Low Temp. Torque, at -54°C (-65°F), Nm (g-cm), Starting, Running, after 1 Hr	ASTM D 1478	0.8 (7600) 0.08 (770)	0.98 (10,000) max 0.098 (1,000) max
Penetration at 25°C (77°F), mm/10, 60 Stoke Worked	ASTM D217	295	265-320
Extended Worked Penetration Stability, mm/10, 100,000 strokes	FTM 791.313	302	350 max
Oil Separation, wt %, 30 Hrs at 177°C	ASTM D 6184	3	10 max
Evaporation Loss, wt %, 22 Hrs at 177°C	ASTM D 2595	6	12 max
Copper Strip Corrosion, 24 Hrs at 100°C	ASTM D 4048	Pass	1b max
Four-Ball Wear, Scar dia, mm	ASTM D 2266	0.55	1.3 min
Load Wear Index, kgf	ASTM D 2596	41	30 min
Rust Protection, > 1mm dia Corrosion Spots, 48 Hrs at 125°F	ASTM D 1743	Pass	0
Water Washout, wt %, 1 Hr at 41°C	ASTM D 1264	1	20 max
High Temperature Performance, Hrs at 177°C	ASTM D 3336	Pass	400 min
Oxidation Stability, pressure drop in kPa, 100 Hrs at 99°C 500 Hrs at 99°C	ASTM D 942	Pass Pass	83 max 172 max
Storage Stability, 6 months at 38°C	FTM 791.3467	Pass	
Dirt Count, Particles/ml, 25-74 Micron 75 Micron or Larger	FTM 791.3005	Pass 0	1000 max 0
Oscillation Friction and Wear, Scar Dia, mm, 35,000 cycles, 90° Angle	Modified ASTM D 3704	Pass	6.35 max
Rubber Swell, L Type Synthetic, vol %, 1 week at 70°C (158°F)	FTM 791.3603	6	10 max

Health and Safety

Based on available information, this product is not expected to produce adverse effects on health when used for the intended application and the recommendations provided in the Material Safety Data Sheet (MSDS) are followed. MSDS's are available upon request through Esso Petroleum Company Ltd., by telephoning 01372 222000. This product should not be used for purposes other than its intended use. If disposing of used product, take care to protect the environment.

Esso Petroleum Oil Company Limited
ExxonMobil House, Ermyn Way
Leatherhead, Surrey, KT22 8UX
Telephone: 01372 22 2000



The ExxonMobil logotype and Mobilgrease are trademarks of Exxon Mobil Corporation, or one of its subsidiaries.

Due to continual product research and development, the information contained herein is subject to change without notification. Typical Properties may vary slightly. All products may not be available locally.

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Mobil DTE[®] Oil Named Series

Circulation Oils

Description

The Mobil DTE Oil Named oils are formulated from premium-quality, highly refined, high VI base stocks. The inherent stability of these base stocks is further enhanced by oxidation inhibitors to resist breakdown at high temperature in the presence of entrained air and catalyzing metals. Defoamants ensure that entrained air is released while metal deactivators prevent corrosion and reduce the catalytic effect of system metals. All grades contain a rust inhibitor selected for its ability to maintain effectiveness over long periods of time. Good demulsibility permits water and other contaminants to separate readily from the oil in the system reservoir.

Application

Bearings and Gears - These oils are intended primarily for the lubrication of plain bearings, anti-friction bearings, and parallel shaft gearing. Selection of the proper viscosity grade should be based on speed, load, temperature, and geometry of the components involved.

Hydraulics - Moderate-pressure hydraulic systems, depending on viscosity requirements, can be lubricated with Mobil DTE Oil Heavy Medium and Heavy. For higher pressure systems which require an oil with both a lower ISO viscosity and antiwear properties, the Mobil DTE 10M and 20 Series oils should be used.

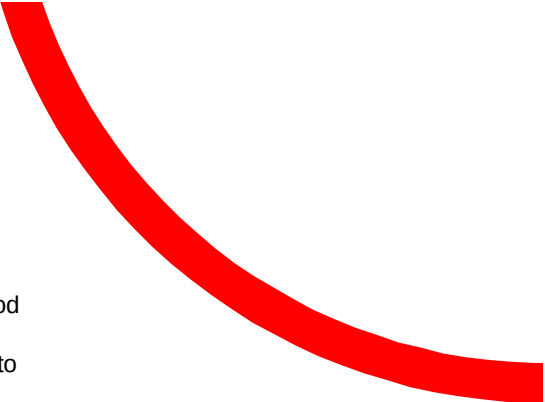
Compressors - Compressors and vacuum pumps handling inert gases generally can be lubricated, depending on viscosity requirements, with Mobil DTE Oil Heavy Medium and Heavy in applications where discharge temperatures do not exceed 150°C. Units in service on air, refrigerants, hydrocarbons, and process gases usually require special engineering consideration.

Typical Properties

Physical characteristics listed in the table are typical and may vary slightly.

Mobil DTE Oil	Heavy Medium	Heavy
Product Number	60016-3	60018-9
Specific Gravity	.877	.881
Pour Point, °C (°F)	-7 (20)	-7 (20)
Flash Point (COC), °C (°F)	204 (400)	210 (410)
Kinematic Viscosity		
cSt at 40 °C	64.6	95
cSt at 100 °C	8.4	11.4
SUS at 100 °F	335	495
SUS at 210 °F	55	65.5
Viscosity Index	95	95
ISO VG	68	100
Color, ASTM	2.5	2.5
Neut. No.	0.30	0.30
Rust Protection, A & B	Pass	Pass
Demulsibility at 54°C	Pass	Pass





Precautions

The Mobil DTE Oil Named oils are manufactured from high quality petroleum base stocks, carefully blended with selected additives. As with all petroleum products, good personal hygiene and careful handling should always be practiced. Avoid prolonged contact to skin, splashing into the eyes, ingestion or vapour inhalation. Please refer to our Imperial Oil Material Safety Data Sheet for further information.

Note: This product is not controlled under Canadian WHMIS legislation.

Due to continual product research and development, the information contained herein is subject to changes without notification. Additional important health and safety information on this product can be found in the Material Safety Data Sheet, which is available online at www.imperialoil.ca/mobil or by contacting Lubricants and Petroleum Specialities, Technical Services, Technical Help Desk: 1-800-268-3183

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Oct-01

600163-00

**600163-00 MOBIL DTE OIL HEAVY MEDIUM
MATERIAL SAFETY DATA BULLETIN**

1. PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME: MOBIL DTE OIL HEAVY MEDIUM
SUPPLIER: EXXONMOBIL OIL CORPORATION
3225 GALLOWES RD.
FAIRFAX, VA 22037
24 - Hour Health and Safety Emergency (call collect): 609-737-4411
24 - Hour Transportation Emergency (Primary) CHEMTREC: 800-424-9300
(Secondary) 281-834-3296
Product and Technical Information: 800-662-4525 703-846-6693
MSDS Fax on Demand: 613-228-1467, other MSDS information: 856-224-4644

2. COMPOSITION/INFORMATION ON INGREDIENTS

CHEMICAL NAMES AND SYNONYMS: PET. HYDROCARBONS AND ADDITIVES
GLOBALLY REPORTABLE MSDS INGREDIENTS:
None.
See Section 8 for exposure limits (if applicable).

3. HAZARDS IDENTIFICATION

Under normal conditions of use, this product is not considered hazardous according to regulatory guidelines (See section 15).
EMERGENCY OVERVIEW: Amber Liquid. DOT ERG No. : NA
POTENTIAL HEALTH EFFECTS: Under normal conditions of intended use, this product does not pose a risk to health. Excessive exposure may result in eye, skin or respiratory irritation.
For further health effects/toxicological data, see Section 11.

4. FIRST AID MEASURES

EYE CONTACT: Flush thoroughly with water. If irritation occurs, call a physician.
SKIN CONTACT: Wash contact areas with soap and water. Remove and clean oil soaked clothing daily and wash affected area. (See Section 16 - Injection Injury)
INHALATION: Not expected to be a problem. However, if respiratory irritation, dizziness, nausea, or unconsciousness occurs due to excessive vapor or mist exposure, seek immediate medical assistance. If breathing has stopped, assist ventilation with a mechanical device or mouth-to-mouth resuscitation.
INGESTION: Not expected to be a problem. Seek medical attention if discomfort occurs. Do not induce vomiting.

5. FIRE-FIGHTING MEASURES

EXTINGUISHING MEDIA: Carbon dioxide, foam, dry chemical and water fog.
SPECIAL FIRE FIGHTING PROCEDURES: Water or foam may cause frothing. Use water to keep fire exposed containers cool. Water spray may be used to flush spills away from exposure. Prevent runoff from fire control or dilution from entering streams, sewers, or drinking water supply.
SPECIAL PROTECTIVE EQUIPMENT: For fires in enclosed areas, fire fighters must use self-contained breathing apparatus.
UNUSUAL FIRE AND EXPLOSION HAZARDS: None.
COMBUSTION PRODUCTS: Fumes, smoke, carbon monoxide, sulfur oxides, aldehydes and other decomposition products, in the case of incomplete combustion.
Flash Point C(F): > 204(400) (ASTM D-93).
Flammable Limits (approx.% vol.in air) - LEL: 0.9%, UEL: 7.0%
NFPA HAZARD ID: Health: 0, Flammability: 1, Reactivity: 0

6. ACCIDENTAL RELEASE MEASURES

NOTIFICATION PROCEDURES: Report spills/releases as required to appropriate authorities. U.S. Coast Guard and EPA regulations require immediate reporting of spills/releases that could reach any waterway including intermittent dry creeks. Report spill/release to Coast Guard National Response Center toll free number (800)424-8802. In case of accident or road spill notify CHEMTREC (800) 424-9300.
PROCEDURES IF MATERIAL IS RELEASED OR SPILLED:

LAND SPILL: Shut off source taking normal safety precautions.
Take measures to minimize the effects on ground water. Recover by pumping or contain spilled material with sand or other suitable absorbent and remove mechanically into containers. If necessary, dispose of adsorbed residues as directed in Section 13.
WATER SPILL: Confine the spill immediately with booms. Warn other ships in the vicinity. Notify port and other relevant authorities.
Remove from the surface by skimming or with suitable absorbents.
If permitted by regulatory authorities the use of suitable dispersants should be considered where recommended in local oil spill procedures.
ENVIRONMENTAL PRECAUTIONS: Prevent material from entering sewers, water sources or low lying areas; advise the relevant authorities if it has, or if it contaminates soil/vegetation.
PERSONAL PRECAUTIONS: See Section 8

7. HANDLING AND STORAGE

HANDLING: No special precautions are necessary beyond normal good hygiene practices. See Section 8 for additional personal protection advice when handling this product.
STORAGE: Keep containers closed when not in use. Do not store in open or unlabelled containers. Store away from strong oxidizing agents and combustible materials. Do not store near heat, sparks, flame or strong oxidants.
SPECIAL PRECAUTIONS: Prevent small spills and leakages to avoid slip hazard.
EMPTY CONTAINER WARNING: Empty containers retain residue (liquid and/or vapor) and can be dangerous. DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION; THEY MAY EXPLODE AND CAUSE INJURY OR DEATH. Do not attempt to refill or clean container since residue is difficult to remove. Empty drums should be completely drained, properly bunged and promptly returned to a drum reconditioner. All containers should be disposed of in an environmentally safe manner and in accordance with governmental regulations.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

OCCUPATIONAL EXPOSURE LIMITS:
When mists/aerosols can occur, the following are recommended: 5 mg/m³

(as oil mist)- ACGIH Threshold Limit Value (TLV), 10 mg/m3 (as oil mist)
- ACGIH Short Term Exposure Limit (STEL), 5 mg/m3 (as oil mist) - OSHA Permissible Exposure Limit (PEL)
VENTILATION: If mists are generated, use adequate ventilation, local exhaust or enclosures to control below exposure limits.
RESPIRATORY PROTECTION: If mists are generated, and/or when ventilation is not adequate, wear approved respirator.
EYE PROTECTION: If eye contact is likely, safety glasses with side shields or chemical type goggles should be worn.
SKIN PROTECTION: Not normally required. When splashing or liquid contact can occur frequently, wear oil resistant gloves and/or other protective clothing. Good personal hygiene practices should always be followed.

9. PHYSICAL AND CHEMICAL PROPERTIES

Typical physical properties are given below. Consult Product Data Sheet

for specific details.

APPEARANCE: Liquid

COLOR: Amber

ODOR: Mild

ODOR THRESHOLD-ppm: NE

pH: NA

BOILING POINT C(F): > 316(600)

MELTING POINT C(F): NA

FLASH POINT C(F): > 204(400) (ASTM D-93)

FLAMMABILITY (solids): NE

AUTO FLAMMABILITY C(F): NA

EXPLOSIVE PROPERTIES: NA

OXIDIZING PROPERTIES: NA

VAPOR PRESSURE-mmHg 20 C: < 0.1

VAPOR DENSITY: > 2.0

EVAPORATION RATE: NE

RELATIVE DENSITY, 15/4 C: 0.875

SOLUBILITY IN WATER: Negligible

PARTITION COEFFICIENT: > 3.5

VISCOSITY AT 40 C, cSt: 68.0

VISCOSITY AT 100 C, cSt: 8.7

POUR POINT C(F): < -6(22)

FREEZING POINT C(F): NE

VOLATILE ORGANIC COMPOUND: NE

DMSO EXTRACT, IP-346 (WT.%): <3, for mineral oil only

NA=NOT APPLICABLE NE=NOT ESTABLISHED D=DECOMPOSES

FOR FURTHER TECHNICAL INFORMATION, CONTACT YOUR MARKETING REPRESENTATIVE

10. STABILITY AND REACTIVITY

STABILITY (THERMAL, LIGHT, ETC.): Stable.
CONDITIONS TO AVOID: Extreme heat and high energy sources of ignition.
INCOMPATIBILITY (MATERIALS TO AVOID): Strong oxidizers.
HAZARDOUS DECOMPOSITION PRODUCTS: Product does not decompose at
ambient temperatures.
HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL DATA

---ACUTE TOXICOLOGY---

ORAL TOXICITY (RATS): Practically non-toxic (LD50: greater than 2000 mg/kg). ---Based on testing of similar products and/or the components.
DERMAL TOXICITY (RABBITS): Practically non-toxic (LD50: greater than 2000 mg/kg). ---Based on testing of similar products and/or the components.
INHALATION TOXICITY (RATS): Practically non-toxic (LC50: greater than 5 mg/l). ---Based on testing of similar products and/or the components.
EYE IRRITATION (RABBITS): Practically non-irritating. (Draize score: greater than 6 but 15 or less). ---Based on testing of similar products and/or the components.
SKIN IRRITATION (RABBITS): Practically non-irritating. (Primary Irritation Index: greater than 0.5 but less than 3). ---Based on testing of similar products and/or the components.
OTHER ACUTE TOXICITY DATA: Although an acute inhalation study was not performed with this product, a variety of mineral and synthetic oils, such as those in this product, have been tested. These samples had virtually no effect other than a nonspecific inflammatory response in the lung to the aerosolized mineral oil. The presence of additives in other tested formulations (in approximately the same amounts as in the present formulation) did not alter the observed effects.

---SUBCHRONIC TOXICOLOGY (SUMMARY)---

No significant adverse effects were found in studies using repeated dermal applications of similar formulations to the skin of laboratory animals for 13 weeks at doses significantly higher than those expected during normal industrial exposure. The animals were evaluated extensively for effects of exposure (hematology, serum chemistry, urinalysis, organ weights, microscopic examination of tissues etc.).

---REPRODUCTIVE TOXICOLOGY (SUMMARY)---

No teratogenic effects would be expected from dermal exposure, based on laboratory developmental toxicity studies of major components in this formulation and/or materials of similar composition.

---CHRONIC TOXICOLOGY (SUMMARY)---

Repeated and/or prolonged exposure may cause irritation to the skin, eyes or respiratory tract. Overexposure to oil mist may result in oil droplet deposition and/or granuloma formation. For

mineral base oils: Base oils in this product are severely solvent refined and/or severely hydrotreated. Chronic mouse skin painting studies of severely treated oils showed no evidence of carcinogenic effects. These results are confirmed on a continuing basis using various screening methods such as Modified Ames Test, IP-346, and/or other analytical methods. For synthetic base oils: The base oils in this product have been tested in the Ames assay and other tests of mutagenicity with negative results. These base oils are not expected to be carcinogenic with chronic dermal exposures.

---SENSITIZATION (SUMMARY)---

Not expected to be sensitizing based on tests of this product, components, or similar products.

12. ECOLOGICAL INFORMATION

ENVIRONMENTAL FATE AND EFFECTS: This product is expected to be inherently biodegradable. Bioaccumulation is unlikely due to the very low water solubility of this product, therefore bioavailability to aquatic organisms is minimal. Available ectotoxicity data (LL50 >1000 mg/L) indicates that adverse effects to aquatic organisms are not expected from this product. When released into the environment, adsorption to sediment and soil will be the predominant behavior.

13. DISPOSAL CONSIDERATIONS

WASTE DISPOSAL: Product is suitable for burning in an enclosed, controlled burner for fuel value. Such burning may be limited pursuant to the Resource Conservation and Recovery Act. In addition, the product is suitable for processing by an approved recycling facility or can be disposed of at an appropriate government waste disposal facility. Use of these methods is subject to user compliance with applicable laws and regulations and consideration of product characteristics at time of disposal.

RCRA INFORMATION: The unused product, in our opinion, is not specifically listed by the EPA as a hazardous waste (40 CFR, Part 261D), nor is it formulated to contain materials which are listed hazardous wastes. It does not exhibit the hazardous characteristics of ignitability, corrosivity, or reactivity. The unused product is not formulated with substances covered by the Toxicity Characteristic Leaching Procedure (TCLP). However, used product may be regulated.

14. TRANSPORT INFORMATION

USA DOT: NOT REGULATED BY USA DOT.
RID/ADR: NOT REGULATED BY RID/ADR.
IMO: NOT REGULATED BY IMO.
IATA: NOT REGULATED BY IATA.
STATIC ACCUMULATOR (50 picosiemens or less): YES

15. REGULATORY INFORMATION

US OSHA HAZARD COMMUNICATION STANDARD: When used for its intended purposes, this product is not classified as hazardous in accordance with OSHA 29 CFR 1910.1200.
EU Labeling: Product is not dangerous as defined by the European Union Dangerous Substances/Preparations Directives. EU labeling not required.
Governmental Inventory Status: All components comply with TSCA, EINECS/ELINCS, AICS, and DSL.
U.S. Superfund Amendments and Reauthorization Act (SARA) Title III: This product contains no "EXTREMELY HAZARDOUS SUBSTANCES".
SARA (311/312) REPORTABLE HAZARD CATEGORIES: None.
This product contains no chemicals subject to the supplier notification requirements of SARA (313) toxic release program.
THIS PRODUCT HAS BEEN AUTHORIZED BY USDA FOR USE UNDER THE FOLLOWING CATEGORY: This product is acceptable as a lubricant where there is no possibility of food contact (complies with earlier USDA guidelines for H-2 lubricant use).
The following product ingredients are cited on the lists below:

CHEMICAL NAME	CAS NUMBER	LIST CITATIONS
TRICRESYL PHOSPHATE (>0.04%)	1330-78-5	22
ZINC (ELEMENTAL ANALYSIS) (<0.01%)	7440-66-6	22
PHOSPHORODITHOIC ACID, O,O-DI	68649-42-3	22

C1-14-ALKYL ESTERS, ZINC SALTS (2:
1) (ZDDP) (0.09%)
--- REGULATORY LISTS SEARCHED ---

1=ACGIH ALL	6=IARC 1	11=TSCA 4	16=CA P65 CARC	21=LA RTK
2=ACGIH A1	7=IARC 2A	12=TSCA 5a2	17=CA P65 REPRO	22=MI 293
3=ACGIH A2	8=IARC 2B	13=TSCA 5e	18=CA RTK	23=MN RTK
4=NTP CARC	9=OSHA CARC	14=TSCA 6	19=FL RTK	24=NJ RTK
5=NTP SUS	10=OSHA Z	15=TSCA 12b	20=IL RTK	25=PA RTK
				26=RI RTK

Code key: CARC=Carcinogen; SUS=Suspected Carcinogen; REPRO=Reproductive

16. OTHER INFORMATION

USE: STEAM TURBINE OIL

NOTE: PRODUCTS OF EXXON MOBIL CORPORATION AND ITS AFFILIATED COMPANIES ARE NOT FORMULATED TO CONTAIN PCBS.

Health studies have shown that many hydrocarbons pose potential human health risks which may vary from person to person. Information provided on this MSDS reflects intended use. This product should not be used for other applications. In any case, the following advice should be considered:

INJECTION INJURY WARNING: If product is injected into or under the skin, or into any part of the body, regardless of the appearance of the wound or its size, the individual should be evaluated immediately by a physician as a surgical emergency. Even though initial symptoms from high pressure injection may be minimal or absent, early surgical treatment within the first few hours may significantly reduce the ultimate extent of injury.

INDUSTRIAL LABEL

Under normal conditions of intended use, this product does not pose a risk to health. Excessive exposure may result in eye, skin or respiratory irritation. Always observe good hygiene measures. First Aid: Wash skin with soap and water. Flush eyes with water. If overcome by fumes or vapor, remove to fresh air. If ingested do not induce vomiting. If symptoms persist seek medical assistance. Read and understand the MSDS before using this product.

**

For Internal Use Only: MHC: 1* 1* 1* 1* 1*, MPPEC: A, TRN: 600163-00,
ELIS: 400304, CMCS97: 970172, REQ: US - MARKETING, SAFE USE: L
EHS Approval Date: 21AUG2001

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640219-00

**640219-00 MOBILTEMP SHC 32
MATERIAL SAFETY DATA BULLETIN**

1. PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME: MOBILTEMP SHC 32
SUPPLIER: EXXONMOBIL CORPORATION
3225 GALLOWES RD.
FAIRFAX, VA 22037
24 - Hour Health and Safety Emergency (call collect): 609-737-4411
24 - Hour Transportation Emergency (Primary) CHEMTREC: 800-424-9300
(Secondary) 281-834-3296
Product and Technical Information: 800-662-4525 703-846-6693
MSDS Fax on Demand: 613-228-1467, other MSDS information: 856-224-4644

2. COMPOSITION/INFORMATION ON INGREDIENTS

CHEMICAL NAMES AND SYNONYMS: SYN. HYDROCARBONS AND ADDITIVES

GLOBALY REPORTABLE MSDS INGREDIENTS:

Substance Name	Approx. Wt%
PENTAERYTHRITOL (115-77-5)	1-5
SODIUM NITRITE (7632-00-0)	1-5
1-NAPHTHYLAMINE, N-PHENYL (90-30-2)	1-5
N-OLEYLSARCOSINE (110-25-8)	<1

OTHER INGREDIENTS:

Substance Name	Approx. Wt%
HYDROXYALKARYL LONG-CHAIN ALKYL ESTER (2082-79-3)	1-5
ALKYLATED DIPHENYL AMINES (68411-46-1)	1-5

See Section 8 for exposure limits (if applicable).

3. HAZARDS IDENTIFICATION

Under normal conditions of use, this product is not considered hazardous according to regulatory guidelines (See section 15).
EMERGENCY OVERVIEW: Red Grease. DOT ERG No. : NA
POTENTIAL HEALTH EFFECTS: Under normal conditions of intended use, this product does not pose a risk to health. Excessive exposure may result in eye, skin or respiratory irritation.
POTENTIAL ENVIRONMENTAL EFFECTS: Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.
For further health effects/toxicological data, see Section 11.

4. FIRST AID MEASURES

EYE CONTACT: Flush thoroughly with water. If irritation occurs, call a physician.
SKIN CONTACT: Wash contact areas with soap and water. Remove and clean oil soaked clothing daily and wash affected area.
INJECTION INJURY WARNING: If product is injected into or under the skin, or into any part of the body, regardless of the appearance of the wound or its size, the individual should be evaluated immediately by a physician as a surgical emergency. Even though initial symptoms from high pressure injection may be minimal or absent, early surgical treatment within the first few hours may significantly reduce the ultimate extent of injury.
INHALATION: Remove from further exposure. If respiratory irritation, dizziness, nausea, or unconsciousness occurs, seek immediate medical assistance. If breathing has stopped, assist ventilation with mechanical device or use mouth-to-mouth resuscitation.
INGESTION: Not expected to be a problem. Seek medical attention if discomfort occurs. Do not induce vomiting.

5. FIRE-FIGHTING MEASURES

EXTINGUISHING MEDIA: Carbon dioxide, foam, dry chemical and water fog.
SPECIAL FIRE FIGHTING PROCEDURES: Water or foam may cause frothing. Use water to keep fire exposed containers cool. Water spray may be used to flush spills away from exposure. Prevent runoff from fire control or dilution from entering streams, sewers, or drinking water supply.
SPECIAL PROTECTIVE EQUIPMENT: For fires in enclosed areas, fire fighters must use self-contained breathing apparatus.
UNUSUAL FIRE AND EXPLOSION HAZARDS: None.
COMBUSTION PRODUCTS: Fumes, smoke, carbon monoxide, sulfur oxides, aldehydes and other decomposition products, in the case of

incomplete combustion.
Flash Point C(F): > 204(400) (ESTIMATED FOR OIL, ASTM D-92 (COC)).
Flammable Limits (approx.% vol.in air) - LEL: NE, UEL: NE
NFPA HAZARD ID: Health: 0, Flammability: 1, Reactivity: 0

6. ACCIDENTAL RELEASE MEASURES

NOTIFICATION PROCEDURES: Report spills/releases as required to appropriate authorities. U.S. Coast Guard and EPA regulations require immediate reporting of spills/releases that could reach any waterway including intermittent dry creeks. Report spill/release to Coast Guard National Response Center toll free number (800)424-8802. In case of accident or road spill notify CHEMTREC (800) 424-9300.

PROCEDURES IF MATERIAL IS RELEASED OR SPILLED:
LAND SPILL: Shut off source taking normal safety precautions.

Take measures to minimize the effects on ground water. Recover by pumping or contain spilled material with sand or other suitable absorbent and remove mechanically into containers. If necessary, dispose of adsorbed residues as directed in Section 13.

WATER SPILL: Confine the spill immediately with booms. Warn other ships in the vicinity. Notify port and other relevant authorities.

Remove from the surface by skimming or with suitable absorbents.

If permitted by regulatory authorities the use of suitable dispersants should be considered where recommended in local oil spill procedures.

ENVIRONMENTAL PRECAUTIONS: Prevent material from entering sewers, water sources or low lying areas; advise the relevant authorities if it has, or if it contaminates soil/vegetation.

PERSONAL PRECAUTIONS: See Section 8

7. HANDLING AND STORAGE

HANDLING: High pressure injection under the skin may occur due to the rupture of pressurized lines. Always seek medical attention. No special precautions are necessary beyond normal good hygiene practices. See Section 8 for additional personal protection advice when handling this product.

STORAGE: Keep containers closed when not in use. Do not store in open or unlabelled containers. Store away from strong oxidizing agents and combustible materials. Do not store near heat, sparks, flame or strong oxidants.

SPECIAL PRECAUTIONS: Prevent small spills and leakages to avoid slip

hazard.

EMPTY CONTAINER WARNING: Empty containers retain residue (liquid and/or vapor) and can be dangerous. DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION; THEY MAY EXPLODE AND CAUSE INJURY OR DEATH. Do not attempt to refill or clean container since residue is difficult to remove. Empty drums should be completely drained, properly bunged and promptly returned to a drum reconditioner. All containers should be disposed of in an environmentally safe manner and in accordance with governmental regulations.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Substance Name (CAS-No.)	Source	---TWA---		----STEL---		NOTE
		ppm	mg/m3	ppm	mg/m3	

PENTAERYTHRITOL (115-77-5)						
Resp Fraction	OSHA		5			
Total Dust	OSHA		10			
	ACGIH		10			

NOTE: Limits shown for guidance only. Follow applicable regulations.
VENTILATION: Use adequate ventilation.

RESPIRATORY PROTECTION: No special requirements under ordinary conditions of use and with adequate ventilation.

EYE PROTECTION: Generally eye contact is unlikely with this type material. If eye contact is likely, safety glasses with side shields or chemical type goggles should be worn.

SKIN PROTECTION: If prolonged or repeated skin contact is likely, oil impervious gloves should be worn. Good personal hygiene practices should always be followed.

9. PHYSICAL AND CHEMICAL PROPERTIES

Typical physical properties are given below. Consult Product Data Sheet

for specific details.

APPEARANCE: Grease

COLOR: Red

ODOR: Mild

ODOR THRESHOLD-ppm: NE

pH: NA

BOILING POINT C(F): > 316(600)

DROP POINT C(F): NE

FLASH POINT C(F): > 204(400) (ESTIMATED FOR OIL, ASTM D-92 (COC))

FLAMMABILITY (solids): NE

AUTO FLAMMABILITY C(F): NA

EXPLOSIVE PROPERTIES: NA
OXIDIZING PROPERTIES: NA
VAPOR PRESSURE-mmHg 20 C: < 0.1
VAPOR DENSITY: NE
EVAPORATION RATE: NE
RELATIVE DENSITY, 15/4 C: 0.952
SOLUBILITY IN WATER: Negligible
PARTITION COEFFICIENT: > 3.5
VISCOSITY AT 40 C, cSt: NE
VISCOSITY AT 100 C, cSt: NE
POUR POINT C(F): NE
FREEZING POINT C(F): NE
VOLATILE ORGANIC COMPOUND: NE
NOTE: MOST PHYSICAL PROPERTIES FOR OIL COMPONENT.
DMSO EXTRACT, IP-346 (WT.%): <3, for mineral oil only
NA=NOT APPLICABLE NE=NOT ESTABLISHED D=DECOMPOSES
FOR FURTHER TECHNICAL INFORMATION, CONTACT YOUR MARKETING
REPRESENTATIVE

10. STABILITY AND REACTIVITY

STABILITY (THERMAL, LIGHT, ETC.): Stable.
CONDITIONS TO AVOID: Extreme heat and high energy sources of ignition.
INCOMPATIBILITY (MATERIALS TO AVOID): Strong oxidizers.
HAZARDOUS DECOMPOSITION PRODUCTS: Product does not decompose at
ambient temperatures.
HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL DATA

---ACUTE TOXICOLOGY---

ORAL TOXICITY (RATS): Practically non-toxic (LD50: greater than 2000
mg/kg). ---Based on testing of similar products and/or the
components.

DERMAL TOXICITY (RABBITS): Practically non-toxic (LD50: greater than
2000 mg/kg). ---Based on testing of similar products and/or the
components.

INHALATION TOXICITY (RATS): Practically non-toxic (LC50: greater
than 5 mg/l). ---Based on testing of similar products and/or the
components.

EYE IRRITATION (RABBITS): Practically non-irritating. (Draize score:
greater than 6 but 15 or less). ---Based on testing of similar
products and/or the components.

SKIN IRRITATION (RABBITS): Practically non-irritating. (Primary
Irritation Index: greater than 0.5 but less than 3). ---Based
on testing of similar products and/or the components.

OTHER ACUTE TOXICITY DATA: Although an acute inhalation study was not

performed with this product, a variety of mineral oils and synthetic base oils, such as those in this product have been tested. These samples had virtually no effect other than a nonspecific inflammatory response in the lung to the aerosolized mineral oil. The presence of additives in other tested formulations (in approximately the same amounts as in the present formulation) did not alter the observed effects.

---SUBCHRONIC TOXICOLOGY (SUMMARY)---

No significant adverse effects were found in studies using repeated dermal applications of similar formulations to the skin of laboratory animals for 13 weeks at doses significantly higher than those expected during normal industrial exposure. The animals were evaluated extensively for effects of exposure (hematology, serum chemistry, urinalysis, organ weights, microscopic examination of tissues etc.).

---REPRODUCTIVE TOXICOLOGY (SUMMARY)---

No teratogenic effects would be expected from dermal exposure, based on laboratory developmental toxicity studies of major components in this formulation and/or materials of similar composition.

---CHRONIC TOXICOLOGY (SUMMARY)---

Repeated and/or prolonged exposure may cause irritation to the skin, eyes or respiratory tract. For mineral base oils: Base oils in this product are severely solvent refined and/or severely hydrotreated. Chronic mouse skin painting studies of severely treated oils showed no evidence of carcinogenic effects. These results are confirmed on a continuing basis using various screening methods such as Modified Ames Test, IP-346, and/or other analytical methods. For synthetic base oils: The base oils in this product have been tested in the Ames assay and other tests of mutagenicity with negative results. These base oils are not expected to be carcinogenic with chronic dermal exposures.

---SENSITIZATION (SUMMARY)---

Not expected to be sensitizing based on tests of this product, components, or similar products.

12. ECOLOGICAL INFORMATION

ENVIRONMENTAL FATE AND EFFECTS:

This environmental assessment was conducted using information on the individual components as no test data was available for this specific formulation.

ECOTOXICITY: This product is expected to be harmful to aquatic organisms. May cause long-term adverse effects in the aquatic environment.

MOBILITY: Not established.

PERSISTENCE AND DEGRADABILITY: This product is expected to be inherently biodegradable, as the principal components have been shown to degrade at slow to moderate rates.

BIOACCUMULATIVE POTENTIAL: Not established.

13. DISPOSAL CONSIDERATIONS

WASTE DISPOSAL: Product is suitable for burning in an enclosed, controlled burner for fuel value. Such burning may be limited pursuant to the Resource Conservation and Recovery Act. In addition, the product is suitable for processing by an approved recycling facility or can be disposed of at an appropriate government waste disposal facility. Use of these methods is subject to user compliance with applicable laws and regulations and consideration of product characteristics at time of disposal.

RCRA INFORMATION: The unused product, in our opinion, is not specifically listed by the EPA as a hazardous waste (40 CFR, Part 261D), nor is it formulated to contain materials which are listed hazardous wastes. It does not exhibit the hazardous characteristics of ignitability, corrosivity, or reactivity. The unused product is not formulated with substances covered by the Toxicity Characteristic Leaching Procedure (TCLP). However, used product may be regulated.

14. TRANSPORT INFORMATION

USA DOT: NOT REGULATED BY USA DOT.
RID/ADR: NOT REGULATED BY RID/ADR.
IMO: NOT REGULATED BY IMO.
IATA: NOT REGULATED BY IATA.

15. REGULATORY INFORMATION

US OSHA HAZARD COMMUNICATION STANDARD: When used for its intended purposes, this product is not classified as hazardous in accordance with OSHA 29 CFR 1910.1200.

EU Labeling: Product is dangerous as defined by the European Union Dangerous Substances/Preparations Directives.
Symbol: Not applicable.
Risk Phrase(s): R52/53.
Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Safety Phrase(s): S61.
Avoid release to the environment. Refer to special instructions/Safety data sheets.

Governmental Inventory Status: All components comply with TSCA, EINECS/ELINCS, and DSL.

U.S. Superfund Amendments and Reauthorization Act (SARA) Title III:

This product contains no "EXTREMELY HAZARDOUS SUBSTANCES".

SARA (311/312) REPORTABLE HAZARD CATEGORIES: None.

This product contains the following SARA (313) Toxic Release

Chemicals:

CHEMICAL NAME	CAS NUMBER	CONC.
SODIUM NITRITE	7632-00-0	1%

THIS PRODUCT HAS BEEN AUTHORIZED BY USDA FOR USE UNDER THE FOLLOWING CATEGORY: This product is acceptable as a lubricant where there is no possibility of food contact (complies with earlier USDA guidelines for H-2 lubricant use).

The following product ingredients are cited on the lists below:

CHEMICAL NAME	CAS NUMBER	LIST CITATIONS
1,3-PROPANEDIOL, 2, 2-BIS(HYDROXYMETHYL)-	115-77-5	1, 10, 20, 21, 23, 25, 26
SODIUM NITRITE	7632-00-0	18, 20, 21, 24, 25

--- REGULATORY LISTS SEARCHED ---

1=ACGIH ALL	6=IARC 1	11=TSCA 4	16=CA P65 CARC	21=LA RTK
2=ACGIH A1	7=IARC 2A	12=TSCA 5a2	17=CA P65 REPRO	22=MI 293
3=ACGIH A2	8=IARC 2B	13=TSCA 5e	18=CA RTK	23=MN RTK
4=NTP CARC	9=OSHA CARC	14=TSCA 6	19=FL RTK	24=NJ RTK
5=NTP SUS	10=OSHA Z	15=TSCA 12b	20=IL RTK	25=PA RTK
				26=RI RTK

Code key: CARC=Carcinogen; SUS=Suspected Carcinogen; REPRO=Reproductive

16. OTHER INFORMATION

USE: HIGH TEMPERATURE GREASE

NOTE: PRODUCTS OF EXXON MOBIL CORPORATION AND ITS AFFILIATED COMPANIES ARE NOT FORMULATED TO CONTAIN PCBS.

Health studies have shown that many hydrocarbons pose potential human health risks which may vary from person to person. Information provided on this MSDS reflects intended use. This product should not be used for other applications. In any case, the following advice should be considered:

INDUSTRIAL LABEL

Under normal conditions of intended use, this product does not pose a risk to health. Excessive exposure may result in eye, skin or respiratory irritation. Always observe good hygiene measures. First Aid: Wash skin with soap and water. Flush eyes with water. If overcome by fumes or vapor, remove to fresh air. If ingested do not induce vomiting. If symptoms persist seek medical assistance. Read and understand the MSDS before using this product.

For Internal Use Only: MHC: 1* 1* 1* 1* 1*, MPPEC: A, TRN: 640219-00,

ELIS: 404060, CMCS97: 970894, REQ: US - MARKETING, SAFE USE: L

EHS Approval Date: 12JUL2002

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Mobiltemp SHC® 32

Grease, Low & High Temperature, Synthetic, ISO 32

Description

Mobiltemp SHC 32 synthetic grease is manufactured from a synthesized hydrocarbon fluid and a nonsoap thickener. It has extreme-pressure characteristics and excellent resistance to water washout. It provides effective lubrication over an extremely wide range of operating temperatures. Mobiltemp SHC 32 synthetic grease is dark red in color and has a smooth, buttery structure with a consistency between an NLGI No. 1 and No. 2 grease. It maintains consistency under conditions of high temperature and high speed. In addition, Mobiltemp SHC 32 has low break-away and running torque values and exhibits excellent low-temperature pumpability.

Application

Mobiltemp SHC 32 synthetic grease is recommended for industrial applications, such as sealed or repackable ball and roller bearings, splines, screws, some enclosed gears, and other mechanisms where high friction reduction, low wear, and low lubricant friction losses are required. Mobiltemp SHC 32 exhibits good pumpability and mechanical stability when used in centralized grease systems. The operating temperature range of Mobiltemp SHC 32 is -54°C (-65°F) to 177°C (350°F).

Combining Mobiltemp SHC 32 with some lithium-type greases may soften the mixture. When replacing any grease, clean or purge the former grease from the bearing.

Advantages

Mobiltemp SHC 32 synthetic grease provides these advantages and benefits.

- Satisfies operating temperatures of -54°C (-65°F) to 177°C (350°F)
- Excellent resistance to corrosion
- Excellent resistance to water washout
- Excellent anti-fretting capability under oscillatory and vibratory conditions
- Outstanding structural stability
- Performs well under heavy loads
- Meets USDA H2 requirements

Typical Characteristics

Physical properties are listed in the table. Except for those shown as minimum or maximum, values are typical and may vary slightly.

	Mobiltemp SHC 32
Product Number	64021-9
NLGI Grade	1.5
Specific Gravity, Calculated	0.9518
Soap Type	Clay
Structure	Smooth
Color	Dark Red
Cone Penetration at 25°C (77°F), Worked, 60 strokes	320
Dropping Point, °C (°F), ASTM D 566	232 (450)
Mineral Oil, %	79
Viscosity of Oil	
ISO VG	32
cSt at 40°	32
cSt at 100°C	5.2
SUS at 100°F	167
SUS at 210°F	44
Low-Temperature Torque, ASTM D 1478,	
-54°C (-65°F) after 100,000 strokes	
Starting, g-cm, max	10,000
Running, g-cm, max	1,000
Copper Strip Corrosion, ASTM D 130, 24 hrs @ 100°C	None
Bomb Oxidation, ASTM D 942, psi loss, 24 hr at 212°F	8
Water Washout, ASTM D 1264, % loss	20
Load Wear Index, kg, min	30
Four Ball Wear Test, Scar, mm, max	1.3

Health and Safety

Based on available toxicological information, it has been determined that this product poses no significant health risk when used and handled properly. Information on use and handling, as well as health and safety information, can be found in the Material Safety Data Sheet which can be obtained from your local distributor; via the Internet on <http://emmsds.ihpspl.com>.

For additional technical information or to identify the nearest U.S. Mobil supply source, call 1-800-662-4525.

Exxon Mobil Corporation TECHNICAL PUBLICATIONS 3225 Gallows Road, Fairfax, Virginia 22037-0001	Mobiltemp SHC® 32 PDS I-34 (5/02) © Copyright 1984-2002 All rights reserved.
Due to continual product research and development, the information contained herein is subject to changes without notification.	



MATERIAL SAFETY DATA SHEET

Zerol 150 Refrigeration Oil

I - PRODUCT IDENTIFICATION

Company Name: Nu-Calgon Wholesaler, Inc.
Address: 2008 Altom Court, St. Louis, MO 63146-4151
Product Name: Zerol 150 Refrigeration Oil
Synonyms: Refrigeration Fluid

Tel No: (314) 469-7000, (800) 554-5499
CHEMTREC: (800) 424-9300
Product Number: 4310

II - HAZARDOUS INGREDIENTS OF MIXTURES

<u>MATERIAL</u>	<u>CAS#</u>	<u>% By Wt</u>	<u>TLV</u>	<u>PEL</u>
Alkyl Benzene	68855-24-3	100	N/A	N/A

All ingredients appear on the Inventory of Chemical Substances published by the EPA under authority of the Toxic Substances Control Act (TSCA)

Approx Dielectric Constant: 2.1 (air=1)(non-additized oil)

III - PHYSICAL DATA

Vapor Pressure: mm Hg: N/A Evaporation Rate: (ether=1): N/A Solubility in H₂O: Insoluble Freezing Point °F: N/AV Boiling Point °F: 315°C Specific Gravity H₂O=1 @25° C: 0.87	Vapor Density (Air=1) 60-90° F: N/A VOC Content (% by wt.): N/A pH @ N/AV Solution: N/AV pH as Distributed: N/AV Appearance: Water white liquid Odor: Characteristic
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IV - FIRE AND EXPLOSION

Flash Point F: (COC) 311°F (MIN) **Flammable Limits:** N/A
Extinguishing Media: CO₂, dry chemical, foam, water fog.
Special Fire Fighting Procedures: For fires involving this material, do not enter any closed or confined fire space without proper protection equipment. This may include self-contained breathing apparatus to protect against the hazardous effect of normal products of combustion or oxygen deficiency. Read entire MSDS.
Unusual Fire and Explosion Hazards: N/AV

V - REACTIVITY DATA

Stability - Conditions to avoid: Stable
Incompatibility: May react with strong oxidizing materials.
Hazardous Decomposition Products: Normal combustion forms carbon dioxide and water vapor; incomplete combustion can produce carbon monoxide.
Conditions Contributing to Hazardous Polymerizations: Will not occur.

VI - HEALTH HAZARD DATA

EFFECTS OF OVEREXPOSURE (Medical Conditions Aggravated /Target Organ Effects),
A. ACUTE (Primary Route of Exposure) EYES: Applications into the eyes of rabbits produced slight membrane irritation without corneal injury. **SKIN:** Product may cause skin irritation on prolonged or frequently repeated contact. Application onto the skin of rabbits produced moderate erythema and edema. **INGESTION:** Not expected to be acutely toxic. (ORAL LD50 Rats >5g/kg). **INHALATION:** Not acutely toxic.
B. SUBCHRONIC, CHRONIC, OTHER: No available information was found.
C. MEDICAL CONDITIONS AGGRAVATED BY OVEREXPOSURE:

VII - EMERGENCY AND FIRST AID PROCEDURES

INHALATION: N/A

EYES: Flush eyes immediately with fresh water for at least 15 minutes while holding eyelids open. If irritation persists, see a doctor.

SKIN: Wash skin thoroughly with soap and water. See a doctor if rash or burn occurs. Launder contaminated clothing.

INGESTION: If swallowed, give water or milk to drink and telephone for medical advice. Consult medical personnel before inducing vomiting. If medical advice cannot be obtained, then take the person and product container to the nearest emergency treatment center or hospital.

VIII - SPILL OR LEAK PROCEDURE

Spill Management: Stop the source of the leak or release. Clean up releases as soon as possible. Contain liquid to prevent further contamination of soil, surface water or ground water. Clean up small spills with absorbent material and remove contaminated soil. Report and respond to larger releases.

Waste Disposal Methods: Place contaminated materials in disposable containers and dispose of in a manner consistent with applicable regulations. Contact local authorities for approved disposal.

IX - PROTECTION INFORMATION/CONTROL MEASURES

Respiratory: None normally required

Eye Protection: Safety goggles **Glove:** Rubber Gloves

Other Clothing and Equipment: Wear impervious clothing.

Ventilation: N/AV

X - SPECIAL PRECAUTIONS

Precautions to be taken in Handling and Storing : Do not weld, heat or drill container. Residue may ignite with explosive violence if heated sufficiently. CAUTION! Do not use pressure to empty drum or explosion may result.

Additional Information: Read and observe all label precautions.

NFPA

Health Hazard.....: 0

Fire Hazard.....: 1

Reactivity.....: 0

Specific Hazard....:

HMIS RATING

Health Hazard.....: 0

Fire Hazard.....: 1

Reactivity.....: 0

Personal Protection....: X (Sec. 9)

Revision Date: 03/21/2002

Seller makes no warranty, expressed or implied, concerning the use of this product other than indicated on the label. Buyer assumes all risk of use and/or handling of this material when such use and/or handling is contrary to label instructions. While Seller believes that the information contained herein is accurate, such information is offered solely for its customers' consideration and verification under their specific use conditions.



Phone or Fax 1 800 554-5499
www.nucalgon.com

PRODUCT BULLETIN

ZEROL[®] Alkylbenzene Refrigeration Oil

3-69

- Excellent miscibility & compatibility with all CFC & HCFC refrigerants.
- Recommended by OEM's for use with HCFC-based service blends.
- Copeland approved (Zerol 200TD).
- Oil of choice for retrofitting with R401a, R401b, R402a, R402b and R123.
- Excellent "clean-up" and replacement oil in burn-outs.

DESCRIPTION

Zerol Refrigeration Oil is a high-quality, premium synthetic alkylbenzene refrigeration oil. It is similar in many ways to natural mineral oils but has superior properties that make it particularly valuable in refrigeration, air-conditioning applications. It is available in three grades: Zerol 150 (150 SUS), Zerol 200TD (200SUS) and Zerol 300 (300 SUS).

Zerol 200TD is a unique formulation of the alkylbenzene in that it contains a special additive package. This package includes an extreme pressure or EP lubricity additive as well as an additive that provides controlled foaming in specific operating conditions. It is preferred by Copeland for field service applications with the HCFC-based service blends.

FEATURES

Zerol Refrigeration Oil is recommended for use with all CFC and HCFC refrigerants as well as with hydrocarbons and ammonia in all refrigeration and air conditioning applications. It has the following distinct features:

- Solvency
- High Miscibility
- Low Foaming
- Excellent Thermal Stability
- Low Floc Point Quality
- Excellent Compatibility

Solvency

As a synthetic alkylbenzene, Zerol has excellent solvency. And because of this solvency, it has the ability to emulsify and clean up mineral oil deposits as well as carbonized oil stemming from a moderate burn-out. As a result, Zerol is an excellent lubricant to use as the first or initial flush oil in HFC-retrofits (see Zerol flush) as well as the flush and replacement oil in burn-outs where naphthenic mineral oils were used. Such burn-outs result in a considerable amount of acid by-products and carbonized or burnt oil, and Zerol emulsifies this matter, enhancing the clean-up.



High Miscibility

The primary reason for improved CFC and HCFC system performance with Zerol is its excellent miscibility with fluorinated refrigerants like R-22 and R-502. Miscibility can be described as the ability of the refrigerant and lubricant to stay together as one homogeneous solution. Zerol has excellent miscibility with these refrigerants; the result is that the oil and the refrigerant will remain as one mixture at wide ranges in temperature and pressure. This is particularly true between Zerol and R-22, explaining why Zerol is the recommended lubricant for retrofitting with the HCFC service blends as they contain a significant amount of R-22.

Low Foaming

The low-foam quality of Zerol, due to its high purity, helps eliminate explosive carryover at startup, a potentially damaging situation caused by foaming and the subsequent oil loss from the crankcase. In the laboratory, Zerol has given outstanding performance vs. competitive oils in compressor foam-out tests. Practical application demonstrates this same low foaming tendency.

In certain specific applications or where it is recommended by the equipment manufacturer, enhanced or controlled foaming can be achieved through the use of Zerol 200TD.

Excellent Thermal Stability

Zerol enhances the life of refrigeration systems by providing better thermal stability in the presence of refrigerants. Laboratory tests and field experience show that Zerol resists change under high temperatures, thus eliminating problems with sludge, acids, and copper plating. A good indicator of a lubricant's stability is the sealed tube or Elsey test.

Low Floc Quality

Because it is synthetic and is highly pure, it contains no paraffin or wax, a substance in mineral oil that normally causes floccing or the formation of waxy deposits that can plug up a system. Strict quality controls from manufacturing through shipping ensure that Zerol maintains its high purity, allowing better efficiency at low temperatures, well below -100°F.

Excellent Compatibility

Converting refrigeration and air conditioning systems to Zerol is easily achieved due to its excellent compatibility. Zerol can be blended with conventional oils, is inert toward motor insulation, has UL approval for use in hermetic systems, and is compatible with most elastomers and additives often used to improve lubricity such as tricresyl phosphate (TCP).

PACKAGING

All Zerol formulations are packaged in 1-gallon bottles (6 per case), and 55-gallon drums. Five-gallon pails are available upon special request.

HANDLING

Contamination problems are extremely critical in refining and handling refrigeration oils. Great care is used to insure that Zerol Refrigeration Oil is free of moisture and noncondensables such as air. The same degree of care must be used by the service professional to make certain that the oil remains clean and dry.

Here are some rules for handling refrigeration oil:

1. Use clean tools – Pumps, funnels, tubing, etc. – while working with the oil.
2. Recap the bottle immediately after use. Refrigeration oil in an open bottle will absorb moisture.
3. Keep refrigeration oil in its original container. If you must pour oil into another container, be sure that the container is clean and dry.
4. Properly dispose of oil that you think dirt or moisture has contaminated. Don't risk using it. It is much less expensive to replace a bottle of refrigeration oil than to repair or replace a compressor.

PRODUCT SPECIFICATIONS

Properties	Test Method	ZEROL 150 Specifications	ZEROL 300 Specifications	ZEROL 200TD* Specifications
Viscosity, SUS, 100°F	ASTM D445, D2161	140-160	290-310	200-210
Color Gardner	ASTM D1544, D1500	2.0 Max.	4.0 Max.	1.0 Max.
Pour Point, °F	ASTM D97	-35 Max.	-30 Max.	-30 Max.
Floc Point, °F	ASHRAE 86	< - 100	< -80	-60 Max.
Gravity, API	ASTM D287	29.5-31.5	29.5-31.5	29.5-31.5
Acid No. mg KOH / gram	ASTM D664, D974	Nil	Nil	Nil
Dielectric Strength, KV	ASTM D877	>25	>25	30 Min.
Flash Point, °F	ASTM D92	310 Min.	310 Min.	310 Min.
Stability, % R22 in 14 days	Elsy Test, 14 Days at 175°C	0.2 Max.	0.2 Max.	0.2 Max.
Specific Gravity, @ 60°F	ASTM D1298			
Refractive Index		1.4865	1.4841	1.4857

PRODUCT CODE NUMBERS

Package	ZEROL 150	ZEROL 300	ZEROL 200TD*
6 x 1 gallon case	4310-07 (43101)	4311-07 (43111)	4308-07 (43080)
55-gallon drum	4310-01 (43108)	4311-01 (43118)	4308-01 (43088)

*USDA Approved

APPLICATIONS

Because of its unique solvency and miscibility properties, Zerol is particularly useful in several specific applications. In burnouts, the solvency of Zerol enables it to be used to clean up the by-products and burnt oil while its exceptional miscibility with R-22 and R-502 makes it the oil of choice with these two refrigerants, especially R-22 low temp applications. And this exceptional miscibility with R-22 is also why Zerol is recommended for use in retrofitting with the HCFC service blends as they are based on R-22.

Recommended for Burn-outs

In systems utilizing mineral oils where a moderate burn-out has occurred, Zerol can be used to flush and clean up the system. Zerol will emulsify the acid by-products, burnt oil and wax that result from such burn-outs, and allow the system to be flushed clean.

Use the following general procedure:

1. Remove burned-out compressor and disconnect electrical leads.
2. Remove expansion valve (or cap tube) and install by-passes.
3. Replace suction and liquid line filter driers.
4. Use connections where burned compressor was located, and push Zerol 150 through the system with a positive displacement pump. Use a volume of Zerol 150 equal to three times the amount (weight) of refrigerant typically charged into the system. Although the system may possibly operate on Zerol 150, Zerol 300 or 200TD, we recommend the use of 150 for the flushing as it is a lighter cut oil and will therefore perform the task more effectively.
5. Circulate for 2-4 hours, change filters often. Then remove Zerol cleaning solution from system.
6. Blow nitrogen through system.
7. Install new compressor as well as other removed components and triple evacuate the system; conduct leak check.
8. Operate the system with fresh refrigerant and Zerol of the proper viscosity (150, 200TD or 300).
9. Check system. If oil discolors, replace oil and filter-driers.

Recommended for Existing R-22 & R-502 Systems

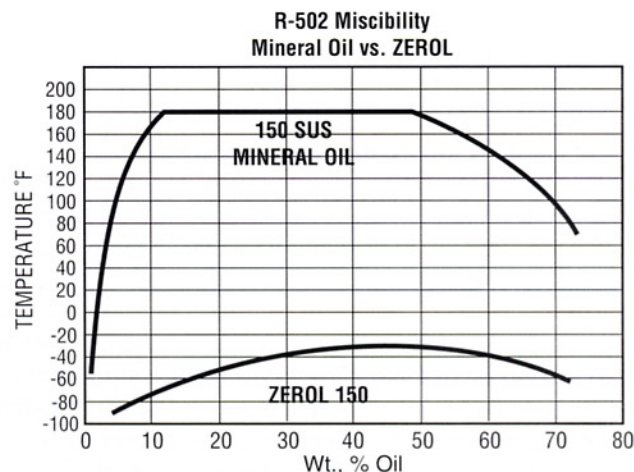
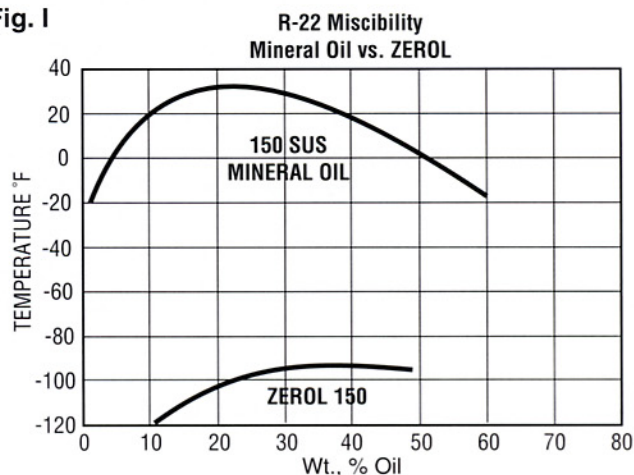
If refrigerants and their oils are not fully miscible at all operating temperatures and pressures, as is the case with mineral oil and R-22 & R-502, they can separate into liquid layers. This results in poor oil return as well as poor heat transfer as the oil logs in the evaporator, an area with the system's lowest temperature. In turn, the poor oil return can result in subsequent lubricity problems.

In addition, refrigerant migration to the compressor during shut down can result in the formation of two layers, with the lighter mineral oil layer on top; this is particularly true with R-502 and mineral oil. At start-up, the heavier refrigerant-rich layer will be lubricating some moving parts, and that is not a good idea. In flooded evaporators, where an oil-rich layer forms on top of the refrigerant, all these problems are magnified.

Of particular concern is the use of mineral oil and R-22 in low temperature applications. At the lower operating temperatures in the evaporator, the immiscibility between the R-22 and the mineral oil can become a significant problem.

As you can see in Fig. 1, even the best naphthenic mineral oil will separate out at about 30°F, where the oil/refrigerant ratio is approximately 20%/80%. Zerol 150, however, will not form two phases until -105°F at the same 20%/80% oil/refrigerant ratio. Similar disparities between these oils when used with R-502 can be seen in the second chart.

Fig. 1



Recommended for Retrofitting with HCFC Service Blends

Because of its exceptional history in working with R-22, due primarily to its excellent miscibility with this refrigerant, Zerol has been identified by compressor and equipment manufacturers as the oil of choice in retrofitting to the interim HCFC-based service blends such as :

R-401a	R-402a	R-408a
R-401b	R-402b	R-409a

The reason for this is simple: these interim refrigerants are blends, and R-22 is a significant component in each of them. It is understood that good refrigerant/lubricant miscibility is one of the performance characteristics that must be maintained as a system is retrofitted from R-12 or R-502 to one of these blends, and the use of Zerol accomplishes this performance level.

Use with HCFC blends.....Cont.

Within the last several years, thousands of systems have been retrofitted with Zerol and the HCFC interim blends. Here is a simplified version of the procedure:

1. Establish and obtain all data pertinent to the systems operation.
2. Evacuate the CFC refrigerant, and recover it appropriately.
3. Remove the mineral oil. In systems with small hermetic compressors, use the Nu-Calgon CanVerter® to extract the oil; in larger systems, use the CanVerter® or similar pump, or drain from the suction line. Utmost care must be exercised to insure that all or nearly all of the mineral oil is removed, particularly in larger systems where components like separators are in use.
4. If compressor was removed in the oil draining process, reinstall it.
5. Measure the volume of mineral oil removed and compare with the systems specification fill to judge how much has been removed. Replace it with a like volume of Zerol of the proper viscosity (ie. Copeland recommends Zerol 200TD).
6. Replace filter drier. Typically for loose filled driers, the XH-9 molecular sieve desiccant or equivalent is recommended. Compacted bead driers may use the XH-9 or XH-6 desiccant or equivalent.
7. Reconnect system and evacuate. Check for leaks.
8. Charge system with appropriate amount of the HCFC service blend refrigerant.
9. Mark system as to refrigerant and lubricant in use. Make appropriate system adjustments.
10. Observe operation. Zerol will tend to clean up old mineral oil deposits, and this matter will be trapped by the drier. Change driers as needed. If the oil loads up with carbon deposits and other matter, this indicates that the system had experienced rather severe depositing of mineral oil over time, and the Zerol should be replaced.

Zerol Flush

As the air conditioning, refrigeration industry continues to move to the increased use of long-term HFC refrigerants, it will be necessary to utilize polyol ester or POE lubricants. These lubricants have been found to provide long term performance advantages with HFC refrigerants. And as existing equipment or systems in the field are retrofitted to these new HFC refrigerants, it will actually be necessary to drain (flush) the system 3-5 times with polyol ester in order to reduce the residual level of the previously used mineral oil to 5% or less. The concern here is once again miscibility as the HFC and mineral oil are not miscible together.

This POE flushing procedure has been successfully accomplished thousands of times in the field. The results are predictable, and the mineral oil residual is effectively reduced to the desired level. Perhaps the only drawback to the procedure is that 2-4 flushes of comparatively more expensive POE oil must be used and discarded.

An attractive and proven alternative is the **Zerol Flush**, a procedure that involves the use of Zerol alkylbenzene lubricant in place of the polyol ester lubricant for the first 1 or 2

oil flushes, followed by 1 or 2 POE flushes. Because Zerol's solvency is similar to the POE, it does an equally effective job of emulsifying mineral oil and cleaning up existing mineral oil deposits. More importantly, it provides for two distinct cost savings:

First, because Zerol is generally one-third the cost of the POE, using it in the first 1 or 2 flushes will result in immediate savings. It also provides the added benefit of being easier to handle since it is not susceptible to moisture contamination like POE's are. This, too, will save money by saving time.

Secondly, because it has been shown that HFC/POE systems can tolerate twice as much residual alkylbenzene as residual mineral oil (see Fig. II), it may be possible to eliminate one of the subsequent POE flushes if the system is initially flushed with Zerol.

Fig. II. The effect on HFC/POE miscibility of residual mineral oil (5%) vs. residual Zerol (10%)

Temperature	5% MO/ 95% POE	5% Zerol 95% POE
35°F	Homogeneous Mixture	Homogeneous Mixture
4°F	Cloudy, with Oil-Rich Layer	Cloudy, Homog. Mixture
-40°F	More Cloudy, Oil Layer Near Solid	More Cloudy, Oil Rich Layer

Tests were conducted with HFC-404a at a ratio of 20% Lubricant and 80% Refrigerant

As you can see in Figure II, we have mixed two equal samples of R-404a with lubricant mixtures, one being 5%/95% Mineral/POE and the other 10%/90% Zerol/POE. At 35°F, both lubricant mixture/refrigerant samples exhibit good miscibility by remaining as homogeneous mixtures. However, at 4°F the sample with 5% mineral oil has turned cloudy and formed an oil-rich layer while the sample with 10% Zerol has only turned cloudy but exhibited no miscibility problems. At -40°F, the oil layer in the 5% mineral oil sample has turned solid while the 10% Zerol sample is only beginning to form an oil rich layer. Simply stated, this shows that twice the residual of Zerol exhibits even less miscibility problems than residual mineral oil.

Here is how to use the Zerol Flush in converting from a CFC to the long term HFC refrigerant:

1. Obtain pertinent system operational data.
2. Drain existing oil and replace with an equal volume of Zerol of the proper viscosity.
3. Operate system for at least 48 hours with CFC in it.
4. Drain lubricant and either repeat steps 2 and 3 one more time, or replace with an equal volume of POE.
5. Repeat steps 3 and 4 using POE until mineral oil residual is below 5% when tested. Use the #4815-0 refractometer for testing.
6. Charge with new HFC refrigerant & label the system.
7. Be sure to change filter driers at appropriate times during the process.



2008 Altom Court • St. Louis, Missouri 63146



SHELL DOLIUM[®] GREASE BRB

For lubrication of high-speed bearings operating at high temperatures

Product Description

Dolium[®] Grease BRB is a specially formulated NLGI No. 2 grade grease consisting of a highly refined mineral oil base stock, a polyurea thickener, and special additives that provide outstanding performance characteristics. **Dolium Grease BRB** effectively resists water washout while offering excellent rust inhibition and oxidation and thermal stability for extended service life. **Dolium Grease BRB** is able to withstand long-term high operating temperatures but will allow bearings to start and run even at low ambient temperatures. It was developed to provide longer service life than soap-based greases for bearings used in equipment operating at elevated temperatures. The grease is available in bulk, 400-pound drums, 120-pound kegs, 35-pound pails and 14.1 oz cartridges.

Applications

- anti-friction bearing such as those found in electric motors on alternators, generators, and air-conditioning units
- high-temperature or long-life applications in industrial plants, including equipment used in or near water
- service applications where accessibility and environment make it impractical to use conventional greases because of their shorter service lives
- lubrication of sealed-for-life bearings
- automobiles, trucks and off-highway equipment, including starters, generators, alternators and water pumps that are subject to temperature extremes and water spray

Features/Benefits

- long bearing life
- superior resistance to rust and corrosion
- superior oxidation stability
- low oil bleed
- excellent resistance to water washout
- excellent shear stability in service

Typical Properties of Shell Dolium Grease BRB

NLGI Grade		2
Product Code	Test Method	71616
Appearance		Purple, Smooth
Base Oil Viscosity: @ 40°C, cSt @ 100°C, cSt	D 445 D 445	129 12.9
Penetration, dmm Worked, 60X	D 217	285
Dropping Point, °F	Mettler	500+
Mechanical Stability, % Change in Penetration after 100,000 strokes	D 217	<10
High Temperature Bearing Test B50, Hrs at 350°F	D 3336	435
FE-9 Bearing Test B50, Hrs at 150°C	DIN 51-821	191
Four-Ball Wear, mm 1 hr/75°C/1200 rpm/40 kgf	D 2266	0.40
Oil Separation, Pressure Bleed, %	D 1742	0.42
Low Temperature Torque @ -30°C, gm-cm: Starting Torque Running Torque	D 4693	15,200 2,700
Rust Prevention, 5% SSW	D 5969	Pass
Copper Corrosion	D 4048	1b
Water Washout, wt% loss @175°F	D 1264	2
Guide To Usable Temperature Min °F Continuous Service, Max, °F Short Exposure, Max, °F		-30 350 400

Handling & Safety Information

For information on the safe handling and use of this product, refer to its Material Safety Data Sheet at <http://www.equivashellmsds.com>. For more information and availability, call **1+800-782-7852** or World Wide Web: <http://www.shell-lubricants.com/>.



SHELL AW HYDRAULIC OILS

Anti-wear hydraulic oils

Product Description

Shell AW Hydraulic Oils are high quality products formulated for systems which require an effective anti-wear hydraulic oil. The product is available in a wide range of viscosity grades (ISO 32 to ISO 100) for use in nearly all machine lubrication and hydraulic fluid applications. **Shell AW Hydraulic Oils** were designed with an effective zinc containing anti-wear agent which minimizes wear in vane, gear and piston pump applications. They are formulated with highly refined paraffinic base stocks which in combination with the specially selected additives, provides excellent performance against thermal and oxidative degradation.

Applications

- hydraulic systems including gear, vane and piston pumps
- circulating oil systems
- general machine lubrication

Features/Benefits

- excellent protection against wear
- fast air release properties
- rapid water separation properties
- excellent hydrolytic stability
- protects against rust and corrosion
- low foaming
- excellent filterability

Approvals and Recommendations

- Denison HF-0
- Vickers M-2950-S (Mobile) and I-286-S (Industrial)
- Cincinnati Machine

C-M SPECIFICATION	SHELL PRODUCT
P-68	Shell AW Hydraulic 32
P-69	Shell AW Hydraulic 68
P-70	Shell AW Hydraulic 46

Typical Properties of Shell AW Hydraulic Oils					
	Test Method	ISO Viscosity Grade			
		32	46	68	100
Product Code		65500	65501	65502	65503
Viscosity:					
@ 40°C, cSt	D 445	32.7	46.3	68.1	100.9
@ 100°C, cSt	D 445	5.4	6.7	8.6	11.3
@ 100°F, SUS	D 88	169	239	353	526
@ 210°F, SUS	D 88	44.4	48.7	55.2	65.0
Viscosity Index	D 2270	98	96	97	98
Flash Point, COC, °F	D 92	415	425	445	450
Pour Point, °F	D 97	-35	-25	-25	-15
Copper Corrosion	D 130	1a	1a	1a	1a
Emulsion Test, @ 130°F, 30 minutes MAX	D 1401	Pass	Pass	Pass	Pass(1)
Rust Test	D 665B	Pass	Pass	Pass	Pass
Turbine Oil Stability Test, hrs	D 943	2000 min	2000 min	2000 min	2000 min

(1) Emulsion @180°F 60 minutes max

Handling & Safety Information

For information on the safe handling and use of this product, refer to its Material Safety Data Sheet at <http://www.equivashellmsda.com>. For more information and availability, call **1+800-782-7852** or World Wide Web: <http://www.shell-lubricants.com/>.

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Shell Lubricants in the USA

MATERIAL SAFETY DATA SHEET

MSDS Number: 65500E - 0

24 Hour Emergency Assistance: CHEMTEL (877) 276-7283

General Assistance Number: (877) 276-7285

SECTION 1 PRODUCT IDENTIFICATION

MATERIAL IDENTITY: Shell AW Hydraulic Oil 32

PRODUCT CODES: 65500

COMPANY ADDRESS: Equilon Enterprises LLC, P. O. Box 4453, Houston, TX.
77210-4453-----
SECTION 2 PRODUCT/INGREDIENTS

CAS#	CONCENTRATION	INGREDIENTS
Mixture	100 %weight	Hydraulic Oil
Mixture	90 - 98.99 %weight	Petroleum distillates
Proprietary	1 - 2.99 %weight	Proprietary Additives

SECTION 3 HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

Appearance & Odor: Amber, clear liquid Mild odor

Health Hazards: No known immediate health hazards. High-pressure injection under the skin may cause serious damage.

Physical Hazards: No known physical hazards.

NFPA Rating (Health, Fire, Reactivity): 0, 1, 0

Hazard Rating: Least - 0 Slight - 1 Moderate - 2 High - 3

Extreme - 4

Inhalation:

Inhalation of vapors (generated at high temperatures only) or oil mist may cause mild irritation of the nose, throat, and respiratory tract.

Eye Irritation:

Lubricating oils are generally considered no more than minimally irritating to the eyes.

Skin Contact:

Lubricating oils are generally considered no more than minimally irritating to the skin. Prolonged and repeated contact may result in defatting and drying of the skin that may cause various skin disorders such as dermatitis, folliculitis or oil acne. Release of the material during high-pressure applications may result in injection under the skin causing possible extensive tissue damage which is difficult to heal.

Ingestion:

Lubricating oils are generally no more than slightly toxic if swallowed.

Signs and Symptoms:

Irritation as noted above. Local necrosis is evidenced by delayed onset of pain and tissue damage a few hours following injection.

Aggravated Medical Conditions:

Pre-existing eye, skin and respiratory disorders may be aggravated by exposure to this product.

For additional health information, refer to section 11.

SECTION 4 FIRST AID MEASURES

Inhalation:

Remove victim to fresh air and provide oxygen if breathing is difficult. Get medical attention.

Skin:

Remove contaminated clothing and shoes and wipe excess from skin. Flush skin with water, then wash with soap and water. If irritation occurs, get medical attention. Do not reuse clothing until cleaned. If material is injected under the skin, transport to the nearest medical facility for additional treatment.

Eye:

Flush with water. If irritation occurs, get medical attention.

Ingestion:

Do not induce vomiting. In general, no treatment is necessary unless large quantities of product are ingested. However, get medical attention.

Note to Physician:

In general, emesis induction is unnecessary in high viscosity, low volatility products such as oils and greases.

SECTION 5 FIRE FIGHTING MEASURES

Flash Point [Method]: 400 °F -440 °F/204.44 °C -226.67 °C [Cleveland Open Cup]

Extinguishing Media:

Material will float and can be re-ignited on surface of water. Use water fog, 'alcohol foam', dry chemical or carbon dioxide (CO2) to extinguish flames. Do not use a direct stream of water.

Fire Fighting Instructions:

Material will not burn unless preheated. Do not enter confined fire space without full bunker gear (helmet with face shield, bunker coats, gloves and rubber boots), including a positive pressure, NIOSH approved, self-contained breathing apparatus.

SECTION 6 ACCIDENTAL RELEASE MEASURES

Protective Measures:

May burn although not readily ignitable.

Wear appropriate personal protective equipment when cleaning up spills. Refer to Section 8.

Spill Management:

FOR LARGE SPILLS: Remove with vacuum truck or pump to storage/salvage vessels.

FOR SMALL SPILLS: Soak up residue with an absorbent such as clay, sand or other suitable material. Place in non-leaking container and seal tightly for proper disposal.

Reporting:

CERCLA: Product is covered by EPA's Comprehensive Environmental Response, Compensation and Liability Act (CERCLA) petroleum exclusion. Releases to air, land, or water are not reportable under CERCLA (Superfund).

CWA: This product is an oil as defined under Section 311 of EPA's Clean Water Act (CWA). Spills into or leading to surface waters that cause a sheen must be reported to the National Response Center, 1-800-424-8802.

SECTION 7 HANDLING AND STORAGE

Precautionary Measures:

Wash with soap and water before eating, drinking, smoking, applying cosmetics, or using toilet. Launder contaminated clothing before reuse. Properly dispose of contaminated leather articles such as shoes or belts that cannot be decontaminated. Avoid heat, open flames, including pilot lights, and strong oxidizing agents. Use explosion-proof ventilation to prevent vapor accumulation. Ground all handling equipment to prevent sparking.

Storage:

Store in a cool, dry place with adequate ventilation. Keep away from open flames and high temperatures.

Container Warnings:

Keep containers closed when not in use. Containers, even those that have been emptied, can contain explosive vapors. Do not cut, drill, grind, weld or perform similar operations on or near containers.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Oil mist, mineral	ACGIH TLV	TWA: 5 mg/m3	STEL: 10 mg/m3
Oil mist, mineral	OSHA PEL	TWA: 5 mg/m3	
Hydrogen sulfide	ACGIH - TLV	TWA: 10 ppm	STEL: 15 ppm
Hydrogen sulfide	OSHA - PEL_IS	TWA: 10 ppm	STEL: 15 ppm
Hydrogen sulfide	Thermal	> 180 Degrees F.	

EXPOSURE CONTROLS

Adequate ventilation to control airborne concentrations below the exposure guidelines/limits.

PERSONAL PROTECTION

Personal protective equipment (PPE) selections vary based on potential exposure conditions such as handling practices, concentration and ventilation. Information on the selection of eye, skin and respiratory protection for use with this material is provided below.

Eye Protection:

Chemical Goggles, or Safety Glasses

Skin Protection:

Use protective clothing which is chemically resistant to this material. Selection of protective clothing depends on potential exposure conditions and may include gloves, boots, suits and other items. The selection(s) should take into account such factors as job task, type of exposure and durability requirements.

Published literature, test data and/or glove and clothing manufacturers indicate the best protection is provided by:
Neoprene, or Nitrile Rubber

Respiratory Protection:

If engineering controls do not maintain airborne concentrations to a level which is adequate to protect worker health, an approved respirator must be worn. Respirator selection, use and maintenance should be in accordance with the requirements of the OSHA Respiratory Protection Standard, 29 CFR 1910.134.

Types of respirator(s) to be considered in the selection process include:

For Mist: Air Purifying, R or P style NIOSH approved respirator. For Vapors: Air Purifying, R or P style prefilter & organic cartridge, NIOSH approved respirator. Self-contained breathing apparatus.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Appearance & Odor: Amber, clear liquid Mild odor

Substance Chemical Family: Lubricants

API Gravity: 30 - - 32.3

Appearance: Amber, clear liquid

Flash Point: 400 °F - - 440 °F [Cleveland Open Cup]

Pour Point: -10 °F - - 0 °F

Specific Gravity: 0.8856

Viscosity: 30 cSt - - 100 cSt @ 40 °C

SECTION 10 REACTIVITY AND STABILITY

Stability:

Material is stable under normal conditions.

Conditions to Avoid:

Avoid heat and open flames.

Materials to Avoid:

Avoid contact with strong oxidizing agents.

Hazardous Decomposition Products:

Thermal decomposition products are highly dependent on combustion conditions. A complex mixture of airborne solids, liquids and gases will evolve when this material undergoes pyrolysis or combustion. Carbon Monoxide, Carbon Dioxide, Hydrogen Sulfide, Methacrylate monomers and other unidentified organic compounds may be formed upon combustion.

SECTION 11 TOXICOLOGICAL INFORMATION

Acute Toxicity

Dermal LD50 >5.0 g/kg(Rabbit) OSHA: Non-Toxic Based on components(s)

Eye Irritation Draize <15 / 110, nonirritating [Rabbit, 24 HOUR(S)]

OSHA: Non-Irritating Based on components(s)

Oral LD50 >5.0 g/kg(Rat) OSHA: Non-Toxic Based on components(s)

Skin Irritation Draize <2.0 / 8.0, nonirritating [Rabbit, 24 HOUR(S)]

OSHA: Non-Irritating Based on components(s)

Carcinogenicity Classification**Hydraulic Oil**

NTP: No IARC: Not Reviewed ACGIH: No OSHA: No

SECTION 12 ECOLOGICAL INFORMATION

Environmental Impact Summary:

There is no ecological data available for this product. However, this product is an oil. It is persistent and does not readily biodegrade. However, it does not bioaccumulate.

SECTION 13 DISPOSAL CONSIDERATIONS

RCRA Information:

Under RCRA, it is the responsibility of the user of the material to determine, at the time of the disposal, whether the material meets RCRA criteria for hazardous waste. This is because material uses, transformations, mixtures, processes, etc. may affect the classification. Refer to the latest EPA, state and local regulations regarding proper disposal.

SECTION 14 TRANSPORT INFORMATION

US Department of Transportation Classification

This material is not subject to DOT regulations under 49 CFR Parts 171-180.

Oil: This product is an oil under 49CFR (DOT) Part 130. If shipped by rail or highway in a tank with a capacity of 3500 gallons or more, it is subject to these requirements. Mixtures or solutions containing 10% or more of this product may also be subject to this rule.

International Air Transport Association

Not regulated under IATA rules.

International Maritime Organization Classification

Not regulated under International Maritime Organization rules.

SECTION 15 REGULATORY INFORMATION

FEDERAL REGULATORY STATUS

OSHA Classification:

Product is hazardous according to the OSHA Hazard Communication Standard, 29 CFR 19.10.1200, because it carries the occupational exposure limit for mineral oil mist.

Ozone Depleting Substances (40 CFR 82 Clean Air Act):

This material does not contain nor was it directly manufactured with any Class I or Class II ozone depleting substances.

Superfund Amendment & Reauthorization Act (SARA) Title III:

There are no components in this product on the SARA 302 list.

SARA Hazard Categories (311/312):

Immediate Health:NO Delayed Health:NO Fire:NO Pressure:NO
Reactivity:NO

SARA Toxic Release Inventory (TRI) (313):

There are no components in this product on the SARA 313 list.

Toxic Substances Control Act (TSCA) Status:

All component(s) of this material is(are) listed on the EPA/TSCA Inventory of Chemical Substances.

Other Chemical Inventories:

Component(s) of this material is (are) listed on the Australian AICS, Canadian DSL, Chinese Inventory, European EINECS, Korean Inventory, Philippines PICCS

State Regulation

The following chemicals are specifically listed by individual states; other product specific health and safety data in other sections of the MSDS may also be applicable for state requirements. For details on your regulatory requirements you should contact the appropriate agency in your state.

SECTION 16 OTHER INFORMATION

HMIS Rating (Health, Fire, Reactivity): 0, 1, 0

Revision#: 0

Revision Date: 01/31/2001

Revisions since last change (discussion): This Material Safety Data Sheet (MSDS) has been newly created to fully comply with the guidance contained in the ANSI MSDS standard (ANSI Z400.1-1998). We encourage you to take the opportunity to read the MSDS and review the information contained therein.

SECTION 17 LABEL INFORMATION

READ AND UNDERSTAND MATERIAL SAFETY DATA SHEET BEFORE HANDLING OR DISPOSING OF PRODUCT. THIS LABEL COMPLIES WITH THE REQUIREMENTS OF THE OSHA HAZARD COMMUNICATION STANDARD (29 CFR 1910.1200) FOR USE IN THE WORKPLACE. THIS LABEL IS NOT INTENDED TO BE USED WITH PACKAGING INTENDED FOR SALE TO CONSUMERS AND MAY NOT CONFORM WITH THE REQUIREMENTS OF THE CONSUMER PRODUCT SAFETY ACT OR OTHER RELATED REGULATORY REQUIREMENTS.

PRODUCT CODES: 65500

Shell AW Hydraulic Oil 32

ATTENTION!

PROLONGED OR REPEATED SKIN CONTACT MAY CAUSE OIL ACNE OR DERMATITIS.
HIGH-PRESSURE INJECTION UNDER SKIN MAY CAUSE SERIOUS DAMAGE.

Precautionary Measures:

Avoid prolonged or repeated contact with eyes, skin and clothing. Wash thoroughly after handling.

FIRST AID

Inhalation: Remove victim to fresh air and provide oxygen if breathing is difficult. Get medical attention.

Skin Contact: Remove contaminated clothing and shoes and wipe excess from skin. Flush skin with water, then wash with soap and water. If irritation occurs, get medical attention. Do not reuse clothing until cleaned. If material is injected under the skin, transport to the nearest medical facility for additional treatment.

Eye Contact: Flush with water. If irritation occurs, get medical attention.

Ingestion: Do not induce vomiting. In general, no treatment is necessary unless large quantities of product are ingested. However, get medical attention.

FIRE

In case of fire, Use water fog, 'alcohol foam', dry chemical or carbon dioxide (CO2) to extinguish flames. Do not use a direct stream of water. Material will float and can be re-ignited on surface of water.

SPILL OR LEAK

Dike and contain spill.

FOR LARGE SPILLS: Remove with vacuum truck or pump to storage/salvage vessels.

FOR SMALL SPILLS: Soak up residue with an absorbent such as clay, sand or other suitable material. Place in non-leaking container and seal tightly for proper disposal.

CONTAINS: Petroleum distillates, Mixture; Proprietary Additives, Proprietary

NFPA Rating (Health, Fire, Reactivity): 0, 1, 0

HMIS Rating (Health, Fire, Reactivity): 0, 1, 0

TRANSPORTATION

US Department of Transportation Classification

This material is not subject to DOT regulations under 49 CFR Parts 171-180.

Oil: This product is an oil under 49CFR (DOT) Part 130. If shipped by rail or highway in a tank with a capacity of 3500 gallons or more, it is subject to these requirements. Mixtures or solutions containing 10% or more of this product may also be subject to this rule.

CAUTION: Misuse of empty containers can be hazardous. Empty containers can be hazardous if used to store toxic, flammable, or reactive materials. Cutting or welding of empty containers might cause fire, explosion or toxic fumes from residues. Do not pressurize or expose to open flames or heat. Keep container closed and drum bungs in place.

Name and Address

Equilon Enterprises LLC
P. O. Box 4453
Houston, TX 77210-4453

TRANSPORTATION EMERGENCY CHEMTEL (877) 276-7283

HEALTH EMERGENCY CHEMTEL (877) 276-7283

ADMINISTRATIVE INFORMATION

COMPANY ADDRESS: Equilon Enterprises LLC, P. O. Box 4453, Houston, TX.
77210-4453

Company Product Stewardship & Regulatory Compliance Contact: Timothy W Childs
Phone Number: (281) 874-7708

MSDS FAX-BACK Phone Number: (877) 276-7285

THE INFORMATION CONTAINED IN THIS DATA SHEET IS BASED ON THE DATA AVAILABLE TO US AT THIS TIME, AND IS BELIEVED TO BE ACCURATE BASED UPON THAT DATA. IT IS PROVIDED INDEPENDENTLY OF ANY SALE OF THE PRODUCT, FOR PURPOSE OF HAZARD COMMUNICATION. IT IS NOT INTENDED TO CONSTITUTE PRODUCT PERFORMANCE INFORMATION, AND NO EXPRESS OR IMPLIED WARRANTY OF ANY KIND IS MADE WITH RESPECT TO THE PRODUCT, UNDERLYING DATA OR THE INFORMATION CONTAINED HEREIN. YOU ARE URGED TO OBTAIN DATA SHEETS FOR ALL PRODUCTS YOU BUY, PROCESS, USE OR DISTRIBUTE, AND ARE ENCOURAGED TO ADVISE THOSE WHO MAY COME IN CONTACT WITH SUCH PRODUCTS OF THE INFORMATION CONTAINED HEREIN.

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34565-11719-100R-2001-01-23 17:30:17



SHELL OMALA[®] FLUIDS HD

Premium quality synthetic gear oils for severe operating conditions

Product Description

Shell Omala[®] Fluids HD are produced using synthetic hydrocarbon base fluids. These base fluids, along with carefully selected additives, offer performance much superior to that of gear fluids blended from conventional petroleum oils.

Applications

- industrial gear reducer systems that encounter severe operating conditions, such as high temperatures, high loading and wide temperature variations
- "sealed-for-life" systems
- ANSI/AGMA 9005-D94:

ISO VG 220	AGMA 5EP
ISO VG 320	AGMA 6EP
ISO VG 460	AGMA 7EP

Features/Benefits

- significantly improved load carrying ability
- excellent rust and corrosion protection
- excellent oxidation and thermal stability
- high viscosity index
- extremely low pour points
- extended lubricant life
- improved system cleanliness
- low maintenance costs
- compatibility with the petroleum oils and elastomers, gaskets, seals and paints used in systems designed for conventional petroleum oils

Typical Properties of Shell Omala Fluids HD

	Test Method	ISO Viscosity Grade			
		150	220	320	460
Product Code		65569	65131	65132	65133
Gravity, °API	D 1298	36.5	35.7	34.4	33.7
Viscosity:					
@ 40°C, cSt	D 445	145	213	304	441
@ 100°C, cSt	D 445	19.3	26.5	34.6	46.1
@ 100°F, SUS	(calc)	754	1106	1632	2051
@ 210°F, SUS	(calc)	98.5	129	171	196
Viscosity Index	D 2270	154	151	159	163
Flash Point, COC, °F	D 92	500	505	510	510
Pour Point, °F	D 97	-45	-40	-40	-35
Rust Test	D 665A D 665B	Pass Pass	Pass Pass	Pass Pass	Pass Pass
Copper Corrosion	D 130	1b	1b	1b	1b
FZG, final stages, pass	DIN 51354	13	13	13	13
Timken OK Load, lbs, pass	D 2782	>80	>80	>80	>80
Four-Ball, Wear Scar Diameter, mm	D 2266	0.34	0.37	0.37	0.37
Four-Ball, Weld Load, kgf	D 2783	250	230	235	240

Handling & Safety Information

For information on the safe handling and use of this product, refer to its Material Safety Data Sheet at <http://www.equivashellmsds.com>. For more information and availability, call **1+800-782-7852** or World Wide Web: <http://www.shell-lubricants.com/>.


[Print](#) [Close](#)

Shell Lubricants in the USA

MATERIAL SAFETY DATA SHEET
EQUILON MSDS: 67554E-03 01/04/99

OMALA(R) HD 320

TELEPHONE NUMBER:

24 HOUR EMERGENCY ASSISTANCE

EQUIVA SERVICES: 877-276-7283

CHEMTREC: 800-424-9300

GENERAL MSDS ASSISTANCE

877-276-7285

NAME AND ADDRESS

EQUILON ENTERPRISES LLC

PRODUCT STEWARDSHIP

P.O. BOX 674414

HOUSTON, TX 77267-4414

SECTION I

NAME

PRODUCT: OMALA(R) HD 320

CHEM NAME: MIXTURE (SEE SECTION II-A)

CHEM FAMILY: SYNTHETIC HYDROCARBON

SHELL CODE: 65132

HEALTH HAZARD: 1 FIRE HAZARD: 1 REACTIVITY: 0

SECTION II-A

PRODUCT/INGREDIENT

NO.	COMPOSITION	CAS NO.	PERCENT
---	-----	-----	-----
P	OMALA HD 320		
1	SYNTHETIC HYDROCARBON BASE OIL	68037-01-4	100
NFPA HAZARD RATING: HEALTH 0 FIRE 1 REACTIVITY 0			

SECTION II-B

ACUTE TOXICITY DATA

NO.	ACUTE ORAL LD50	ACUTE DERMAL LD50	ACUTE INHALATION LC50
---	-----	-----	-----
P	NOT AVAILABLE	NOT AVAILABLE	NOT AVAILABLE

SECTION III

HEALTH INFORMATION

THE HEALTH EFFECTS NOTED BELOW ARE CONSISTENT WITH REQUIREMENTS UNDER THE OSHA HAZARD COMMUNICATION STANDARD (29 CFR 1910.1200).

EYE CONTACT: BASED ON COMPONENT INFORMATION, THIS PRODUCT IS EXPECTED TO BE NO MORE THAN MINIMALLY IRRITATING TO THE EYES.

SKIN CONTACT: BASED ON COMPONENT INFORMATION, THIS PRODUCT IS EXPECTED TO BE NO MORE THAN MINIMALLY IRRITATING TO THE SKIN. PROLONGED OR REPEATED CONTACT MAY CAUSE VARIOUS SKIN DISORDERS SUCH AS DERMATITIS, FOLLICULITIS OR OIL ACNE.

INHALATION: INHALATION OF VAPORS (GENERATED AT HIGH TEMPERATURES ONLY) OR OIL MISTS MAY BE MILDLY IRRITATING TO THE NOSE, THROAT AND RESPIRATORY TRACT.

INGESTION: BASED ON COMPONENT INFORMATION, PRODUCT IS GENERALLY CONSIDERED TO HAVE A LOW ORDER OF ACUTE ORAL TOXICITY.

SIGNS AND SYMPTOMS: IRRITATION AS NOTED ABOVE.

AGGRAVATED MEDICAL CONDITIONS:

NONE IDENTIFIED.

OTHER HEALTH EFFECTS:

THIS PRODUCT AND ITS COMPONENTS ARE NOT CLASSIFIED AS CARCINOGENS BY INTERNATIONAL AGENCY FOR RESEARCH ON CANCER (IARC), NATIONAL TOXICOLOGY PROGRAM (NTP) OR OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA).

SECTION IV

OCCUPATIONAL EXPOSURE LIMITS

COMP NO. PEL/TWA	OSHA PEL/CEILING	TLV/TWA	ACGIH TLV/STEL	OTHER
---	-----	-----	-----	-----
P NONE ESTABLISHED				

SECTION V EMERGENCY AND FIRST AID PROCEDURES

EYE CONTACT: FLUSH EYES WITH PLENTY OF WATER FOR 15 MINUTES WHILE HOLDING EYELIDS OPEN. GET MEDICAL ATTENTION.

SKIN CONTACT: REMOVE CONTAMINATED CLOTHING/SHOES AND WIPE EXCESS FROM SKIN. FLUSH SKIN WITH WATER. FOLLOW BY WASHING WITH SOAP AND WATER. IF IRRITATION OCCURS, GET MEDICAL ATTENTION. DO NOT REUSE CLOTHING UNTIL CLEANED.

INHALATION: REMOVE VICTIM TO FRESH AIR AND PROVIDE OXYGEN IF BREATHING IS DIFFICULT. GET MEDICAL ATTENTION.

INGESTION: DO NOT INDUCE VOMITING. IN GENERAL, NO TREATMENT IS NECESSARY UNLESS LARGE QUANTITIES OF PRODUCT ARE INGESTED. HOWEVER, GET MEDICAL ADVICE.*

NOTE TO PHYSICIAN: *IN GENERAL, EMESIS INDUCTION IS UNNECESSARY IN HIGH VISCOSITY, LOW VOLATILITY PRODUCTS, I.E., MOST OILS AND GREASES.

SECTION VI SUPPLEMENTAL HEALTH INFORMATION

NONE IDENTIFIED.

SECTION VII PHYSICAL DATA

BOILING POINT (DEG F): N/AV	SPECIFIC GRAVITY (H2O = 1): 0.8493	VAPOR PRESSURE (MM HG): <0.1
MELTING POINT (DEG F): -50 (POUR POINT)	SOLUBILITY IN WATER: NEGLEGIBLE	VAPOR DENSITY (AIR = 1): N/AV
		VISCOSITY: 296 (CS @ 104 DEG F)

EVAPORATION RATE (NORMAL BUTYL ACETATE = 1):N/AV

APPEARANCE AND ODOR:PALE YELLOW LIQUID, MILD ODOR

PHYS/CHEM PROPERTIES: SEE ABOVE FOR DETAILS

SECTION VIII FIRE AND EXPLOSION HAZARDS

FLASH POINT AND METHOD: 515 DEG F (COC)

FLAMMABLE LIMITS/PERCENT VOLUME IN AIR: LOWER: N/AV HIGHER: N/AV

EXTINGUISHING MEDIA:
USE WATER FOG, FOAM, DRY CHEMICAL OR CO2. DO NOT USE A DIRECT STREAM OF WATER. PRODUCT WILL FLOAT AND CAN BE REIGNITED ON SURFACE OF WATER.

SPECIAL FIRE FIGHTING PROCEDURES AND PRECAUTIONS:
MATERIAL WILL NOT BURN UNLESS PREHEATED. DO NOT ENTER CONFINED FIRE SPACE WITHOUT FULL BUNKER GEAR (HELMET WITH FACE SHIELD, BUNKER COATS, GLOVES AND RUBBER BOOTS), INCLUDING A POSITIVE PRESSURE NIOSH APPROVED SELF-CONTAINED BREATHING APPARATUS. COOL FIRE EXPOSED CONTAINERS WITH WATER.

UNUSUAL FIRE AND EXPLOSION HAZARDS:
NONE IDENTIFIED

SECTION IX REACTIVITY

STABILITY: STABLE HAZARDOUS POLYMERIZATION WILL NOT OCCUR

CONDITIONS AND MATERIALS TO AVOID:
AVOID HEAT, FLAME AND CONTACT WITH STRONG OXIDIZING AGENTS.

HAZARDOUS DECOMPOSITION PRODUCTS:
THERMAL DECOMPOSITION PRODUCTS ARE HIGHLY DEPENDENT ON THE COMBUSTION CONDITIONS. A COMPLEX MIXTURE OF AIRBORNE SOLID, LIQUID, PARTICULATES AND GASES WILL EVOLVE WHEN THIS MATERIAL UNDERGOES PYROLYSIS OR COMBUSTION.

CARBON MONOXIDE AND OTHER UNIDENTIFIED ORGANIC COMPOUNDS MAY BE FORMED UPON COMBUSTION.

SECTION X	EMPLOYEE PROTECTION
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RESPIRATORY PROTECTION:

NO EXPOSURE LIMITS HAVE BEEN ESTABLISHED FOR SYNTHETIC OILS. RESPIRATOR USE IS RECOMMENDED IF MISTS OR VAPORS, GENERATED BY HIGH TEMPERATURES, COULD BE PRESENT. USE A NIOSH APPROVED AIR-PURIFYING RESPIRATOR FOR ORGANIC VAPORS AND PARTICULATES OR ATMOSPHERE-SUPPLYING RESPIRATOR.

PROTECTIVE CLOTHING

AVOID CONTACT WITH EYES. WEAR SAFETY GLASSES OR GOGGLES AS APPROPRIATE. AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR CHEMICAL RESISTANT GLOVES AND OTHER CLOTHING AS REQUIRED TO MINIMIZE CONTACT.

ADDITIONAL PROTECTIVE MEASURES:

NONE IDENTIFIED

SECTION XI	ENVIRONMENTAL PROTECTION
------------	--------------------------

SPILL OR LEAK PROCEDURES:

MAY BURN ALTHOUGH NOT READILY IGNITABLE. USE CAUTIOUS JUDGMENT WHEN CLEANING UP LARGE SPILLS. *** LARGE SPILLS *** WEAR RESPIRATOR AND PROTECTIVE CLOTHING AS APPROPRIATE. SHUT OFF SOURCE OF LEAK IF SAFE TO DO SO. DIKE AND CONTAIN. REMOVE WITH VACUUM TRUCKS OR PUMP TO STORAGE/SALVAGE VESSELS. SOAK UP RESIDUE WITH AN ABSORBENT SUCH AS CLAY, SAND OR OTHER SUITABLE MATERIAL; DISPOSE OF PROPERLY. FLUSH AREA WITH WATER TO REMOVE TRACE RESIDUE. *** SMALL SPILLS *** TAKE UP WITH AN ABSORBENT MATERIAL AND DISPOSE OF PROPERLY.

SECTION XII	SPECIAL PRECAUTIONS
-------------	---------------------

WASH WITH SOAP AND WATER BEFORE EATING, DRINKING, SMOKING, APPLYING COSMETICS, OR USING TOILET FACILITIES. LAUNDRY CONTAMINATED CLOTHING BEFORE REUSE. STORE IN A COOL, DRY PLACE WITH ADEQUATE VENTILATION. KEEP AWAY FROM OPEN FLAMES AND HIGH TEMPERATURES.

SECTION XIII	TRANSPORTATION REQUIREMENTS
--------------	-----------------------------

DEPARTMENT OF TRANSPORTATION CLASSIFICATION:

NOT HAZARDOUS BY D.O.T. REGULATIONS

DOT PROPER SHIPPING NAME: NOT APPLICABLE

OTHER REQUIREMENTS: NOT APPLICABLE

SECTION XIV	OTHER REGULATORY CONTROLS
-------------	---------------------------

THE COMPONENTS OF THIS PRODUCT ARE LISTED ON THE EPA/TSCA INVENTORY OF CHEMICAL SUBSTANCES.

PROTECTION OF STRATOSPHERIC OZONE (PURSUANT TO SECTION 611 OF THE CLEAN AIR ACT AMENDMENTS OF 1990): PER 40 CFR PART 82, THIS PRODUCT DOES NOT CONTAIN NOR WAS IT DIRECTLY MANUFACTURED WITH ANY CLASS I OR CLASS II OZONE DEPLETING SUBSTANCES.

IN ACCORDANCE WITH SARA TITLE III, SECTION 313, THE ATTACHED ENVIRONMENTAL DATA SHEET (EDS) SHOULD ALWAYS BE COPIED AND SENT WITH THE MSDS.

SECTION XV	STATE REGULATORY INFORMATION
------------	------------------------------

STATE LISTED COMPONENT	CAS NO	PERCENT	STATE CODE
------------------------	--------	---------	------------

BASED ON INFORMATION AVAILABLE, THIS PRODUCT DOES NOT CONTAIN ANY CHEMICAL SUBSTANCE REGULATED BY A SPECIFIC STATE LIST.

SECTION XVI	SPECIAL NOTES
-------------	---------------

PRODUCT NAME CHANGE; FORMERLY HYPERIA(R) S 320.

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ENVIRONMENTAL DATA SHEET
EQUILON EDS: 67554E

OMALA(R) HD 320

TELEPHONE NUMBER:

24 HOUR EMERGENCY ASSISTANCE

EQUIVA SERVICES: 877-276-7283

CHEMTREC: 800-424-9300

GENERAL MSDS ASSISTANCE

877-276-7285

NAME AND ADDRESS

EQUILON ENTERPRISES

PRODUCT STEWARDSHIP

P.O. BOX 674414

HOUSTON, TX 77267-4414

PRODUCT CODE: 65132

SECTION I PRODUCT COMPOSITION

NO.	COMPOSITION	CAS	PERCENT
P	OMALA HD 320		
1	SYNTHETIC HYDROCARBON BASE OIL	68037-01-4	100

SECTION II SARA TITLE III INFORMATION

NO.	EHS RQ (*1)	EHS TPQ (*2)	SEC-313 (*3)	313 CATEGORY (*4)	311/312 CATEGORY (*5)
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BASED ON THE DATA AVAILABLE, THIS PRODUCT IS NOT REGULATED BY SARA, TITLE III.

*1 = REPORTABLE QUANTITY OF EXTREMELY HAZARDOUS SUBSTANCE, SEC 302

*2 = THRESHOLD PLANNING QUANTITY, EXTREMELY HAZARDOUS SUBSTANCE, SEC 302

*3 = TOXIC CHEMICAL, SEC 313

*4 = CATEGORY AS REQUIRED BY SEC 313 (40 CFR 372.65 C), MUST BE USED ON TOXIC RELEASE INVENTORY FORM

*5 = CATEGORY (FOR AGGREGATE REPORTING REQUIREMENTS UNDER SARA 311, 312)

HEALTH: H-1 = IMMEDIATE (ACUTE) HEALTH HAZARD

H-2 = DELAYED (CHRONIC) HEALTH HAZARD

PHYSICAL: P-3 = FIRE HAZARD

P-4 = SUDDEN RELEASE OF PRESSURE HAZARD

P-5 = REACTIVE HAZARD

SECTION III ENVIRONMENTAL RELEASE INFORMATION

THIS PRODUCT IS NOT COVERED BY EPA'S COMPREHENSIVE ENVIRONMENTAL RESPONSE, COMPENSATION AND LIABILITY ACT (CERCLA) PETROLEUM EXCLUSION, HOWEVER THERE ARE NO SUBSTANCES IN THIS PRODUCT THAT HAVE REPORTABLE QUANTITIES (RQ). UNDER EPA'S CLEAN WATER ACT (CWA), THIS PRODUCT IS CONSIDERED AN OIL, AND, AS SUCH, SPILLS INTO OR LEADING INTO SURFACE WATERS THAT CAUSE A SHEEN MUST BE REPORTED TO THE NATIONAL RESPONSE CENTER, 800-424-8802. THIS PRODUCT IS AN OIL UNDER 49 CFR (DOT) PART 130. IF SHIPPED BY RAIL OR HIGHWAY IN A TANK WITH A CAPACITY OF 3,500 GALLONS OR MORE, IT IS SUBJECT TO THE REQUIREMENTS OF PART 130. MIXTURE SOLUTIONS IN WHICH THIS PRODUCT IS PRESENT AT 10% OR MORE MAY ALSO BE SUBJECT TO THIS RULE.

SECTION IVRCRA INFORMATION

IF THIS PRODUCT BECOMES A WASTE, IT WOULD NOT BE A HAZARDOUS WASTE BY RCRA CRITERIA (40 CFR 261). PLACE IN AN APPROPRIATE DISPOSAL FACILITY IN COMPLIANCE WITH LOCAL REGULATIONS.

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KAREN G. HAYNES

EQUIVA SERVICES LLC
P.O. BOX 674414
HOUSTON, TX 77267-4414

FOR ADDITIONAL INFORMATION ON THIS ENVIRONMENTAL DATA PLEASE CALL
(877) 276-7285

FOR EMERGENCY ASSISTANCE PLEASE CALL

EQUIVA SERVICES LLC: (877) 276-7283
CHEMTREC: (800) 424-9300

}



Shell Turbo T Oils - Premium Turbine Oils 3.03.08

Shell Turbo T Oils are mineral oil based, rust and oxidation (R&O) inhibited oils.

Recommended Applications:

- Steam, water and hydraulic turbines in industrial and marine use
- Stationary, frame type gas turbines requiring a mineral oil lubricant
- Circulating systems requiring R&O oils
- Hydraulic systems requiring an R&O type oil (low pressure)
- AGMA rust and oxidation inhibited gear lubricants (non-EP)
- Plain and rolling element bearings in pump crankcases
- For heavier viscosity grades see Shell Morlina Oils

Advantages:

- Outstanding oxidation resistance
- Specially processed and handled to enhance water separation and air release
- Excellent rust protection

Approvals:

- Meets Hydro Quebec PAEM 89201
- Meets Ontario Hydro M-332M-B5
- Meets General Electric GEK-32568C
- Meets Westinghouse 551125 AC
- Shell Turbo T32 meets ASEA Brown Boveri (ABB) spec VT13200-3/1994/03
- Meet Cincinnati Milacron "Rust and Oxidation Inhibited Hydraulic Oils"
 - CM spec P-38 Shell Turbo T 32
 - CM spec P-54 Shell Turbo T 68
 - CM spec P-55 Shell Turbo T 46
- Approved by the CFIA (Canadian Food Inspection Agency) for industrial food plant use
- Approved for the requirements of CFIA (Canadian Food Inspection Agency), Type n1, for general use; avoid food contamination during application and storage.



Properties	SHELL TURBO T OILS					
	32	46	68	78	100	
Grades						
Product Code	407-952	407-953	407-954	407-955	407-956	
Viscosity, cSt 40°C 100°C	32	46	67	78.6	99	
	5.4	6.8	8.6	9.68	11.4	
Viscosity Index	102	101	100	101	100	
Pour Point, °C	-30	-27	-27	-24	-24	
Turbine Oil Stability Test, hr.	+9,000	+6,500	+6,500	+6,500	+6,500	



SAFETY DATA SHEET

according to EEC Directives 91/155/EEC and
93/112/EEC

Product name: LGHQ 3
Last revised date: 99-03-26

Page: 1/4
Print date: 99-05-11
SDS-ID: GB/IRL/3.0

1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND COMPANY/UNDERTAKING

Product name: LGHQ 3
Application: SKF high temperature grease Container size: -
Supplier: SKF Maintenance Products B. V.
Postbus 134
NL-3640 AC Mijdrecht
The Netherlands
Tel.: +31 297 290390
Fax.: + 31 297 281156
Contact person: Magnus Rydin
Email: magnus.rydin@skf.com

2. COMPOSITION/INFORMATION ON INGREDIENTS

The product contains: mineral oil, lithium soap, and additives.

<u>Chemical name:</u>	<u>CAS-No.:</u>	<u>%:</u>	<u>Hazard classification:</u>	<u>Notes:</u>
Baseoil - unspecified	-	-	-	DMSO-content < 3%

3. HAZARDS IDENTIFICATION

Man: Prolonged or repeated contact with skin may cause redness, itching, irritation, eczema/chapping and oil acne.

Environment: Greases are generally hazardous to the environment.

4. FIRST-AID MEASURES

Inhalation: Move into fresh air and keep at rest. In case of persistent throat irritation or coughing: Seek medical attention and bring along these instructions.

Skin contact: Remove contaminated clothing immediately and rinse skin with rinsing cream. After this, apply a fatty cream. In case of rashes, wounds or other skin disorders: Seek medical attention and bring along these instructions.

Eye contact: Immediately flush with plenty of water for up to 15 minutes. Remove any contact lenses and open eyelids widely. If irritation persists: Seek medical attention and bring along these instructions.

Ingestion: Immediately rinse mouth and drink plenty of water. Keep person under observation. If uncomfortable: Seek hospital and bring along these instructions. Do not induce vomiting. If vomiting occurs, keep head low.

5. FIRE-FIGHTING MEASURES

Extinguishing media: Extinguish with foam, carbon dioxide, dry powder or water fog. Do not use water jet as an extinguisher, as this will spread the fire.

Specific hazards: By heating and fire, harmful vapours may be formed.

SAFETY DATA SHEET

according to EEC Directives 91/155/EEC and 93/112/EEC

Product name: LGHQ 3

Page: 2/ 4

Last revised date: 99-03-26

Print date: 99-05-11

SDS-ID: GB/IRL/3.0

6. ACCIDENTAL RELEASE MEASURES

Personal precautions: Avoid contact with skin and eyes. In case of spills, beware of slippery floors and surfaces. For personal protection, see point 8.

Environmental precautions: Avoid discharge into drains, water courses or onto the ground.

Methods for cleaning up: Absorb spillage with oil-absorbing material. Clean contaminated area with oil-removing material.
For waste disposal, see point 13.

7. HANDLING AND STORAGE

Safe handling advice: Avoid prolonged and repeated contact. Observe good industrial hygiene practices.

Technical measures: Work practice should minimise contact.

Technical precautions: When working with heated grease, mechanical ventilation may be required.

Technical measures for safe storage: No special precautions.

Storage conditions: Store in a cool and well-ventilated place.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Engineering measures: Provide adequate ventilation. Observe Occupational Exposure Limits and minimize the risk of inhalation of vapours from heated grease. Provide access to washing facilities incl. soap, skin cleanser and fatty cream.

<u>Chemical name:</u>	<u>Exposure limits:</u>	<u>Type:</u>	<u>Notes:</u>	<u>References:</u>
Oil mist, mineral	- 5mg/m ³	OES	-	EH 40/99

Respiratory equipment: In case of inadequate ventilation or when the product is heated, use suitable respiratory equipment with gas filter (type A2).

Hand protection: Wear protective gloves. Suitable gloves can be recommended by the glove supplier.

Eye protection: Risk of contact: Wear goggles/face shield.

Skin protection: Wear apron or protective clothing in case of splashes.

Hygiene measures: Remove contaminated clothing and wash the skin thoroughly with soap and water after work. Wash contaminated clothing before reuse.

SAFETY DATA SHEET

according to EEC Directives 91/155/EEC and 93/112/EEC

Product name: LGHQ 3

Page: 3/ 4

Last revised date: 99-03-26

Print date: 99-05-11

SDS-ID: GB/IRL/3.0

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: blue paste
Odour: mild
pH: not relevant
Melting point: 300°C
Boiling point: not available
Flash point: >200°C
Explosive properties: not available
Relative density: 0,92
Solubility: insoluble in water

10. STABILITY AND REACTIVITY

Stability: Stable under normal temperature conditions.

Conditions/
materials to avoid: Avoid exposing to heat and contact with strong oxidizing substances.

Hazardous
decomposition products: By heating and fire, harmful vapours may be formed.

11. TOXICOLOGICAL INFORMATION

Inhalation: Not relevant at normal room temperatures. Inhalation of oil mist or vapours formed during heating of the product will irritate the respiratory system and provoke coughing.

Skin contact: Prolonged or frequent contact may cause redness, itching, irritation, eczema/chaps and oil acne.

Eye contact: Direct contact may irritate.

Ingestion: Not likely, due to the form of the product. However, ingestion may cause irritation and malaise.

12. ECOLOGICAL INFORMATION

Mobility: The product is insoluble in water.

Degradability: The product is slowly degradable.

Ecotoxicity: The product may be hazardous to the environment.

13. DISPOSAL CONSIDERATIONS

Dispose of waste and residues according to agreement with local authorities.

Waste from residues: EWC-code: 13 02 02

Contaminated packaging: Dispose of contaminated packings as residue.

SAFETY DATA SHEET

according to EEC Directives 91/155/EEC and 93/112/EEC

Product name: LGHQ 3

Page: 4/ 4

Last revised date: 99-03-26

Print date: 99-05-11

SDS-ID: GB/IRL/3.0

14. TRANSPORT INFORMATION

The product is not covered by international regulation on the transport of dangerous goods (IMDG, IATA, ADR/RID).

15. REGULATORY INFORMATION

Labelling: Based on information from the manufacturer regarding the chemical composition of the product, an assessment by the Danish Toxicology Centre has shown that according to EEC Directives 67/548/EEC and 88/379/EEC, the product is not liable to classification and labelling.

National regulation: EC Directives 67/548/EEC on dangerous substances and 88/379/EEC on dangerous preparations (up to and inclusive of the 23rd adaptation).
The Chemicals (Hazard Information and Packaging for Supply) Regulations 1994 (SI 1994/3247), with Amendment Regulations 1996 (SI 1996/1092), 1997 (SI 1997/1460), 1998 (SI 1998/106) and 1999 (SI 1999/197).
Occupational Exposure Limits EH 40/98 from the Health and Safety Executive.
SI 819: The Collection and Disposal of Waste Regulations 1988.

16. OTHER INFORMATION

The user must be instructed in the proper work procedure and be familiar with the contents of these instructions.

The information on this data sheet represents our current data and is accurate to the best of our knowledge as to the proper handling of this product under normal conditions and in accordance with the application specified on the packaging and/or technical guidance literature. Any other use of the product which involves using the product in combination with any other product or any other process is the responsibility of the user.

Made by DANISH TOXICOLOGY CENTRE, Kogle Allé 2, DK-2970 Hørsholm, Denmark. Telephone +45 45 76 20 55 Telefax +45 45 76 24 55

Customer No: 20917

Filing No: -5994

MATERIAL SAFETY DATA SHEET

P/N: MSDS-7200015

REV. A

REF: P/N 72-00015 &
72-00016

Page 2 of 7

ISSUE DATE: 3/22/99

WRITTEN BY: P. Falcone

SUPERSEDES: Rev. none, June 1993

SECTION 1: Product Information

Product Description: Conductive Silver Filled Silicone Grease.**Company/undertaking identification:**

Tecknit Europe Ltd, Swingbridge Road, Grantham, Lincs, NG31 7XT

Telephone No: 44 1476 590600 Fax No: 44 1476 591600

Dept Responsible for information (technical information) Telephone No: 44 1476 590600**Emergency Information**

Telephone No: 44 1476 590600

Emergency Number

Telephone No: 44 1476 590600

Product Information Number: (908)-272-5500**Emergency Overview:**

May cause mild eye irritation on contact. Prolonged skin contact with product may cause discoloration.

SECTION 2: Composition

<u>Ingredient</u>	<u>Cas No.</u>	<u>Wt. %</u>
dimethyl polysiloxanes	63148-62-9	15 - 25
soda lime glass	65997-17-3	22 - 28
silver	7440-22-4	50 - 55

(Note: See Section 8 for Exposure Guidelines)

SECTION 3: Hazards Identification

Route(s) of Entry: Inhalation? Yes Skin? Yes Ingestion? Yes**Health Hazards (Acute and Chronic):**

This product contains Methyl polysiloxane which can generate Formaldehyde at approximate temperature of 300°F(150°C) and above, in atmosphere which contains oxygen.

Formaldehyde is a known skin and respiratory sensitizer, eye and throat irritant, an acute toxicant, and a potential cancer hazard. Prolonged or repeated exposure to silver compounds can lead to gray/blue discoloration of skin known as argyrosis.

MATERIAL SAFETY DATA SHEET

P/N: MSDS-7200015	REF: P/N 72-00015 & 72-00016	Issue Date: 3/22/99
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SECTION 3: Hazards Identification (Continued)

Carcinogenicity: NTP? NO IARC Monographs? NO OSHA Regulated? NO

Signs and Symptoms of Exposure:

May cause mild eye irritation. Prolonged skin contact with silver compounds may discolor skin.

Medical Conditions Generally Aggravated by Exposure: None known.

SECTION 4: First Aid Measures

Ingestion: N/A

Inhalation: N/A

Skin Contact: Wash with soap and water.

Eye Contact:

In case of contact, immediately flush eyes with plenty of water for at least 15 minutes and get medical attention if irritation persists.

SECTION 5: Fire-Fighting Measures

Flash Point (Method Used): 301.6°C (575°F) PMCC

Flammability Limits: LEL N/A UEL N/A

Extinguishing Media: All standard fire fighting media

Special Fire-Fighting Procedures:

Evacuate all non-essential personnel. Self-contained breathing apparatus and appropriate protective clothing should be worn.

Unusual Fire and Explosion Hazards:

At temperature above 300°F (150°C) or in a fire this material may generate formaldehyde. Silver dust is moderately flammable when exposed to flame or chemical oxidizers.



MATERIAL SAFETY DATA SHEET

P/N: MSDS-7200015	REF: P/N 72-00015 & 72-00016	Date: 3/22/99
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SECTION 6: Accidental Release Measures

Wipe, scrape or soak up in an inert material and put into an appropriate container for disposal. Wash walking surfaces with detergent and water to reduce slipping hazard.

SECTION 7: Handling and Storage

Wash hands with soap and water after handling this product. Avoid generating silver dusts. Avoid storing near strong oxidizing agents, nitric acid, acetylene and hydrogen peroxide. Store in tightly closed containers in a dry area.

SECTION 8: Exposure Controls

Exposure Guideline

<u>Ingredient</u>	<u>OSHA</u>		<u>ACGIH</u>		<u>Other</u>		<u>Units</u>
	<u>TWA</u>	<u>STEL</u>	<u>TWA</u>	<u>STEL</u>	<u>TWA</u>	<u>STEL</u>	
dimethylpolysiloxane	N/E	N/E	N/E	N/E	N/E	N/E	N/A
silver	0.01	N/E	0.1	N/E	N/E	N/E	(mg/m ³)
glass beads	15	N/E	5	N/E	N/E	N/E	(mg/m ³)

SECTION 8: Exposure Controls (Continued)

Respiratory Protection:

None normally required. However, if airborne dusts of silver are generated then a NIOSH approved dust mask or respirator should be used.

Ventilation:

General room ventilation should be sufficient but local exhaust is recommended if use of this product is likely to generate airborne silver dusts or fumes.

Protective Gloves: Vinyl or rubber gloves recommended.

Eye Protection: Safety glasses

Other Protective Clothing or Equipment: None

SECTION 9: Reactivity

Stability: Stable: X
Unstable:



MATERIAL SAFETY DATA SHEET

P/N: MSDS-7200015	REF: P/N 72-00015 & 72-00016	Date: 3/22/99
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SECTION 9: Reactivity Continued

Conditions to Avoid / Incompatibility:

Silver is incompatible with acetylene, ammonia, hydrogen peroxide, ethyl alcohol with nitric acid, oxalic acid.

Hazardous Decomposition or Byproducts:

Carbon dioxide, carbon monoxide, formaldehyde, silver and silver oxide fumes.

Hazardous Polymerization: May occur:

Will not occur: X

SECTION 10: Physical and Chemical Properties

Boiling Point: N/E

Specific Gravity: $2.5 \pm .3$ %

Vapor Pressure (mm Hg): Negligible

Melting Point: N/A

Vapor Density (air = 1): N/A

Evaporation Rate (butyl acetate = 1): < 1

Solubility in Water: Insoluble

% Volatile: N/E

VOC: N/D

Appearance and Odor: Silver/tan colored, odorless liquid.

SECTION 11: Toxicological Information

Dimethylpolysiloxane

Acute oral LD50 (mg/kg): > 40,000

Acute dermal LD50 (mg/kg): none found

Acute inhalation LC50 (mg/l): > 535 (rat)

Other: Non Irritating to rabbit skin.

Ames Test: unknown

SECTION 12: Ecological Information

No information available

SECTION 13: Disposal Considerations

Dispose of this product in accordance with local, state and federal regulations or send to a refinery for silver recovery.



MATERIAL SAFETY DATA SHEET

P/N: MSDS-7200015	REF: P/N 72-00015 & 72-00016	Date: 2/19/98
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SECTION 14: Transport Information

Proper Shipping Name: None
Technical Name: None
Hazard Class: Not DOT regulated
NA Number: None
Packing Group: None

There are no special shipping requirements for this product.

SECTION 15: Regulatory Information

SARA Title III:

Section 302 Extremely Hazardous Substances:

<u>Ingredient</u>	<u>CAS No.</u>	<u>Wt. %</u>	<u>RQ of Product</u>
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No Product ingredients listed.

Section 304 CERCLA Hazardous Substances:

<u>Ingredient</u>	<u>CAS No.</u>	<u>Wt. %</u>	<u>RQ</u>
Silver	7440-22-4	50 - 55	1000

Section 313 Supplier Notification:

This product contains the following chemical(s) subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know act of 1986 and of CFR 372.

<u>CAS No.</u>	<u>Chemical Name</u>	<u>WT. %</u>
7440-22-4	silver	45-60

This information must be included in all MSDS(s) that are copied and distributed for this material

TSCA Inventory Status:

This product is a mixture and is not listed in the TSCA inventory. However, the individual components which make up this product are listed where applicable.





MATERIAL SAFETY DATA SHEET

P/N: MSDS-7200015	REF: P/N 72-00015 & 72-00016	Date: 2/19/98
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SECTION 15: Regulatory Information Continued

State Regulatory Information:

California proposition 65:

No Product ingredients listed.

New Jersey State Compliance Information

<u>CAS No</u>	<u>Chemical Name</u>
7440-22-4	silver

Massachusetts State Compliance Information

<u>CAS No</u>	<u>Chemical Name</u>
7440-22-4	silver

Pennsylvania State Compliance Information

<u>CAS No.</u>	<u>Chemical Name</u>
7440-22-4	silver

Section 16: Other

Hazards Rating Systems:

HMIS	Health: 1	Flammability: 0	Reactivity: 1
NFPA	Health: N/E	Flammability: N/E	Reactivity: N/E

To the best of our knowledge the information contained herein is accurate. However, TECKNIT does not assume any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards which exist.

N/A - Not Applicable
N/D - Not Determined
N/E - Not Established





PneuLube SYNTHETIC OIL

PneuLube synthetic oil from Tuthill Vacuum & Blower Systems is specifically blended to tackle your toughest rotary blower applications. As a result of this specific proprietary formulation, PneuLube can be utilized in low temperature as well as high temperature ambient conditions, providing a single, all-purpose lubricant for any rotary blower application.

Extended Oil Life

The special formulation of PneuLube quadruples oil life expectancy as compared to conventional mineral oils, requiring fewer oil changes and allowing for longer maintenance intervals.

Superior Performance

PneuLube is formulated with synthesized hydrocarbon fluids and selected additives to provide superior resistance against oxidation, and maximum protection from wear, rust, corrosion and foaming. PneuLube provides unsurpassed thermal and oxidation stability as well as increased protection from viscosity loss as compared to conventional mineral oils. The high viscosity index of PneuLube provides higher viscosities and greater film strength at higher temperatures and lower viscosities at low temperatures for minimum friction and better lubrication than other synthetic blower lubricants on the market today.

Your Local Tuthill Vacuum & Blower Systems Sales Professional:

Compatibility Assured

PneuLube is compatible with essentially all seal materials, elastomers and paints, including:

- Buna-N acrylonitrile
- Neoprene
- Viton®
- Teflon®
- Polyethylene
- Polyurethane ether
- Fluorocarbon
- Polysulfate
- Polyacrylate
- Ethylene acrylic
- Epoxy
- Plastisol
- PVC
- Acrylic paint
- Lacquer

PneuLube is also fully compatible with hydrogen service applications. PneuLube does not thicken or congeal in the presence of hydrogen.

PneuLube Physical Properties:

Viscosity Grade: ISO 100

Specific Gravity at 16° C (62° F): 0.859

Viscosity at 40° C (104° F): 91.8 cSt:

Viscosity at 100° C (212° F): 13.1 cSt:

Viscosity Index: 142

Pour Point: -51° C (-60° F)

Flash Point: -246° C (475° F)

Copper Corrosion Rating: 1A

PneuLube is available in:

- Individual pints, quarts, half-gallons and gallons
- Case lots of all of the above sizes
- 5-gallon pails and 55-gallon barrels

Contact your local Tuthill Vacuum & Blower Systems Sales Professional to start using PneuLube in all of your rotary blowers today.

LEADING THE SEARCH FOR NEW SOLUTIONS



TUTHILL
Vacuum & Blower Systems



4840 West Kearney Street, P. O. Box 2877
Springfield, Missouri USA 65801-2877
Tel 417 865-8715 800 825-6937 Fax 417 865-2950

<http://pneumatics.tuthill.com>

0902

TUTHILL PNEUMATICS GROUP

MATERIAL SAFETY DATA SHEET

1. CHEMICAL PRODUCT and COMPANY IDENTIFICATION

PRODUCT NAME: PneuLube Synthetic Oil (TPG Part number 16444-100- **)

CHEMICAL FAMILY:

CHEMICAL NAME: Mixture

** Blank is placeholder for size of container:

P=pint; Q=quart; 2Q=2-quart; G=gallon;
5G=5-gallon pail; B=55-gallon barrel

MANUFACTURER:

Tuthill Pneumatics Group
4840 West Kearney Street
P. O. Box 2877
Springfield, MO 65801-2877
(417) 865-8715 (800)825-6937
Preparation Date: 06/18/2002

EMERGENCY TELEPHONE NUMBERS:

CHEMTREC: 1-800-424-9300

HAZARD RATING: NFPA HMIS

HEALTH: 0 0

FIRE: 1 0

REACTIVITY: 0 0

2. COMPOSITION/INFORMATION

ON INGREDIENTS

PneuLube Synthetic Oil
Mixed Synthetic Aliphatic Hydrocarbons
Minor Additives

GAS

Mixture
Mixture
Mixture

CONCENTRATION(%)

100
95
5

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

POTENTIAL HEALTH EFFECTS:

ROUTES OF ENTRY

INHALATION

Inhalation, skin contact/absorption, eye contact, ingestion.

Inhalation of vapors are expected to cause only mild irritation in the upper respiratory tract.

SKIN CONTACT

No more than mildly irritating to the skin. Injection of oil into skin may cause local necrosis.

EYE CONTACT

Product may cause mild irritation to the eyes.

INGESTION

Based on component information, product has a low order of acute toxicity.

CARCINOGENICITY

This product and its components are not classified as carcinogens by IARC, NTP or OSHA.

4. FIRST AID MEASURES:

EYES

Flush eyes with water. If irritation occurs, get medical attention.

SKIN

Remove contaminated clothing/shoes, wipe excess from skin. Wash with soap and water. If irritation occurs, get medical attention. If material is injected under skin, get medical attention promptly to prevent serious damage; do not wait for symptoms to develop.

INHALATION

Remove victim to fresh air and provide oxygen if breathing is difficult.

Get medical attention.

INGESTION

Do not induce vomiting. Get medical attention immediately.

PRODUCT TRADE NAME: PneuLube Synthetic Oil (TPG Part number 16444-100- **)

Page 2 of 3

** Blank is placeholder for size of container:

P=pint; Q=quart; 2Q=2-quart; G=gallon;
5G=5-gallon pail; B=55-gallon barrel

5. FIRE FIGHTING MEASURES

FLASH POINT: 375-550 Deg. F (COC)

FLAMMABLE LIMITS / % VOLUME IN AIR:

LOWER N/D

UPPER N/D

EXTINGUISHING MEDIA: Use water fog, foam, dry chemical or CO₂.

SPECIAL FIRE FIGHTING PROCEDURES AND PRECAUTIONS: Use NIOSH/MSHA approved self-contained breathing apparatus where this material is involved in a fire. Cool containers with H₂O and remove from fire exposure.

UNUSUAL FIRE HAZARD: Empty containers could contain flammable vapors.

6. ACCIDENTAL RELEASE MEASURES

SPILL OR LEAK PROCEDURES: Wear protective clothing as appropriate. Shut off source of leak if safe to do so. Dike and contain.

LARGE SPILLS: Remove with vacuum trucks or pump to storage/salvage vessels. Soak up residue with an absorbent material such as clay, sand or other suitable material; dispose of properly.

SMALL SPILLS: Take up with an absorbent material and dispose of properly.

7. HANDLING AND STORAGE

For general personal hygiene, wash hands thoroughly after handling material. Avoid contact with skin and eyes.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

EYE PROTECTION REQUIREMENTS:

Splash goggles.

SKIN PROTECTION REQUIREMENTS:

Impervious gloves and protective clothing as required to minimize skin contact.

RESPIRATOR/VENTILATION REQUIREMENTS: If overheated, use approved respiratory protective equipment. Local mechanical exhaust.

EXPOSURE LIMITS:

Not established.

9. PHYSICAL AND CHEMICAL PROPERTIES

PHYSICAL FORM: Liquid

EVAPORATION RATE N/D

WATER SOLUBILITY: Slight

ODOR Mild Gear Oil Odor

SPECIFIC GRAVITY: <1.0@60°F

PERCENT VOLATILE N/D

VAPOR PRESSURE: >1mm Hg

VAPOR DENSITY N/D

10. REACTIVITY

STABILITY:

Stable

HAZARDOUS POLYMERIZATION

Will not occur

INCOMPATIBILITIES

Strong oxidizing materials

DECOMPOSITION PRODUCTS

Burning produces oxides of carbon.

** Blank is placeholder for size of container:

P=pint; Q=quart; 2Q=2-quart; G=gallon;
5G=5-gallon pail; B=55-gallon barrel

11. TOXICOLOGICAL INFORMATION

Effects of overexposure – Low oral and dermal toxicity. Prolonged or repeated exposure may cause eye irritation, nausea and vomiting.

Toxic Substance Inventory Control Act – All components are included on the TSCA Inventory and are in compliance with the TSCA.

12. ECOLOGICAL INFORMATION

Report spills as required to appropriate authorities. U.S. Coast Guard regulations require immediate reporting of spills that could reach any waterway including intermittent dry creeks. Report spill to Coast Guard Toll Free Number (800) 424-8802. In case of accident or road spill, notify Chemtrec (800) 424-9300.

13. DISPOSAL CONSIDERATIONS

Waste Management – Dissolve waste in a solvent and dispose by supervised incineration in compliance with applicable laws and regulations.

14. TRANSPORTATION INFORMATION

D.O.T. SHIPPING NAME:	Not applicable
TECHNICAL SHIPPING NAME:	Not applicable
D.O.T. HAZARD CLASS:	Not hazardous by D.O.T. regulations.

15. REGULATORY INFORMATION

SARA SECTION 313 SUPPLIER INFORMATION

This product contains the following Toxic Chemicals subject to the reporting requirements of SECTION 313 of the EMERGENCY PLANNING and COMMUNITY RIGHT-TO-KNOW ACT of 1986 and of 40 CFR 372.

None

This information is furnished without warranty, expressed or implied, except that it is accurate to the best knowledge of TUTHILL PNEUMATICS GROUP. The data on this sheet is related only to the specific material designated herein. TUTHILL PNEUMATICS GROUP assumes no legal responsibility for use or reliance upon these data.

SECTION 14 MACHINE COMPONENT SETTING UP INSTRUCTIONS

This section of the manual describes important notes on how components and systems supplied on the machine have been calibrated and configured. The principles of operation are also explained for some items of equipment.

Not all the equipment listed in this section of the manual is applicable to this machine. The following table has been compiled by the test engineer who tested the machine to indicate the equipment fitted upon it.

Section	Equipment	Applicable to this machine
Section 14.1	ABB Load Cell System Calibration	YES
Section 14.2	Aluminium Wire Feeder System	YES
Section 14.3	Removal And Fitting Of Pipework Isolation Valves	YES
Section 14.4	Vacuum Gauge Maintenance	YES
Section 14.5	Pressure Switch Setting Adjustment	YES
Section 14.6	Machine Services Interlock Pressure Switches	YES
Section 14.7	Coating Shutter Operation	YES
Section 14.8	Diffusion Pumps	YES
Section 14.9	Chamber - Cart Alignment	YES
Section 14.10	Fine Valves (VV7) - Pneumatic	YES
Section 14.11	Inspection of High Vacuum Valve O-ring Seal	YES
Section 14.12	Cryogenic System – Cryo-Generator Units	YES
Section 14.13	Evaporator Power Supply Thyristor Units	YES
Section 14.14	Evaporator Power Supply Transformers	YES
Section 14.15	Evaporator System Busbar System	YES
Section 14.16	Winding System Drive System	YES
Section 14.17	Electrical Drawings	YES
Section 14.18	Machine Data Communication System	YES
Section 14.19	Winding Cart Movement	YES
Section 14.20	Electrical Leadthroughs	YES
Section 14.21	Vacuum "O Ring"	YES
Section 14.22	Zoning Seals	YES
Section 14.23	Bearing, Coverings & Vacuum Seal Replacement On Rollers	YES
Section 14.24	Tuning Non Contact DCM Probes	YES
Section 14.25	Installing or Removing DCM Resistance Monitor Beam	YES
Section 14.26	Removing And Refitting Non Contact DCM Probes	YES
Section 14.27	Non Contact Resistance Monitor Probe Calibration	YES
Section 14.28	Simple Troubleshooting Of DCM Beams	YES
Section 14.29	Water Supply Strainers/Filters	YES
Section 14.30	Repairing Cork Covered Rollers	YES
Section 14.31	Gravity Fill Oil Bottles	YES
Section 14.32	Mechanical Rotary Leadthroughs	YES
Section 14.33	Process Chiller System	YES
Section 14.34	Nip Roller Checks/Alignment	YES
Section 14.35	Layarm Roller Assembly Alignment	YES
Section 14.36	Lay Arm Roller Operation	YES
Section 14.37	Tendency Drive System	YES
Section 14.38	Inspections Of Electrical Panels	YES
Section 14.39	Removing Drive Belt Guard Panels	YES
Section 14.40	Timing Belt Inspection	YES
Section 14.41	Bearings	YES
Section 14.42	Vee Belts	YES
Section 14.43	Hose And Tubing Installation	YES
Section 14.44	Pneumatic Pipework	YES
Section 14.45	Taper Lock Bushes	YES
Section 14.46	Rotary Vacuum Rated Shaft Seals	YES

Section	Equipment	Applicable to this machine
Section 14.47	Unwind/Rewind Chucks and Reel Shaft System	YES
Section 14.48	Adhering Structures To Machine	YES
Section 14.49	Sight-Glass Windows	YES
Section 14.50	AC Vector Drive Motors (Winding Mechanism)	YES
Section 14.51	Siemens MICROMASTER 440 - Booster Pumps	YES
Section 14.52	Siemens Combimaster Drive - Cart Travers	YES
Section 14.53	Coating Shield - Kinertrrol Unit	YES
Section 14.54	Restoring Software	YES
Section 14.55	Gas Ballast Valve Maintenance – Rotary Pump	YES
Section 15.56	Air Admittance Valve Maintenance	YES
Section 14.57	Cable Coiler Maintenance	YES
Section 14.58	Rotary Union Removal	YES
Section 14.59	Replacing Drum Lip Seals	YES
Section 14.60	Bellows Installation	YES
Section 14.61	Evaporation Source Insulators	YES
Section 14.62	Calibration And Care Of The Touch Screen	YES
Section 14.63	Process Gas Wedge	YES
Section 14.64	Substrate Threading Device	YES
Section 14.65	Source Rotary Window	YES
Section 14.66	Installing or Removing Optical Monitor Beams	NO
Section 14.67	Optical Monitor Probe Calibration	NO
Section 14.68	Plasma Treater	YES
Section 14.69	Mass Flow Controllers (MFC's)	YES
Section 14.70	Baratron Pressure Gauge	YES
Section 14.71	Plasma System Basic Safety Interlocks Troubleshooting	YES
Section 14.72	Motorised Spreader Roller Set-Up	NO
Section 14.73	Layarm Spreader Roller Apex Position Adjustment	YES

Table 14.0 - 1 Machine Components

14.1 ABB Load Cell System Calibration

The load cell system consists of number sets of load cell transducers, cables, and leadthroughs and amplifier cards. The transducers are mounted in pairs at each end of a lightweight roller. They are wired via a vacuum leadthrough to the amplifiers, which are located on the faceplate, on the outside of the chamber.

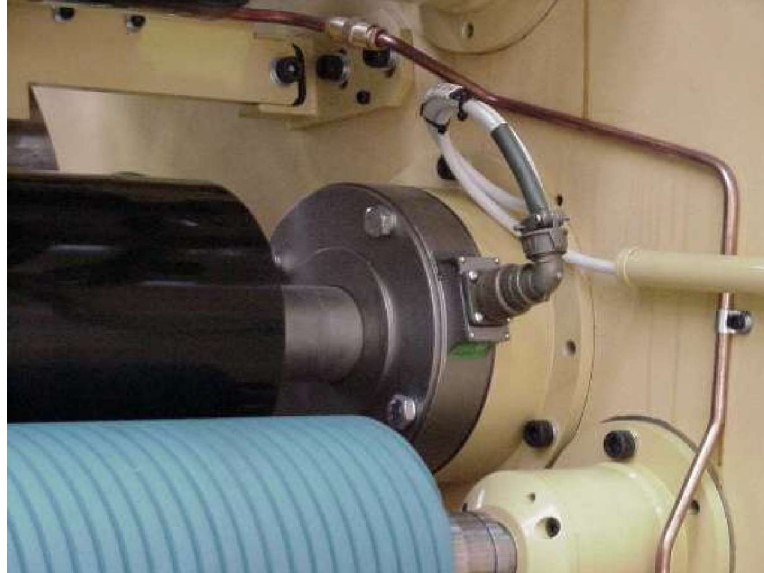


Figure 14.1 - 1 Photograph of Load Cell Transducers in a Machine

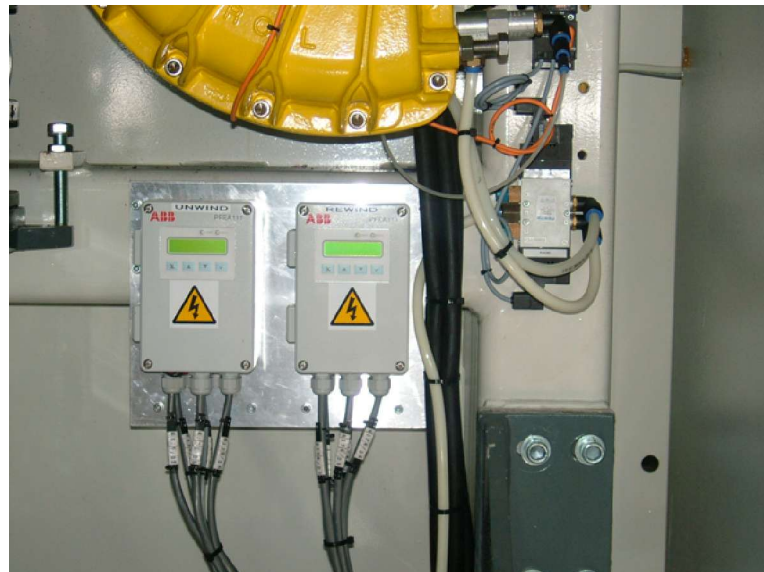


Figure 14.1 - 2 Photograph of Load Cell Amplifiers

Load Cell Adjustments

Load cells and their associated amplifiers normally remain in calibration for long periods of time. However, should the process roll condition harden or soften more than expected, whilst running at the normal tension settings, it is advisable to check the load cell system calibration.

The output signal is fed to the drive system to provide indication of the tension on the substrate and also to control the unwind and rewind drives.

Calibration

The calibration of the system generally follows the following procedure.

Choose either the unwind or rewind load cell system. The amplifiers are identified.

These instructions are a general guide and the manual supplied by the load cell manufacturer should be referred to before and during the calibration procedure.

Using Fast Set Up Feature for recalibration

Display	Button	How to use it
		Steps back to previous menu. Sometimes you must press this button two or more times to get back to the desired menu.
		Steps up in a list.
		Steps down in a list.
		Steps to next main menu.
		OK (acknowledgment) button. Confirms a selection or a parameter setting.

Figure 14.1 - 3 ABB Load Cell Amplifier Interface Panel

Locate the amplifiers on the winding cart, as shown in Figure 14.1 – 2.

Performing a Fast Setup

The fast setup guides you through a minimum number of steps to set up the tension electronics.

You are asked to answer some questions and to enter desired values. These selections and parameter settings must be made to get the tension electronics ready for measurement.

Only a limited number of selections and parameter settings are set in the fast setup. All other parameters are set as factory default values.

The fast setup can be made in two ways, depending on how the wrap gain is set.

The wrap gain can be set by selecting “HangWeight” or “EnterWrapGain”. In our case we always use “Using Hanging Weights” method.

The simplest method to set the wrap gain in this tension system is to use a known weight that loads the center of the roll with a rope that follows the web path exactly. When going over rubber rollers, the load should be spread using cardboard.

Caution !	
	When using heavy weights, take necessary precautions to support them and to minimise the possibility of damage, if they are dropped.

Below is the instruction set from the vendors manual.

For recalibration, all the following values are known.

Step 4 Two Cells, (at this stage, just press the Ok button)

Step 5 As per the transducers information, (at this stage, just press the Ok button)

Step 6 With only the dead weight load of the roller, carry this out.

Thread a flat tape around the rolls (See Figures 14.1 – 5 & 14.1 - 6) so that a spring balance or weight can be used to apply load in the same manner as the substrate. Ensure the tape is in the middle of the roller.

Step 7 The value being applied by the weights should be entered here.

Step 8 Select Voltage

Step 9 The full range value as per section 7 of this manual.

Step 10 Leave at 15 mS

Step 11 Select No

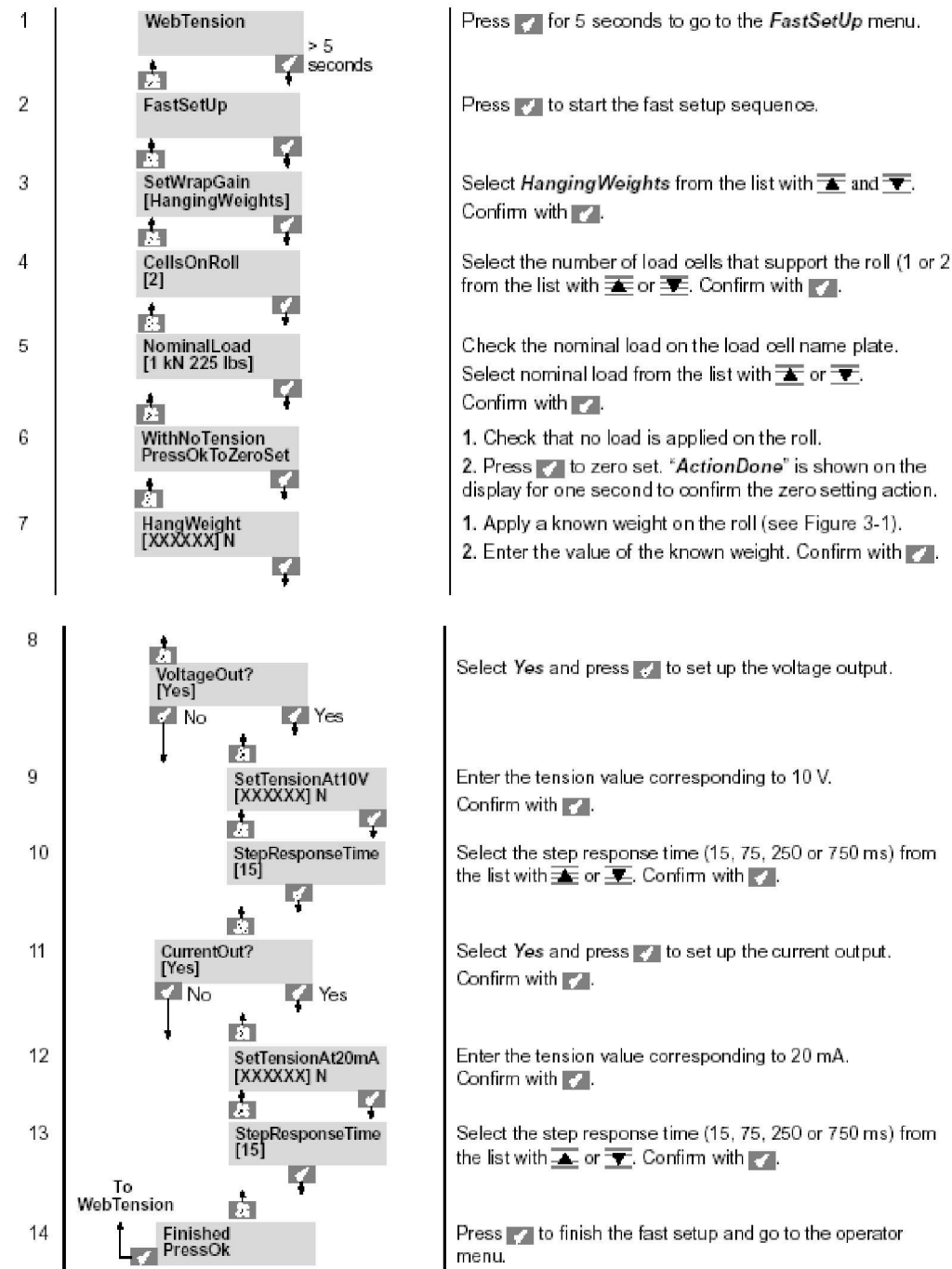


Figure 14.1 - 4 Performing a Fast Setup Procedure for ABB Load Cell System

In the case of a machine that has a different wrap angle for under/overwind web paths on the unwind, always use the underwind path as shown in Figure 14.1 - 5.

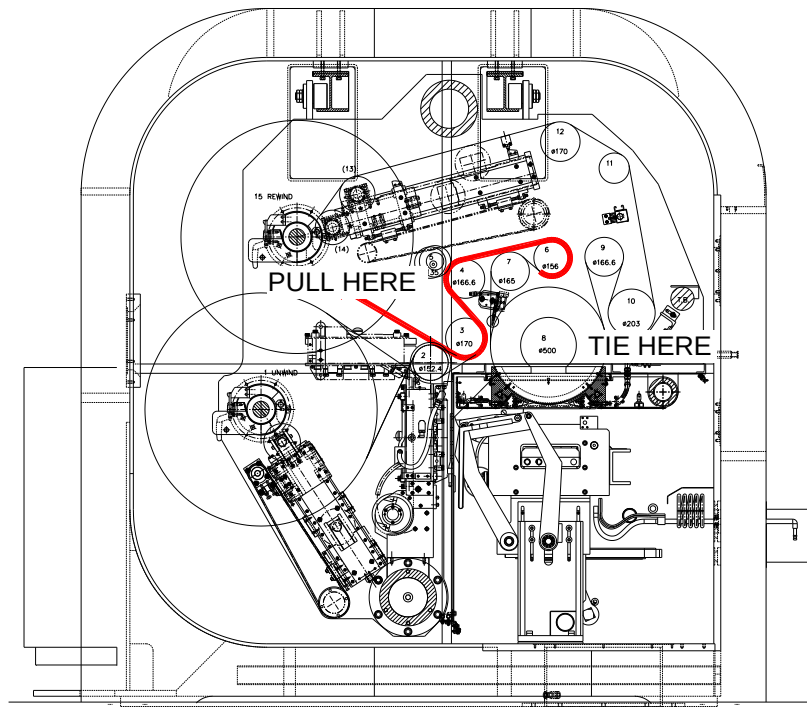


Figure 14.1 - 5 Unwind Load Cell Calibration

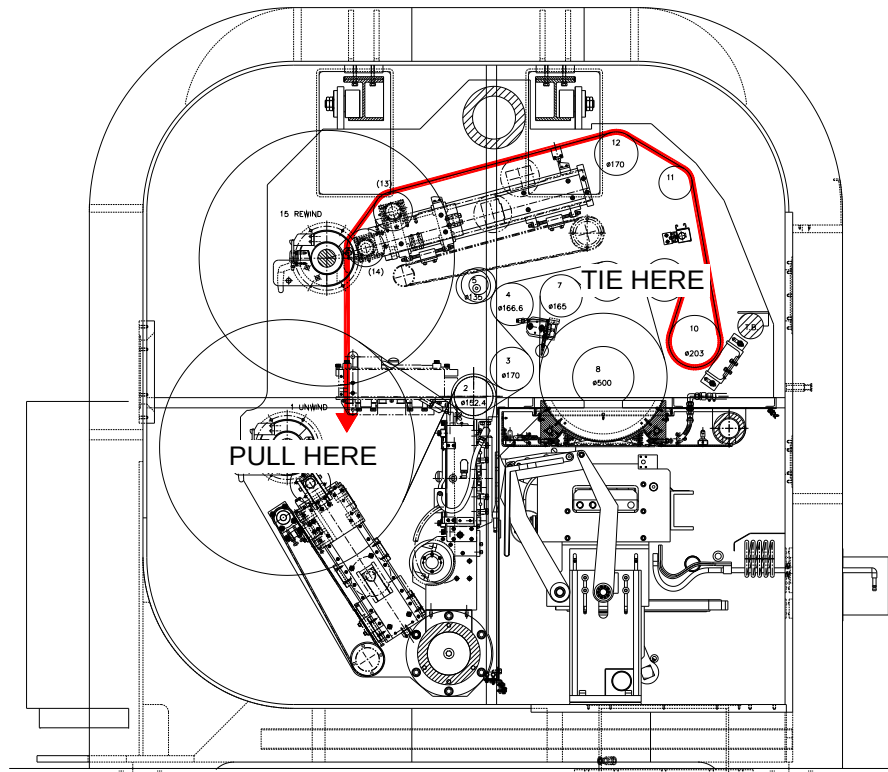


Figure 14.1 - 6 Rewind Load Cell Calibration

14.2 Aluminium Wire Feeder System

Removal and Maintaining Wire feeders

The aluminium wire feeders and associated stepper motors can be removed for inspection and service.

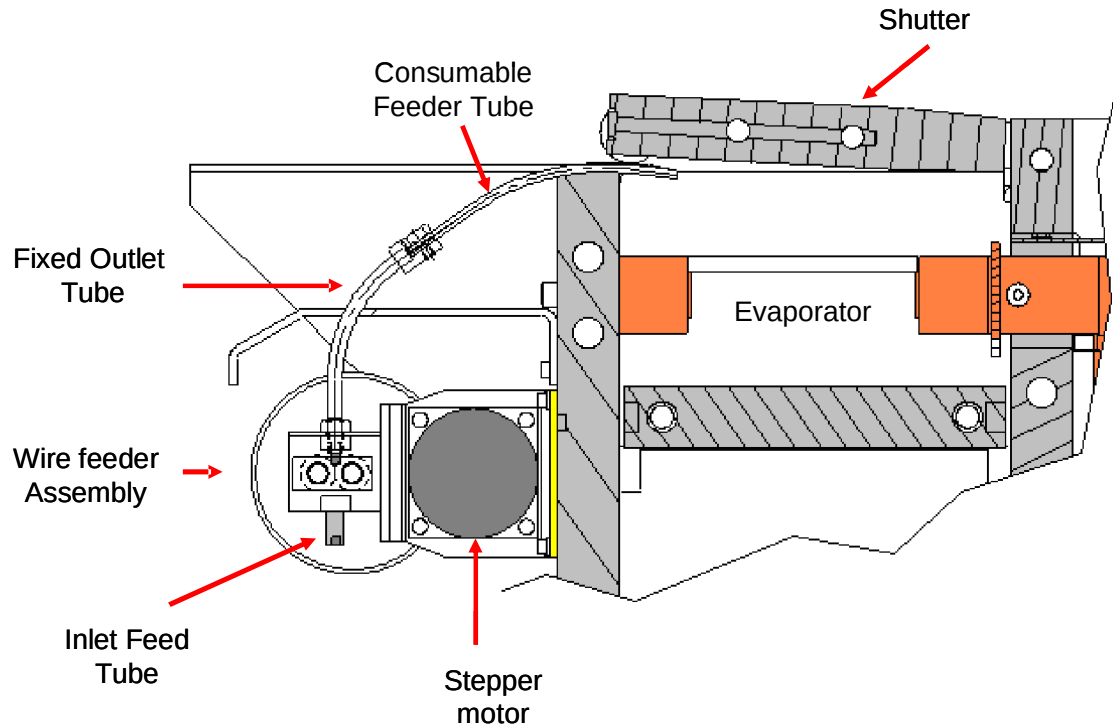


Figure 14.2 - 1 Wire Feeder Layout

Depending upon the type of feeder supplied, refer to the assembly drawing supplied and the following instructions:

1. Remove top copper wire feed tube.
2. Remove Stainless Steel Guard plate from above feeder
3. Loosen and remove wire feed fixing bolts.
4. Lift wire feeder away from its support beam.
5. Disconnect the plug and socket arrangement on the stepper motor
6. Loosen and remove the fixing bolts
7. Remove the stepper assembly.

To replace both items, install in reverse order (i.e. 7-1).

NOTE: Normal wear parts are:

- Non lubricating plastic bushes.
- Wire feeding toothed rollers.
- Wire feed tubes



Figure 14.2 - 2 Wire Feed Wear Parts

The wire feed tubes are considered consumable parts and they are manufactured from half hard copper or stainless steel tube.

	Caution !
	The stepper motors are a self- contained unit and should not be disassembled, as the performance will be affected.

The wire feeders can be disassembled easily to gain access at the internal driving gears and bushes.

To prolong a long useful life from of the wire feeders, they should be greased on a weekly basis.

They should also be completely stripped and cleaned as required by the preventive maintenance schedule.



Figure 14.2 - 3 Stripping a Wire Feeder

Operation of Wire Feed Control System

The wire feeders are driven by individual stepper motors. These motors are calibrated to provide an accurate wire feed rate to the evaporators. The maximum feed rate that the system is calibrated to provide is 200 Cm/min.

To provide a reasonable evaporator boat life, it is recommended that a feed rate of no more than 110 cm of 2.0 mm wire be fed.

There is an insulating gear wheel located between the stepper motor and wire feeder. Its purpose is to ensure that there is not a ground path for the wire that touches the evaporator (which is at a voltage potential higher than ground).

Controls

The stepper motors are controlled by their control units. These units are located within the source panel. These in turn are controlled from the machine computer via a Profibus communication link.

From the substrate's width information provided by the operator, the computer will send control signals only to the wire feeders that are required to coat substrate of the selected width.



Figure 14.2 - 4 Photograph Showing Location of Control System In Source Panel

The operator will also set up a base speed range within the computer. This range allows a minimum and maximum wire feed rate to be set up. For example, if a wire feed rate of 50 cm/min. is set up, the computer will send a signal to the steppers to allow a maximum wire feed rate of 100 cm/min. (at 200% output)

It is normal to set the wire feed rate individual outputs to approximately 100 % of this rate. If the computer requires more wire across the whole of the substrate's width, for example, if the speed is increased, this range can be increased from the PROCESS screen.

The setting of this base speed is dependent upon the coating speed, target deposition and substrate.

The wire feeders are switched on by enabling the stepper motor controllers touching the WIRE FEED On button on the PROCESS screen.

With the winding cart withdrawn from the chamber, the stepper units can be enabled. However, when closed, they can only be enabled on when the BOAT power is applied.

Testing of Stepper Motors

The motors can be tested with a electrical multi-meter to check the continuity between the connections. The following procedure should be followed;

1. Switch the stepper power off.
2. Remove the motor off the machine by unplugging it. Note that each individual motor and feed lead have numbers identifying them.

3. Check the electrical wiring schematic and check the resistances across the following;
 - a) A+ to A- = 2 Ohms (approximately)
 - b) B+ to B- = 2 Ohms "
 - c) A phase to B phase = Infinite
 - d) B phase to the motor case = Infinite
 - e) A phase to the motor case = Infinite
4. Turn the motor shaft manually, there should not be any friction.

If these tests are OK, suspect the drive controller as the problem

A quick test that can be carried out if you suspect that there is a faulty motor, swap its connections with the adjacent motor and try to operate it.

Drive Unit Maintenance and Installation Notes

Stepper Modules

These modules use a standard PLC output type device.

These units are used in systems using Profibus communication protocol network.

They are located in an air-cooled rack to provide a compact application. There are indicator lights to make users aware of any potential problems.

	Caution ! If any of the units are changed, ensure that the cooling fan is re-installed onto the unit.
--	--

Copies of the vendor's manuals are supplied in the proprietary manual binders.

Calibrating Aluminium Wire feeders

The wire feeders are driven by precision micro-stepping motors. Once they have been commissioned, there should be no requirement to adjust their calibration factor. To check the calibration;

1. Adjust the base speed of the feeders being tested.
2. Switch the wire-feeder power supply on with the "Wire feeder On" button
3. Place the control mode into the "Warm Up" mode,
4. Check that the actual wire fed from the feeders agrees with the value displayed on the output screen.

If for any reason that the gears used to drive the wire feeders are changed, the calibration value in the configuration file will require changing. It is advised that GENERAL VACUUM EQUIPMENT be contacted for instructions to carry out this procedure.

14.3 Removal and Fitting of Vacuum Pipe Work Isolation Valves

Isolation valves are provided in the vacuum system pipe work and are known as butterfly valves.

These vacuum isolation valves are used for their robustness and reliability features.

In the case where a valve is being installed or being removed, the following instructions should be followed.

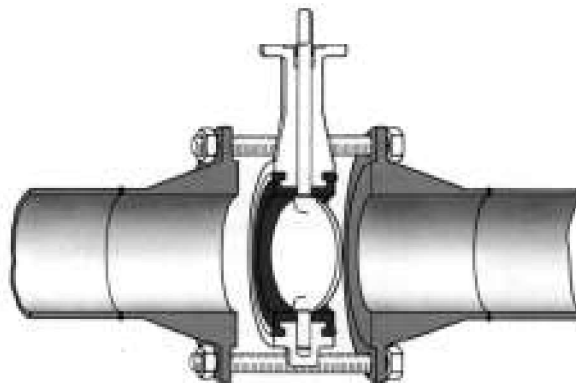
Removal of the Valves

Turn the disc to the nearly closed position. Loosen all flange bolts and remove those, which prevent removal of the valve. Use suitable tooling to spread the flanges and then remove the valve.

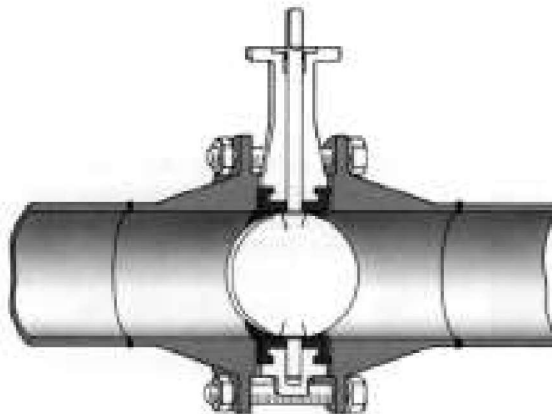
Insertion into an Existing System:

Use adequate tooling to spread the flanges for easy insertion of the valve. Close the valve so that the disc edge is at least 10mm within the body. Insert the valve between flanges, centre the valve body and insert the flange bolts.

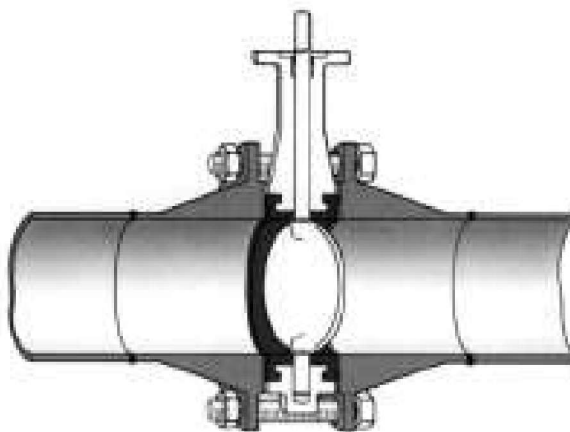
Open the valve completely. (The disc stem is in line with valve shaft parallel flats or keyway). Hand-tighten the bolts. Slowly close the valve clockwise to ensure disc clearance with pipe's inside diameter. Return the disc to the full open position and cross-tighten all bolting to the proper torque.



1. Spread the flanges with the adequate tooling. Insert some flange bolts to bear the valve.



2. Open the valve and remove the flange spreads.



3. Close the valve clockwise, return to open position and cross-tighten all bolting.

Figure 14.3 - 1 Installation of Isolation Valves

Operation

The valves are supplied with air at a pressure of 80 PSI to overcome an internal spring/cam mechanism.

The air is supplied to the valve via a spool valve located externally on the valve body. The spool is moved by a solenoid coil, which is in turn supplied with a control signal from the ASI network.

	Warning ! Do not use the override device during the normal operation of the valve in the machine's normal vacuum cycle. Ensure that the pressure in the pipe work either side of the valve is equal before operating.
---	--

There is a manual override on the spool valve, but this should be only used when testing the operation of the valve with equal pressure in the pipe work, each side of the valve.

ASI Systems

On the equipment supplied, ASI networking has been used on the pumping group. This allows simpler wiring of the various components on the machine.

The valves are supplied with a device called an IND sensor. These are proximity sensor devices that are triggered by an actuator that is sat on top of the valve's shaft. This system is very reliable in indicating the operation of the valve in both the open and closed positions. A data sheet has been supplied in the vendor binders to explain the principle in a little more depth.

14.4 Vacuum Gauge Maintenance

Dial Gauge

Dial gauges require little maintenance and should only be cleaned as required.

Maintenance

The gauge can be cleaned carefully of dust by dismantling it off the machine, remove the locking ring and remove the glass. The mechanism can be removed, but care must be taken not to alter the calibration of the gauge.

Care must be taken in during this procedure. Once it is reinstalled, the component should be leak checked.

Calibration

Calibration of the instrument must be undertaken only by the manufacturer and therefore should be returned to GENERAL VACUUM EQUIPMENT.

Edwards Active Penning Gauge

This gauge is only operational when the high vacuum valve (VV7) is opened.

The gauge is only used for monitoring purposes of the vacuum level in the evaporation zone. It is not used for any interlock circuits.

If the vacuum reading from the Penning head is erratic the problem is usually due to contamination. To clean the gauge head, follow the instructions in the vendor manual supplied under section 5.5.



Figure 14.4 - 1 Penning Gauge

MKS Pirani Gauge

If the vacuum reading from the Pirani gauge is erratic and the gauge cannot be calibrated, the problem is can due to contamination. Clean the case and replace the sensor tube as described in the vendor manual.

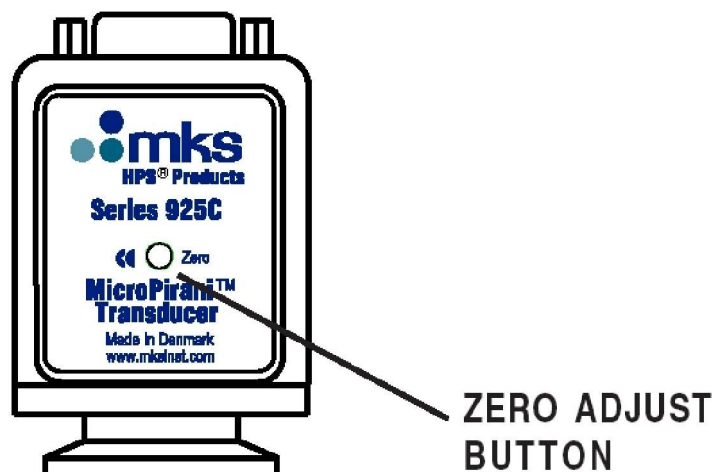


Figure 14.4 - 2 MKS Pirani Gauge

Maintenance

The maintenance and calibration of the gauge is described in detail within the manufacture's manual.

Calibration

To carry out the calibration at the lower vacuum end, it is suggested that the gauge be placed within a diffusion pump at the gauge point.

To carry out the calibration, the following procedure should be followed;

	Warning ! Do not vent the diffusion pump to atmosphere, as a potential explosion hazard can occur.
---	--

- a) The diffusion pump should have been in operation for at least 1 hour.
- b) Close the hand valve fully between the pump and the existing pirani gauge.(located in the pump's inlet)
- c) Remove the lead from the gauge.
- d) The power to the pump will be switched off at this time automatically.
- e) Remove the gauge and replace it with the gauge to be calibrated.
- f) Reconnect the lead back into the gauge.
- g) Carry out the atmosphere calibration at this point as per the manufactures instructions. Monitor at the voltage being feedback on the VACUUM SCREEN under the GAUGES POP UP SCREEN Monitor at the PLC terminals.
- h) Open the valve between the gauge and pump.
- i) Ensure that the pump is switched back on and leave for 15 minutes
- j) Carry out the vacuum calibration as per the manufactures' instructions. Monitor at the voltage being feedback on the VACUUM SCREEN under the GAUGES POP UP SCREEN Monitor the voltage at the PLC input terminals.
- k) Repeat the steps a) through i) for any other gauges to be calibrated

Note: If the pump is not allowed to warm up, the gauge may be incorrectly calibrated, which will lead to problems in the pump's operation. This is because the computer system recognises that if the voltage signal falls to below 1.8 volts, it assumes that the lead has been removed and the pump will be switched off. The gauge should only be calibrated when the pressure in the pump is below 1×10^{-4} mbar

Location of Gauges on Machine

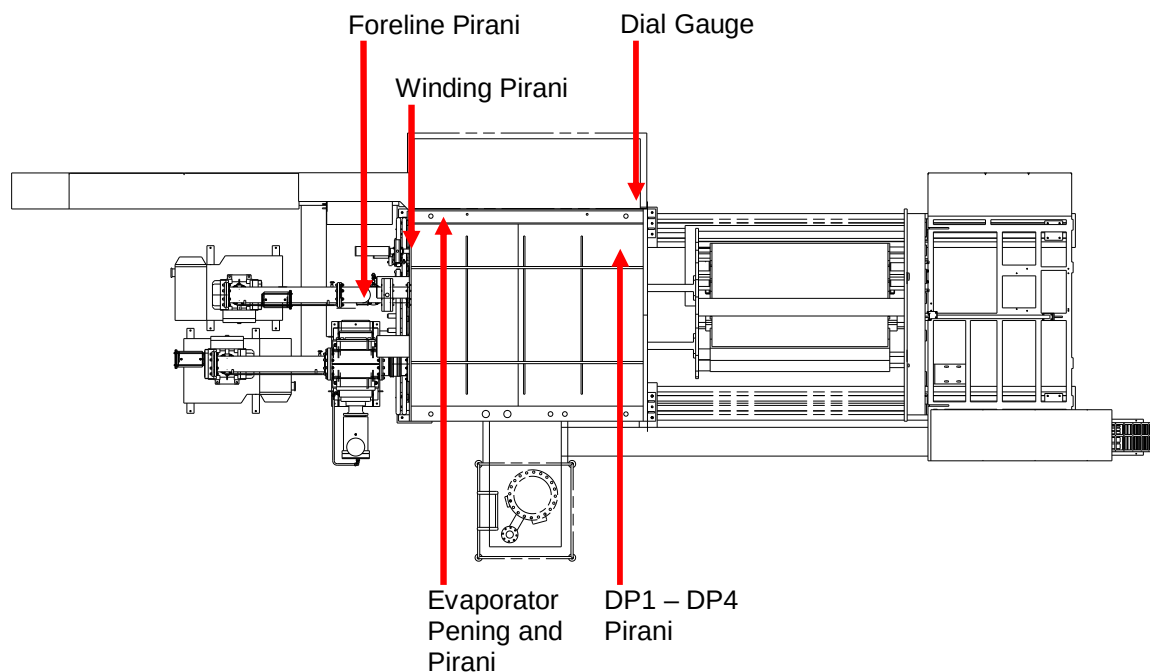


Figure 14.4 - 3 Plan View of Machine Indicating Location of Vacuum Gauges

14.5 Pressure Switch Setting Adjustment

The gauges provide 2-10 DC volt signals to the Siemens PLC analogue input modules that are relative to the vacuum level at the point in the system that the gauge is mounted.

The signals are converted and they are transferred to the computer via a Profibus communications Network.

There is a configuration page in the machine's software, which allows setting up of different switch points for interlocking of the machine. This should only be altered by the General Vacuum Equipment Engineer, and the values comply with component vendors requests.

The computer compares the incoming signals with the preset levels and when they are the same, it will send a signal to allow the interlocked operation to take place.

The software has the ability to have a differential function, by having a switch and then break pressure level.

If the cable between the gauge and the PLC module is removed, the pressure switch will fail to an open situation, and a reading of 0.0×10^{-4} mbar, or lower would normally be seen on the VACUUM screen

There is calibration factors set up in the computer to provide vacuum pressure read outs on various screens.

When purchasing new gauges, ensure that the type bought are the same as the ones on the machine, otherwise they may have a slightly different calibration curve.

14.6 Machine Services Interlock Pressure Switches

Water Pressure Monitoring Sensors

At the main water manifold, a number of pressure monitoring sensors are located in the pipe work to monitor the main inlet and outlet water pressure levels.

There are also additional sensors in individual pipe work that can be isolated.

The manifold is located in the area shown on the machine plan below;

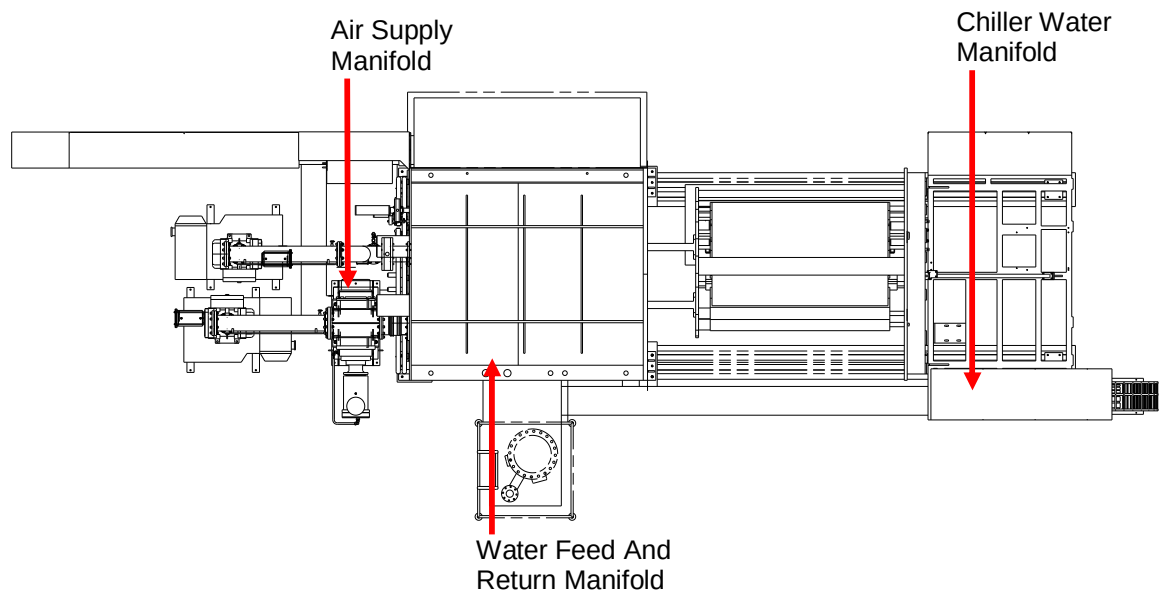


Figure 14.6 - 1 Plan View of Machine Indicating Water Manifold Location



Figure 14.6 - 2 Photograph Showing Main Inlet Manifold with Pressure and Temperature Sensors

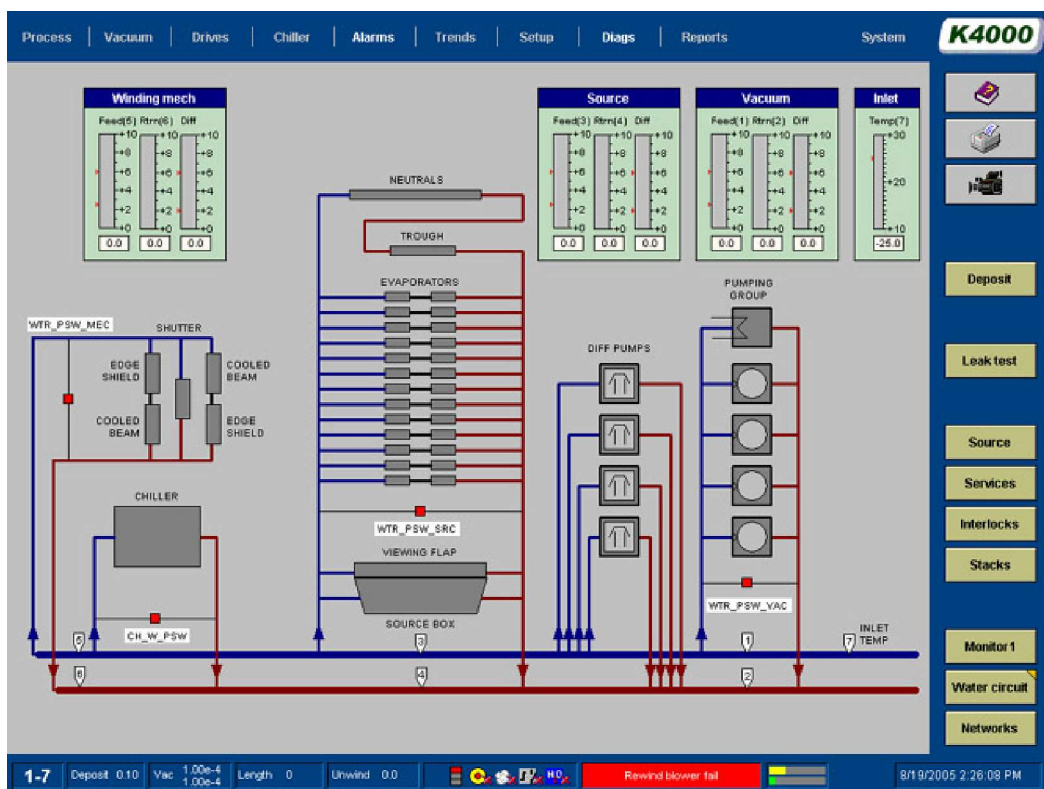


Figure 14.6 - 3 MMI Screen indicating Sensor Readings

This screen is provided to allow the user to check on the status of the water system on the machine.

	Warning !
	Do not modify any circuits on the machine without written consent of General Vacuum Equipment.

The incoming supply temperature is also monitored and in cases when the temperature rises above the alarm set point, an alarm will be given.

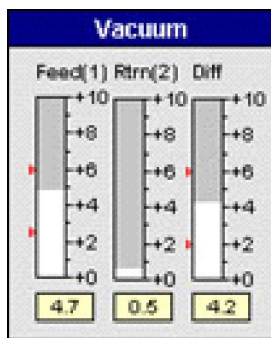


Figure 14.6 - 4 Water Pressure Diagnostic Section of Screen

The pressure monitoring part of the screen has three indicators.

- Inlet (Feed) pressure
- Outlet (Return) pressure
- Differential.

The red arrow heads on the outside of the columns indicates the preset operating levels that the system is to work within.

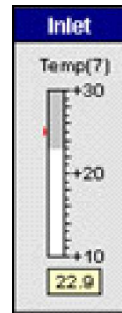


Figure 14.6 - 5 Water Pressure Diagnostic Section of Screen

The inlet temperature is also monitored not to rise above a predetermined level, as indicated by the red arrow head.

These preset positions are preset in General Vacuum Equipment's factory and cannot be altered.

The operating levels are set to indicate if the inlet pressure is less than 70 Psi and if the outlet pressure rises higher than the 10-psi inlet pressure. These switches are interlocked into the computer to protect the various equipment that is cooled from the relevant manifolds.

There is a mechanical pressure gauges provided to indicate the inlet pressure.

There is time delays programmed in the PLC program on each of these circuits to allow for slight instantaneous pressure losses in the supply.

Pneumatic Pressure Switch

An air pressure switch is located on the panel where the regulator is located. It is set to indicate a problem if the pneumatic supply pressure falls below 80 Psi.

To allow for short instantaneous losses in the supply, a timer is programmed in the Soft PLC program to ensure that false indication is not given. This can occur when multiple valves open at one time.

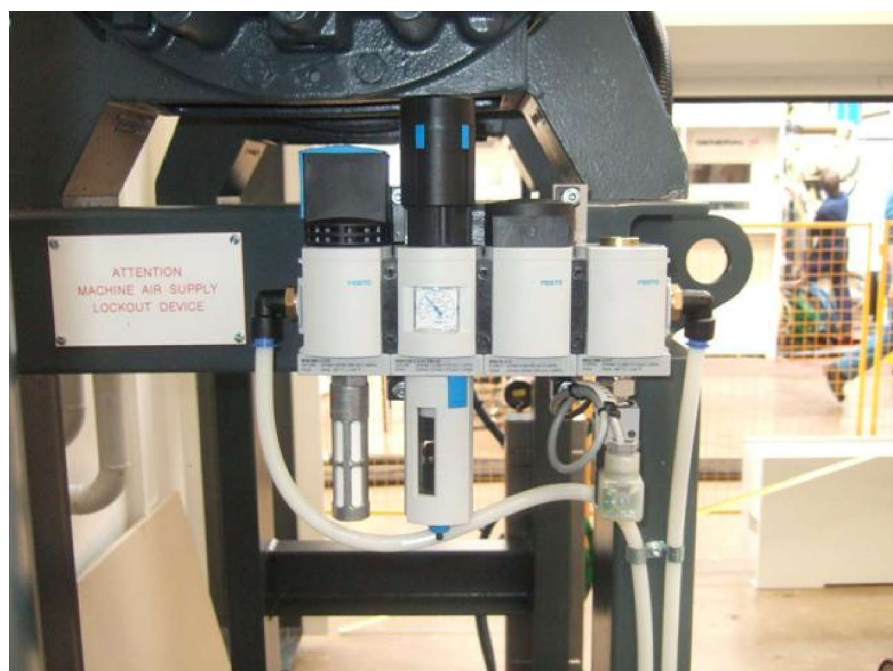


Figure 14.6 - 6 Photograph of Pneumatic Manifold Showing Pressure Switch



Figure 14.6 - 7 Photograph of Nip Roller Pneumatic Isolation Regulator Locked Out

14.7 Coating Shutter Operation

A water-cooled shutter (moving shield) is located above the evaporation source. This is moved by external pneumatic cylinder via a rotational drive assembly. The cylinder is located on the atmospheric side of the chamber.

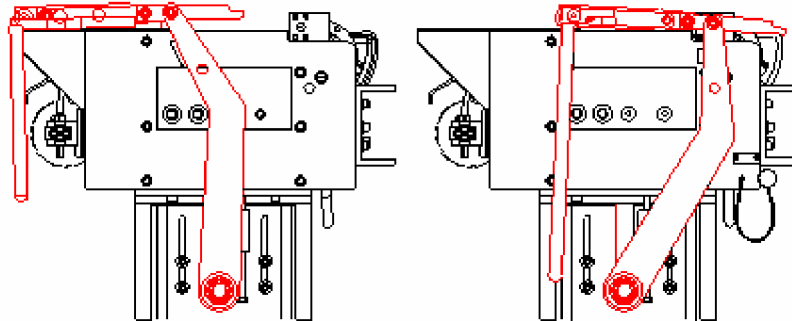


Figure 14.7 - 1 View of a Shutter Assembly Layout

The system is designed to have the shutter closed at all times unless the coating of the substrate is taking place. If power is lost to the machine, the shutter will close.

The shutter is interlocked to only open when the computer receives the web-running signal and the operator touches the open push-button on the PROCESS screen.

Alternatively, the operation can be automated by selecting the auto operation on the SETUP - MODES screen.

Enable and Disable Circuit

There is an enable or disable circuit on the shutter cylinder to ensure that it is safe to work in the shutter operating area while the machine is opened.

There are activation solenoids that are energised by the cart fully closed limit switch.

These solenoids activate spool valves that either exhaust the cylinder or allow air to be injected into it.

The winding cart cannot be driven into the chamber unless the shutter is closed.

There is a magnetic reed switch that is activated when the shutter is closed in its normal plane of operation. The switch is located on the shutter movement cylinder.

Maintenance

There is no specific daily maintenance required for this system.

The spool valves should be checked for leaks and the internal bearings in the shutter movement assembly, greased as per the preventive maintenance schedule.

The Lifting handle and locator pin should be checked on a weekly basis to ensure that normal operation is ok.

14.8 Diffusion Pumps

Diffusion pumps are supplied with this machine and reference to the manufacture's supplied manuals must be made before any work is carried out upon them.

Power Requirements and Testing

Diffusion pumps are a very reliable piece of equipment, but from time to time, faults may occur.

One of these faults is heater failure. The heaters are located on the base of the pump and can be checked quite quickly by measuring the current being supplied to them.

Use an ammeter to check the amps being drawn on each of the three supply wires. They should be approximate equal and to the value specified in the supplier's manual when measured. Any large differences usually indicate either a failed heater or a bad connection. This can be carried out in the vacuum panel.

Further checks can be made with the power switched off the pump. See the vendor proprietary manual for details.

	DANGER
	Beware Live Electrical Circuits, Isolate before working in this area

Check the condition of the wiring and plated terminals at the heater terminals. Replace with the correct replacements if required.

Another potential hardware fault is repetitive tripping of the protective thermostats. These thermostats are located in the following places;

- Oil Temperature Thermostat - on side
- Water Temperature Thermostat - on side

The settings of these thermostats are listed in the proprietary manual.

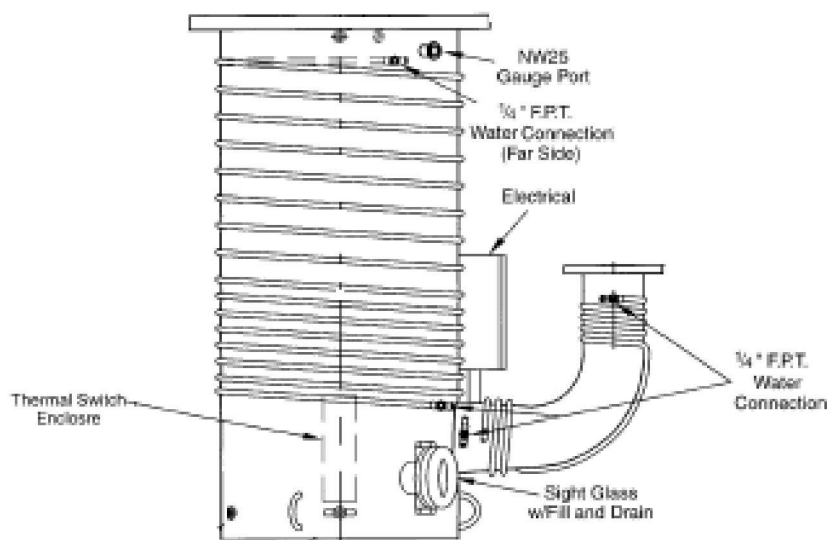


Figure 14.8 - 1 Varian HS20 Diffusion Pump

Diffusion Pump Inspection and Changing Fluids

	Warning !
	Do not attempt to vent the pump to atmosphere until it has cooled down. The pump is extremely hot and care must be taken when attempting to inspect the oil condition.

	DANGER
	Use only silicon based oil in the diffusion pump. Do not use hydrocarbon based oil. If in doubt consult General Vacuum Equipment.

It is most important that the manufacturer's manual is read and understood before any work is carried out on the pump.

	Note!
	Refer to maintenance and service section in the vendor manual

The oil supplied in the pump by Valmet General is CVC4. Other suppliers of the oil can be found locally. If a replacement oil type is to be used, ensure, it meets the same specification as CVC4. A specification sheet is supplied in section 4 of this manual.

It is recommended that replacement o-rings are purchased to allow them to be changed during the procedure (this is a recommended part change by the vendor when changing the oil).

The following steps should be used in draining and replacing the oil in pumps.

1. Switch the main machine isolator off and lock it out as per your company's Lock Out/Tag Out procedures.
2. Allow the pump to cool for a minimum of 2 hours.
3. Close the hand valve between the Pirani gauge and the pump.
4. Remove the Pirani gauge and open the hand-valve to allow air into the pump.
5. Place a container under the pumps drain port to collect the oil.
6. Use a correctly sized spanner (wrench) to release the drain plug in an anti-clockwise direction.



Figure 14.8 - 2 Photograph Removing the Drain Plug from a Diffusion Pump

7. Allow all the oil to drain out of the pump.
8. Remove the filler plug using the same spanner (wrench)
9. Replace the o-rings in both the drain and the fill blanks.
10. Place a little of the diffusion pump oil on the o-ring for lubrication
11. Place the drain plug back onto the pump
12. Measure the oil into a container and fill the oil back into the pump

Note: The identification plate on the pump will state the amount of oil that the pump requires.

All units are in US quarts or Gallons.█



Figure 14.8 - 3 Photograph Showing Oil Being Put Into a Diffusion Pump

13. Place a little of the diffusion pump oil on the o-ring for lubrication
14. Place the fill plug back onto the pump

	Caution !
	The oil used in this pump can be recycled by specialised companies. If advice is required, please contact General Vacuum Equipment.

The pump has to be evacuated using the pumping system.

15. Replace the pirani gauge back onto the hand valve (leave the valve open)
16. Switch on the power to the machine by unlocking the main isolator
17. Allow the computer to "Boot Up" and then select the VACUUM screen.
18. Touch the PUMPS button and the rotary pumps will start. (All services must be present and correct)
19. The relevant VV6 isolation valve will open automatically allowing the pump to be evacuated.

20. Allow the diffusion pump to be evacuated for a few minutes and check that the displayed vacuum level reading changes from 1×10^{-3} mbar to a level less than 1×10^{-1} mbar. If not there is probably a leak on the hand-valve connection or the filler or drain caps.
21. If there is a problem select the SHUTDOWN button and investigate.
22. Carry out the automatic evacuation once again.
23. Monitor the vacuum level on the VACUUM screen for the diffusion pump. After one hour, the level should be approximately 10^{-3} mbar

Oil Condition

The oil should be clear to slightly yellowish colour in its normal state. If the oil is a brownish colour or is contaminated with debris, it must be changed.

Normally with machine's processing film only, the oil should last more than 12 months. On paper machines, this period may be less. We know of existing user's, who use special lacquers, etc., who have to change the oil every six months and clean the pump stack.

When solvents are used in coating of substrates, the solvents can be drawn out of the substrate and can locate in the diffusion pump. This leads to a coating on the jet assembly, which will cut the efficiency of the pumps.

If the pumps are loosing oil check the relevant trouble shooting section in component malfunctions.

	Caution !
	The oil used in this pump can be recycled by specialised companies. If advice is required, please contact General Vacuum Equipment.

Pumping Efficiency

To check pumping efficiency of individual diffusion pumps:

1. Let the pump cool down. Allow 2 hours to allow the oil to stop outgassing.

	Warning !
	Do not attempt to vent the pump to atmosphere until it has cooled down. The pump is extremely hot and care must be taken when attempting to inspect the oil condition.

2. Ensure that the backing line's valves are closed and then vent the pump to atmosphere by removing a Klein fitting's blank.
3. Place a penning gauge direct in to a Klein fitting at the top of the diffusion pump (Do not use a hand valve).
4. Evacuate the diffusion pump by following the procedure in the previous section.
5. Measure the vacuum level after one hour and it should be low 10^{-5} mbar range or mid to high 10^{-6} mbar range.

If vacuum level is only in 10-4 mbar range:

1. Switch off and cool pump again.
2. Vent pump as previously described.
3. Open the relevant right angle valve from pneumatic panel, using the manual override.
4. Inspect the condition of pump stack. If a brown coloured contamination is seen on the stack, it requires cleaning (It is normal that a surface coating of oil is present on the stack).

Contaminated Pump

If the high vacuum valve is opened, the inside of the diffusion pump can be inspected. Normally, the inside of the pump is clean.



Figure 14.8 - 4 Photograph of Badly Contaminated Pump

In the case where the pump is badly contaminated, it has to be removed for cleaning.

14.9 Chamber - Cart Alignment

Winding Cart - Chamber

At the time of installation of the machine, the chamber to winding cart gap was checked for alignment.

If it is suspected that there is some misalignment of the cart in relationship to the chamber, the following should be checked.

Close the machine fully, and check that the reference lines on the chamber's band face and winding cart are aligned horizontally.



Figure 14.9 - 1 Photograph Showing Alignment Reference Lines On a Chamber

If they are not aligned, either add or remove shim from below the cart. This is located between the wheel blocks and the band face.

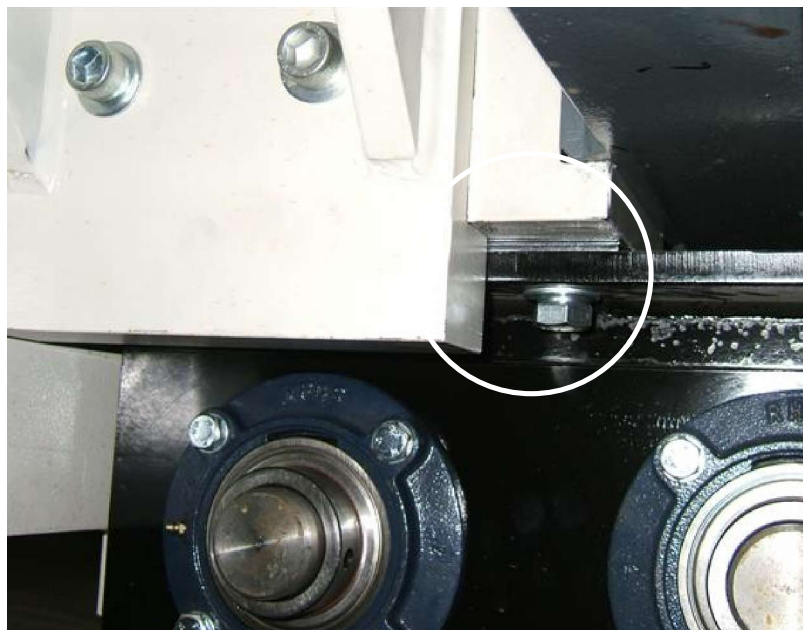


Figure 14.9 - 2 Photograph Showing Location of Shims Located at Wheel Blocks

	Note!
	It is not normal that these faces become misaligned, but it has been found that sometimes the floor that the machine is located on sinks slightly.

If there is a suspicion that there is misalignment of the winding mechanism within the machine, it is suggested that you arrange for a visit from an engineer from General Vacuum Equipment to carry out further tests and corrective action. They will check the winding mechanism, and make modifications to the internal castor location and check for parallelism within the mechanism.



Figure 14.9 - 3 Photograph Showing Internal Castors Showing Jacking Bolts

14.10 Fine Valves (VV7) - Pneumatic

This is the descriptive name given to the valves located between the diffusion pumps and the chamber. They are also sometimes referred to as "High Vac Valves".

Operation

There is a valve assembly located above the diffusion pumps.

This is activated from a pneumatic cylinder on the outside of the chamber

The cylinder is activated automatically as required by the machine in its normal operating cycle. All the required interlocks are in the correct state, for the valves to be opened.

The design of the valve assembly allows the valves to be either fully opened or fully closed. They will close in the case of a shutdown of the machine's power system and leave them in a safe condition.

There are reed (magnetically activated) switches located at either end of the cylinder to indicate if the valves are open or closed.

The operating spool valve is located at the base of the cylinder. The control signal is via the ASI network. The VACUUM screen on the computer will indicate if the valve is open or closed.



Figure 14.10 - 1 Photograph of Fine Valves That Locate Over the Diffusion Pumps as Viewed Under the Evaporation Source

There is a coloured section that represents the individual valves.

- Grey indicates that the valve is closed.
- White indicates that the valve is open

If the colour is red, this indicates that the valve has been commanded to open or close, but has not reached its fully open position.

The valves can be opened by overriding the control spool valve for maintenance purposes.

	Warning !
	Do not use the manual override unless the pipe work is at atmosphere on both sides.

	Caution !
	The diffusion pump should have been switched off for a minimum of 4 hours and vented to atmosphere to allow all internal gases to be disapated..



Figure 14.10 - 2 Photograph of the High Vacuum Valves

Lubrication

There is a gravity feed oil bottle that lubricates the vacuum oil seal on rotary lead-trough.

The level of oil in these bottles must be monitored daily. Major losses of oil will indicate failure of the seal arrangement and the need to change the seal.

The oil bottles are mounted above the cylinder.

On some of these gravity bottles, air bubbles can be seen. These are caused by a slight pneumatic leak across the cylinder seal, but this does not cause a problem with the operation of the seals.

The design of the seal and wiper arrangement of these seals can be better understood by referring to the assembly drawing supplied.

There are grease points on the location slides, that should be greased.

14.11 Inspection of High Vacuum Valve O-ring Seals



Warning !

Do not attempt to vent the pump to atmosphere until it has cooled down.

The pump is extremely hot and care must be taken when attempting to inspect the oil condition.

The valve assembly is designed to allow the valve lids to be opened for inspection



Figure 14.11 - 1 Location of location bolt valve

Remove the wire feed spool holder after cutting all the wires as described in another section of this manual.

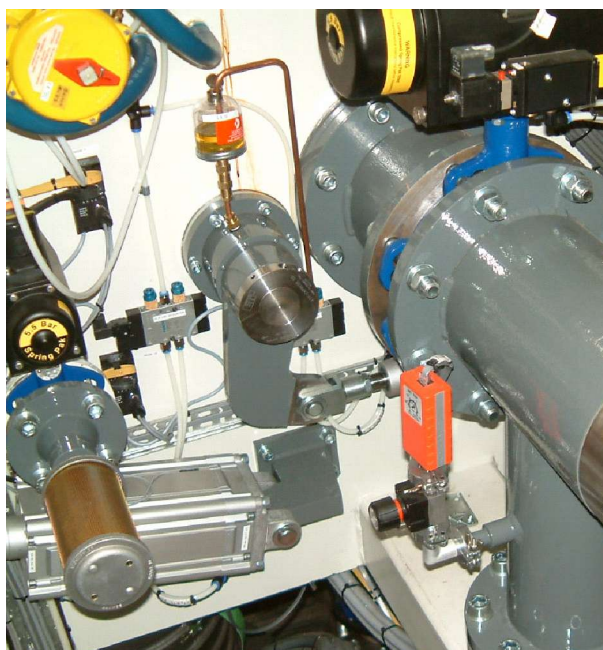


Figure 14.11 - 2 Photograph of Location of control spool valve on valve

Once this has been completed, the movement actuation via the cylinder can take place.

To raise the valve seat assembly, isolate the air to the machine and switch over the air- lines between the spool valve and the cylinder

Then switch on the main air supply and the assembly will move as the cylinder operates.

This operation can only be carried out when the pressure level has been equalised on either side of the valve's plate. This can be carried out when the pump is cool and isolated from the "backing pumps" and the pump vented to atmosphere.

Venting takes place by removing the Pirani gauge and letting air into the pump.

	Caution !
	The diffusion pump should have been switched off for a minimum of 4 hours and vented to atmosphere to allow all internal gases to be dissipated..

When the movement assembly is raised, it is suggested that a piece of timber or another item is used to support the assembly. This can be placed between the cross member and the chamber floor as shown below.



Figure 14.11 - 3 Photograph VV7 Valve in Open Position

It is strongly recommended that any person inspecting this area does not place their body between the valve and the diffusion pump.

The o-ring can be cleaned or replaced at this time.

Take care not to drop any objects into the diffusion pump, while the lids are removed.

Reed Switch Operation

The fine valve cylinder is fitted with two magnetically operated reed switches, to indicate when the valve is open or closed.

A magnetic strip is attached to the piston and therefore moves as the piston moves within the cylinder. When this magnet strip locates opposite the reed switch, its magnetic force pulls the switch's contacts together and makes the switch.

In the time between the valve being fully opened or closed, the position indicator on the VACUUM screen will red

The reed switches are connected to a local ASI interface unit.

14.12 Cryogenic System - Cryo-Generator Units

Cryo-generator® cryogenerator units are provided, to provide cryogenic cooling of the coils in different zones of the chamber.

The units are freestanding unit and are supplied with cooled water, and electrical power. They are interfaced to the computer to operate in the required modes of operation as required.

Two types are supplied from different manufacturers, Polycold and Lauda.

It is suggested that a log is kept of operating pressures on these units and a sample sheet can be found in the manual supplied.

	DANGER
	Extreme cold surfaces are to be found within the enclosures and care needs to be taken in not touching these surfaces.

An in-depth maintenance and troubleshooting section can be found in the manufacturer's manuals supplied

	DANGER
	The refrigerant gases contained within these units are pressurised and should only be added to or released in the prescribed manner as the manufacturers manuals.

14.13 Evaporator Power Supply Thyristor Units

The evaporators are fed AC power from thyristor units via transformers. The thyristors are located in the source panel and are mounted in sets of four in a large heatsink/enclosure.

The control of these thyristors is by a Profibus communication link from the computer. This allows the set point of the individual thyristors to be controlled. It also allows the computer to monitor the various power, voltage and current levels.

Operation

The auxiliary power to the thyristor units is supplied at all times when the main isolator on the source cabinet is engaged. This supply is indicated by the Green LED upon the top controller pcb on the left-hand side.

	DANGER Beware Live Electrical Circuits, Isolate before working in this area
---	---

The thyristor "stacks" are force cooled by fans located below them. The supply to these fans is normally supplied from the isolator cabinet. Before working upon the thyristor units, remove this supply.

To switch the on, the BOATS button on the PROCESS screen must be touched. The switch will move to the on position to indicate that the power circuit is enabled. This can only take place when all the interlocks are correct to allow the evaporators to be switched on. Such interlocks are;

- Source and Winding Cart water supply OK
- Chamber closed
- VV7 valve Open

The control signal that operates the individual thyristors is fed from the computer when the BOATS circuit is energized. The power level will rise to a preset level and the evaporators will then allow aluminium to be evaporated from them.

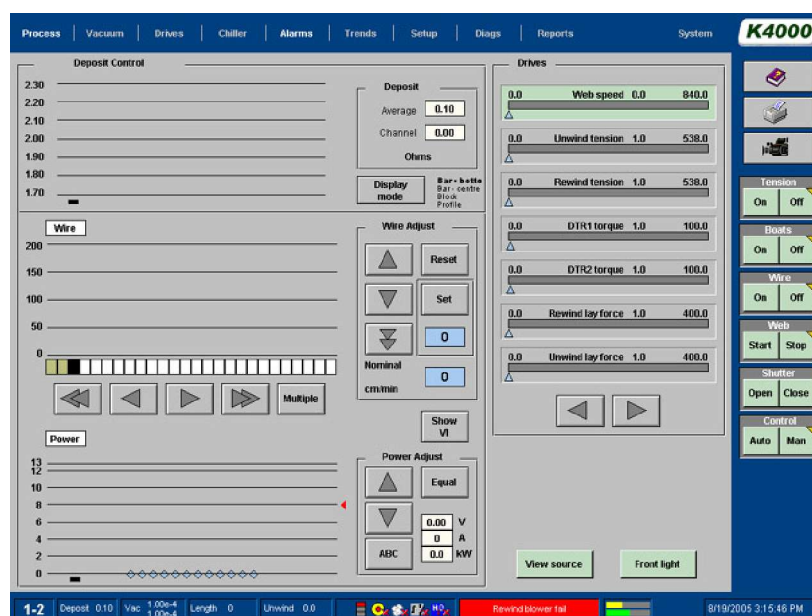


Figure 14.13 - 1 PROCESS Screen

The power level, voltage and current of an individual evaporator can be indicated on the computer on the PROCESS screen.

A feature of this machine is the ability to increase or decrease the power level automatically as the wire rate increases or lowers. The computer will increase or decrease the power levels at a configured predetermined rate.



Figure 14.13 - 2 Photograph Of Thyristor Units In Source Panel

If a fault occurs, a red LED, on the left-hand side of the thyristor unit, will illuminate to indicate the channel that is faulty. The most common fault that has been monitored, is a "partial load failure" fault that occurs when there is bad contact between the evaporators and the contact blocks.

The fault condition can only be reset when the thyristors are switched off and then on by the BOATS button on the PROCESS screen.

A diagnostic unit is available from the manufacture and can be used to troubleshoot the equipment. It is recommended that end users purchase the diagnostic unit, type Eurotherm 260.

For further information on the thyristor units, please see the relevant proprietary manuals supplied

	Note!
	The units have software installed in them to allow ramping of power to the evaporators. If any channel does not record current being drawn from it when 3 volts is applied, it will shut down. This can occur when the evaporator is not located correctly between the contact posts.

Set Up Configuration of New or Replacement Units

The following procedures should be used when replacing the thyristor units.

	DANGER
	Beware Live Electrical Circuits, Isolate before working in this area

1. Isolate the power supplies to the thyristor units. This includes, main, auxiliary and fan power supplies.
2. Remove the field wiring, noting where the cables were connected to.
3. Remove the unit from the enclosure.
4. Check that the voltage supplies stated on the thyristor label agree with those on the General Vacuum Equipment electrical drawings, i.e.; power and auxiliary supplies.



Figure 14.13 - 3 Photograph Of Location Of Identification Label On Unit.

The following work can be carried out on a workbench away from the machine.

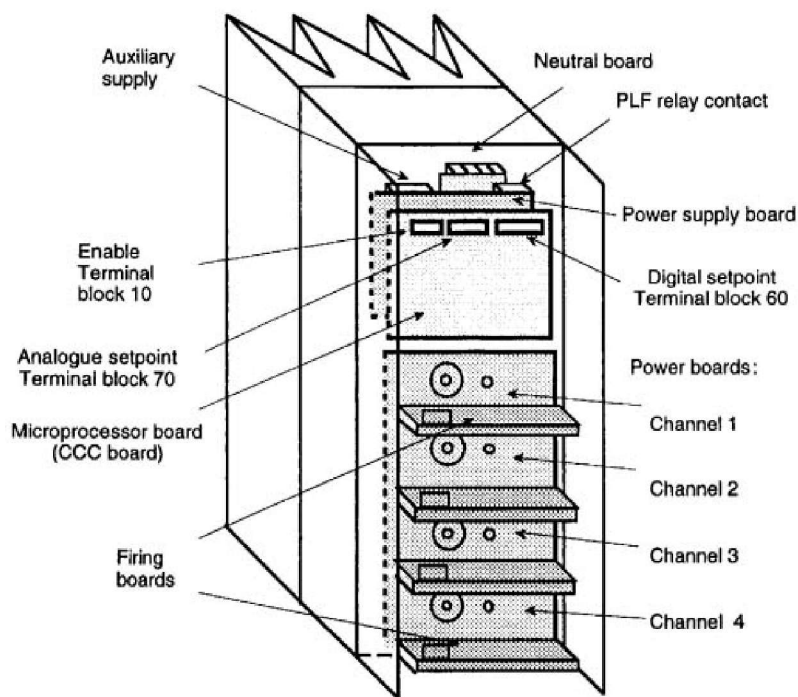


Figure 14.13 - 4 Identification Of PCB Boards In Unit

Power Supply Board Links

This is the board located at the top of the unit behind the microprocessor board.

Check that the jumper ST1 on the power supply board is set to the correct position, i.e. 230 volts

Eurotherm normally sets this up during assembly. See board layout for jumper location. This will allow the unit to use a 240 Volt AC control power supply. Check that link ST2 is present; Eurotherm should have set them.

Check that the thermal switch loom is connected to ST3.

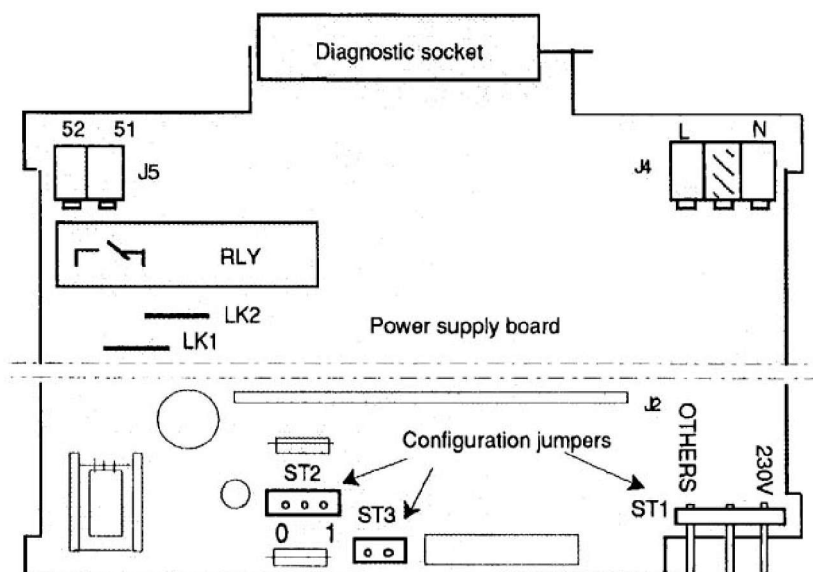


Figure 14.13 - 5 Power Supply Board Layout

Microprocessor Board

There are a number of links upon the microprocessor board and should be set as shown below on all the units used. There are the communication links that will differ from stack to stack (ST11-16).

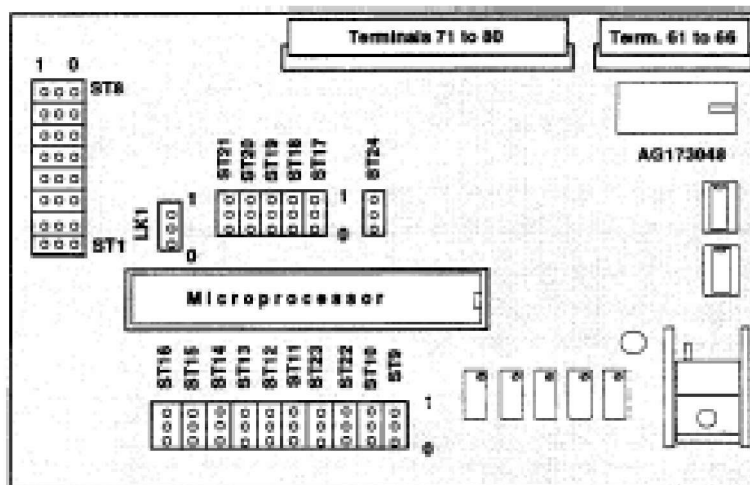


Figure 14.13 - 6 Microprocessor Board Layout

The communications links are arranged as listed to allow the unit to communicate over the Profibus network.

Thyristor Stack Identification	Profibus Address
Stack 1	Address 4
Stack 2	Address 5
Stack 3	Address 6
Stack 4	Address 7
Stack 5	Address 8
Stack 6	Address 9
Stack 7	Address 10

Table 14.13 - 1 Communications Links

Binary code is used to set up the addressing code; 128,64,32,16,8,4,2,1,0

The switches are to be as shown in the table.

Decimal	Binary							
	ST16	ST15	ST14	ST13	ST12	ST11	ST23	ST22
Stack 1	0	0	0	0	0	1	0	0
Stack 2	0	0	0	0	0	1	0	1
Stack 3	0	0	0	0	0	1	1	0
Stack 4	0	0	0	0	0	1	1	1
Stack 5	0	0	0	0	1	0	0	0
Stack 6	0	0	0	0	1	0	0	1
Stack 7	0	0	0	0	1	0	1	0

Table 14.13 - 2 Switches

The jumpers are set as below table.

ST1 -0	ST7 -0	ST13 - see above	ST 19 -0
ST2 -0	ST8 -0	ST14 - see above	ST20 -0
ST3 -0	ST9 -1	ST15 - see above	ST21 -0
ST4 -0	ST10 -0	ST16 - see above	ST22 - see above
ST5 -0	ST11 - see above	ST17 -0	ST23 - see above
ST6 -0	ST12 - see above	ST18 -1	ST24 -0

Table 14.13 - 3 Jumpers

The Link 2 is set to the - 0 - position

Firing Boards

The mini switches are configured as shown below

Channel	SW1-1	SW1-2	SW1-3	SW1-4	SW1-5	SW1-6	SW1-7	SW1-8
1	On	Off	On	Off	On*	On*	On	Off
2	On	Off	On	Off	On*	On*	Off	Off
3	On	Off	On	Off	On*	On*	Off	Off
4	On	Off	On	Off	On*	On*	Off	Off

Table 14.13 - 4 Mini Switches

Note: switches indicated with * are set for 50 Hz operation, change to Off position for 60 Hz

Note!	
	The individual boards are addressed using the jumpers KD1 to KD8

Channel	KD1	KD2	KD3	KD4	KD5	KD6	KD7	KD8
1	Yes				Yes			
2		Yes				Yes		
3			Yes				Yes	
4				Yes				Yes

Table 14.13 - 5 Individual Boards

Check that the power supply phase wiring agrees to the wiring drawing. This will ensure that the "phasing" of the units are correct.

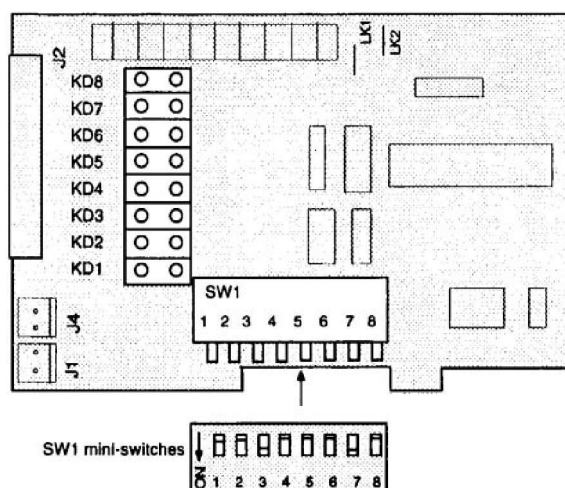


Figure 14.13 - 7 Firing Board Layout

Power Boards Settings

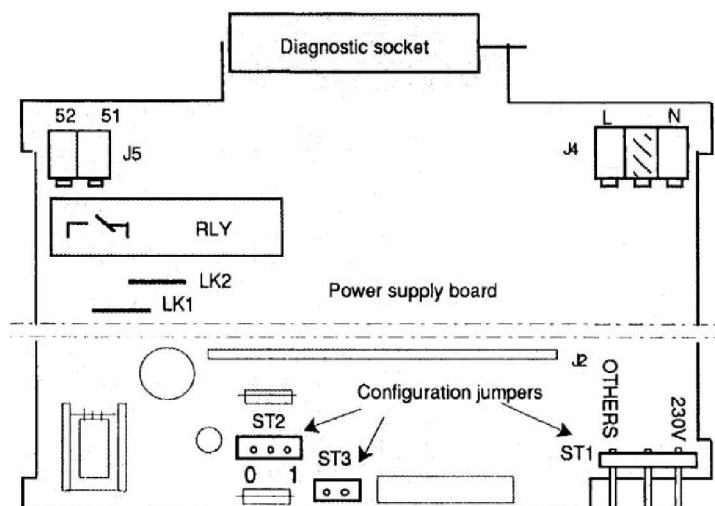


Figure 14.13 - 8 Power Board Layout

When installing a new stack or changing the power board, make sure that the jumpers KP1 and KP2 are installed correctly for the local voltage conditions.

Nominal Voltage Setting (Volts)	Calibrated Value (Volts)		Position of Jumpers	
	Minimum	Maximum	KP1	KP2
100	85	115	0	0
120	102	132	0	0
200	170	220	1	0
240	204	264	1	0
380	323	418	0	1
415	353	456	0	1
440	374	484	1	1
500	425	550	1	1

Table 14.13 - 6 Jumper Voltage Settings

External Wiring

Ensure that there is a wire link between terminals 73 and 74 at the top of the units.

Install the replacement unit within the electrical enclosure. Ensure that the earth cable is fastened to the heatsink chassis.

After installing the new unit ensure that all the field wiring except the load cables are connected to the unit. Refer to the wiring drawings to identify the load wiring to the transformers. Insulate the cables that are not connected.

Switch on the auxiliary supply to the units and check for a healthy green led.

Ensure that the forced air-cooling fans are running. Located at the bottom of the "stacks"

Connect the diagnostic unit to the individual stacks and set up the local line voltage setting.

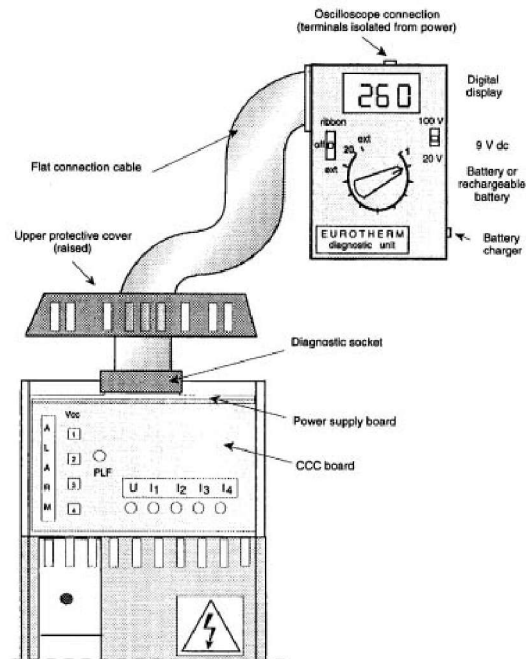


Figure 14.13 - 9 Picture Of Diagnostic Tool Being Used

This is carried out by monitoring point 15 and adjusting potentiometer P2 to give 4.0 volts. The main supply to channel one has to be on to allow this parameter to be set up.

After the unit has been calibrated, isolate the supply voltage and reconnect the LOAD cables.

Set up the current limit values on each of the four thyristors using potentiometers P3-P6 and monitoring the diagnostic points 16,17,19 and 20 respectively. The potentiometers should be adjusted to give a reading of 3.2 volts.

Switch the evaporator's power supply on with no evaporators present and switch the boats on from the PROCESS screen on the computer. Monitor the voltage screen to watch the voltage rise. When the voltage reaches approximately 3.5 volts, the thyristors will close down. This takes place because no current is being taken by the load (evaporators). A red fault Led should be lit for each of the faulty stacks.

Note: If version 619 EPROM's have been fitted to the stack, the voltage will not stop ramping.

Switch the main supply off and the stacks should be reset and the red LED's will be extinguished.

Change the width in the initialise page and monitor that the correct thyristors are being switched on.

Thyristor to Computer Communications

A Profibus communications protocol is used to communicate between the thyristors and the computer.

There is an additional pcb fitted onto the microprocessor board to allow Profibus communications to be used.

The board has three diagnostic LED's that can be used for diagnostic purposes.

The green led will indicate that the processor is in a data exchange state. The other two LED's are used for troubleshooting after the initialisation period.

Red Led	Orange Led	Reason
Off	On	Normal Operation
On	Flashing (0.5 Hz)	External Communications Fault
On	Flashing (1Hz)	Internal Communications Fault
On	Off	Fatal Error
Off	Off	No Power or major hardware Fault

Table 14.13 - 7 LED Troubleshooting

Refer to the Eurotherm Manual TU/TC Range Profibus DP Interface (HA175215 ENG Issue 1 04/97)

Repairing Thyristor stack Components

Before attempting any repairs to the thyristor stacks, ensure that the power is isolated from them.

Follow your company 's lock out procedures.

	DANGER
	Beware Live Electrical Circuits, Isolate before working in this area

Using the comprehensive troubleshooting guide in section one of this manual, determine if any parts of the thyristor stack have failed.

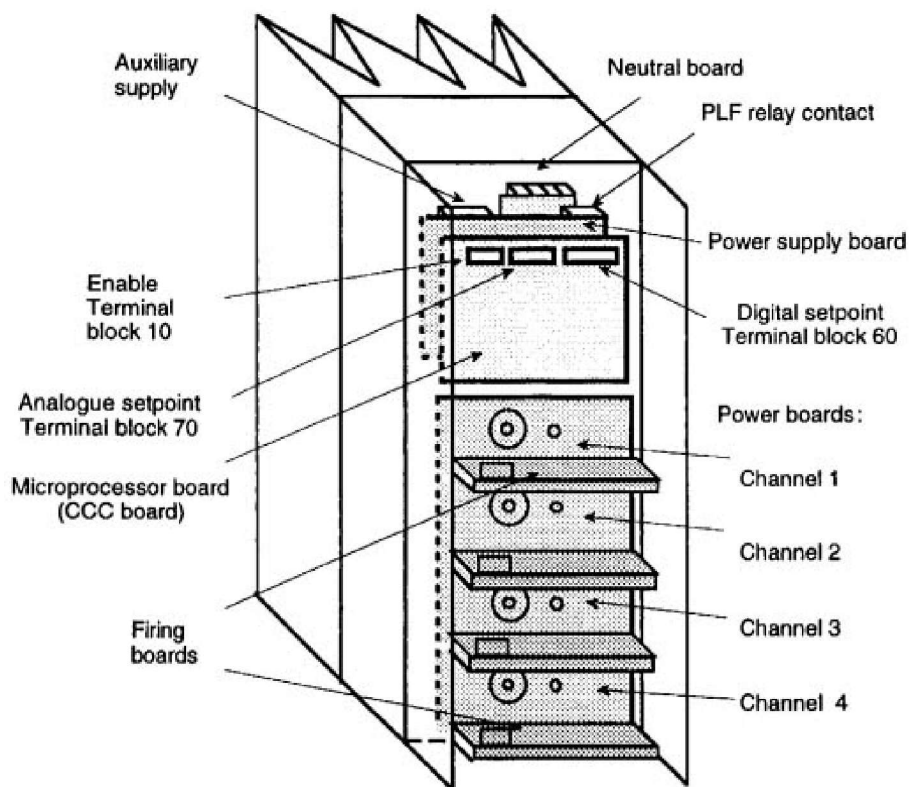


Figure 14.13 - 10 Layout Of Components

	Note!
	The Neutral board identified above is also referred to as the “ Synchronisation Voltage Board”

	DANGER
	Beware Live Electrical Circuits, Isolate before working in this area

It is easier to remove the complete stack from the source cubicle to carry out this work.

Removing Complete Stack

1. Remove the power cables from the bottom of the stack, noting their positions
2. Remove the cooling fan connections carefully
3. Remove the communication cables from the top of the unit, firstly releasing the clamps and then pulling the plugs out of their sockets.
4. Remove the plugs from the Synchronisation Voltage Board
5. Remove the earth cable from the side of the stack.
6. Loosen the two screws at the lower part of the stack. These are the ones holding the stack against the back plate.
7. Remove the two upper screws while supporting the stack at the top.



Figure 14.13 - 11 Loosening The Upper Fixing Screws

8. Lift out the stack cleanly.



Figure 14.13 - 12 Photograph Of Stack Being Removed From The Source Cubicle

Equipment Required to Replace Components

- Set of Metric Allen Keys
- Set of Metric Torx Screwdrivers including T8 and T10 drivers
- A pair of long nosed pliers for removing links
- A large flat (10mm) blade screwdriver
- A medium flat (4mm) blade screwdriver

Place the thyristor unit on a flat surface with a clear area around it.



Figure 14.13 - 13 Photograph Of Thyristor Unit On Flat Surface

Removing Profibus Board

Remove the pcb board in front of the microprocessor board by loosening the two cap head screws using a 4mm Allen key.



Figure 14.13 - 14 Removing Front Cover

Remove the board to gain access to the microprocessor board and the Profibus pcb.

The Profibus board is attached to the right-hand side of the microprocessor board by a plug and socket arrangement.

Lightly pull the Profibus board away, holding it in the corners.



Figure 14.13 - 15 Removing Profibus Board From Microprocessor Board

Removing the Microprocessor Board

The microprocessor board has to be removed to gain access at the unit's power supply board and synchronisation boards.



Figure 14.13 - 16 Removing Torx Screws

2. Remove the Torx screws at the top of the board.
3. Remove the power distribution cable plug on the left-hand side of the board.
4. Remove the two-spacer pillars using a small spanner (wrench)
5. Pull the board forward away from the nylon spacer pillars, on which it rests. Be careful not to damage the long connection pins as they are removed from the power supply unit behind the board.

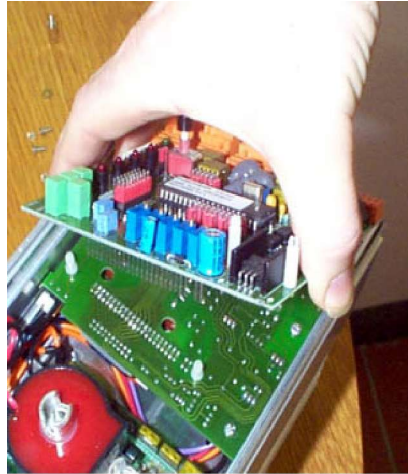


Figure 14.13 - 17 Removing The Microprocessor Board

Replacing Microprocessor Board

1. Install the board carefully back onto the nylon pillars. Initially ensure that the long power distribution pins are entering the power supply board holes without any strain.
2. Push the board down firmly onto the pillars.
3. Install the two-spacer pillars using a small spanner (wrench)
4. Install the power distribution cable plug on the left-hand side of the board.
5. Install the Torx screws at the top of the board

Check that the jumpers on the new board are set to the same positions as the board that has been replaced.

Removing Power Supply Board

1. Remove the Screen clamp assembly by using a Torx screwdriver through the holes provided. Remove away from the unit.

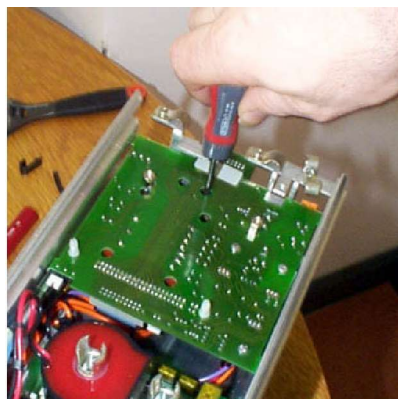


Figure 14.13 - 18 Loosening the Electrical Screen Clamp Bracket

2. Loosen the four screws using a correctly sized flat screwdriver, so that the power supply can be lifted.



Figure 14.13 - 19 Loosening The Holding Down Screws

3. Disconnect the ribbon cable and orange cable plugs from the sockets on the rear of the power supply.
4. Lift the power supply unit out of the stack.



Figure 14.13 - 20 Removing The Power Supply Board

To re-install, use the above instructions in the reverse manner

Check that the voltage jumpers on the new board are set to the same positions as the board that has been replaced.

Removing Synchronisation Voltage (Neutral) Board

1. Remove the Torx screws using a suitable screwdriver.
2. Remove the cables from the board, noting the positions on the terminal

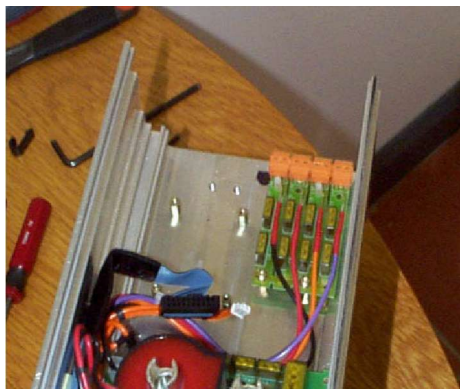


Figure 14.13 - 21 Photograph Of Synchronisation Voltage (Neutral) Board

Replace the board in the reverse order.

Accessing and Removing Firing Board

1. Loosen and remove all the large brass screws at the front of the unit. Use the correctly sized screwdriver, so that damage does not occur to the screws. remove the white plastic screw covers by pulling on them.

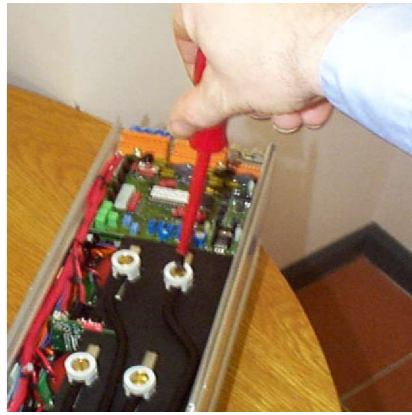


Figure 14.13 - 22 Removing Clamping Screws From The Stack

2. Loosen the cable connections at the bottom of the stack.
3. Remove the cable carefully ensuring that the crimps do not come off the cables.

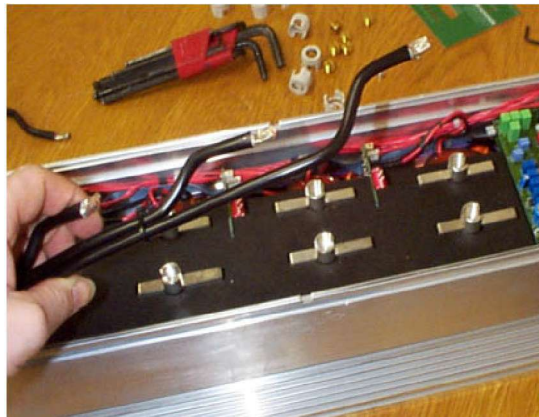


Figure 14.13 - 23 Removing The Cables From The Stack

4. Remove the insulation panel from the front of the stack by tilting it to one side.

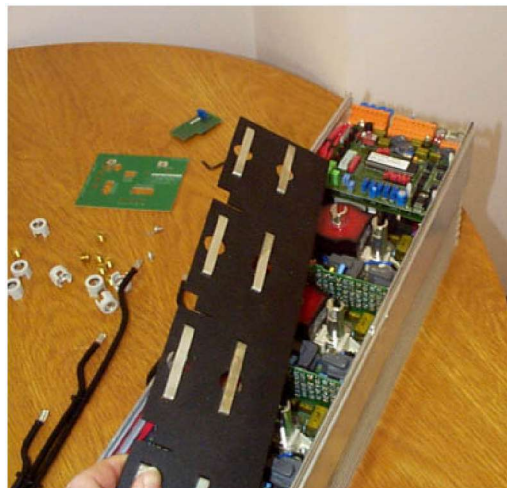


Figure 14.13 - 24 Removing The Insulating Cover

5. The firing boards can now be accessed.

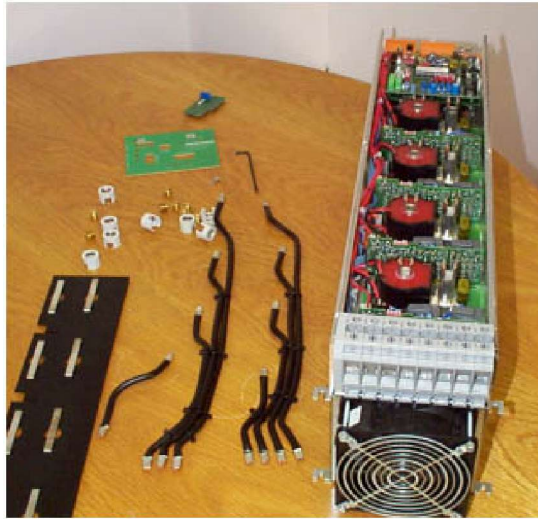


Figure 14.13 - 25 Photograph Of Unit Showing Access To Firing Boards

6. Remove the ribbon cable to the left-hand side of the firing board.
7. Remove the red power distribution cable from the firing board
8. Remove the CT coil



Figure 14.13 - 26 Photograph - Removing The CT Coil - Thyristor Stack

9. Remove the cables connected to the board and then pull the firing board away from the power board behind it.

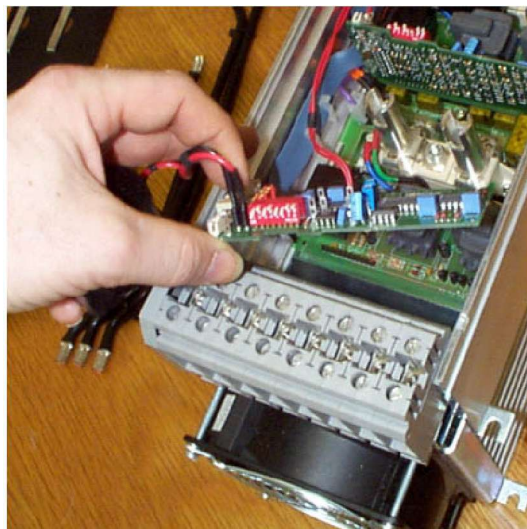


Figure 14.13 - 27 Removing The Firing Board

Replacement Board

Check that the jumpers and dipswitches on the new board are set to the same positions as the board that has been replaced. Install the board and connect the cables

Removing and replacing the Power Board

After removing the firing board, the power board can be removed.

1. Using a Torx driver, remove the holding screw at the top-middle section of the board.



Figure 14.13 - 28 Loosening The Torx Holding Screws

2. Remove the green and blue cables from the connections J2 and G2



Figure 14.13 - 29 Loosening The Torx Holding Screws

3. Using a 4mm Allen key loosen the three screws holding the board onto the SCR below the board.
4. Carefully lift the board away from the SCR.



Figure 14.13 - 30 Lifting The Power Board Away From The SCR

Check that the jumpers on the new board are set to the same positions as the board that has been replaced.

Replacing the SCR

1. Remove the SCR by loosening the two cap head screws using a 4mm Allen key.



Figure 14.13 - 31 Releasing The SCR From The Heatsink

2. Remove the blue and green cables noting their connecting terminal locations.
3. Ensure heat transfer paste is applied to the rear of the unit.
4. Fasten the SCR back on to the heat sink.

Testing the Thyristor Stacks Off the Machine

The thyristor stacks can be tested away from the machine as described in the vendor manual supplied using either an analogue or digital test procedure. To have the ability to carry out the digital testing of the equipment, special test equipment is required, as described in the vendor manual. This software and interface unit is available from General Vacuum Equipment.

14.14 Evaporator Power Supply Transformers

To allow the evaporators to be heated up with a low voltage (11-13 Volts) and high current supply, step down transformers are used.

The air-cooled transformers are located below the operator desk. This area should be kept clean at all times-

The transformers are rated for 13 KVA power output for 40 minutes and 20 minutes off.

	Warning !
	Strong electromagnetic - magnetic fields are generated by the transformers anybody wearing a heart pace-maker should avoid standing close to the machine.

They are generally rated to reduce the voltage by a factor of 38, that is 400 volts to 13 volts. At the same time increase the current taken by the evaporators by the same factor.

Connections



Figure 14.14 - 1 Photo of Connections To The Evaporator Power Supply Transformers

The connections to the primary side of the transformers are accessible by removing the panels from in front of them. There are three connections T1, T2 and T3. This allows the primary power from the thyristors to be connected to either;

Refer to the transformer information, on the electrical drawings, for the exact secondary outputs.

	DANGER
	Beware Live Electrical Circuits, Isolate before working in this area

The secondary side of the transformer makes use of copper links plates and flexible cables to allow connection to the busbars.

If the connections are changed, the volts and amps display on the PROCESS screen have to be re-calibrated. This is carried out in the Configuration screen.

These connections should be checked for tightness on a regular basis as scheduled by the preventive maintenance schedule.

Access to the transformers is by removing the access panels below the operator desk.

14.15 Evaporator System Busbar System

The connections to the evaporators from the transformers are by thick copper conductors.

The connector that passes through the chamber wall is a thick walled copper tube that is cooled by water.

Over a long period of time, through heating and cooling of the connections by the power passing through them, the brass bolts may become loose.

Check as recommended in the preventive maintenance schedule.



Figure 14.15 - 1 Checking The Evaporator Busbar System Brass Bolt Connections



Warning !

Strong electromagnetic - magnetic fields are generated by the bus bars. Anybody wearing a heart pace-maker should avoid standing close to the machine.

14.16 Winding System Drive System

The substrate winding mechanism is rotated by a number of AC motors located on the winding cart.

The motor controllers are mounted in the drive cubicle located on the winding cart. The user interfaces with the drive system through the computer to alter various set points.

The drive control system on this machine uses the field proven technology from Siemens.

They combine hardware and software to form a reliable drive system.

Basic Operation

The drive system is based around a distributed control system. This system is designed for the management and control of the various components and drive motors that are used on the machine

The system incorporates many components, some are listed below:

- Drive motor controllers, which can control the speed and torque levels of the individual motors.
- Operator interface module, which allow the operator to adjust and monitor parameters.
- PLC processor racks containing the following;
 - o Analogue signal input/output modules, which allow interfaces with analogue signals such as load cells, servo position potentiometers.
 - o Digital signal input/output modules, which allow interfaces with digital signals such as push buttons, limit switches, etc.
- Communications interface module, which allow Profibus communications with auxiliary equipment.

Copies of the system's software configuration files are supplied on a back up disk.

General description of the supplied equipment is provided below.

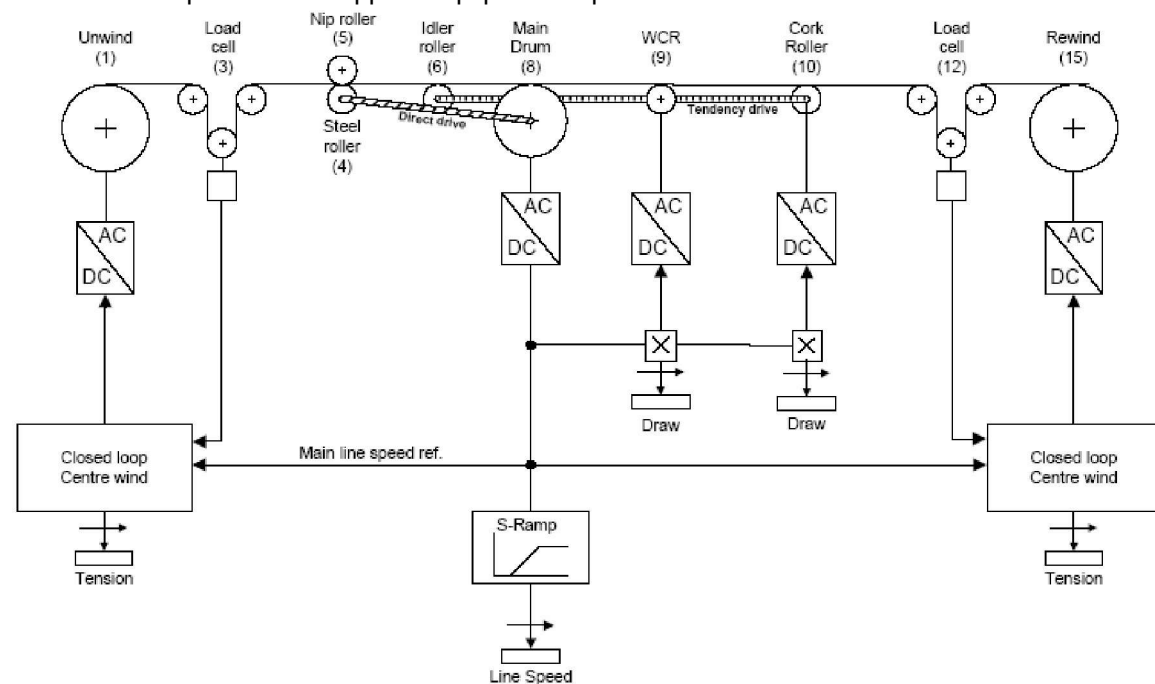


Figure 14.16 - 1 Diagram showing the web transport drive system configuration

The machine drive sections are described below:-

Unwind (1):

The unwind drive will operate as a closed loop centre winder with load cell feedback (3). Diameter and inertia compensation will be provided. Because the wrap angle changes on the Unwind load cell roller – according to whether overwind or underwind mode is selected – compensation will be made in the drives PLC to factor the tension feedback accordingly.

The unwind load cell roller will need to be calibrated for the maximum wrap angle on machine set-up.

Full inertia compensation is included to achieve acceptable machine performance during machine speed changes.

NOTE – The load cell roll including sensors must not introduce resonance frequencies into the tension feedback of greater than 5% FSD or less than 150Hz. During the start up routine, unwind diameter will be determined by using the layon arm.

Main coating drum – DRUM (8):

The main coating drum will operate as the line speed reference master. In order to affect accurate line speed control, the drive controlling the main drum is also direct-driven to a steel roller (4) that will be held in good contact with the web material through a nip roller (5) - an idler roller.

Water cooled roller – DTR1 (9):

A selectable torque control mode will be provided for DTR1.

No tension transducers are provided between DTR2 and DTR1. Therefore this roller will be controlled in speed control, with a +/- 5% trim factor applied from the main drum set point. Steady state speed holding will be 0.01% of set point.

In the case of torque control, DTR1 will be over-speeded by 10% of maximum line speed at all times.

NOTES – Speed ratio control will NOT operate correctly if the web cannot slip on the water cooled roller steel surface.

Cork Covered Roller – DTR2 (10):

A selectable torque control mode will be provided for DTR2.

No tension transducers are provided between the DTR2 and DTR1. Therefore this roller will be controlled in speed control, with a +/- 5% trim factor applied from the DTR1 roller set point. Steady state speed holding will be 0.01% of set point.

In the case of torque control, the cork roller speed feedback will be used as set point to the water covered roller.

Rewind (15):

The rewind will operate as a closed loop centre winder with load cell feedback (12). Diameter and inertia compensation will be provided. The rewind has a taper tension facility and this will be applied to the rewind tension loop. The taper tension profile will be calculated outside the drive control system, in the drives PLC.

Full inertia compensation is included to achieve acceptable machine performance during machine speed changes.

NOTE – The load cell roll including sensors must not introduce resonance frequencies into the tension feedback of greater than 5% FSD or less than 150Hz.

During the start up routine, rewind diameter will be determined by using the layon arm.

The front sheet lists in the binder all the drawings in the drawing set.

1	TITLE PAGE	3129	P.LC INPUT MODULE	WIRING DIAGRAM	8096	RECTIFIER	WIRING DIAGRAM
2	INDEX	3130	P.LC INPUT MODULE	WIRING DIAGRAM	8097	DRIVE BRAKE RESISTOR/CHOKER	WIRING DIAGRAM
20	REFERENCE	3131	P.LC INPUT MODULE WIRING DIAGRAM		8098	UNIMAD PUMP	WIRING DIAGRAM
21	REFERENCE	3132	P.LC POWER SUPPLY	WIRING DIAGRAM	8099	DRIVE MOTOR	WIRING DIAGRAM
22	TECHNICAL NOTES	3136	P.LC OUTPUT MODULE	WIRING DIAGRAM	8100	OTRI DRIVE WIRING DIAGRAM	
26	TECHNICAL NOTES	3138	P.LC POWER MODULE	WIRING DIAGRAM	8101	DRIVE MOTOR	WIRING DIAGRAM
27	TECHNICAL NOTES	3139	P.LC INPUT MODULE	WIRING DIAGRAM	8102	REWIND DRIVE WIRING DIAGRAM	
40	POWER SCHEMATIC	3140	P.LC OUTPUT MODULE	WIRING DIAGRAM	8103	REWIND DRIVE WIRING DIAGRAM	
41	OUTBOARD COMMUNICATIONS	3141	P.LC POWER MODULE	WIRING DIAGRAM	8104	UNIMAD LAUNDRY WIRING DIAGRAM	
48	ASL SCHEMATIC	3142	P.LC OUTPUT MODULE	WIRING DIAGRAM	8105	ROTARY PUMP	WIRING DIAGRAM
100	MACHINE LAYOUT	3144	P.LC POWER MODULE	WIRING DIAGRAM	8107	P.LC LAYOUT [ALL OPTIONS SHOWN]	
306.22	TECHNICAL SPECIFICATION	3146	P.LC OUTPUT MODULE	WIRING DIAGRAM	8108	P.LC INTERFACE MODULE	WIRING DIAGRAM
306.23	VACUUM SCHEMATIC	3150	P.LC POWER MODULE	WIRING DIAGRAM	8109	ANALOGUE INPUT MODULE	WIRING DIAGRAM
306.24	WATER SUPPLEMENT SCHEMATIC	3151	P.LC OUTPUT MODULE	WIRING DIAGRAM	8111	P.LC INPUT MODULE	WIRING DIAGRAM
306.25	PHUWATUNG SCHEMATIC	3152	P.LC OUTPUT MODULE	WIRING DIAGRAM	8112	P.LC POWER MODULE	WIRING DIAGRAM
306.26	VACUUM SUPPLEMENT SCHEMATIC	3153	P.LC OUTPUT MODULE	WIRING DIAGRAM	8125	P.LC POWER MODULE	WIRING DIAGRAM
3011	CONNECTION WIRING DIAGRAM	3155	P.LC OUTPUT MODULE	WIRING DIAGRAM	8135	P.LC POWER MODULE	WIRING DIAGRAM
3019	TRANSFORMER WIRING DIAGRAM	3156	P.LC INPUT MODULE	WIRING DIAGRAM	8136	P.LC POWER MODULE	WIRING DIAGRAM
3020	24V DC POWER SUPPLY	3157	P.LC OUTPUT MODULE WIRING DIAGRAM		8153	P.LC POWER MODULE	WIRING DIAGRAM
3024	FLITER	3158	P.LC POWER MODULE	WIRING DIAGRAM	8156	P.LC OUTPUT MODULE	WIRING DIAGRAM
3026	30V AC POWER SUPPLY	3159	P.LC OUTPUT MODULE	WIRING DIAGRAM	8157	P.LC OUTPUT MODULE	WIRING DIAGRAM
3027	48V POWER SUPPLY	3160	STEPPER DRIVE 1 (2,3,4)	WIRING DIAGRAM	8170	ASL WIRING DIAGRAM	
3028	500V POWER SUPPLY	3181	STEPPER DRIVE 1 (1,2,3,4,5,6,7,8)	WIRING DIAGRAM	8171	ASL WIRING DIAGRAM	
3029	12V DC POWER SUPPLY	3182	STEPPER DRIVE 1 (1,2,3,4,5,6,7,8)	WIRING DIAGRAM	8175	UNIMAD LOADESS	WIRING DIAGRAM
3030	SAFETY CIRCUIT	3183	STEPPER DRIVE 1 (1,2,3,4,5,6,7,8)	WIRING DIAGRAM	8177	WIRING DIAGRAM	
3051	SAFETY CIRCUIT	3184	STEPPER DRIVE 1 (7,8)	WIRING DIAGRAM	8181	SPREADER ROW ADJUST WIRING DIAGRAM	
3052	SAFETY CIRCUIT	3185	STEPPER DRIVE 1 (1,2,3,4,5,6,7,8)	WIRING DIAGRAM	8182	PL MOTORION DRIVE	WIRING DIAGRAM
3060	AIR CONDITIONING	3186	STEPPER DRIVE 2 (2,3,4,5,6,7,8)	WIRING DIAGRAM	8183	LEADTHROUGH	
3062	VESSEL LIGHT	3188	ASL WIRING DIAGRAM		8184	LEADTHROUGH	
3071	ROTARY PUMP 1 & 2	3170	ASL WIRING DIAGRAM		8185	ESOL LAYOUT	
3072	ROTARY PUMP 3	3171	ASL WIRING DIAGRAM		8186	ESOL LAYOUT	
3073	ROTARY PUMP 4	3172	ASL WIRING DIAGRAM		8187	ESOL LAYOUT	
3075	BOOSTER PUMP 1	3173	ASL WIRING DIAGRAM		8190	LAWREN PULBUSTATION STATION LAYOUT	
3076	BOOSTER PUMP 2	3174	ASL WIRING DIAGRAM		8191	ASL DETAIL	
3077	BOOSTER PUMP 3	3180	VIDEO CAMERA WIRING DIAGRAM		8192	TECHNICAL STANDARDS	
3078	BOOSTER PUMP 4	3181	FLA NEW WIRING DIAGRAM		7016	PUMP ISOLATION	WIRING DIAGRAM
3079	BOOSTER PUMP 5	3184	LEADTHROUGH		7017	CONNECTION WIRING DIAGRAM	
3080	BOOSTER PUMP 6	3185	LEADTHROUGH		7018	WATER PUMP	WIRING DIAGRAM
3081	BOOSTER PUMP 7	3186	P.LC / SECRET	WIRING DIAGRAM	7101	INTERFACE MODULE	WIRING DIAGRAM
3082	BOOSTER PUMP 8	3193	LAYOUT		7102	ANAL P.LC INPUT	WIRING DIAGRAM
3083	BOOSTER PUMP 9	3194	LEADTHROUGH		7103	POWER MODULE	WIRING DIAGRAM
3084	BOOSTER PUMP 10	3195	LEADTHROUGH		7111	P.LC INPUT	WIRING DIAGRAM
3085	BOOSTER PUMP 11	3196	LEADTHROUGH		7112	P.LC INPUT	WIRING DIAGRAM
3086	BOOSTER PUMP 12	3197	LEADTHROUGH		7113	P.LC INPUT	WIRING DIAGRAM
3087	BOOSTER PUMP 13	3198	LEADTHROUGH		7114	P.LC INPUT	WIRING DIAGRAM
3088	BOOSTER PUMP 14	3199	LEADTHROUGH		7115	P.LC INPUT	WIRING DIAGRAM
3089	BOOSTER PUMP 15	3200	LEADTHROUGH		7116	P.LC INPUT	WIRING DIAGRAM
3090	BOOSTER PUMP 16	3201	LEADTHROUGH		7117	P.LC INPUT	WIRING DIAGRAM
3091	BOOSTER PUMP 17	3202	LEADTHROUGH		7118	P.LC INPUT	WIRING DIAGRAM
3092	BOOSTER PUMP 18	3203	LEADTHROUGH		7119	P.LC INPUT	WIRING DIAGRAM
3093	BOOSTER PUMP 19	3204	LEADTHROUGH		7		

Sl	Est	Date	Name	Date	Name
		21.09.05	P.D.B		
		Checked			

GENERAL 30 INDEX

Project Name	224G781	Master Drawing	0	TECHNICAL SECTION
Job Number	Y04G022	Project name	2	Page
				2

Figure 14.17 - 2 Electrical Drawings Front Sheet Example

Identification of Components

The machine components that have any electrical power or control circuits connected to them will be generally identified with a reference back to the page number in the electrical drawing set.

Component Naming – BS EN61346-1:1996

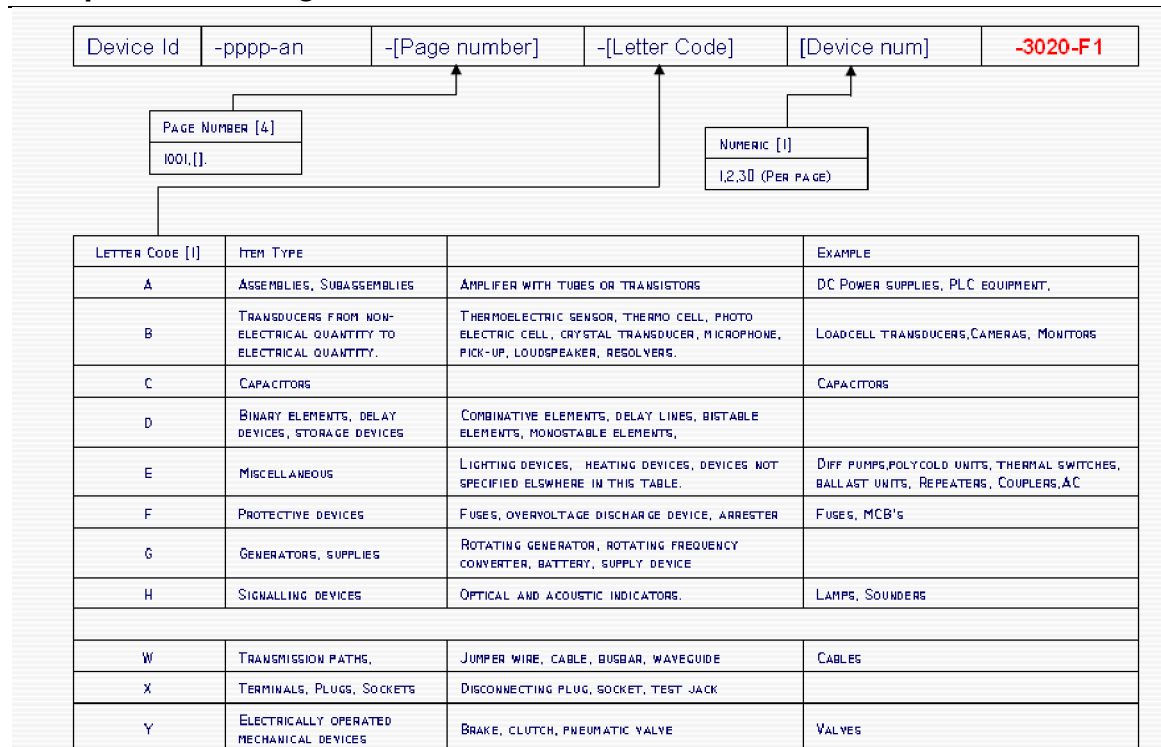


Figure 14.17 - 3 Electrical Drawings – Component Naming Reference

Terminal Strips

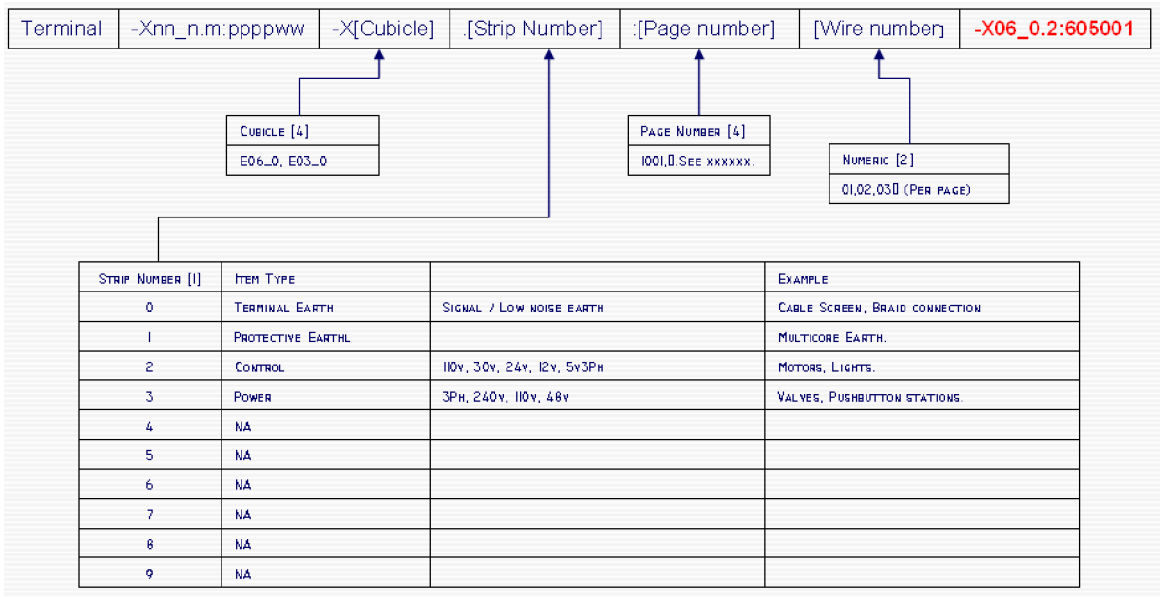


Figure 14.17 - 4 Electrical Drawings – TerminalComponent Naming Reference

How to Read and Understand the Electrical Drawings

Initially, reference should be made to the front sheet to allow the user to locate the part of the equipment that they are investigating. A description of various control circuits is listed with the relevant page number. Once identified, the page can be selected

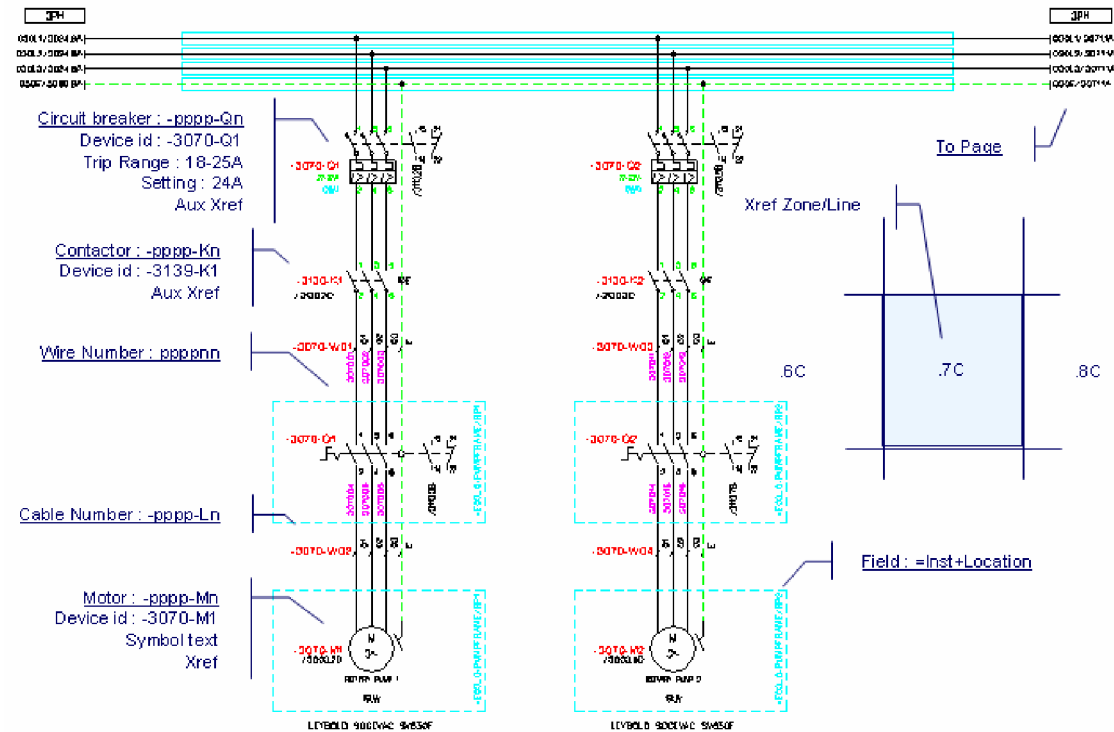


Figure 14.17 - 5 Cross Reference Identification Example- Electrical Drawings

Each page in the drawings has columns identified from 1-8 from left to right across the top of the page. These are used with the page number to allow cross-referencing between pages.

For example on the page shown, at the top right-hand side, there is connections going to the sheet 3022. The two connections have been identified as 3022.8A. This indicates that the other end of this connection will be on page 3022 - column 8, row A.

Another example is the symbolic contactor contacts shown below the 3025-M1 contactor.

The indication is that the contactors " contacts" 1-2, 3-4 and 5-6 are all used on a circuit that is shown on page 3125 column 4, row D. Contacts' 13-14 are used in a circuit on page 3052 column 3, row B.

It can be seen that full descriptive names are located at the top of the drawing, which are related to the component. i.e. –3025-M1 is the Rotary pump 1 contactor.

Further Documentation – Cable List

Date	Name										
29/07/04	R.H	1									
Cable ID	From	To	Cable type	Cable				L	Supplier	Part Number	From - To
-3011-W01	=T03_12	-3011-F1	4 Core 35mm PVC/PVC YYNR cable	35mm				15	ANNIXTER		Vessel Cubicle - Pit Jb
-3011-W02	=T03_12	-3011-F2	4 Core 50mm PVC/PVC YYNR Cable	50mm				15	ANNIXTER	A4AGE-C004NR / CBEU29	Vessel Cubicle - Pit Jb
-3011-W03	=T03_12	-3011-F3	4 Core 16mm PVC/PVC YYNR Cable	16mm				15	ANNIXTER	A4AGC-C004NR / CBEU22	Vessel Cubicle - Pit Jb
-3011-W11	=T03_12	-X6_0.2	4 Core 50mm HF TMPUPVC/PVC YYNR Cable	50mm				20	ANNIXTER		Pit Jb - Drive Cubicle
-3011-W12	=T03_12	-X9_0.2	4 Core 50mm HF TMPUPVC/PVC YYNR Cable	50mm				15	ANNIXTER		Pit Jb - Chiller Cubicle
-3011-W13	=T03_12	-X9_2.2	4 Core 16mm HF PVC/PVC YYNR Cable	16mm				15	ANNIXTER	4946653	Pit Jb - Chiller Cubicle
-3054-W1	=E03_0	-3054-Y1	3 Core 1mm PVC/PVC YYNR Cable	1mm				5	ANNIXTER	A4AE-C003 /CBEU00B	Vessel Cubicle - Shutter Enable Solenoid
-3054-W2	=E03_0	-3054-Y2	3 Core 1mm PVC/PVC YYNR Cable	1mm				5	ANNIXTER	A4AE-C003 /CBEU00B	Vessel Cubicle - Shutter Disable Solenoid

Figure 14.17 - 6 Electrical Drawings – Cable Lists

Documentation – Wire list

Device position/ Location code	Device part/ Terminal strip	DEVKE PIN	CORE	WIRE NUMBER	WIRE NO.	INS T	CABLE ID	CABLE TYPE	CABLE SIZE	WIRE NO.	WIRE NO.	INS T	CORE	DEVKE PIN	DEVICE PART/	DEVKE POSITION/
=T03_12 -JB	-X3_12.3		PVC/PVC YYNR				-3011-W01	4 Core 35mm PVC/PVC YYNR cable	35mm					=E03_0 -CUBICLE	-3011-F1	
		:DL1	01	DL1									01			
		:DL2	02	DL2									02			
		:DL3	03	DL3									03			
		:E	E	E									E	:E		
=T03_12 -JB	-X3_12.3		PVC/PVC YYNR				-3011-W02	4 Core 50mm PVC/PVC YYNR Cable	50mm					=E03_0 -CUBICLE	-3011-F2	
		:C1L1	01	C1L1									01			
		:C1L2	02	C1L2									02			
		:C1L3	03	C1L3									03			
		:C1E	E	E									E	:E		
=T03_12 -JB	-X3_12.3		PVC/PVC YYNR				-3011-W03	4 Core 16mm PVC/PVC YYNR Cable	16mm					=E03_0 -CUBICLE	-3011-F3	
		:C2L1	01	C2L1									01	:2		
		:C2L2	02	C2L2									02	:4		

Figure 14.17 - 7 Electrical Drawings – Wire Lists

Cubicle (Panel) Specifications

The panels are built to the required standards depending upon on the site location. The build standard description is listed in the drawing package.

Cubicle (Panel) Names:

Each drawing will have a descriptive name in the bottom right-hand corner. The names are given to the cubicles, panels or junction boxes that contain the section of wiring shown on the drawing.

- Cubicles are prefixed Enn_n
Example : E06_0 = Drive Cubicle
- Junction boxes are prefixed Tnn_n
Example : T06_13 = Loadcell Jb
- Terminals are Xnn_n.m:
Example : X06_0.2:605002

Cubicle Reference	Description	Location	Terminal Id		Page number
Technical		Section	NA	Project information, Project schemes, page lists, reference material.	1-999
e01_0	NA		X01_0.		1000-1199
E02_0	Isolator	Cubicle	X02_0.	Main Power Distribution	2000-2199
e03_0	Vacuum / Vessel	Cubicle	X03_0.	Pumping Group / Source control	3000-3199
e04_0	Source	Cubicle	X04_0.	Source Control	4000-4199
e05_0	Desk	Cubicle	X05_0.	Operator desk	5000-5199
e06_0	Drive	Cubicle	X06_0.	Winding / Drive control	6000-6199
e07_0	Plasma	Cubicle	X07_0.	Plasma control	7000-7199
e08_0	NA		X08_0.		8000-8199
e09_0	Chiller 1	Cubicle	X09_0.	Chiller control	9000-9199
e09_2	Chiller 2	Cubicle	X09_2.	Chiller control	9200-9399

Figure 14.17 - 8 Electrical Drawings – Panel Names

The location of the cubicles or panels are shown on the drawing below;

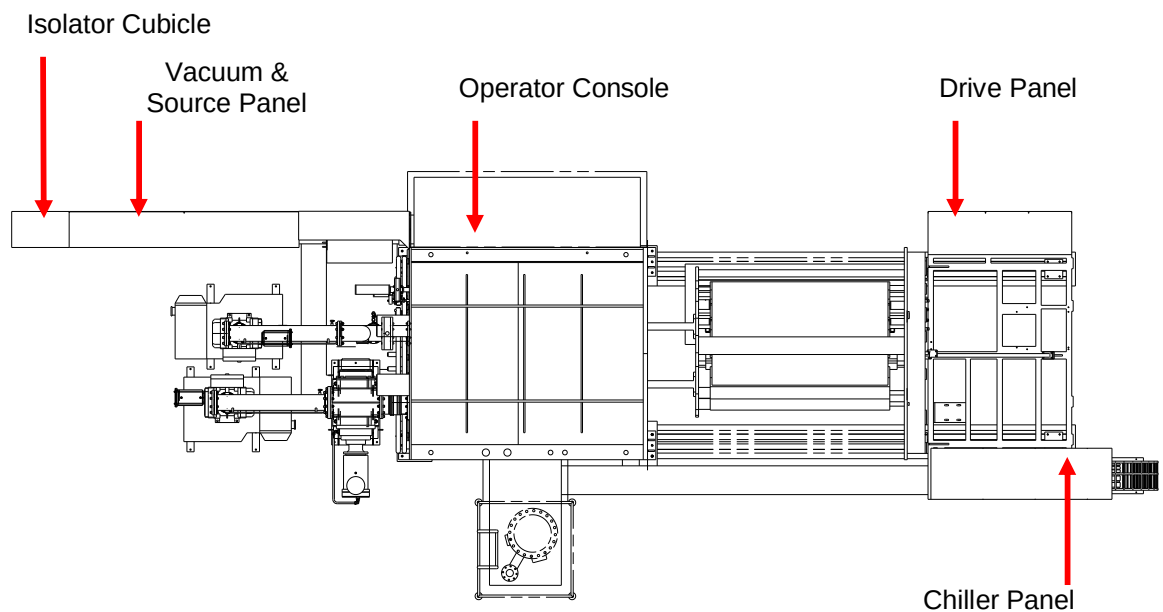


Figure 14.17 - 9 Plan View of Machine Showing Location of Electrical Panel

- **Vacuum/Source Panel** Contains control equipment for supplying power to the various vacuum system components including vacuum pumps. Also contains control equipment for supplying parts of the Evaporation system including evaporator control thyristors,
- **Operator Console** The operator console contains the MMI interface screen..
- **Chiller Panel** This is a junction box inside the chiller unit that contains plc interface units and power control items that are used in operating the chiller unit.
- **Drive Panel** This panel contains the mechanism transport drive system equipment and other control equipment required for the winding cart.

Equipment Set Up and Addressing Settings

It will be seen on some components, within the drawing set, that, setting switches and jumpers are shown to indicate to the user which settings are used for the communication and voltage rating purposes, etc.

Machine Wiring

General Vacuum Equipment makes use of multi-core cables to carry out wiring to motors and other components on the machine.

The cables have been terminated to comply with the latest industry standards and should not be modified.

The cables to the components are identified with a number at the side as shown below;

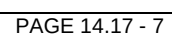


Figure 14.17 - 10 Multi-Core Electrical Cable Identification

The designation 3025-V2 can be referenced to the cable listing at the front of the drawing set. The details of the routing and type of the cable can be found in the wire list, there is a cross reference to the page number in the wire list.

14.18 Machine Data Communication System

The communications network on this machine uses a PROFIBUS DP and ASI systems, which are standard industrial communication systems.

What is PROFIBUS DP

PROFIBUS DP (PROcess Field BUS Distributed Periphery) is a field level network for small amounts of data to be transferred at high speed. It is designed as a remote input/output system.

Profibus has an open protocol, which means that it is not tied to any one electronic control's manufacturer. In principle, it allows units from one manufacturer to operate with units from another, without any problems or special interfaces.

Typical Profibus units (slaves) can be;

- PLC Input modules
- PLC Output Modules
- Drive Systems
- Thyristor Stacks.
- Serial Interface communications.

Further information on the Profibus and ASI network communication systems can be found on the user bodies' web sites at <http://as-interface.com> and www.profibus.com

Cabling

The Profibus Network uses a 2-wire cable for data transmittance between a single master unit and several slaves using RS485. The cable is basically twisted pair screened cable. The transmit speed is between 9,600 to 12,000,000 baud over the network.

Main ASI - A nodes			Main ASI - B nodes			Profibus		
Node	Fault	Device	Node	Fault	Device	Node	Fault	Device
1A	☐	VV1	1B	☐	VV1B	3	☐	ASI GATEWAY
2A	☐	VV2	2B	☐	VV2B	4	☐	THYRISTOR 1
3A	☐		3B	☐		5	☐	THYRISTOR 2
4A	☐	VV4	4B	☐	VV8	6	☐	THYRISTOR 3
5A	☐		5B	☐		7	☐	THYRISTOR 4
6A	☐	VV6A	6B	☐	VV6B	8	☐	THYRISTOR 5
7A	☐	VV7	7B	☐		9	☐	THYRISTOR 6
8A	☐	DRIVE GUARD STOP	8B	☐		10	☐	THYRISTOR 7
9A	☐	DRIVE ESTOP	9B	☐		18	☐	WIRE FEED 1
10A	☐	VV10A	10B	☐	VV10B	20	☐	WIRE FEED 2
11A	☐	VESSEL ESTOP	11B	☐		26	☐	BOOSTER PUMP 1 INVERTOR
12A	☐		12B	☐		27	☐	BOOSTER PUMP 2 INVERTOR
13A	☐	SOURCE VIEWING FLAP	13B	☐		28	☐	BOOSTER PUMP 3 INVERTOR
14A	☐	VV6C	14B	☐	VV6D	29	☐	BOOSTER PUMP 4 INVERTOR
15A	☐	SHUTTER	15B	☐		32	☐	VACUUM PANEL
16A	☐	WATER TEMPERATURE	16B	☐		46	☐	MASS FLOW CONTROLLER
17A	☐	WATER TEMPERATURE	17B	☐		52	☐	PLASMA TREATER
18A	☐	DESK ESTOP	18B	☐		54	☐	DEPOSITION MONITOR COMMS
19A	☐	VESSEL GUARD STOP	19B	☐		60	☐	DRIVE PANEL
20A	☐	MECH OPEN/CLOSE SAFETY	20B	☐		64	☐	CHILLER PANEL 1
21A	☐		21B	☐		66	☐	CHILLER PANEL 2
22A	☐	NIP	22B	☐		94	☐	SOURCE VIEWING STROBE
23A	☐	HINGED SHIELD	23B	☐		110	☐	DRUM DRIVE
24A	☐	ESTOP/GUARD STOP RESET	24B	☐		111	☐	UNWIND DRIVE
25A	☐		25B	☐		112	☐	REWIND DRIVE
26A	☐	MECH ISOLATOR	26B	☐		113	☐	REWIND LAY ARM DRIVE
27A	☐	ZONE SEAL	27B	☐		114	☐	DTR1 DRIVE
28A	☐	MECH CLOSED LIMIT SWITCH	28B	☐		115	☐	DTR2 DRIVE
29A	☐	MECH DOOR SWITCH	29B	☐		116	☐	UNWIND LAY ARM DRIVE
30A	☐	MECH OPEN/CLOSE PB	30B	☐				
31A	☐	TRAVERSE SAFETY	31B	☐				

Figure 14.18 - 1 Communication Diagnostic Interface

Profibus Slaves

Each Profibus slave unit must have its own unique address on the network. It can be seen from the network schematic drawing that General Vacuum Equipment has designated the addresses;

Drive Units - Winding Mechanism	-	110 to 116
Drive Units – Pumping System	-	27-30
Thyristor Stacks	-	4 to 10
Wire feeder Controllers	-	18 & 20
Plc Modules	-	32, 54, 60, 64, 66
ASI Interface	-	3

The master unit is the computer, whose address is 1

Each slave unit has its own built in watchdog timer that checks the data received by the slave and it can shut down if there are network or master unit failures.

The addressing of the slaves' units can be carried out by different methods depending upon the hardware supplied. Reference must be made to the equipment manual supplied for further information.

What is ASI

The ASI (Actuator Sensor Interface) is a device level network for small amounts of data to be transferred at high speed.



It is used as a local network on sections of the machine.

Cabling

The cable used for the network uses 2 wires to both data and power supplies to units attached to it.

The ASI slave units are connected to the cable (yellow) by clamping the two halves of a slave unit to it. Pins built into the slave unit pierce the cables outer sheath and insulation to connect the network. The cable is specially shaped to ensure correct connection to the + and - supplies.

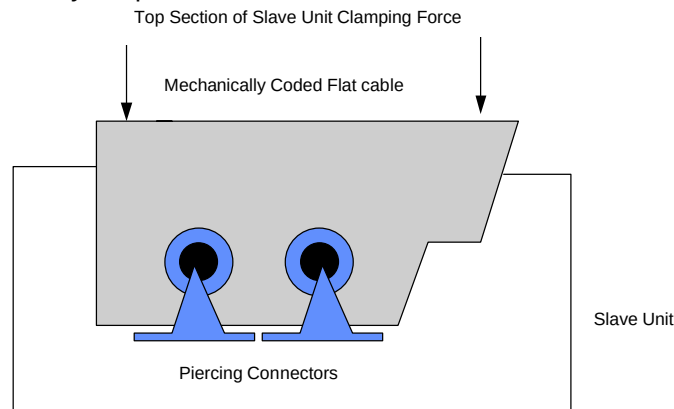


Figure 14.18 - 2 ASI Cable

The cable is self-healing and it allows slave units to be moved and relocated without losing any of its properties.

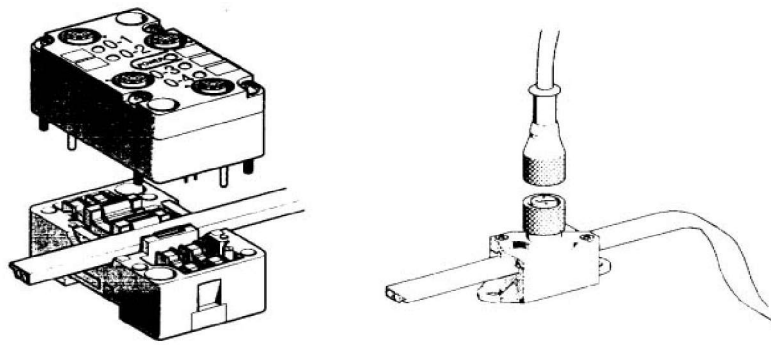


Figure 14.18 - 3 Diagram of ASI Slave Units Attached To the Cable

What is ASI Safety Network

The Asi Bus safety network is a reliable way to implement safety devices on a communication bus. It can also be used to reliably activate outputs, such as locking solenoids. Data from all connected devices is transmitted through a two-wire cable to a safety controller. This will turn off hazardous machine functions in the event of an unsafe condition.

A number of safety monitors devices are located in different panels around the machine.

These are pre-programmed by General Vacuum Equipment to monitor various safety equipment. The program is password protected to stop changes that could make system unsafe. Configuration and authorisation files are kept by General Vacuum Equipment for future reference.



Figure 14.18 - 4 Photograph of Asi-safe Monitor Unit

A Windows compatible software package is available for easy configuration, monitoring and diagnostics. This software is supplied and can be installed on the user's computer.

Back ups of the configurations are also supplied to allow installation onto replacement units and for diagnostic purposes.

Control Equipment Troubleshooting and Diagnostics

There are a large number of components used in the control systems that have individual diagnostics. Not all of these are installed on this machine.

- Eurotherm Thyristor Units
- Reo Thyristor Units
- Siemens Drive System
- Siemens Micromaster Units
- Siemens PLC System
- IFM Asi Interface Units
- Strogra Drive Units
- Parker Stepper Drives
- B&R Stepper Drives

Eurotherm Thyristor Stack Unit - Profibus Communications Status

There is an electronic board fitted onto the right-hand side (as viewed) to allow Profibus communications to be used.



Figure 14.18 - 5 Photograph of Profibus PCB - Eurotherm Stack

The board has three diagnostic LED's that can be used for diagnostic purposes.

The green led will indicate that the processor is in a data exchange state. The other two LED's are used for troubleshooting after the initialisation period.

Red Led	Orange Led	Reason
Off	On	Normal Operation
On	Flashing (0.5 Hz)	External Communications Fault
On	Flashing (1Hz)	Internal Communications Fault
On	Off	Fatal Error
Off	Off	No Power or major hardware Fault

Table 14.18 - 1 Eurotherm Thyristor Module LED Troubleshooting Look Up Table

Refer to the Eurotherm Manual TU/TC Range Profibus DP Interface (HA175215 ENG Issue 1 04/97)

Siemens Drives - Profibus Communications Status

A CB1 technology board has been supplied in each of the main winder drives and a CBP board in the Layarm drive.

Both types of boards have status LEDs.

CB1 Board Status

Green LED	Red LED	Yellow LED	Meaning
Flashing	Flashing	Flashing	Normal Condition when transferring data. All LEDs are lit in synchronism and for the same time period. CB1 is OK and dual port RAM interface is OK
On	Flashing	Off	Awaiting parameterisation by converter (CU)

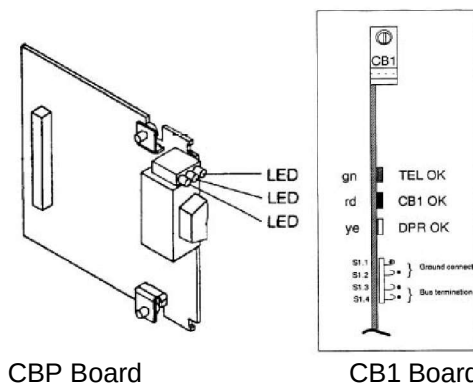
Green LED	Red LED	Yellow LED	Meaning
Flashing	On	Off	waiting for the end of parameterisation
Off			No useful data being transferred from Profibus Network
	Flashing		Bus connector- wiring fault, check terminations
		Flashing	Node is not addressed correctly

Table 14.18 - 2 Siemens CB1 Technology Board LED Troubleshooting Look Up Table

CBP Module Status

Green LED	Red LED	Yellow LED	Meaning
Flashing	Flashing	Flashing	- CBP operating OK, - power is on, - Fault free data exchanges taking place
On or Off	On or Off	On or Off	- Voltage supply to CBP missing, replace the unit. - Data exchange with drive faulty, replace CBP or drive. - Transfer to network faulty, cabling faulty or network down

Table 14.18 - 3 Siemens CBP Technology Board LED Troubleshooting Look Up Table



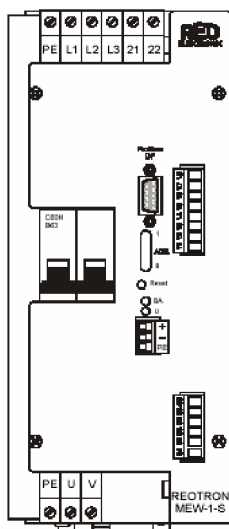
CBP Board

CB1 Board

Figure 14.18 - 6 Diagrams of Siemens drive Unit Technology Boards

REO Thyristor Stack Unit - Profibus Communications Status

This unit allow Profibus communications to be used.



There is a "BA" led that flashes when communications is taking place.

If communications is not taking place, check the dip switch settings for the correct addressing.

Figure 14.18 - 7 Diagram of a REO Stack

Siemens Micromaster Units

These units are used on some machines to control the speed of the vacuum booster pumps.



Figure 14.18 - 8 Photograph of Siemens Interface Unit for M440 Units

With the BOP, individual parameter settings can be made. Values and units are shown on a 5-digit display. A BOP can be used for several inverters. It is directly mounted on the inverter front



Figure 14.18 - 9 Photograph of Siemens Profibus Unit for M440 Units

The function of the profibus communication board is to provide a link between drives of the drive and the rest of the machine components

The board is connected to the profibus system via bus connector. It is a module which can be snapped onto the drive inverter.

The three-coloured LED display is on the front panel of the PROFIBUS-DP communication board. It provides instantaneous information about the status of the board.

Possible LED displays are explained in the table below.

LED	Diagnostic information
off	No power supply
red, flashing fast	Invalid PROFIBUS address on DIL switch (126/127 is invalid) or hardware fault or software error
red on	Start-up and no communication (yet) with the inverter or new communication board configuration, after modification of a board parameter If this status is steady, then the inverter or PROFIBUS optional board is defective

LED	Diagnostic information
Orange flashing	Communication link to inverter has been established No connection to PROFIBUS, e.g. PROFIBUS connector is not inserted or PROFIBUS master disconnected.
orange on	Communication link to inverter and connection to PROFIBUS have been established, but no cyclical data exchange is taking place.
Green flashing	Cyclical process data exchange in progress, but set points invalid (control word = 0), e.g. because SIMATIC master is in "Stop" state green on Cyclical process data exchange in progress and o.k.

Table 14.18 - 4 Siemens Micromaster M440 Unit LED Troubleshooting Look Up Table

Siemens PLC System

Siemens IM153-1 Processor LED Troubleshooting



Figure 14.18 - 10 IM153-1 Processor LEDs

LEDs			Meaning	What to Do
SF	BF	ON		
Off	Off	Off	There is no voltage applied to the IM 153-1, or the IM 153-1 has a hardware defect.	Switch on the power supply module or change the IM 153-1
*	*	On	There is voltage applied to the IM 153-1.	
* On On			The IM 153-1 is incorrectly parameterized – there is no data transfer between the DP master and the IM 153-1. Causes: Configured and set PROFIBUS address is incorrect Faults on the bus	- Check the IM 153-1. - Check the configuration and parameter assignment. - Check the PROFIBUS address on the IM 153-1 and in the STEP 7 project. - Check the cable length with respect to the baud rate. - Check the terminating resistor settings.
*	On	On	No connection to the DP master (baud rate detection) Causes: The bus communication via PROFIBUS-DP to the IM 153-1 has been interrupted.	Check the bus setup. - Check that the bus connector is correctly inserted. - Check if the interconnecting cable to the DP master has been interrupted. - Switch the 24 V DC switch on the power supply module off and then on again.

LEDs			Meaning	What to Do
SF	BF	ON		
On	Flashes	On	The configured setup of the ET 200M does not match the actual setup of the ET 200M.	Check the PROFIBUS address and the ET 200M configuration, to find out if a module is missing or defective or whether an unconfigured module is inserted. Check the configuration (for example with <i>COM PROFIBUS</i> or <i>STEP 7</i>).
On	Off	On	Invalid PROFIBUS address Is the SF LED of an SM/FM also on? If so: There is an error or diagnosis in an S7-300 module If not: The IM 153-1 is defective.	Set a valid PROFIBUS address (1 to 125) on the IM 153-1. Check the SM/FM with the diagnosis. Change the S7-300 module or the IM 153-1 or contact your Siemens partner.
Off	Off	On	Data transfer is taking place between the DP master and the ET 200M. The desired configuration and actual configuration of the ET 200M match.	–

Table 14.18 - 5 Status and Error LEDs On The IM 153

Siemens CPU 317 LED Troubleshooting

Refer to section 10.6 in suppliers manual S7-300_IHBx_e.pdf

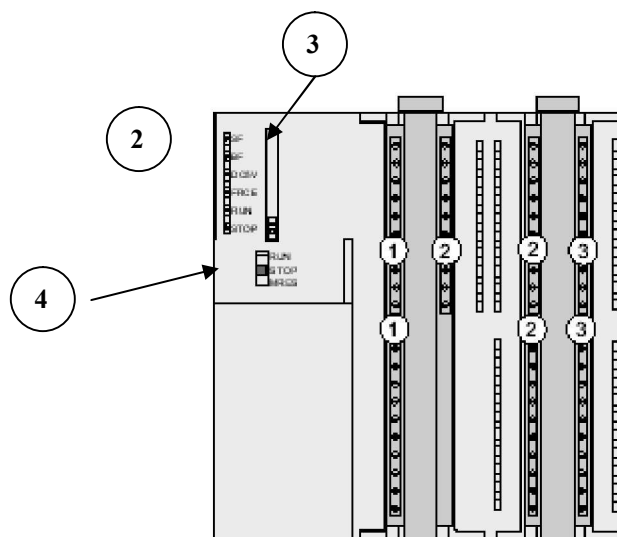


Figure 14.18 - 11 Siemens CPU 317 Processor LEDs

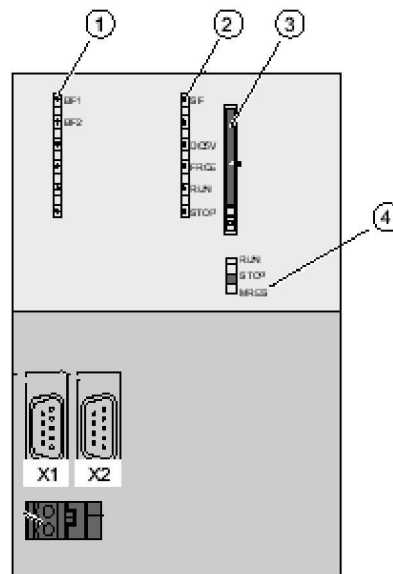


Figure 14.18 - 12 Siemens CPU 317-DP Processor LEDs

The figures above show the following CPU elements

- (1) Bus error indicator
- (2) Status and error displays
- (3) Slot for the Micro Memory Card (MMC), incl. the ejector
- (4) Mode selector switch

Status and Error Displays of CPUs

LED					Meaning
SF	DC5V	FRCE	RUN	STOP	
Off	Off	Off	Off	Off	CPU power supply missing : Remedy: Check whether the power supply module is connected to mains and switched on.
Off	On	X (see the description)	Off	On	The CPU is in STOP mode. Remedy : Start the CPU
On	On	X	Off	On	The CPU is in STOP mode as a result of an error. Remedy : refer to this table
X	On	X	Off	Flashes (0.5Hz)	The CPU requests memory reset
X	On	X	Off	Flashes (2Hz)	The CPU executes memory reset
X	On	X	Flashes (2Hz)	On	The CPU is in start up mode
X	On	X	Flashes (0.5Hz)	On	The CPU was halted by a programmed break point. Refer to vendor manual
On	On	X	X	X	The user has enabled the FORCE function. Refer to vendor manual
X	X	Flashes (2Hz)	X	X	Node flashing test was activated

Table 14.18 - 6 Status and Error Displays On the CPUs

Notes: X means the status is irrelevant for the current CPU function

If the SF is shown all the time, it indicates a software fault. Refer to the vendor manual for help

Status and Error Displays of CPUs with DP Interface

Description of the BF, BF1 and BF2 LED's

LED					Meaning
SF	DC5V	BF	BF1	BF2	
On	On	On/flashes			Profibus DP interface error. See table below.
On	On		On/flashes		Error at the first profibus DP interface. See table below.
On	On			On/flashes	Error at the first profibus DP interface. See table below.

Table 14.18 - 7 Status and Error Displays Of CPUs With DP Interface

When BF is lit,

Possible Errors	CPU reaction	Possible remedies
- Bus fault (Hardware). - DP interface error. - Different transmission rates in multiple DP master mode. - In the DP slave/master interface is active: short circuit on the bus. - With passive DP slave interface: transmission rate search, i.e. there are no other active nodes on the bus.	Call OB 86 (when CPU is in RUN mode). CPU switches to STOP, if OB86 is not loaded.	- Check the bus cable for short circuit or breaks - Analyse the diagnostic data. Edit the configuration. Refer to GVE

Table 14.18 - 8 Troubleshooting When BF Is Lit

Which functions do the LED have on the RS485 Repeater?

The RS485 Repeater with order number 6ES7 972-0AA01-0XA0 has three LEDs.

The **DC 24V** LED indicates whether or not the Repeater has power.

The two LEDs **DP1** and **DP2** indicate bus activity. When these LEDs light it means that there is a lot of message traffic on the bus. The bus activity is high. When the LEDs blink or flicker it means that there is less message traffic (low bus activity).

Note:

The LEDs DP1 and DP2 cannot indicate any error on the bus.

AS-Interface Safety monitor Indicator LEDS

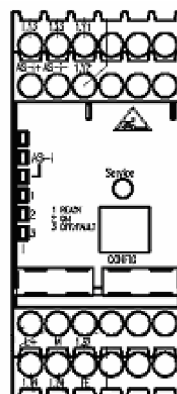
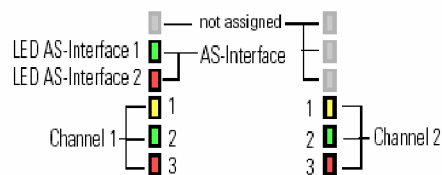


Figure 14.18 - 13 ASI Master Unit

- IFM Unit Status LED's

The IFM master ASI unit has the following LEDs to indicate the status of the network

LED	Color	Meaning
AS-Interface 1	Off	No AS-Interface supply
	Green steady	AS-Interface supply present
AS-Interface 2	Off	Normal operation
	Red steady	Communication error
1 Ready (per channel)	Off	—
	Yellow steady	Starting / restarting lockout active
	Yellow flashing	External test required
2 ON (per channel)	Off	Contacts of safety output open
	Green steady	Contacts of safety output closed
	Green flashing	Delay time running stop category 1
3 ON / FAULT (per channel)	Off	—
	Red steady	Contacts of safety output open
	Red flashing	Fault (of conncted sensors)


Switching on

All LEDs light up apart from AS-Interface 1.

Configuration mode

As long as configuration mode is active, LEDs 1 - 3 of channel 1 are activated successively (running light). This also applies if the monitor is brand-new without a valid configuration.

Fault

LEDs 1 - 3 flash rapidly.

Figure 14.18 - 14 Master ASI Unit LEDs

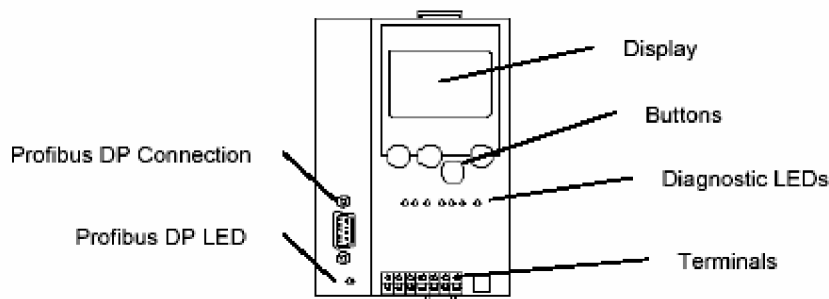
IFM Asi/DP Interface Units

Unit Type AC1335/AC1305/AC1326

The three diagnosis LEDs on the front of the AS-i DP -Gateway give information concerning the state of the AS-i master and the connected AS-i system. The LEDs have the following meaning:

LED	Status	Meaning
PWR/COM	On	AS-i voltage (Power) present and communication with at least one slave has been established
	Flashing	AS-i voltage (Power) present but no slave communication established
P R O J	On	Projection mode active, configuration monitoring not active
	Flashing:	Projection mode active, switch to protected mode not possible because a slave having address 0 is connected
CONF / PF	On:	Projected and current configuration not the same
	Flashing	Periphery fault on at least one of the connected slaves

Table 14.18 - 9 Status and Errors Displays on the AC1335/AC1305/AC1326

AC1326:


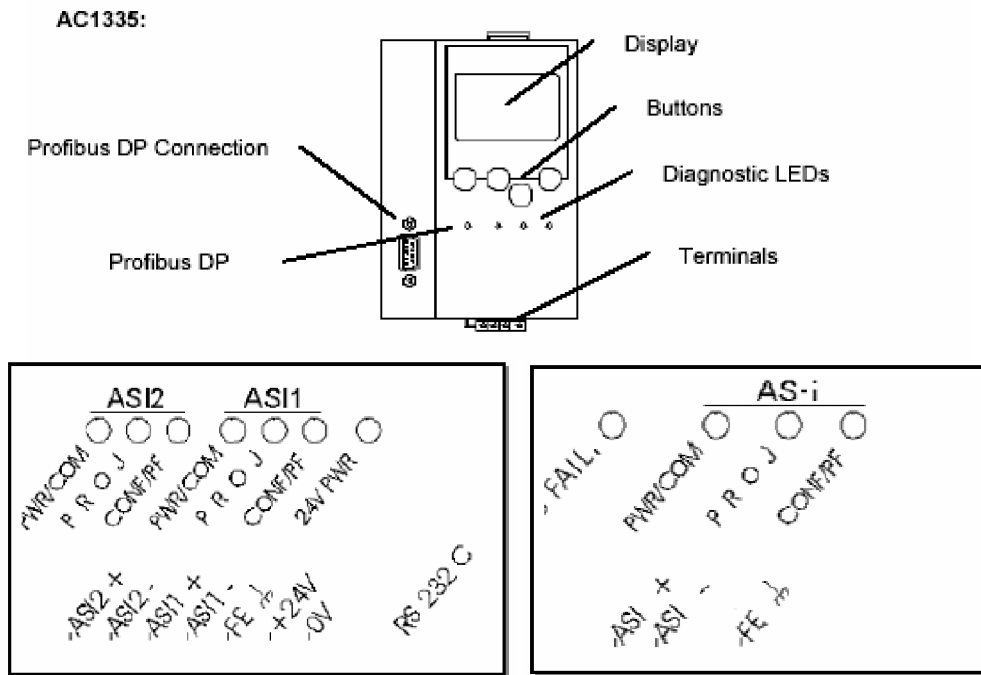


Figure 14.18 - 15 Layout of IFM Asi DP Interface Modules

For further details see the vendor manuals supplied.

Unit Type AC1005

The IFM Profibus DP to ASI gateway is a Profibus DP slave and an ASI master. Each slave on the ASI network has 4 bits of input and 4 bits of output assigned in the Profibus input/output image. The units are orange in colour and are located inside one of the electrical cubicles.

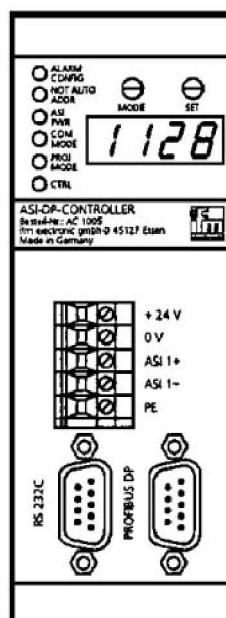


Figure 14.18 - 16 IFM ASI Master Unit

- IFM Unit Status LED's

The IFM master ASI unit has the following LEDs to indicate the status of the network

Status LED	Colour	Meaning
ALARM/CONFIG	RED	Actual and preset configuration of slaves is different.
NOT AUTO ADDR	RED	Automatic addressing is not enabled.
ASI PWR	GREEN	ASI voltage is OK.
COMM MODE	GREEN	Normal operation mode.
PROJ MODE	YELLOW	Projecting mode, all slaves activated.
PROJ MODE	YELLOW	Slave 0 detected, switching into flashing projected mode not possible.
CTRL	YELLOW	Controller in run mode

Table 14.18 - 10 IFM Profibus LED Troubleshooting Look Up Table

To assist in setting up and troubleshooting there is also a seven segment LED display, which will indicate operational conditions.

1st digit Mode
 2nd digit Master 1 or 2 (note on the single master version used at VG (type AC1005)
 Selecting master 2 will produce an error.
 3rd+4th digit Slave address, error number or operation specific display.

- Replacing a faulty AFM unit.

Refer to the manual supplied and set the operational mode to MODE A -Set Profibus address.

This allows the unit to address of the gateway on the Profibus Network.

Refer to the Communication Network Schematic drawing for the address to be used.

1. Press SET to select the digit for setting.
2. Press MODE to set the digit as required.
3. Repeat until all digits are set.

AFM Error Messages

Normally the display is blank.

The following error messages can be displayed by the unit.

1106 Display of missing or faulty slave(s)
 Digit 1 = Mode 1
 Digit 2 = Master number
 Digit 3&4 = Address of slave(s)

See the vendor manual for a complete description.

E-50 to E-59	Internal error	E-70	No entry in list of detected slaves
E-60	Master 2 not connected.	E-71	No entry in list of projected slaves.
E-72	No entry in list of activated slaves	E-73	No free slave address available.
E-90	Slave 0 not connected.	E-91	Target address is already allocated
E-92	New address cannot be allocated.	E-94	Actual configuration is not the same as projected configuration.
E-95	ID code wrong - does not correspond to projected value. (Has slave been replaced with wrong type?)	E-96	Addressing is not possible since automatic addressing is not enabled.

Table 14.18 - 11 AFM Error Messages

IFM - ASI Slave Units

Operation Indication

Each of the Siemens units has indication LEDs. There is a yellow Led that is lit when power is supplied to the unit.

There are also green LEDs that are illuminated when the relevant switch or interlock is operated.

Faulty Units

If the Profibus Network is not operating and the IFM unit indicates a faulty ASI slave unit, it should be replaced.

To replace a faulty slave unit on the winding cart or vacuum pumping group, the following procedure should be used;

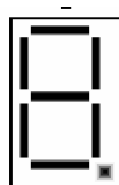
Note: All new slave units are supplied from the manufacturer with a 0 address

1. Remove the faulty slave unit from the network.
2. Connect the new slave (with address 0) to the network.
3. The master unit will automatically re-address the new slave with the address of the old one as long as they are the same type of module.

Strogra Drive Units

These drive units are used in a number of applications on the machine, dependant upon the type and options selected.

Operational status and error indication is via 7-segment display, which indicates the actual status of the SERS.



Error and status indications are indicated by constant illuminating characters. Warnings are indicated by blinking characters.

Following table shows all possible indications:

Indication	Description	comment
-	initialisation phase	Is indicated 1 second after a power-on-reset
4	ready for switching phase current ON	
5	phase current of motor is ON positioning jobs can be executed	
7	Check sum error of dates in E2Prom	Overwrite EPROM-parameters with backed up parameters (with SERS-software) or select standard parameters (with P1004=3 - see P1004)
8	over temperature - from 85 °C (10%) on the power amplifier	Error check cooling of SERS – forced draft (fan and 24 VDC in ELK) o.k.?
9	error under voltage - see 2.1.4	check power supply
8 (blinking)	warning over temperature from 75 °C (10%) on the power amplifier	
9 (blinking)	warning under voltage - see 2.1.4	

Indication	Description	comment
A (blinking)	warning position overflow – positioning job is reaching position limit	Parameter W (P47) to big-check positioning mode (for endless positioning P1014=1 see P1014)
c	Limit switch is open	check limit switches at machine and limit switch inputs of SERS
E	Short circuit in motor or at power amplifier board when installing the motor check phase connections	
F	error step angle control - the motor could not follow the position command value (only with option step angle control) – see P1029 in manual	check encoder connections-acceleration to high- external load to big (not enough motor torque)
F (blinking)	warning step angle control – description as F – see P1029 in manual	See comment for F
H (blinking)	warning program error – executable program stopped because of an error in the program	check parameters and labels in program(use SERS-software for debugging)
L (blinking)	Warning drive reached software limit switch.	See parameters P1039 and P1040 in manual

Table 14.18 - 12 Status And Error Displays On The Strogra Drive Units

Parker Stepper Drives

Troubleshooting

IMPORTANT NOTE - ensure that power is turned off before any connections are removed or changed. Removing a drive with power applied can turn a recoverable fault situation into a major problem.

Drive LED indications

In the event of any system malfunction, the first thing to check is the status of the LEDs on the front of the drive module. If no LEDs are illuminated, verify that power is turned on and that the correct supply (see Table 3-1) is present at the X1 connector. If it is, but there are still no LEDs illuminated, the drive is faulty and needs to be returned for repair.

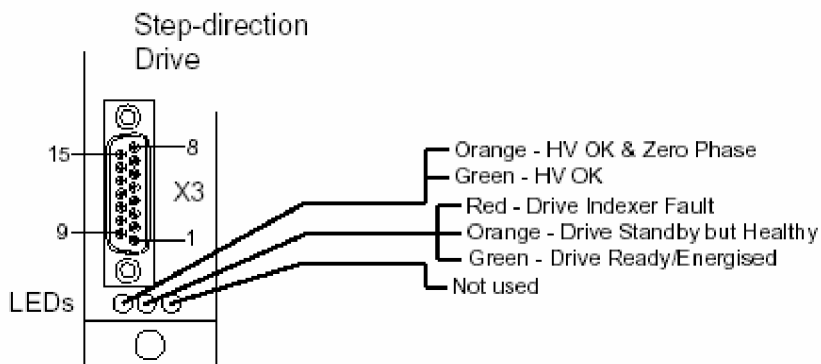


Figure 14.18 - 17 Parker OEM Stepper Drive Status LEDs

The drive LEDs are 3-colour devices and provide information on the state of the drive

Note: Standby means the drive is shutdown or de-energised.

Power (left) LED

In normal operation this LED is either green or orange. When **green** it indicates that the power supply is healthy; **orange** shows both that power is healthy and that the drive is in its zero phase state. The LED should show orange when the drive is first powered up and no step pulses have been applied.

Drive Status (middle) LED

In normal operation this LED should be **green**, showing that the drive is healthy and is energised, i.e. not in a standby condition.

When this LED is **orange**, the drive is in a standby or de-energised state but is otherwise healthy. In this condition there will be no holding torque from the motor and the drive will not respond to step pulses. If this is the cause of the problem, check the status of the differential Shutdown inputs (X2 pins 5 and 9) and also the auxiliary Shutdown input on X3 pin 13.

If the centre LED is **red** it indicates that the drive has shut down due to a fault condition.

This may be due to any of the following causes:

1. The power input voltage is either too high or too low. Check that the supply is within the limits specified in the Hardware Reference Section.
2. The X1 connector may not be fully inserted into the drive - check that the wiring is not preventing the connector from being pushed fully home.
3. There is a short circuit across two of the motor connections or between one of the motor connections and ground. Disconnect power, remove X1 and check the motor connections with a meter. This condition may also arise if a very low inductance motor is used – the preferred range is 0.8mH to 10mH.
4. The drive has shut down due to excess temperature rise. Check that the ventilation slots at the top and bottom of the housing are not obstructed and that there is free flow of air through the drive. Make sure that the drive is not mounted directly above a source of heat such as a power supply, and if possible check the ambient temperature in the vicinity of the drive casing. Allow the drive to cool down and re-check. If the motor has ample torque for the application, try reducing the current setting on the bit switch which will reduce the drive dissipation.
5. There is an internal fault in the drive or with the internally-generated logic supply. In either case the drive will need to be returned for repair.

Right LED

This LED has no function in the XL25/XL50drives

Canbus Interface

This device allows communications between the profibus network and the Parker stepper drives

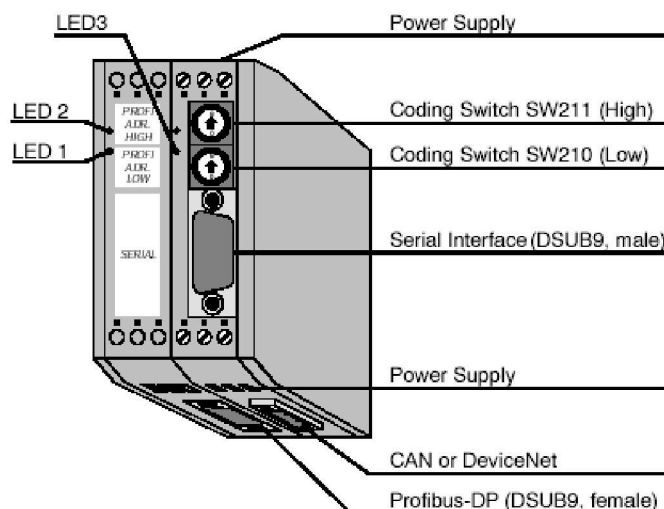


Figure 14.18 - 18 Canbus Unit Drawing

The two led indicate transmit and receive signals.

AnyBus® Communicator for Profibus

This device, which is generally located in the drive panel, allows communications from the profibus network to the deposition monitors.

#	State	Status
1 - Fieldbus Online	Off	Not online
	Green	Online
2 - Fieldbus Offline	Off	Not offline
	Red	Offline
3 - (Not used) - -		
4 - Fieldbus Diag	Off	No diagnostics present
	Red, flashing 1Hz	Error in configuration
	Red, flashing 2Hz	Error in user parameter data
	Red, flashing 4Hz	Error in initialization
5 - Subnet Status ^a	Off	Power off
	Green, flashing	Initializing and not running
	Green	Running
	Red	Stopped or subnet error, or timeout
6 - Device Status	Off	Power off
	Alternating Red/Green	Invalid or missing configuration
	Green	Initializing
	Green, flashing	Running
	Red, flashing	If the Device status LED is flashing in a sequence starting with one or more red flashes, please note the sequence pattern and contact the HMS support department

Table 14.18 - 13 Status And Error Displays On The Anybus Communicator For Profibus

Note: This led turns green when all transactions have been active at least once. This includes any transactions using “change of state” or “change of state on trigger”. If a timeout occurs on a transaction, this led will turn red.

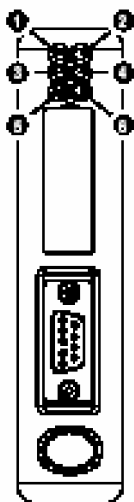


Figure 14.18 - 19 Drawing of Anybus Module

IFM - ASI Slave Units – Keystone Valves

Operation Indication

Each of the Siemens units has indication LEDs. There is a yellow LED that is lit when power is supplied to the unit.

There are also green LEDs that are illuminated when the relevant switch or interlock is operated.

Faulty Units

If the Profibus Network is not operating and the IFM unit indicates a faulty ASI slave unit, it should be replaced.

To replace a faulty slave unit on the winding cart or vacuum pumping group, the following procedure should be used;

Note: All new slave units are supplied from the manufacturer with a 0 address

1. Remove the faulty slave unit from the network.
2. Connect the new slave (with address 0) to the network.
3. The master unit will automatically re-address the new slave with the address of the old one as long as they are the same type of module.

B&R Stepper Drives LED indications

There are 2 yellow LEDs on the processor card that will flash a regular time interval as long as communications are taking place.



Figure 14.18 - 20 Photograph of Stepper Modules

What Happens when a communications fault occurs?

When a fault on the communications network occurs, the usual outcome is that the machine will stop responding to commands from the computer and will shut down. If this occurs, open up the Network screen and look for indication of a faulty network component.

How to detect a communications fault

In the case of a communications fault, the first thing to do is to check the status of the network. This is achieved by opening the network diagnostic screen.

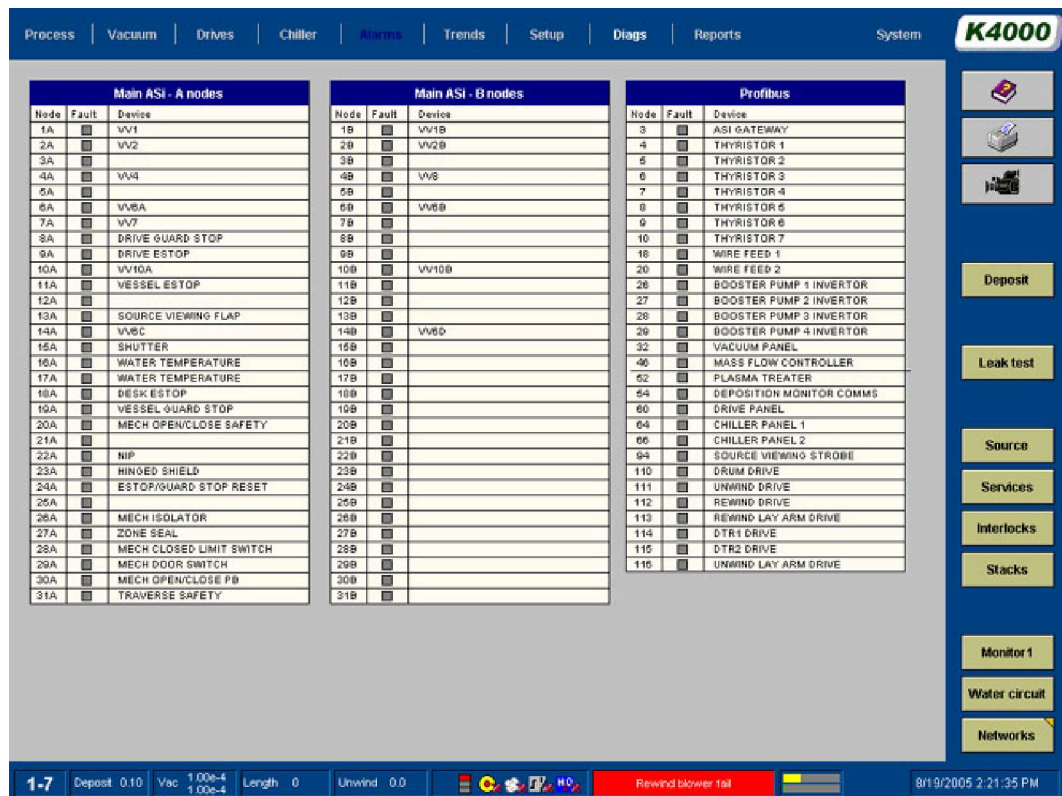


Figure 14.18 - 21 Network Diagnostic Screen

This screen will give immediate indication of a fault by showing a red module, if one of the modules has a fault.

If all the nodes are red, this would be an indication of a problem with the wiring in the computer or the interface card at the computer.

Reasons for a Communication Fault

To diagnose the reason for the fault, the following procedure is suggested;

- Check that the power healthy LED is illuminated on all Profibus slave units.
- Check for error message on the IFM unit, which will indicate a problem with one of the ASI slaves.
- Check the interconnection wiring between all units.
- Check the status of all the communication status LEDs on all Profibus slave units. Replace any units that are showing internal faults.
- Check the status of the switches on the connection plugs between modules. With 2 cables connected, the switch should be set to the OFF position. With a single cable, the switch should be set to the ON position.

Faulty wiring is the most common of these faults. It is very rare that there is a software fault and if a fault is suspected, contact General Vacuum Equipment for advice.

14.19 Winding Cart Movement

The winding cart is moved in and out of the chamber using a motor. It is located under the floor of the cart.

The linear drive motor used to drive the cart into the chamber is an AC type, which is controlled by an inverter, which is attached to the motor.

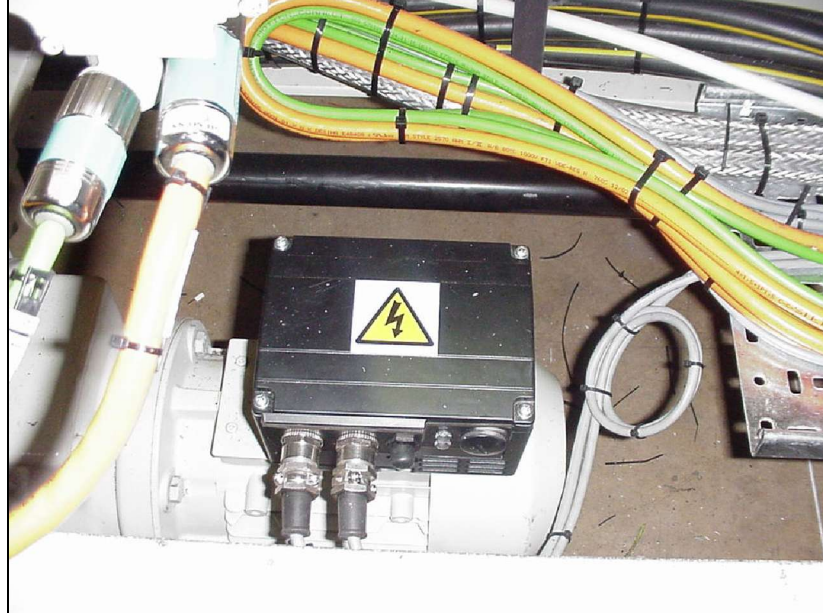


Figure 14.19 - 1 Photograph of Motor located under floor level

The users of the machine should check for small obstructions on the tracks that will stop any movement.

There are interlocks within the plc ladder software that will stop the winding cart moving, and the users of the equipment should become familiar with them.

To comply with safety regulations, a LOCKOUT ISOLATOR is provided on the machine to isolate this motor when the winding cart is withdrawn from the chamber.

	Warning !
	Always lockout the isolator when working in the area between the winding cart and chamber.

The locking of this isolator will disable the linear drive motor. It should be locked out by all personnel that are entering the chamber, or working in the winding mechanism area.

Control Sequence

The control circuit uses the limit switches located below the winding cart to activate stages in the control sequence. These switches are energized by the striker plates on the floor tracks.

Moving Out of the Chamber

With the required machine interlocks energized the winding cart can be retracted. The "Mechanism Open" (arrow) and enable push-buttons must be pressed together for 5 seconds, then released for 2 seconds and then pressed again for the period of movement.

The traverse contactor in the drive panel will be energised. This can only occur if the E-Stop circuit has been reset and the movement lockout isolator has been returned to the on position.

The speed of the movement is controlled by the status of the limit switches located under the cart.

When the motor is requested to move, a digital set speed signal is applied to the motors on board inverter unit. This digital signal is generated from the computer and transferred to the motor via the plc analogue card. The speed set point can only be changed in the CONFIGURATION screen.

As the winding cart approaches the end of the tracks, one of the limit switches is activated and the speed set point changes to allow the movement to be slower.

When the cart reaches the end of the track, another limit switch is activated to indicate to the computer that the winding cart is fully withdrawn from the chamber.

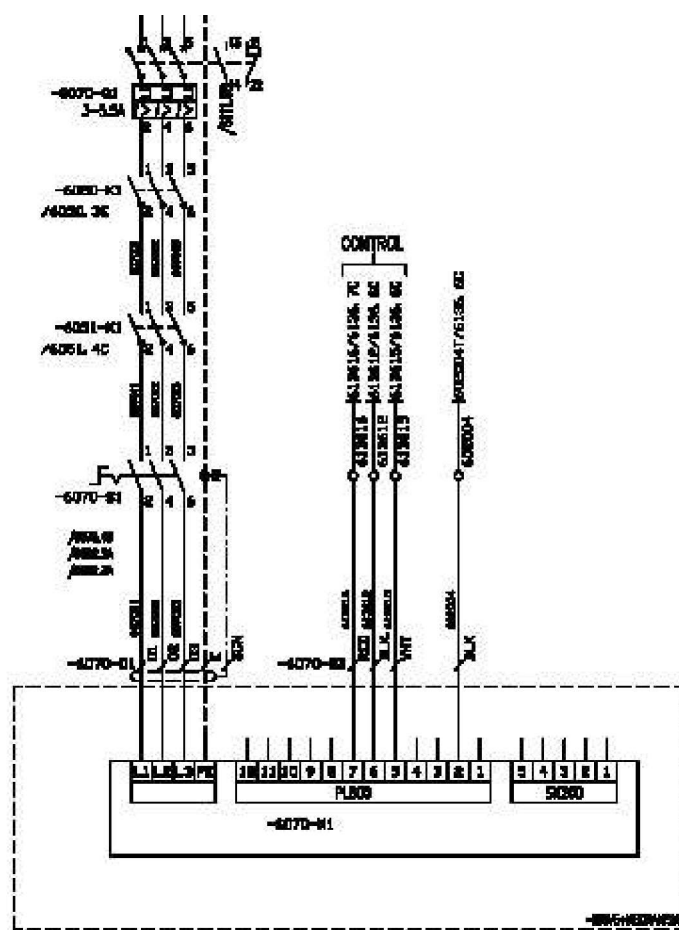


Figure 14.19 - 2 Electrical Schematic Drawing Of Traverse Movement Motor

Moving Into the Chamber

With the required machine interlocks energized the winding cart can be retracted. The "Mechanism Close" (arrow) and enable push-buttons must be pressed together for 5 seconds, then released for 2 seconds and then pressed again for the period of movement.

The traverse contactor in the mech. cubicle will be energised. This can only occur if the E-Stop circuit has been reset and the Movement Lockout isolator has been returned to the on position.

This can only occur if the E-Stop circuit has been reset and the Movement Lockout isolator has been returned to the on position.

As the winding cart approaches the chamber, one of the limit switches is activated and the speed set point changes to allow the movement to be slower.

When the cart reaches the chamber, another limit switch is activated to indicate to the computer that the winding cart is fully closed to the chamber.

There are a number of interlocks within the computer software, which will stop the winding cart moving, and the users of the equipment should become familiar with them. Refer to the operation manual for more details.

Motor Diagnostics

The motor has two LEDs on the outside of the case. These are used to indicate the status of the motor.

Green LED	Yellow LED	Meaning
On	On	Mains power is on, Motor in Standby
On	Off	Motor running as per control signals
Flashing	Flashing	Current Limit Warning
Flashing	On	Overtemperature of inverter
On	Flashing	Overtemperature of motor
Off	On	Other Fault - i.e. trip - see Siemens Combimaster manual
Off	Flashing	Main power under voltage
Off	Off	Main power off

Table 14.19 - 1 Combimaster LED Troubleshooting Look up Table

Castors

The cart is supported on castor wheels that run on the floor rails. These castors also act as a guide for locating the cart. The castors are to be greased on a monthly basis.



Figure 14.19 - 3 Photograph of Mechanism Carriage Castors

Chain Drive

A triplex chain is used to transfer the driving motion between the motor and the cart drive shaft. An adjustable positioning sprocket is provided to keep the chain under tension. The tension of the chain should be checked regularly, and tensioning sprocket adjusted.

This chain is located under the floor plate, which should always be replaced after maintenance.

Lubricate the chain as per the maintenance schedule.

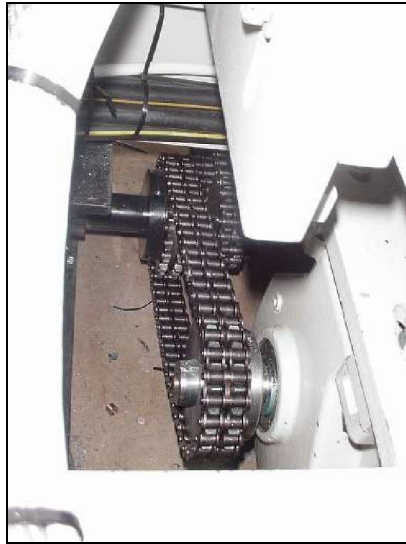


Figure 14.19 - 4 Photograph of Cart Triplex Drive Chain

14.20 Electrical Leadthroughs

Electrical leadthroughs are provided on the machine to allow both power and control circuits to be continued from the atmospheric side of the chamber into the vacuum side.

The leadthroughs generally are supplied in a 6, 12 or 25 pin configuration with different power ratings. The individual leadthroughs on the winding cart are mounted upon flanges that are removable. The leadthroughs are "built up" from a plug, socket and bulkhead fittings. The bulkhead fittings are manufactured to carry the rated power or control signals, while also being sealed for a vacuum environment.

	DANGER
	Beware Live Electrical Circuits, Isolate before working in this area



Figure 14.20 - 1 Photograph Showing Electrical leadthroughs

A listing of the leadthrough types and numbers are listed in the electrical drawing set. This shows the type of leadthrough being used and also the pins used and those spare.

Under normal circumstances, these leadthroughs will operate for the life of the machine. Care must be taken not to add or modify circuits on the leadthrough to limits beyond its rated capacity.

14.21 Vacuum "O Ring"

Winding Cart to Chamber

The chamber to "faceplate" o-ring seal is located within a "Dovetail" groove on the faceplate. It is manufactured from a length of 1/2" OD Nitrile cord and is jointed with a standard o-ring jointing kit.

The joint position is located at a 3 o'clock" position on the band face, in the winding zone.

If for any reason this o-ring is damaged, it can be replaced by pulling it out of the groove and inserting a new piece of cord.

If a complete new o-ring is to be installed, it should be made about 150 mm longer than the circumference of the groove.

It should be placed around the winding mechanism and jointed. Grease the cord with the Apizon M grease and insert it in the groove in a "serpentine" pattern. Use a rubber mallet to "knock" the o-ring into position, ensuring that there is no damage or pinch points.

Valves and Other Fittings

Isolation Valves

There are o-rings within the shaft assembly, which can be supplied in a spares kit by the valve manufacturer. The valve seat is manufactured with a rubber section and no o-ring is used. If the seat is damaged, replace the whole valve body as the seat is bonded to it.

Vacuum Pipework

Where the sections of pipework are jointed together with flanges, an o-ring will be used to ensure the vacuum integrity of the joint. These O-rings are manufactured by General Vacuum Equipment out of rubber cord and are jointed with a standard sealing kit.

Diffusion Pump(s)

The o-rings in the diffusion pump are supplied by the pump manufacturer and are available from them or General Vacuum Equipment. Care must be taken in the type of o-ring material selected.

Klein Fittings

The various gauge points on the machine are designed to accept fittings manufactured to the "Klein" standard. These fittings will be either 16, 25 or 40 mm diameter and are referred to as 16KF, for example. The o-rings used to seal these joints use cantering rings for support.

Fine Valve Plates

Access to these o-rings contained to the plate valves has been described previously.

Care must be taken in raising the plates.

The o-rings are manufactured by General Vacuum Equipment from o-ring cord.

14.22 Zoning Seals

Flexible Seals

To allow for relative pressure differentials between the various zones, a number of "zoning" devices are used.

There are silicone seals used for flexible joints on the contact points between the chamber and the winding cart.

- Lubrication and inspection

Check the walls of the chamber to check that the end the zone seals are sealing. If they are not, deposits will be seen where the seal has broken down.



Figure 14.22 - 1 Photograph Showing Zoning Seal attached to coating shield



Figure 14.22 - 2 Photograph Showing Zoning Seal attached Winding Mechanism

	Warning ! Under no circumstances must any person enter the coating shield area unless they have locked out the movement isolator.
---	---



To lockout the movement lockout isolator, turn it through 90 degrees and place a padlock on the switch.

Figure 14.22 - 3 Photograph Showing User Locking Out Of Movement Isolator

Low Conductance Zoning Seals

There is a seal assembly built on either side of the drum, which allow the substrate to enter the evaporation zone. These are called the low conductance seals. Their purpose is to stop gas molecules from the winding zone entering the evaporation zone without affecting the substrate.

Accessing the Seals

On this machine, the zone sealing arrangement is attached to the drum coating shield.

To access the zone sections away from the drum, the coating shield is lowered.

Always check for debris on these shields, as any can cause scratches, as it builds up

The normal position for these seals to be in is close to the drum. Leaving the shield lower than its designated closed position during production cycles will affect the performance of the pumping system.

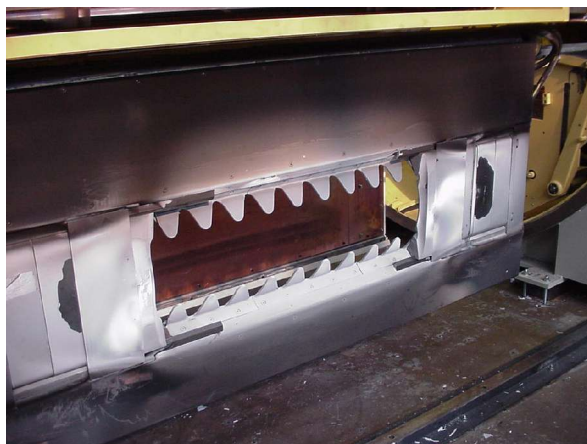


Figure 14.22 - 4 Coating Shield in lowered position

14.23 Bearing, Coverings and Vacuum Seal Replacement on Rollers

Specifications

The specifications of material type and size of all bearings and seals can be found in the bill of materials supplied with the machine.

Replacement of Bearings on Driven Rollers

The rollers, i.e. process drums and draw rollers are designed to be left in position within the winding mechanism while bearing/seal changes take place.

The seals/bearings are located in manufactured cartridges at either end, which can be removed.

Always support the roller while carrying out these repairs and refer to the drawings provided at all times.

Replacement of Bearings on Idler/Spreader Rollers

The rollers have to be removed from the winding mechanism to replace the bearings

Material Specifications

It is recommended that before having any of the above rollers recoated, that the customer contacts General Vacuum Equipment for the type and hardness of the rubber to be used.

If fitted, the "Bow" roller is manufactured by Mount Hope and should be only repaired by the manufacture. Please contact General Vacuum Equipment, if you have problems with locating a repair agent.

Replacement

General Vacuum Equipment can supply replacement rollers, but due to the different width of machines supplied by General Vacuum Equipment, due notice is required. Please be aware that certain rollers are designed and manufactured by third parties.

14.24 Tuning Non Contact DCM Probes

If a hardware fault is indicated during a self-test sequence, check the voltage levels on the VOLTS screen on the computer screen. If one of the probes shows a low voltage, attempt to adjust the centre frequency as described below;

Procedure for tuning individual probes

1. In the SETUP screen, create a Recipe called "Calibration", with maximum width and /sq value of the calibration sheet that will be used and down load it into the program.
2. Load the SETUP parameters
3. Allow at least 30 minutes from power up for the DCM Interface Buffer box & Beam to stabilize.



Figure 14.24 - 1 Photograph of Buffer Box Location

4. Enter the Supervisor login name and password to allow access to monitor diagnostic screen, MONITOR 1

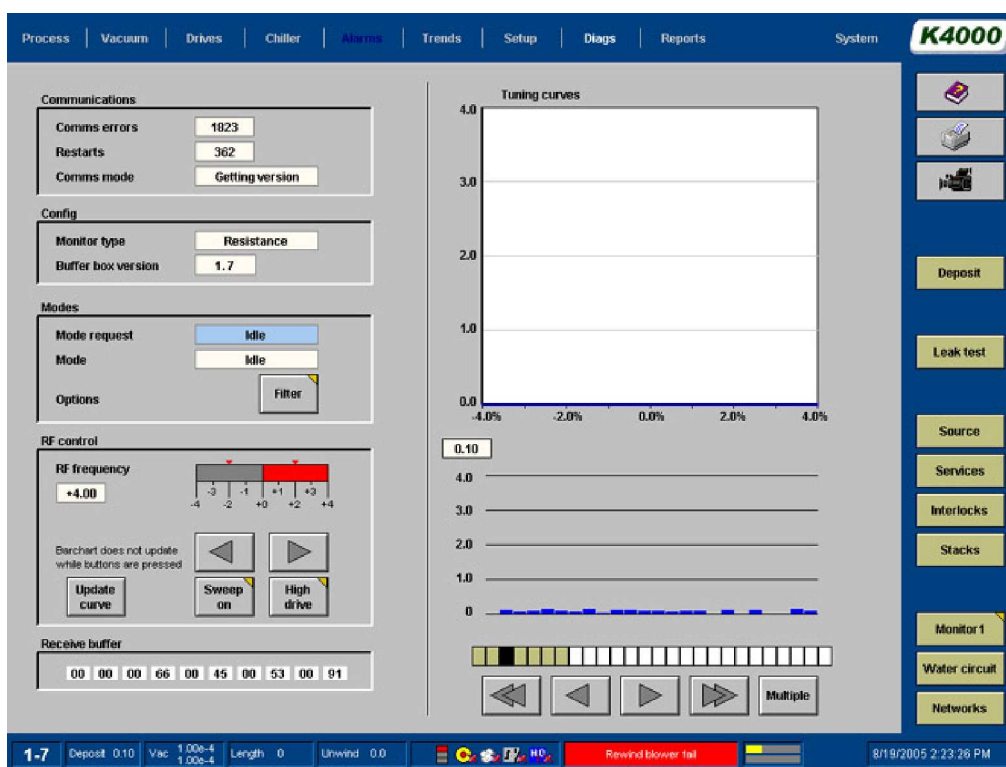


Figure 14.24 - 2 MONITOR 1 Screen Indicating Feedback

5. Perform a self-test by selecting the feature from the MODE REQUEST drop down listing

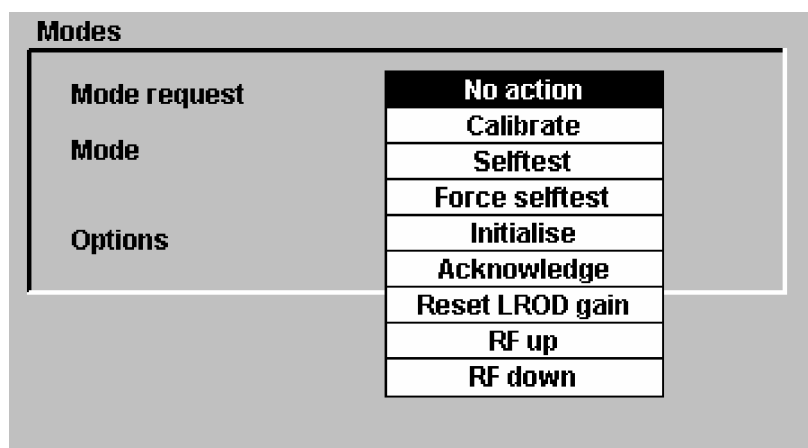


Figure 14.24 - 3 MONITOR 1 Screen Indicating Mode Request



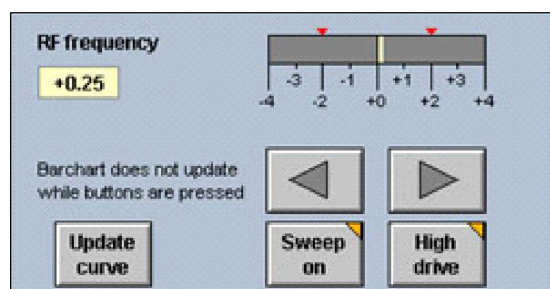
Figure 14.24 - 4 Self-Test Pop up Screen

6. Once the self-test is completed (bar in box as shown on the screen has travelled fully from left to right) observe the self-test results on the left hand side of the CALIBRATION screen, all of the frequency results should be within +/- 0.25% of the centre frequency.

SELFTEST RESULTS			
PROBE	0.00%	ZERO	OFFSET
3	+0.00%	3.252V	0.040V
4	+0.00%	3.243V	0.036V
5	-0.25%	3.240V	0.039V
6	+0.00%	3.232V	0.043V

Figure 14.24 - 1 Self-Test Results Section of Screen

7. On average the centre beam frequency should be set by the computer between -1.00% to +1.00%.
8. If an individual probe has a low voltage reading <2.8 volts, or has a high reading compared to the rest of the probes 3.4 volts, then it will need to be tuned.



9. Return to the MONITOR 1 Diagnostic screen and turn the RF Sweep off by touching the SWEEP button once and the action indicator will be removed. It should also be noticed that all the probe voltages drop by up to a volt

Figure 14.24 - 5 RF Frequency Section of Screen.

10. Using the UP/DOWN centre frequency bottoms, adjust the centre frequency by $\pm 0.25\%$ to re-maximize output voltages.
11. On the monitor beam, locate the internal ferrite core through the dedicated hole in the monitor beam.

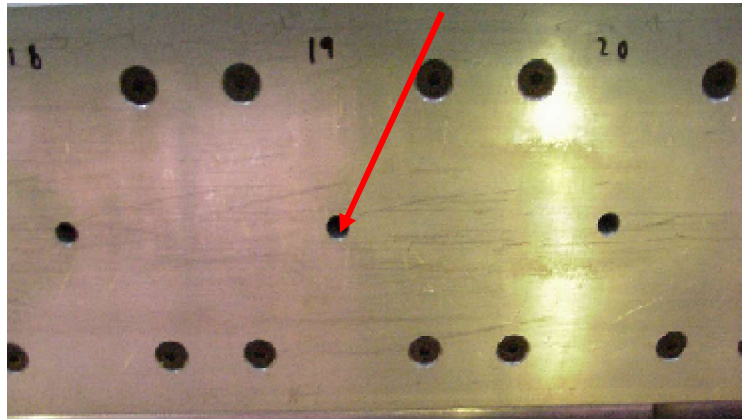


Figure 14.24 - 6 Access Hole in DCM Beam

12. Turn the internal ferrite core clockwise and/or anti-clockwise to increase and/or decrease the feedback voltage as required to maximise the output.
13. NB using a normal screwdriver (ferrous) will effect the voltage reading even when the core has not been moved (usually it increases the voltage value). It is recommended to use a plastic non-ferrous electronic trimmer tool with a brass head.
14. Each suspect voltage level should be adjusted till the maximum output is reached.(the point where value will start to decrease)
15. After all adjustments have taken place, switch the RF Sweep back on and perform a self-test observing that it completed successfully.

Buffer Box

1. If no voltage feedback is evident on the CALIBRATION or DIAGNOSTIC screen, then check that the buffer box is being supplied with a 24V power supply. Use a meter to measure across Pins 5 (+24V) and 9 (0V).

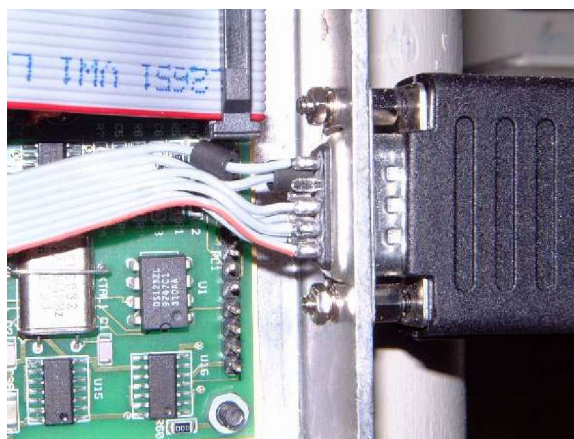


Figure 14.24 - 7 Location of pins in DCM Buffer Box

2. If the RF Sweep frequency is out of the range (-1.00% to $+1.00\%$)
3. Set the centre frequency to zero by pressing either the up or down buttons.
4. Then open the buffer box's cover and locate the potentiometer on the left-hand side of the card.

Potentiometer

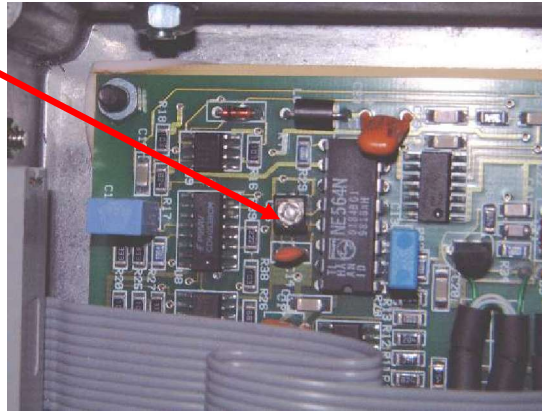


Figure 14.24 - 8 DCM Tuning potentiometer

1. Turn the RF Sweep off again, turn the Pot very slowly in small amounts until the all the probe voltages reach maximum value.
2. Once this voltage value has been achieved, turn the RF Sweep back on and perform a self-test once again.

14.25 Installing or Removing DCM Resistance Monitor Beam

The monitor beam should not normally require removal, unless a probe is damaged for example, or the cork is to be recovered on the DTR2 roller.

Removal:

Before any attempts to remove the beam are made, the power to the system should be removed.

	Note! It is suggested that the main isolator to the machine be switched off to ensure that the whole system is isolated.
---	--

At the end of the beam, there is a control cable connection, which should be unplugged, and the cable withdrawn to a safe position, where it will not become damaged.



Figure 14.25 - 1 Photograph Showing DCM Beam cable Connection

The beam is located on support brackets. There are two cap-head screws in each end of the beam that should be loosened and removed, the beam will not fall as they are removed.

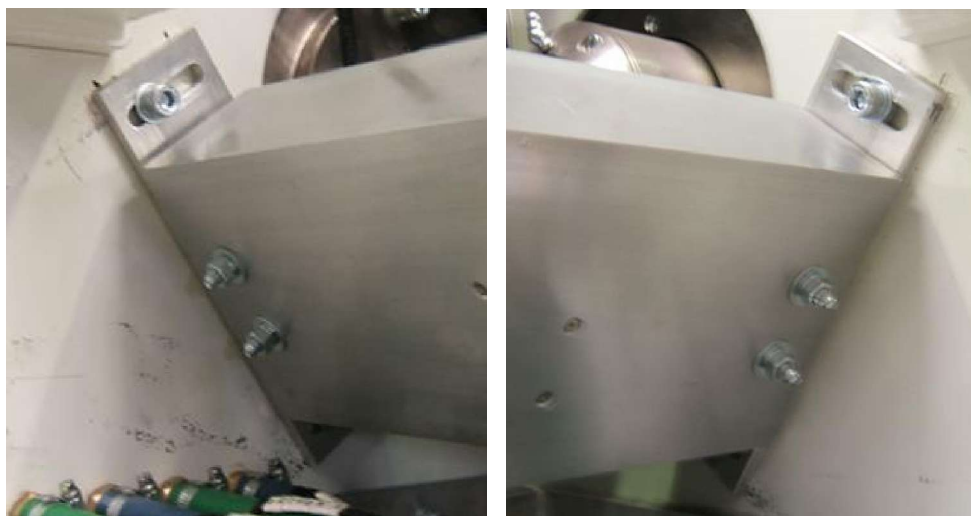


Figure 14.25 - 2 Photograph Showing DCM Beam Support Brackets - End

On some machines, there is also a third support beam that is located in the centre of the beam. This has to be removed to allow the beam to be removed from the machine.

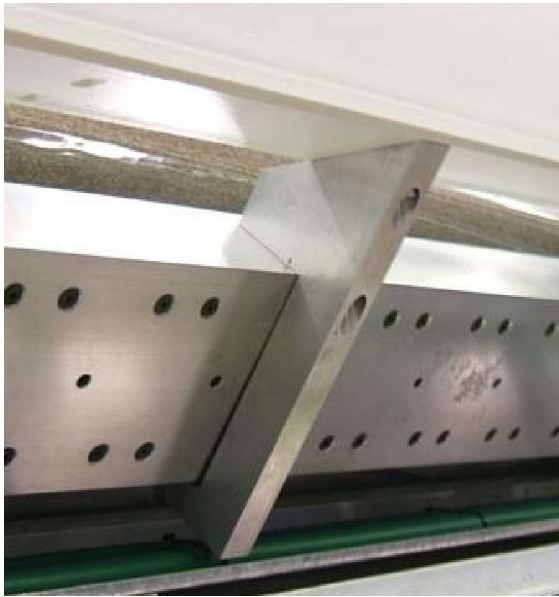


Figure 14.25 - 3 Photograph Showing DCM Beam Support Bracket - Centre

After removal of the screws, the aluminium beam can be lifted out of the machine.

Installation:

The beam can be re-installed after any work has been carried out on it.

Replace the beam in the reverse sequence to that mentioned above.

Checking Distance between Probes and Roller:

The beam should be checked to have a gap of 1.8mm (maximum 2.0mm) between the individual probes and the roller, to allow accurate deposition monitoring. This can be carried out by using plastic shim material placed between the roller and the probes.

If individual probes are not to the required distance, the screws, holding the probes, can be adjusted.



Figure 14.25 - 4 Photograph Showing DCM Probe to Roller Distance Checks

After replacing the beam, calibrate all the probes.

14.26 Removing and Refitting DCM Non Contact Probes

To replace probes, the beam has to be removed as described in section 14.25 of this manual

To change the probe, the following procedure should be used;

1. Switch off the machine power supply.
2. Disconnect the buffer box to beam cable.
3. Disconnect the monitor beam and remove it from the machine.

	Note!
	It is suggested that the main isolator to the machine be switched off to ensure that the whole system is isolated.

4. Unscrew the four countersunk setscrews and withdraw the suspect probe as far as the cable will allow.
5. Pull of the connector gently from the probe.
6. Withdraw the probe completely away from the cable.

Installing new Probe:

7. To refit, check that the retaining screws are in place. Tape over the screws heads so that the screws cannot be pushed out of the beam.
8. Hold the D connector and push the probe onto it within the beam.
9. Rest the probe on the screw threads and secure it in place by taping over the probe and onto the beam. The beam may now be turned over with the probe in place and the screws tightened.
10. Tighten the screws so that the new probe is level with the adjacent probes. Check by laying a ruler across both sides of the black part of the probe. Apply a drop of thread locking compound to each screw thread in the probe. E.g. Loctite 242e.

Re-install the beam and check for the probe to roller distance.

11. Switch on the machine power supply
12. Switch the deposition monitor computer/rack on and select the voltage screen.
13. Check that all the probes are giving a reading. Note the self-test on power up may fail due to the new probe being un-tuned.
14. Allow 20 minutes for the monitor hardware to warm up.
15. Tune the replacement probe as per section 24 of this manual.

14.27 Non Contact Resistance Monitor Probe Calibration

Equipment Required

A "swatch" of metallized material should be laminated. The size of the "swatch" should be 8.5x11 inches or equivalent to an A4 sheet. The deposit of copper on the "swatch" of material should be measured using the test method described in the AMICAL Technical Manual. The swatch should have locating datum's marked upon it as shown in figure below;

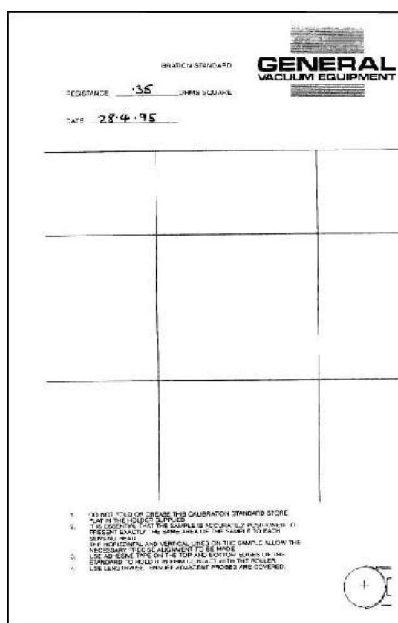


Figure 14.27 - 1 Calibration Swatch

Procedure

1. Switch the system on and allow the system to warm up over 27 minutes.
2. Clean the insulated cork covered roller as described in the operation manual supplied.
3. Within the SETUP screen, select the maximum width of the machine and enter the calibration "swatch" resistance value. Load the set-up information.

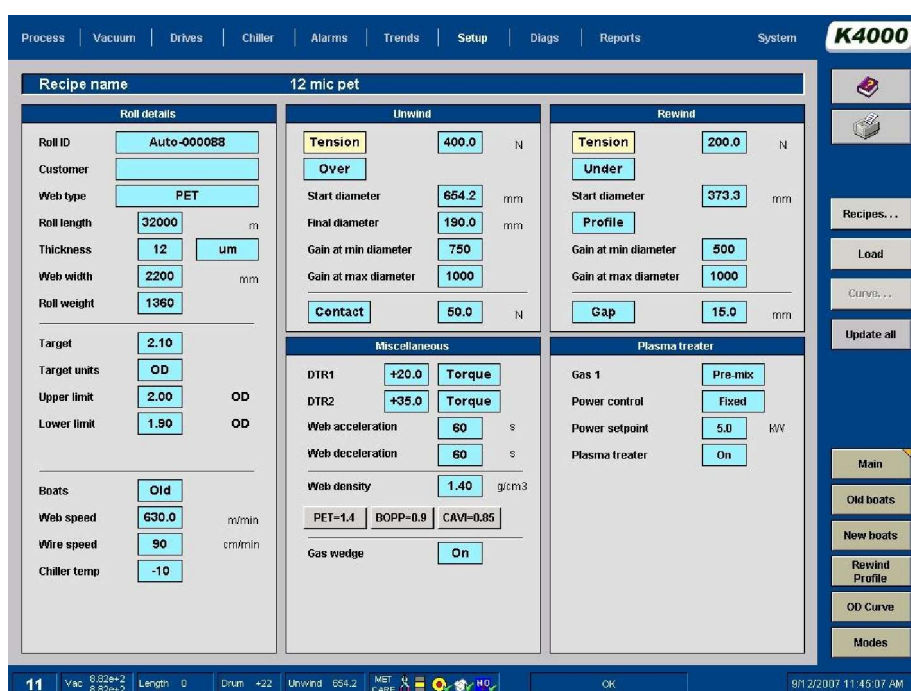


Figure 14.27 - 2 SETUP Screen

4. Log onto the machine at the supervisory level of operation to allow access to the CALIBRATION screen.
5. Select the CALIBRATION screen

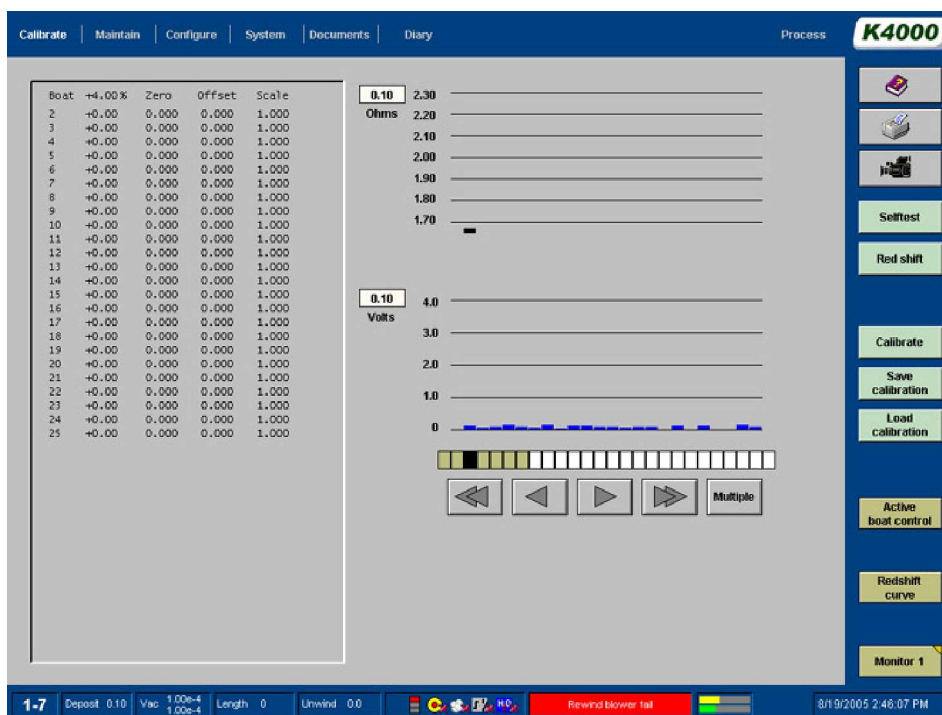


Figure 14.27 - 3 CALIBRATION Screen

6. Tape the "swatch" on to the insulated roller, in front of probe 2, as shown. Align the vertical guidelines on the outside edges of the probe.
7. Rotate the insulated roller to place the swatch in front of probe 2, as shown, so that the horizontal line can be positioned at a known position.



Figure 14.27 - 4 Taping Calibration Swatch to the DTR2 Roller

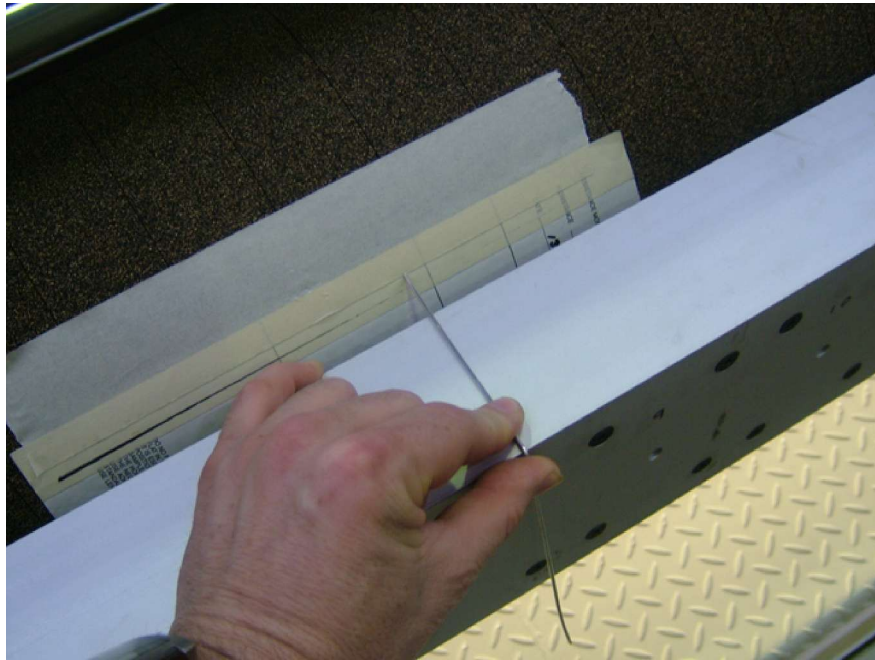


Figure 14.27 - 5 Aligning the Swatch Horizontally

8. It is important that the "swatch" is placed in the same position in front of every probe. As shown a piece of aluminium wire can be used to check the depth of alignment.
9. The cursor on the Calibration screen should be moved to probe 2 position using the left and right direction cursors. The reading given on the left hand side is the deposition rate measured by the probe.
10. The value should be adjusted by pressing the calibration function button.
11. Repeat the above steps with the "swatch" located in front of each of the probes as described above.
12. After all the probes have been checked/calibrated, press any of the other screen tabs and the calibration values will be saved.

Reasons for Calibration Differences

It has been found that the difference in calibration between regular checks is usually negligible. However, sometimes under the following circumstances, calibration is definitely required;

- Removal of Monitor Beam
- Replacement of faulty Probe

It has been found that the following can cause the measurements to differ from that required;

1. Placing the system in a "Self Test" when light depositions are present on film in front of the probes, i.e.; 15 ohms and above.
2. Build up of coating medium dust around the probes.
3. Accidental changes in calibration file.

14.28 Simple Troubleshooting Of DCM Beams

To check on the operation of the monitor beams quickly, the voltage signals from the probes can be monitored on the CALIBRATION screen as shown below;

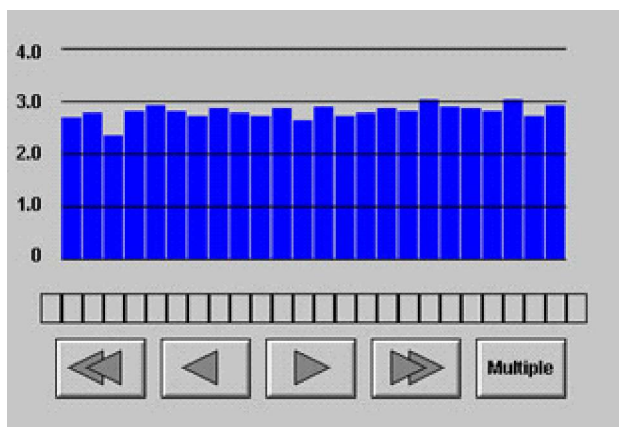


Figure 14.28 - 1 Repair Screen

With no substrate in front of the probes, the voltages should be similar at a level above 3.0 volts, but not exceeding 4.0 volts. Placing metallized substrate in front of the probes, will have the effect of dropping the voltage to about 2.0 volts.

If there are no volts shown on this display (as shown above), check the power supply to the interface box on the back of the winding cart, for the correct supply voltages.

Also check that the interconnecting cables are engaged, if not power down the machine totally before re-installing them.

If an individual probe 's feedback voltage is low, attempt to retune it as described earlier in this manual.

14.29 Water Supply Strainer/Filter

The water supply strainer (filter) is located in the main incoming water supply line and is supplied to stop contamination of the machine by substances and particles in the water supply.

The machine is supplied with a water system diagnostic screen which should be used to monitor the performance of the machine.

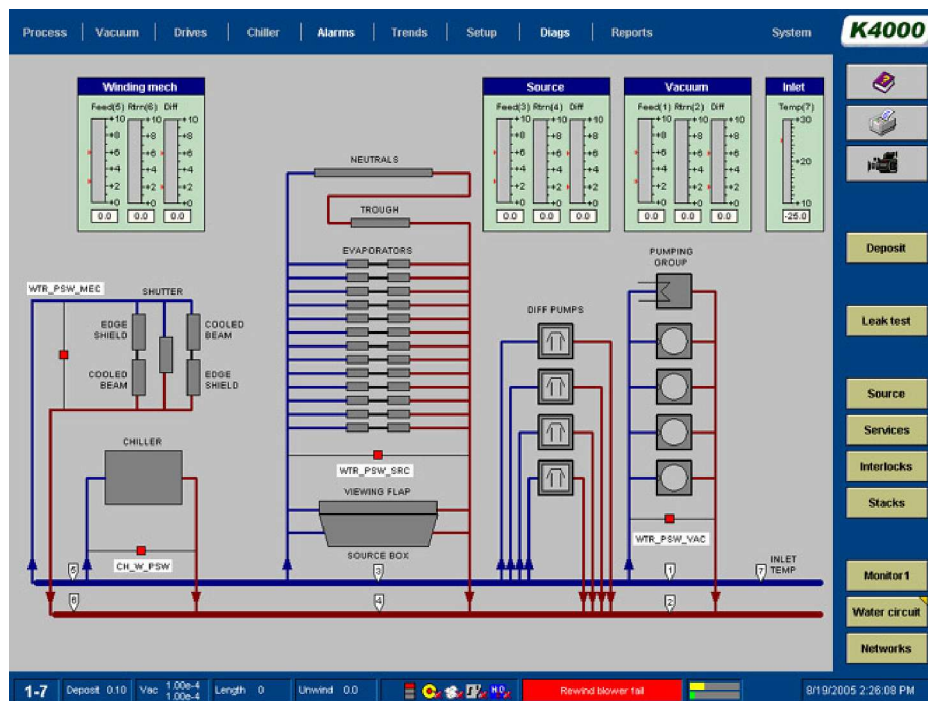


Figure 14.29 - 1 Water Diagnostic Screen

Dependant upon the water supply, the strainer should be cleaned as recommended in the preventive maintenance schedule.

	Caution !
	Do Not Remove Any Mesh Strainer From This Unit Permanently. Only Remove For Cleaning

The machine has been built to a design specification based upon a certain cleanliness of the water supply. If this specification is not met, damage could occur to the machine.

Location

The strainer is located on the rear of the machine.

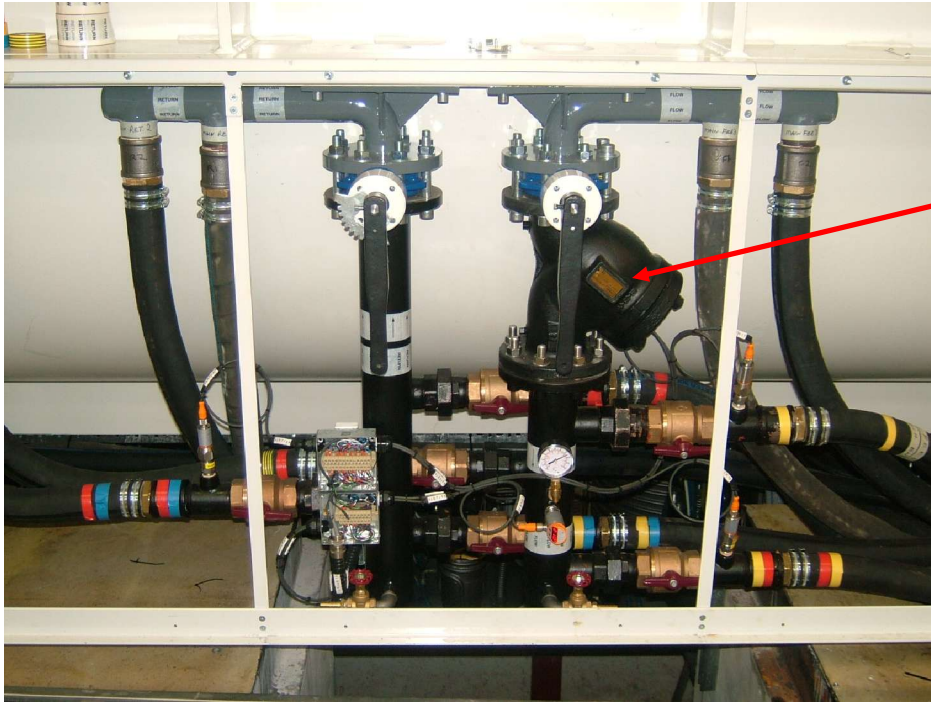


Figure 14.29 - 2 Water Strainer/Filter Location

14.30 Repairing Cork Covered Rollers

The cork covering on rollers within the winding mechanism, which are damaged or badly contaminated, can cause substrate defects. It should be replaced with a 50mm wide tape of the same quality as used originally. This is available from General Vacuum Equipment.

Tools and Materials Required

The following tools and equipment are required;

- Sharp knife.
- Pencil
- Electrical Insulation tape.
- Tape ruler
- Cleaning fluid to remove adhesive.
- Armstrong adhesive backed cork tape
- Set of stepladders
- Piece of pipe to span between the stepladders.

Removing the Old Cork Tape:

The following procedure should be used;

1. Remove the electrical insulation tape from either end of the roller.
2. Find one end of the cork and ease it away from the roller.
3. Pull the cork off the roller core, rotating the roller by the pulling action.
4. Place the old cork in a suitable disposal container and remove it from site.

We are not aware of any special conditions that are required for disposing of the material.

Cleaning the Base Roller Core/shaft:

The adhesive remaining on the roller shaft should be removed before any further steps are taken.

This can be completed using a suitable degreasing solvent and acetone.

Setting Up to apply the new cork tape:

To allow the tape to be applied at the correct angle and with ease, the following procedure should be used;

1. Use the pencil to apply two parallel marks on the circumference of the roller at a gap of 200mm.
2. At the second mark if viewed in a clockwise direction a further cross mark should be placed on the roller at 20mm from the edge of the roller.
3. A line should be drawn between the cross mark at the second position and the first mark on the edge of the roller.
4. This is the angle that the cork will be applied

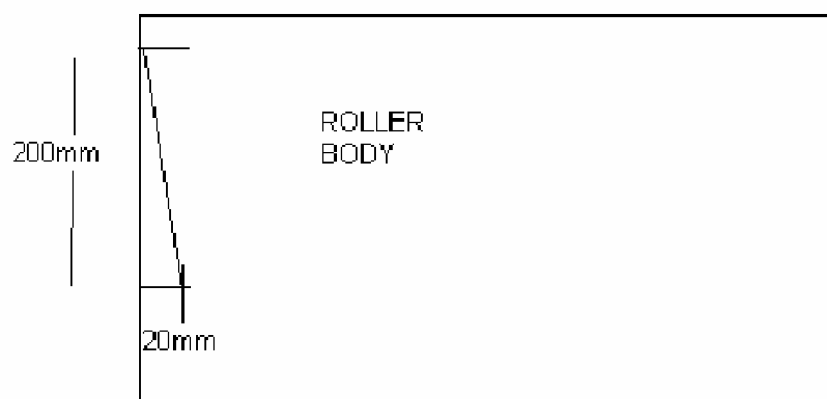


Figure 14.30 - 1 Cork- Roller Datum

It is suggested that the cork roll is placed between the steps on a piece of bar to allow it to be easily pulled off. This reduces the chances of the cork twisting as it is applied to the roller.

Applying the New Cork Tape:

1. Remove the staple from the roll of cork.
2. Ensure that the length of cork is long enough to cover the roller.
3. Remove the backing cover off the tape for a distance of approximately 100mm.



4. Place the cork down on the roller along the outside of the line that has been drawn on it. The cork will overhang the edge of the roller.

Figure 14.30 - 2 Starting to Cork the roller

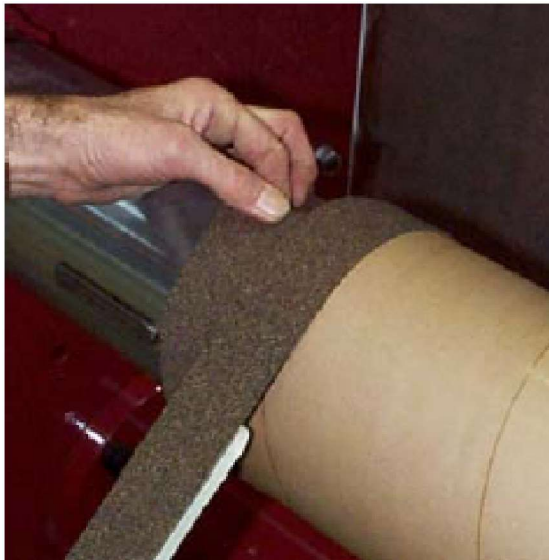


Figure 14.30 - 2 Corking the roller – Step 2

5. Slowly rotate the roller and place the next layer of the cork butted up to the first layer. It is important to keep the same amount of force on the cork while rotating the roller evenly.



Figure 14.30 - 4 Corking the roller – Step 3



Figure 14.30 - 5 Corking the roller – Step 4



Figure 14.30 - 3 Corking the roller – Step 5

9. Apply electrical insulation tape to the edge of the cork and onto the first 10mm of the cork to ensure that the ends of the cork do not become loose.



Figure 14.30 - 4 Smoothing The Edges of the Electrical Insulating tape

The rolls of tape that can be purchased from General Vacuum Equipment are 200 foot long and have an adhesive backing.



Figure 14.30 - 5 Finished Cork Covered Roller

14.31 Gravity Fill Oil Bottles

All the high-speed rotary leadthroughs provided on this machine comprise of special oil seals that are lubricated by oil. The oil is fed to the seals by gravity from bottles located close to the seals.

The seal assembly has a feed and a return line to the bottles. The bottle is placed higher than the seal arrangement to ensure that all air is removed from the seal area.

The same arrangement can be found on the

The bottles are useful in detecting faults upon the seal arrangement. As the oil seals begin to fail, the oil from the bottle will either be drawn into the chamber, or leak outside the rotary seal arrangement.

	Caution !
	The oil used in the various bottles may differ on some parts of the machine. Reference must be made to the lubrication section of this manual.



Figure 14.31 - 1 Gravity Fill Oil Bottles

	Caution !
	The lubricating oils from the seals are potentially harmful to the environment. Never allow oil or parts dipped in oil to penetrate the supply of drinking water. Lubricating oil (even small amounts!) must be disposed of as hazardous waste.

14.32 Mechanical Rotary Leadthroughs

Low Speed

There are some rotary leadthroughs on the machine, which are classified as low speed. These are listed below;

- Shutter Movement leadthrough
- Drum/Coating Shield raise /lower leadthrough

They are filled with the recommended grease when being assembled. Over a long period of time, the components in the leadthrough will eventually require replacing. At that time, the grease should also be replaced totally with the correct recommended grease.

	Caution !
	Do not use grease types that are not on the recommended list, as this could lead to failure of components upon the machine.

High Speed

As previously mentioned, the high speed leadthroughs are oil lubricated from gravity fed oil bottles. These leadthroughs are used for;

- Booster pump shaft seals
- Driven roller on the winding mechanism
- Evaporation Source Flap Valve Leadthrough
- High Vacuum Valve Leadthrough

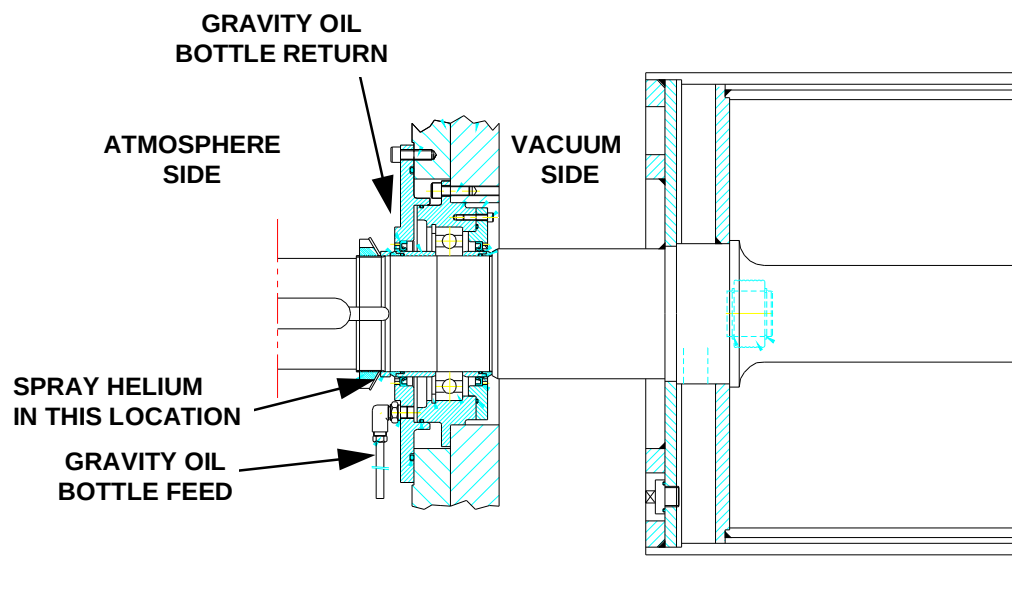


Figure 14.32 - 1 Rotary Leadthrough on Drum

	Caution ! Do not use oil types that are not on the recommended list, as this could lead to failure of components upon the machine.
	Note! Contaminated components and operating media must be disposed of in accordance with local government regulations where the machine is operated.
	Warning ! Lubricating oil contaminated with hazardous substances can be hazardous to your health. Skin or eye contact with oil and inhaling oil mists can be hazardous to your health. Wear relevant personal protective equipment when draining oil from vacuum pumps, which can potentially contain harmful gases or other harmful substances, and when disposing of exhaust condensation or filters.
	Caution ! The lubricating oils or greases are potentially harmful to the environment. Never allow oils/greases or parts dipped in oil/grease to penetrate the supply of drinking water. Oil/grease (even small amounts!) must be disposed of as hazardous waste.

14.33 Process Chiller System

The process chiller system consists of a refrigeration unit and auxiliary storage tank, valves and transfer pump.

The refrigeration unit manufacturer supplies a fully documented manual that should be referred to for reference.

There are monitoring links between the chiller system and the computer to allow the user to monitor the operation and control the chiller unit.

The CHILLER interface and ALARM screens will indicate when there is a fault on the chiller unit. Indication of a fault occurs when;

- There is a loss of refrigerant gas (Low Gas Pressure)
- There is a loss of cooling medium (High Gas Pressure)
- Low refrigerant oil pressure or flow
- Circuit breakers tripped
- Over temperature situation
- Low cooling medium flow to the process.
- Loss of condenser water to the chiller

All these conditions are explained in the operation manual.

The system uses a mixture of ethylene glycol and water as its coolant.

Testing the Coolant

The coolant, when measured with a hydrometer must give a specific gravity of 1.080 at 68°F (20°C), which gives a freezing point of -30°C. This is the normal; mixture for chillers rated to -15 lowest temperature.

A specific gravity less than 1.080 may cause the coolant to freeze and damage the chiller.

A specific gravity higher than 1.080 will delay the cooling process to taking place and in fact the process may not reach its operating temperature.

The coolant mixture should be tested as per the recommendation in the preventive maintenance schedule. It should be tested in the following manner.

1. Fill a container with a sample of the coolant taken from the cooling system. This can be drained from one of the two drains found under the chiller, near to the process in/out valves.. The container should be inches (254 mm) deep and at least 3 inches (75mm) wide.
2. Place a Hydrometer in the sample and allow it to settle. The specific gravity level can be read off the scale. Where the liquid bisects the scale is the value.
3. If the value is greater (i.e. 1.130) add water to the top up gravity fed tank. Recirculate the coolant and retest a sample.
4. In fact it is very rare to find a mixture change from its original specific gravity value.
5. If the number is lower (i.e. 1.040) add Ethylene Glycol to the mixture.



Caution !

The chiller heat exchange will be damaged if the mixture freezes due to incorrect coolant mixtures.

If the cool down time of the chiller is excessive it could be due to the mixture going 'mushy' prior to freezing.

Draining the Process Lines from the Chiller to the Rollers

The cooling medium can be drained from the cooling system by opening valves located under the cart floor.

The amount of cooling medium in the system will vary from machine to machine. It is usually in excess of 50 litres and therefore containers are required when draining the complete system.

It is suggested that the medium should also be removed from here when a sample is required.

Glycol Specification

The chiller unit uses a medium made of a mixture of ethylene glycol and water. The quality of the ethylene glycol is specified in the consumable section of this manual.

Filling the Chiller and Bleeding

It is necessary to prime the water / glycol system in order to purge all of the air.

Recommended Priming Procedure:

1. Check that the entire system has been tested for leaks.
2. Ensure that both the outlet ball valve (V4) and inlet ball valve (V1) are closed
3. Fill the tank by feeding water / glycol in through its lid (located underneath the service panel at the manifold end of the unit). Tank should be filled to within 100mm of the top
4. Bleed the pump (P1) via its bleed valve.
5. Start the priming procedure by setting the prime pumps button on the CHILLER screen. The pump will start and the water / glycol will circulate around the tank loop
6. Check that the pressure transducer on the outlet (Glycol pump) is reading between 1 and 2 bar
7. Open outlet ball valve (V4) slowly.
8. Crack open inlet ball valve (V1) to allow air to slowly be purged from the drum and roller.
Note: On some machines there are two circuits.
9. Continue to monitor the pressure transducer on the outlet (Glycol pump). If it falls to zero, the pump has lost its prime and will require re-priming. Stop prime pumps function on the CHILLER screen. Shut outlet ball valve (V4) and inlet ball valve (V1) and bleed and prime the pump (P1) using the same procedure as step 4. Restart the priming procedure.
10. Continue this process until there is a steady stream of water / glycol entering the tank.

Any bubbles indicate that the system is not properly primed

Testing the Heaters

The heaters should be checked for equal current being drawn on all three phases using a clamp ammeter. If the currents are not even, prepare to replace a heater.

Losses of Coolant From the System

There will be a natural evaporation of the coolant over a long time period. Checks should be made weekly on the level in the tanks. Addition of liquid must be made as required.

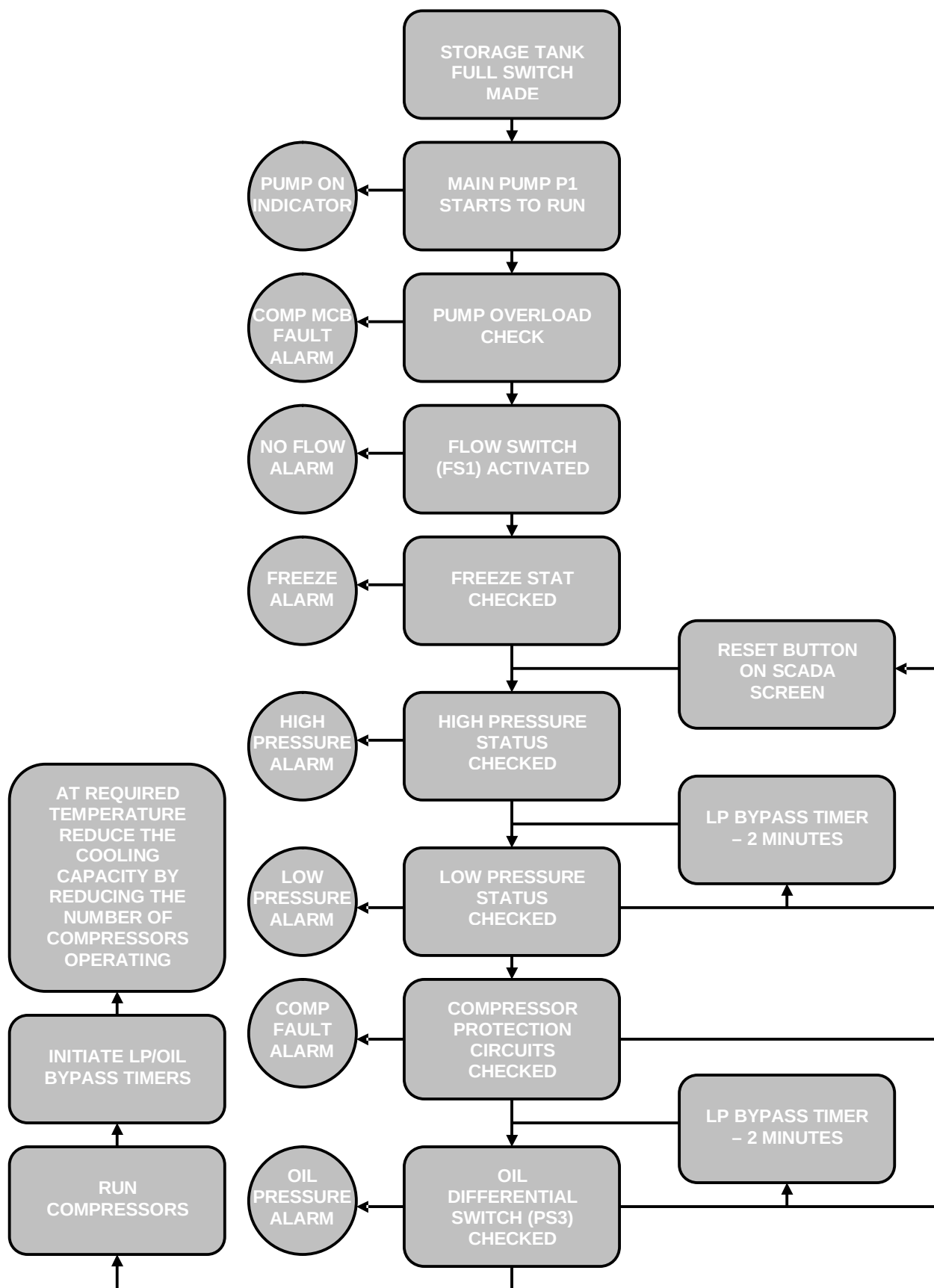
If the coolant volume changes for any reason and the upper level switch in the storage tank is not energised, the main circulation pump will not operate.

Checking for normal operation of the direction valve

The in-line valve should operate as shown in the operation flow chart.

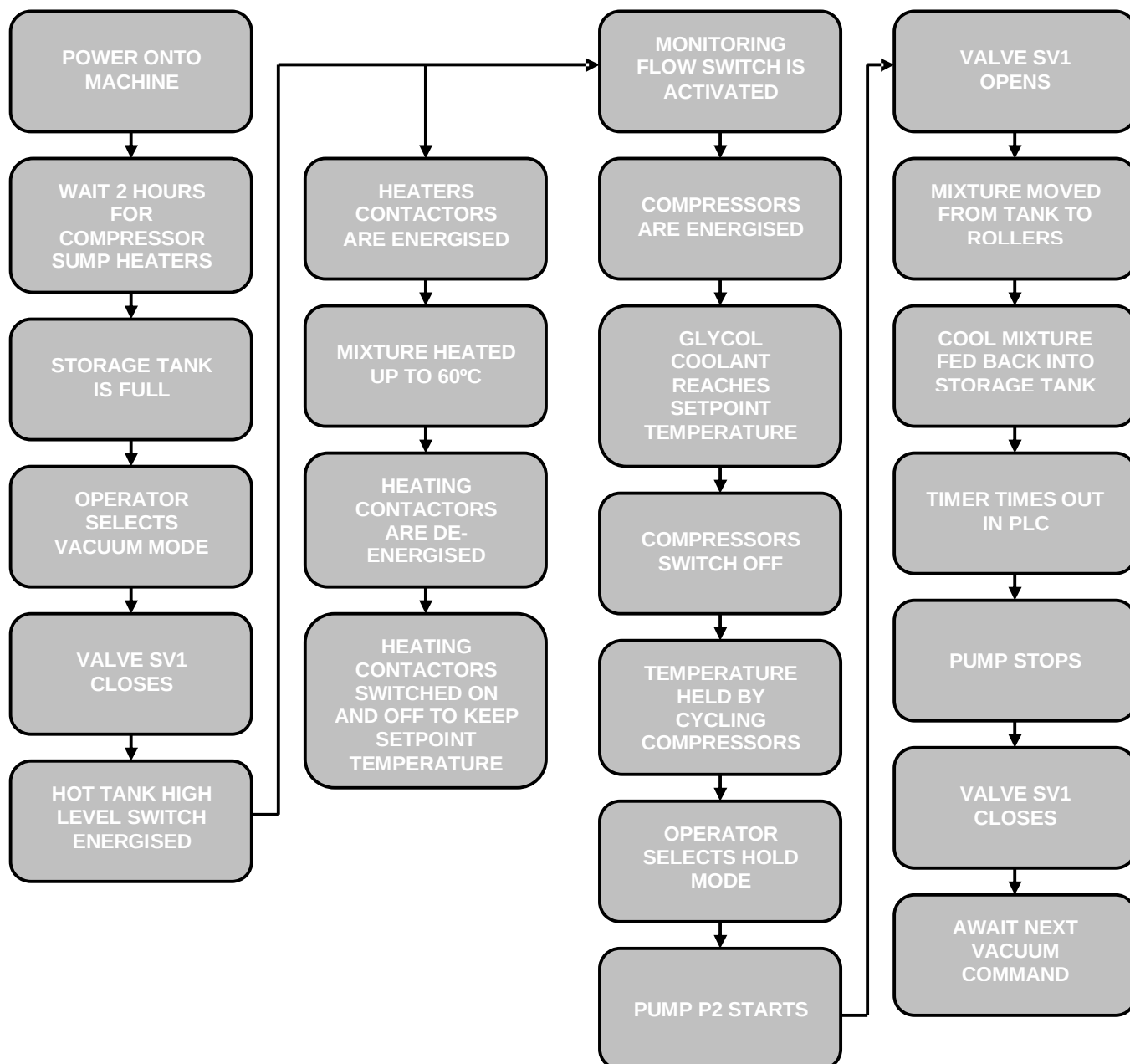
If a problem is suspected, a manual override is provided to check the operation.

Normal Operation Flow Chart: Refrigeration Unit



Flow Chart 14.33 - 1 Normal Operation Flow Chart: Refrigeration Unit

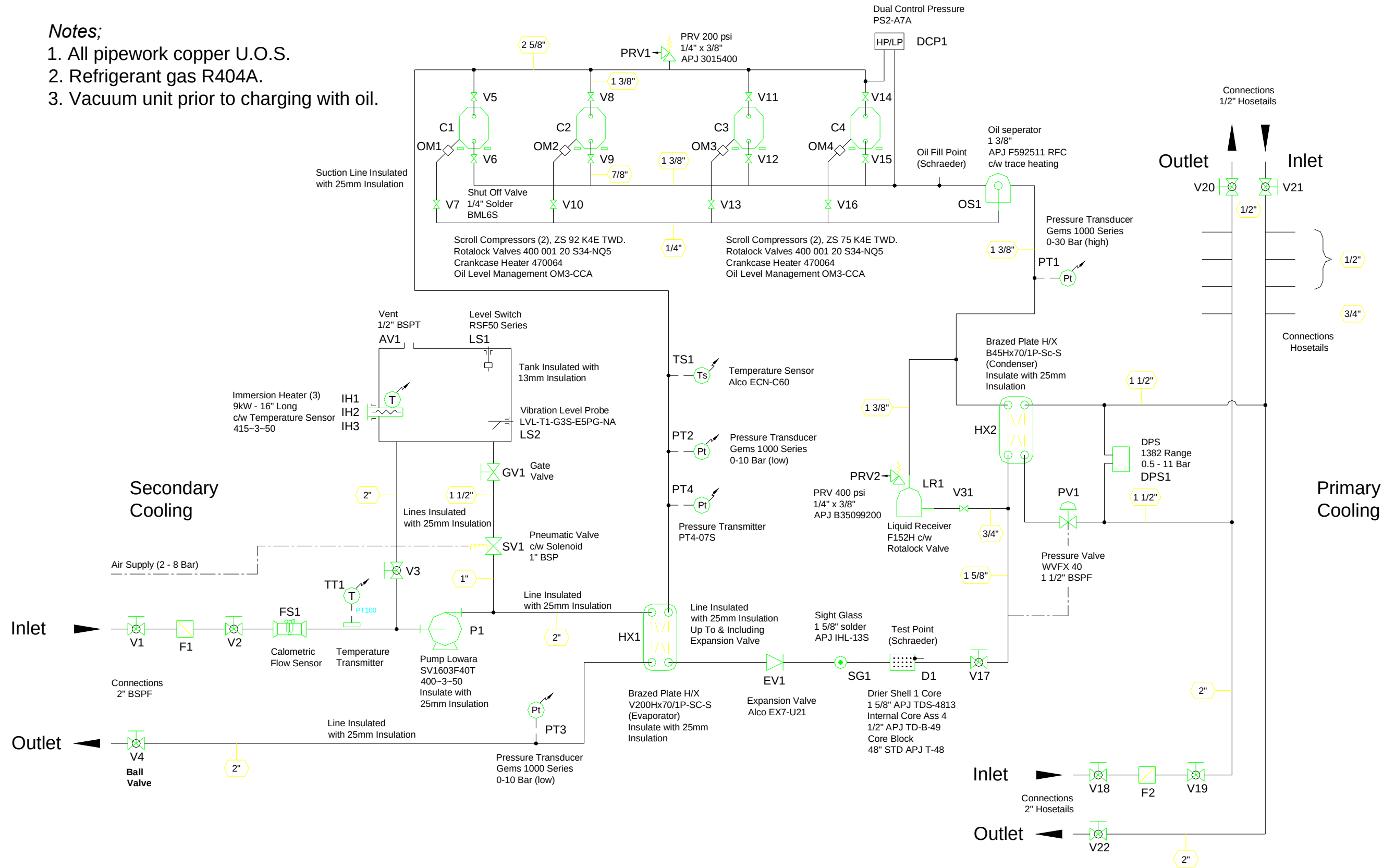
Normal Operation Flow Chart - Chiller System



Flow Chart 14.33 - 2 Normal Operation Flow Chart - Chiller System

Notes;

- 1. All pipework copper U.O.S.
- 2. Refrigerant gas R404A.
- 3. Vacuum unit prior to charging with oil.



Flow Chart 14.33 - 3 Chiller Flow Diagram

1. Identification of the substance/preparation and company/undertaking.**NAME** Mono Ethylene Glycol**S.D.S. No.** M027**Synonyms** Ethylene glycol * MEG * Ethane-1,2-diol ***Supplied by:** Univar Ltd46 Peckover Street
BRADFORD
West Yorkshire
BD1 5BDTel: 01274 377000
Fax: 01274 737058
Emergency No. (Office hours): 01274 377070
Emergency No. (24 hours): 01865 407333**2. Composition/information on ingredients.**

Contains:

<u>Chemical</u>	<u>% Conc</u>	<u>Classification</u>	<u>Exposure</u>	<u>CAS</u>	<u>EINECS</u>
Ethylene glycol	>90	Xn;R22	10 mg/m ³	107-21-1	203-473-3

3. Hazards identification.

Ingestion of excessive doses may cause central nervous system effects, cardiopulmonary effects and kidney failure. Estimated lethal dose for an average person is 100ml. Excessive exposure may cause irritation to upper respiratory tract.

4. First Aid measures.

<u>Exposure Route</u>	<u>Symptom</u>	<u>Treatment</u>
Inhalation	Irritation of breathing passages.	Remove from exposure, rest and keep warm. In severe cases, or if recovery is not rapid or complete seek medical attention.
Skin Contact	Mild irritation.	Drench the skin with plenty of water. Remove contaminated clothing and wash before reuse. If large areas of the skin is damaged or if irritation persists seek medical attention.
Eye Contact	Mild irritation	Irrigate thoroughly with water for at least 10 minutes. Obtain medical attention.
Ingestion	Irritation of upper digestive tract nausea and vomiting.	Wash out mouth with water. Do not induce vomiting. If patient is conscious, give water to drink. If patient feels unwell seek medical attention.

Immediate Treatment / Antidote: Supportive care. Treatment based on judgement of the physician in response to reaction of the patient

S.D.S. No M027**Page** 1 of 4

S.D.S. No M027

NAME: Mono Ethylene Glycol

5. Fire Fighting measures.**Suitable Extinguishers** Water fog, alcohol resistant foam, CO₂, dry chemical.**Hazardous Combustion Products** Oxides of carbon.**Special Equipment for Fire Fighting** Self contained breathing apparatus and full protective clothing.6. Accidental Release measures.**Safety Precautions** Wear appropriate PPE - See section 8**Environmental Precautions** Avoid entry into sewers or water courses.**Clean up Procedure** Absorb or bund spillage, transfer to suitable containers for waste disposal.7. Handling & Storage.**Handling** **Ventilation** Good general ventilation.**Recommended procedures & equipment** Avoid skin and eye contact. Prevent accidental ingestion.**Storage** **Temperature range** Ambient**Keep away from** See section 10**Suitable storage Media** Store in original containers.**Precautions against static discharge**8. Exposure Control/personal protection.

Exposure Limits	10 mg/m ³ 8Hr TWA particulates 52 mg/m ³ 8Hr TWA vapour 104 mg/m ³ STEL	Type	EH40
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Monitoring Method as Ethane-1,2-diol**Protective Measures****Respiratory:** Type approved RPE for organic vapours and mists if required.**Hand:** Impervious gloves**Eye:** Goggles**Skin:** Overalls**Hygiene Measures** Always wash thoroughly after handling chemicals.

S.D.S. No M027

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S.D.S. No M027

NAME: Mono Ethylene Glycol

9. Physical & Chemical Properties.

Appearance	Clear, colourless liquid.
PH	6-8
Boiling Point/range	198°C
Melting Point/range	-13°C
Flash point	116°C closed cup
Autoignition temperature	400°C
Explosive limits	3.2% - 15.3% v/v
Vapour Pressure	< 10 Pa @ 20°C
Relative density	1.115
Solubility	Completely miscible in water
Partition coefficient	-1.36 (ethylene glycol)

10. Stability & Reactivity.**Stability** Stable**Known Hazardous reactions** Violent reactions with oxidizing agents.**Materials to avoid** Oxidizing agents.**Hazardous decomposition products** Oxides of carbon.11. Toxicological Information.**Effects**

Excessive exposure may cause Central Nervous System effects. Lethal dose in humans is approximately 1.6g/Kg.

LD₅₀ 6000-13000 mg/Kg **oral-rat**12. Ecological Information.**Environmental Effects****Mobility** Liquid with low volatility**Degradability** BOD₂₈/COD = 56 - 64%; COD = 1.29 gO₂/g**Aquatic Toxicity** LC₅₀, goldfish, 24 h static > 5000 mg/l**Bioaccumulative potential** Low.

S.D.S. No M027

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14.34 Nip Roller Checks/Alignment

A rubber covered roller is provided to "nip" the substrate at a driven roller. The roller is directly driven from the drum by a mechanical belt and pulleys.

The nip roller is driven into position on a "Cam roller", which is turned by a pneumatic cylinder, located on the outside of the winding mechanism. This cylinder is activated from a button on the DRIVES screen.

The nip roller covering is a 30-60-shore hardness neoprene rubber. Check the mechanical drawing for exact value.

	Caution ! Do not apply excess air pressure to the cylinder.
---	---

Testing the Nip Roller Contact

In cases where creases can be seen on the process drum, there are a number of possible reasons, one of these could be incorrect nip roller operation.

The nip roller has to be parallel in all planes to its contact point and the nip roller pressure must be even.

All nip rollers should be set up as described in the following procedure.

1. Visually check the roller for damage or flat spots.
2. Rotate the roller to ensure that it turns freely.
3. Set the mechanical stop that the cylinder pulls against. This will determine the amount of contact when the roller is engaged.

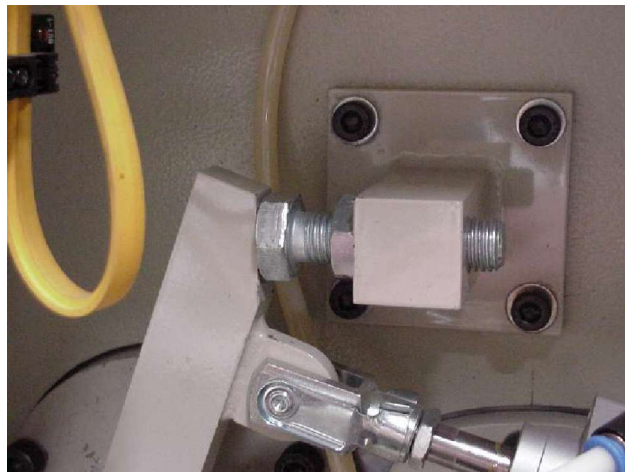


Figure 14.34 - 1 Photograph showing contact mechanical stop.

4. Set the air pressure on the local regulator to 2 bar. This regulator is located on the winding cart.

**Note!**

Do not exceed the 2 bar pressure limit. It is normal for the regulator to have a padlock to stop changes to this value.

5. Check that the area around the nip roller is clear and manually actuate (Nip on) the spool valve allowing the cylinder to retract.
6. Check if the nip roller is touching the driven roller that it locates against.
 - 6a. If yes, adjust the mechanical stop to allow light to be seen between the two rollers surfaces.
 - 6b. If not, Place strips of the thinnest gauge substrate that will be used on the machine (Example: 12 micron PET or 20 micron BOPP), as shown below.

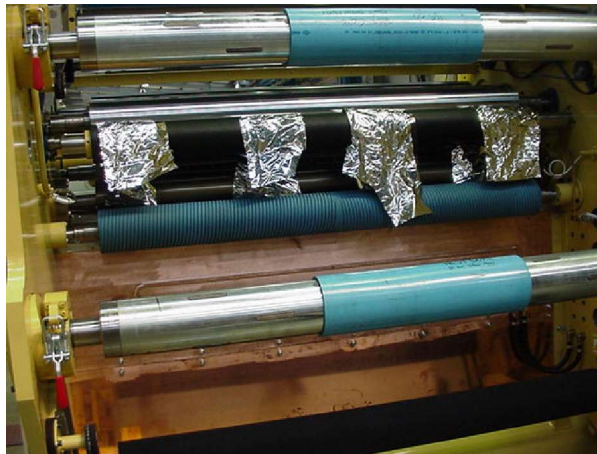


Figure 14.34 - 2 Photograph showing film placed between the nip and driven rollers

7. Turn the rotation limit bolt in until the nip roller contacts all areas of the driven roller, and the film cannot be pulled from between the nip and the driven rollers, without rotating the roller.

**Caution !**

Allowing too much surface contact can lead to damage to the nip roller surface



Figure 14.34 - 3 Photograph showing film placed between the nip and driven rollers being checked for slippage

8. Fasten up the lock nut to ensure that the bolt does not move.

With the nip roller disengaged, there should be enough room to allow the substrate to be passed between the nip and driven roller.

To check for even contact across the full width of the roller, the following test is suggested;

1. Lift the nip roller away from the driven roller using the control circuit
2. Place two pieces of carbon paper on normal paper between the driven roller and the nip roller.
3. Engage the nip roller, and then disengage.
4. Check the paper for even witness marks.

Reasons For Damage To Nip Rollers

- Excess contact - check and adjust mechanical stop
- Wear - Check the circumference of the roller
- Incorrect cleaning fluids Used
- Damage by operators by careless use of knives.

Running machine with Nip Roller Disengaged

There can be problems when the winding system is ran in this manner. The drive system requires the substrate to travel in contact with the drum and therefore at the same speed.

If slippage occurs between the drum and the substrate, the diameter calculator in the drive system can be incorrect. This will lead to incorrect auto-stopping functions, and layarm positioning problems.

	Note!
	If the nip roller is disengaged, while tension is applied, the unwind and rewind tension setpoints will be equalised.

14.35 Layarm Roller Assembly - Mechanical Alignment

Rewind

To check mechanical alignment between the last rollers in the lay arm assembly to the rewind roll, the following procedure is suggested;

1. Place a reel bar in the rewind position, without a core upon it.
2. Drive the lay arm to a distance from the reel bar, for example 100mm.
3. Use "Stick Micrometer" to measure the gap at both ends of the gap.
4. Modify the distance, if required, by adjusting the belt tensioners



Figure 14.35 - 1 Photograph of Mechanic Checking Roller Alignment

The distance should not normally change under normal operation, however, it has been found that incorrect loading/unloading procedures can cause the belt to jump the sprockets.

Ensure that the positioning belts have even tension on both sides.

If a "Stick Micrometer" is not available, a alternative method is to use a circumference measuring tape-rule.

Unwind

The unwind layarm's roller is located on a pivot assembly to allow for differences in the unwind roll circumference.

To check mechanical alignment of the assembly, the following procedure is suggested;

5. Allow the roller to drive to it's fully out parked position
6. Ensure that the belts are tensioned correctly.
7. Check that the clamp plate (M010) is fully clamped onto the belt.
8. Loosen nuts (S012) on either side of the locating bracket (M016)
9. Adjust bolts (B102) to move the position of the arm, so it is touching the mechanical stop on both sides.

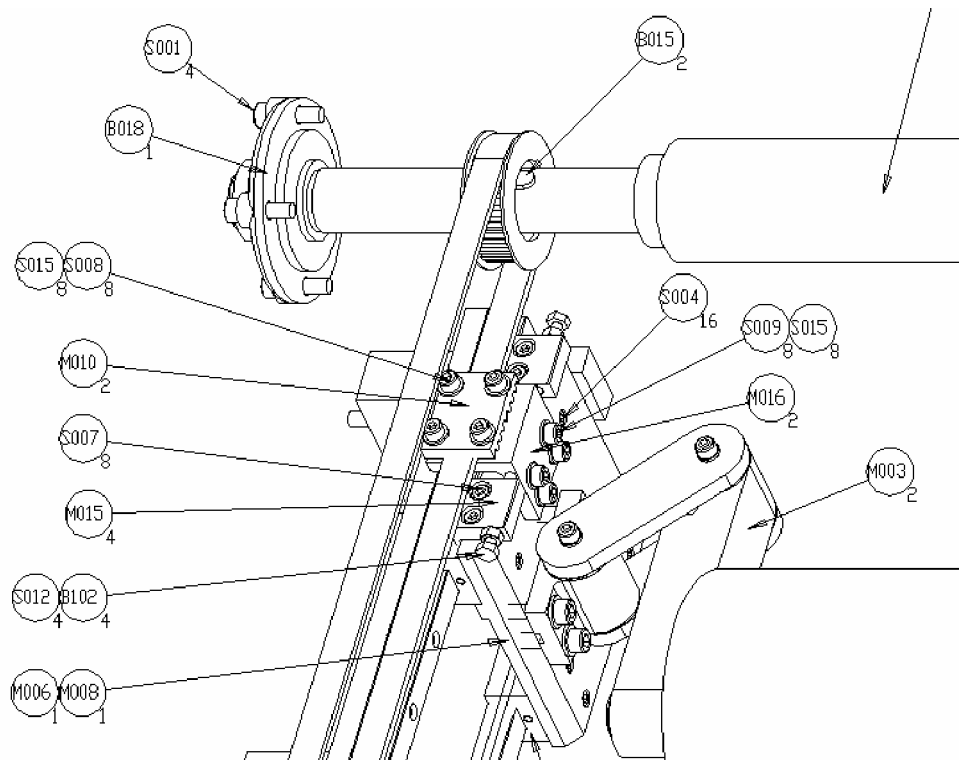


Figure 14.35 - 2 Drawing of Unwind Layarm Pivot assembly Roller Alignment

The distance should not normally change under normal operation, however, it has been found that incorrect loading/unloading procedures can cause the belt to jump the sprockets.

Ensure that the positioning belts have even tension on both sides.

14.36 Lay Arm Roller Operation

The lay arm roller assembly is positioned upon a slide arrangement, which travels towards and away from the rewind roll. The assembly is moved by a timing belt/sprocket/shaft mechanism.

The shaft that drives the assembly passes through the chamber wall, through a rotary vacuum leadthrough, and is attached to an AC vector drive motor.

The drive motor moves at a very accurate speed in either direction and is controlled from a drive interface unit in the drive cubicle.

Principle of Operation

The drive system calculates the diameter of the rewind roll by using internal software and feedback signals from the master and winder motors. It then produces a reference value for the rewind diameter. This value is sent to the computer.

The layarm motor sends signals from its encoder back to the drive unit. In turn this signal is converted and sent to the computer.

These signals are used by the computer and a control signal is sent to the drive unit to rotate the motor to a position set by the user.

Setting the position of the Layarm

The rollers on the layarm are used to support the substrate between the rewind load cell roller and the rewind roll. Therefore it is required to travel closely to the outside circumference of the rewind roll.

The amount of gap can be set by the operator from the DRIVE screen. There are two modes of operation;

Lay Off (Gap) Mode

In this mode, the assembly is calibrated to allow the last roller to be touching, with no force, the rewind roll. The operator has the ability to set a gap between the roller and rewind roll on the drives interface unit. This should normally be set to less than 12mm

As the roll diameter increases, the layarm assembly will move away from it keeping the same gap.

Lay On (Contact) Mode

In this mode the arm is pressed against the roll. The amount of force being applied is achieved by reducing the amount of torque in the rewind layarm motor. Increasing the amount of torque in the unwind layarm motor will increase the pressure.

This is a maximum of XX newtons. This is dependant upon the weight of the roller assembly for the rewind and the amount of torque available to the unwind

Note: 1KG = 10 Newtons = 2.2 lbs

Manual Operation

There are push buttons located at the rear of the machine for lowering and raising the layarm assemblies. These are only active when the machine is fully open.

- Rewind down arrow - moves the arm to the roll
- Rewind up arrow - moves the arm away from the roll
- Unwind down arrow - moves the arm to the roll
- Unwind up arrow - moves the arm away from the roll

Note: the pushbuttons only operate when the winding cart is fully retracted.

Limit Switch

There is a limit switch located at the fully retracted position of the slide rails to switch the motor off when it touches the switch. The switch is wired across normally open contacts.

This switch is also used by the drive system to indicate that the layarm is in its "home" position for use, when calculating the location of the assembly, by use of the encoder.

Layarm Drive Set Up and Calibration Procedures

The layarm calibration normally takes place after the AC drive has been auto-tuned (refer to drive manufacturers information) and the encoder scaling factors are correctly set. All other calibrations are carried out in the configuration screen.

This can only be accessed at the engineer level, which is normally limited to General Vacuum Equipment staff.

	Assumption!
	It is assumed that the layarm system has/is operational, but there are concerns about the accuracy of the positioning in gap control or layon pressure in contact mode.

Motor Direction Checking

Use the raise and lower push buttons located to the rear of the machine, to check for the correction direction of the motor.

The direction required is dependant upon the location of the lay-on assembly in relationship to the drive belt.

- For unwind lay arm, Raise button (towards core/shaft) should turn the motor/gearbox pulley. Depressing the raise button will turn the coupling in a clockwise direction as viewed from the behind the motor.
- For the rewind lay arm, Depressing the raise button (away from core/shaft) will turn the coupling in a clockwise direction as viewed from the behind the motor. Remember if the home switch has been contacted, the motor will not start.

If the motor is going in the incorrect direction, power down the dc bus and change two phases over at the drive.

Tension null	5500	Bits
Maximum force	400.0	N
Maximum force	6000	Bits
Layon null	8200	Bits
Lower speed	-1500	Bits
Raise speed	2000	Bits
Diameter 1	165.0	mm
Diameter 2	1131.0	mm
Position 1	-18125	Bits
Position 2	-1921	Bits
Maximum gap	50.0000	mm
Torque limit for raise	10000	Bits

If there are initial problems with rotation, check that the value in the torque limit for raise normally set between 8000 and 10000 bits.

Once the motor has turned the correct way the coupling can be reconnected. Locate the correct alignment using the raise and lower buttons and then lock out the traverse isolator to prevent any accidents.

The raise / lower speed set points can be checked at this point on the configuration screen

Lower value is normally set at to between -1500 (RW) and -2000 bits, while the raise is normally set to 2000 bits.

Figure 14.36 - 1 Layarm Configuration Screen 1

If it found that the movement is sluggish with the arm connected to the motor/gearbox, check for mechanical alignment and binding.

'Gap' Mode

1. Check is the fully home limit switch operation on both lay arms.
2. Ensure zero reading is being displayed in SCADA lay arms actual gap position in configs pages, when the lay arm stops on the limit switch.

Actual position +0 Bits

3. The span calibration of the lay arm travel can now be done at this stage, place the reel shaft in the machine, if it is not already with a core on it.

Figure 14.36 - 2 Layarm Configuration Screen 2

4. With the reel bar core diameter value down loaded into the drives, and 'gap' mode selected, Bring the lay arm onto the core using local pushbuttons at the rear of the machine.
5. Press the 'raise' button momentarily to take backlash out of mechanical assembly. The lay arm roller may move away from the core by 1-2 mm.
6. Add this value to the core diameter value and enter the value into placed in lay arm configuration as '**Diameter 1**'. Read the value of the displayed actual gap position enter this value in '**Position1**'
7. Move the lay arm to within 50mm of the 'home' limit switch, physically measure the diameter of this position between the core and the layon roller.

8. Enter the measured value into in lay arm configuration as '**Diameter 2**'. Read the value of the displayed actual gap position enter this value in '**Position2**'

9. Finally set a gap of approximately 10mm between the core, and the lay arm roller, measure this gap (suggest using a set of metric allen keys as a slip gauge) it should agree with gap reading being displayed in SCADA. This completes the set up of the lay arm in 'gap' mode.

'Layon Null'

1. The 'layon null' value needs to be set at this point. This factor controls the torque limit for moving the assembly to the core on the unwind section and away on the rewind section.
2. Start with a nominal value of 10,000 in the configuration. Use the pushbuttons at the rear of the machine to move the layon roller to the core. With this value, it should move freely without any problems. Reduce the value to about 8200 bits, and check if the assembly still moves. Reduce to a level where the assembly is balanced.

Normal values are approximately 8200 bits for a K4000 2450 rewind and 7900 for the unwind layarm.

'Contact' Mode

This mode of operation is also called 'layon'.

'Contact' Mode - Rewind

This is best carried out with web in the machine and tension applied. The line-speed set-point has to be set to zero mechanism closed limit switch over-ridden.

1. Apply tension to the web at for example 100 newtons on both the unwind and rewind winders of the machine. Place the lay arm into 'contact' mode (zero newtons force), **ENSURE ZERO SET POINT** and then select line run from the drive screen – 'web start'
2. The drive system is now fully enabled at a standstill condition. Change the contact force to a value of 10 newtons. At this, the layarm should be quite easily pushed away from the core.
3. change the contact force to a value between 50 and 100 newtons, check force required to push arm away as it increases proportionally,
4. At this time, the lay on 'tension null' value can be set by reducing and increasing tension i.e., min to max level, no appreciable difference in push off force should be noticed.
5. However by increasing tension the layarm is pushed away or becomes more difficult to push away when higher or lower tension is applied, the 'tension null, value will have to be adjusted in the configuration. Note: The maximum contact force on the rewind roller is the full weight of the lay arm assembly, ie brake off zero torque demand at motor.

Approximate lay arm weights for EHF4, M1 and K4000, are

- 180n (18kg) for 1250 mm wide machine
- 300n (30kg) for 1650
- 400n (40kg) for 2450

'Contact' Mode - Unwind

1. Unlike the rewind lay arm, the unwind lay arm drives onto the roll in 'contact' mode. To check and set up the maximum amount of contact force, a spring balance has to be used, which is tied between the lay arm roller and the lower mechanism tie bar.

Note this task is carried out without the drive system being in either the tension or run modes and preferably with the unwind chucks empty.

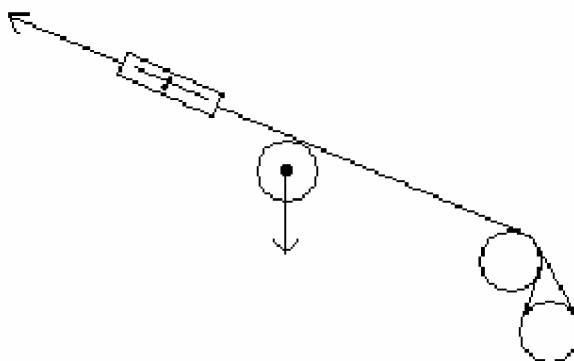


Figure 14.36 - 3 Sketch Showing Unwind Contact Setup

Nominally, the maximum force should be set to the same value as the previously set rewind layarm.

Tension null	5500	Bits
Maximum force	400.0	N
Maximum force	6000	Bits
Layon null	8200	Bits
Lower speed	-1500	Bits
Raise speed	2000	Bits
Diameter 1	165.0	mm
Diameter 2	1131.0	mm
Position 1	-18125	Bits
Position 2	-1921	Bits
Maximum gap	50.0000	mm
Torque limit for raise	10000	Bits

Figure 14.36 - 4 Layarm Configuration Screen 3

- Set the maximum contact force by reducing the value in 'torque limit for raise' to a value of 6000 bits, and press the raise button at rear of machine noting the value in Kg on the spring balance.
- Set the required 'maximum force' value required to the same value as the rewind layarm
- Adjust the 'torque limit for raise' value in small steps until repeat step 2 until the required value is achieved.
- Finally entering this 'torque limit for raise' value in the 'maximum force' value.
- Return the 'torque limit for raise' to its original value,

14.37 Tendency Drive System

The winding mechanism consists of a number of rollers that are "tendency driven". This term describes where a roller shell is located on a shaft, which is rotated. The source of rotation is through a pulley and timing belt arrangement on the outboard plate of the winding mechanism.

The purpose of this arrangement is to allow the roller shell to rotate with a minimum amount of friction.

Also this arrangement allows for slight speed discrepancies between the substrate and a roller if it was driven at line speed with all other rollers.

The pulleys and belts are located on the outside of the winding mechanism plate. A guard is placed over the belts and pulleys for safety purposes. Visual inspection can take place through the mesh section of the guard.

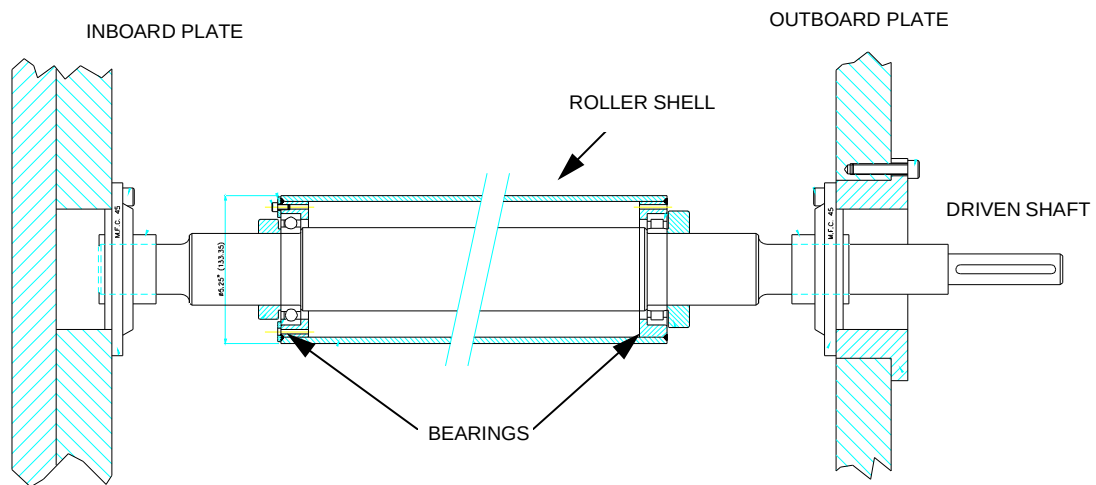


Figure 14.37 - 1 Tendency Roller Arrangement Cross Section

Pulleys

The pulleys are mounted on the end of the shafts. The shafts pass through the wall of the winding mechanism. Generally the pulleys use a taper lock and key-way to fasten them onto the shaft.

Belts

The timing belts supplied will be manufactured to a high grade and are easily replaced from most stockists. These belts may be single sided or double sided, depending upon the design requirements.

"Jockey Pulleys"

This is the name given to the tensioning pulleys in the tendency drive system. They have been installed to allow correct tensioning of the timing belts.



Warning !

Lockout the movement isolator and apply lock before attempting to work in this area.



Figure 14.37 - 2 Photograph Showing an Adjustable Jockey Pulley

Checks must be made on a regular basis of the condition of the tendency drive system, and greasing should be carried out as advised in the preventive maintenance schedule.



Figure 14.37 - 3 Photograph Showing a Jockey Pulley being adjusted to tighten up the belt

Warning !	
	After carrying out work in this area, Reinstall the guard covering the belts, before removing the lock from the isolator.

14.38 Inspections of Electrical Panels

A regular inspection of the electrical wiring will reduce the risk of costly production time losses due to avoidable faults. When inspections are made within the electrical enclosures, the following points should be noted;

- Keep doors closed at all times.
- Check that all cable connections on terminal blocks are tight
- Check that all enclosures are suitably ventilated and free from build-ups of deposits or condensation.
- If air conditioning units have been fitted, check that they are functioning correctly and that the thermostat is set correctly.
- Replace any spent lamps or bulbs
- Check the mechanical operation of all components

	DANGER
	Beware Live Electrical Circuits, Isolate before working in this area

Air Conditioner Units

The air conditioners fitted to the electrical cabinets have been selected to ensure that the components within the enclosure are kept within their operating temperature range. The units should be checked for operation, correct temperature setting <30 degrees Centigrade, and that the drain is free.

14.39 Removing Drive Belt Guard Panels

Where timing belts are used on the machines, as on the winding cart for driving the unwind and rewind rolls, guarding is supplied.

These guards are used to protect personnel from the rotating belts and should be located in place while the belts are being rotated.

Warning labels are supplied to make users aware that the guards have to be fitted to the machine.

	Warning !
	Never operate the machine without the guard panels in place and secured. Failure to follow this warning could result in serious injury.

Occasionally, it may be necessary to remove the guarding to gain access for maintenance purposes. The guards' panels can be easily removed in sections by removing the appropriate screws that locate them.

The panel over the internal winding mechanism tendency drive system has a mesh to allow operators to visually inspect the belts and pulleys.

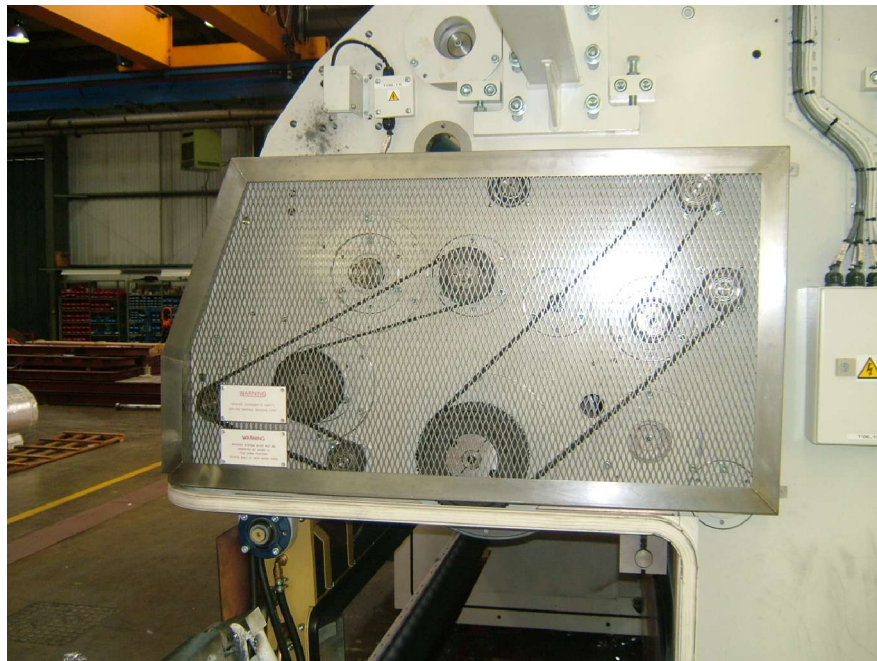


Figure 14.39 - 1 Photograph of the Tendency Drive Guard



Figure 14.39 - 2 Photograph of the winding Cart Guarding Panels

14.40 Timing Belt Inspection

Timing belts are used extensively throughout the machine. Regular inspection and correct tensioning of the belts will ensure their long life and reliability.

Should a timing belt show signs of wear, it should be replaced immediately. Use the following table to diagnose the cause of the failure and prevent recurrence.

The table shows that the most cause of belt failure is over-tensioning. This cause of failure can be eliminated with careful belt installation.

SYMPTOMS	CAUSE OF FAILURE	CORRECTIVE ACTION
Excessive edge wear	Bent pulley flange	Straighten Flange
Wear on pressure face side of belt tooth	Excessive belt tensioning	Reduce belt tension
Excessive wear between belt teeth (exposed tension members)		
Excessive drive noise		
Cracks in neoprene backing	Exposure to excessively low temperatures (-40 °C)	Prevent exposure to low temperatures
Softening of neoprene backing	Exposure to excessive heat and or oil	Prevent exposure to heat or oil

Table 14.40 - 1 Timing Belt Faults

Belt Handling and Installation

To prevent belt failure, the following guidelines should be followed;

- Belts should not be excessively bent or creased in storage.
- Do not subject the belts to extreme heat, low temperature or high humidity
- Never attempt to force the belt over a pulley. Always slacken the take up pulley, taper lock or motor position to allow the belt to be slipped over them.

Adjusting one of the following sets the belt tension;

- Take up "jockey" pulley
- The motor on it's mounting base
- Refitting the taper lock assembly.

14.41 Bearings

The rollers within the winding mechanism rotate at slow speeds when compared to many other bearing applications.

A routine check of all bearings is recommended. Re-lubrication of the bearings should be carried out as per the preventive maintenance schedule in section 12 of this manual.

On this machine, within the winding mechanism, there are a number of special SKF CARB bearings. A manufacturers application booklet is included in the vendors manuals supplied.

For further information we recommend the user to visit the SKF Web Site at:-

http://www.skf.com/portal/skf/home/products?lang=en&maincatalogue=1&newlink=1_0_112

	Caution !
	Over greasing as well as under greasing can lead to failure. Avoid over lubrication Use the correct grease, as described in section 13 of this manual

Routine Checks

The operating performance of a rolling bearing is signalled by changes in temperature, noise or vibration. It should be possible to bear the temperature of the housings and feel only modest vibration.

Should the condition of a bearing change, reference should be made to the following table.

If a bearing is removed for inspection, it should be cleaned with a suitable solvent, and be re-greased immediately to prevent corrosion.

Work the grease into the spaces between the rings and the cage taking care to keep clear of contamination. Generally the space in the housing each side of the bearings should be 50-60 % filled.

Ensure that there is enough space for the surplus grease that will be expelled from the initial rotation. Use the appropriate grease as described in section 4 of this manual.

Note: As a general guide to the quantity of grease to be used, use the formula below.

GREASE QUANTITY (grams) = $1/2 \times \text{Bearing OD (mm)} \times \text{Bearing Width (mm)}$

	Note!
	Contaminated components and operating media must be disposed of in accordance with local government regulations where the machine is operated.

MODIFIED CHANGE	PROBABLE CAUSE	ACTIONS
Temperature increase	Too much grease	Remove the bearing and wash out all the old grease - half fill the bearing with fresh grease and refit
Temperature still increase	Wrong grease	Check the grease type and replace if necessary
Temperature & vibration increase	Too little grease	Add more grease of the same brand and type
	Grease life exhausted	If grease is contaminated, wash out the old grease and regrease and re-install the bearing
	Bearing fatigue	If raceways, rolling elements or cage are worn, replace the bearing
Vibration Increases	Fatigue Failure	Replace bearing
	Contamination from external source	Check lubricant for outside contamination. Check or replace seals. Refurbish shaft contact surface if worn. If bearing has not failed, it can be run after cleaning and re-greasing until a new bearing can be obtained

Table 14.41 - 1 Bearing Faults

	Caution !
	Do not use oil types that are not on the recommended list, as this could lead to failure of components upon the machine.
	Warning !
	Lubricating grease contaminated with hazardous substances can be hazardous to your health. Wear relevant personal protective equipment when draining oil from vacuum pumps, which can potentially contain harmful gases or other harmful substances, and when disposing of exhaust condensation or filters.
	Caution !
	The lubricating greases are potentially harmful to the environment. Never allow greases or parts dipped in oil/grease to penetrate the supply of drinking water. Grease (even small amounts!) must be disposed of as hazardous waste.

14.42 Vee Belts

Vee belts are used on some pumps on the machine, the following table is to be used as a guide;

TROUBLESHOOTING CHART - VEE BELTS		
PROBLEM	POSSIBLE CAUSE	CORRECTION
Belt slipping	- Not enough tension - Drive overload - Oily drive	- Increase tension - Increase drive capacity - Correct oil leakage
Belt Squealing	- Drive overload - Heavy Starting load - Small arc of contact	- Increase drive capacity - Increase tension - Lengthen centre distance
Turned over belts	- Broken cords from running belts onto pulleys - Not enough tension - Impulse loads	- Reduce pulley centres when installing belts - Increase tension - Provide pivoted motor base or spring loaded tensioner roller
Repeated belt breakage	- Shock loads - Broken cords from running belts onto pulleys - Foreign objects	- Increase tension or increase drive capacity - Reduce pulley centres
Rapid wear	- Worn pulley grooves - Pulley too small - Misalignment - Drive overloaded - Belts rubbing guards	- Increase tension or increase drive capacity - Reduce pulley centres
Underside cracked or belt hardened	Excessive heat	
Excessive whipping	- Drive centres too long - Pulsating load	

Table 14.42 - 1 Vee Belt Fault Chart

V-BELT FAILURE CHECK LIST	
APPEARANCE	FAILURE CAUSE
Worn Belt Sides	Normal wear Misalignment, grit or dust
Underside Cracked and Belt Hardened	Excessive Heat
Narrow Spots In Belt	Broken Cords from Prying or Running Belts into Pulley Grooves
Swollen, Spongy and/or Cover Stack Loose	Detonation from exposure to oil, grease, or chemicals
Spot Wear or Burn Marks	Slippage when starting or at peak loads
Ply Separation	Belt running over too small a pulley
Belt Excessively Stretched	Broken internal cords and /or Excessive Tensioning
Frayed or Gouged Belt Sides	Misalignment, Rubbing Damaged sheaves (pulleys)

Table 14.42 - 2 Vee Belt Fault Diagnostic Chart

14.43 Hose and Tubing Installation

Some pneumatic and lubrication system failures can be the result of a mechanical failure of tubing, hoses and connectors. Such failures can be held to a minimum if certain simple rules of installation are followed. The following is a summary of these rules and constitutes good piping practice in the installation of piping systems.

Tube Bending

When solid tubing is bent for an installation, it is good practice to always using a bending tool made for the purpose. For smaller diameter tubing, this may consist of a spring type of bender, which has is an outer sleeve made from a wound spring. This prevents the tubing from kinking while being bent. For large tubing a lever type former is usually used.

In general, when bending tubing, a safe rule to observe is to make the radius approximately three times the outside diameter of the tube. Commercially available tube benders have the standard following bending radii and are acceptable by all piping codes.

Tube Outside Diameter	Standard radius "R"
1/8"	3/8"(10mm)
3/16"	7/16"(11mm)
1/4"	9/16"(14mm)
5/16"	11/16"(18mm)
3/8"	15/16"(24mm)
1/2"	1-1/2"(40mm)
5/8"	2"(50mm)
3/4"	2-1/2"(65mm)
7/8"	3"(75mm)
1"	3-1/2"(90mm)

Table 14.43 - 1 Tube Bending Minimum Radius Chart

Installing the Solid Tubing

When installing tubing, any cuts that are made in the tube must be square with the centreline of the tubing. Both the outside and inside walls must be deburred. The tube should be cleaned to ensure that no debris is left in the tube.

If flared fittings are being used, always be sure that that the correct tool is used to form the flare. Never attempt to use a copper tubing flaring on steel pipe for example.

When assembling the pipework, never spring it to make up a connection to compensate for incorrect bending. Always correct the bend in order to make a connection that is free of stresses.

When connecting adjacent units, never join the pipe in a straight line. A bend will generally eliminate a strain by absorbing any vibrations and by compensating for contraction and expansion.

Hose Installation

The hoses supplied on the machine can be connected generally in two ways, rattle nozzle and hose clamp, or manufactured connectors.

Manufactured Connections

The manufactured connectors should be loosened and tightened using correctly sized wrenches. Do not use an extender to over tighten the fittings.

See the figure below giving recommended methods of connection of this type of hose.

Riffle Nozzles

These hose connections are used in a number of locations on the machine. This includes

- Internal evaporation source area.
- Internal coating shield area
- All external lines except where vendors require straight push on connections

The hose is pushed over the riffle fitting and hose clamps are used to tighten the hose in position.

The inside wall of the hose will grip on the "barbed" sections of the fitting.

Always ensure that the hose and fittings are sized correctly for the application.

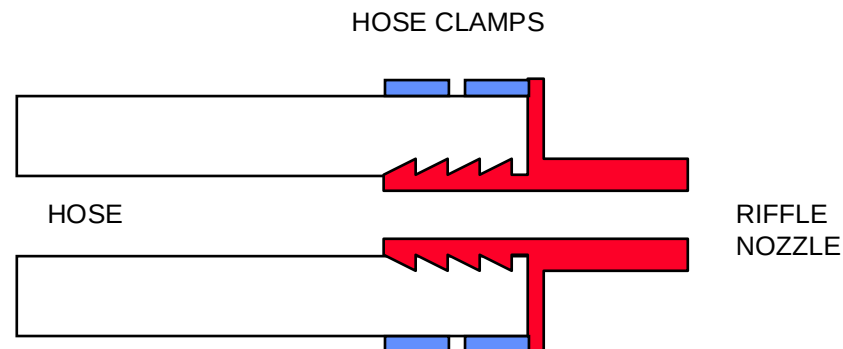


Figure 14.43 - 1 Cross Section Of A Riffle Nozzle Fitting

Swagelok Push On Hose Connectors

This type of connectors are used widely on the machine.



Figure 14.43 - 2 Cross Section Of Swagelok Push On Hose

Multipurpose Push-On Hose Assembly



1. Cut a clean, square edge on end of hose.



2. Cover first barb of end connection with hose. (Lubricating fitting stem with light oil [10W40] may ease assembly.)



3. Hold end connection against flat surface. Grip hose and push with a steady force until hose seats into blue finishing cap.

Do not use a hose clamp.

Figure 14.43 - 3 Instructions for Assembling Swagelok Push on Hose Fittings

Multipurpose Push-On Hose Disassembly



1. Slit hose lengthwise from blue finishing cap to end of barb. Do not damage barbs.



2. Bend hose to expose barbs, then remove hose with a quick pull.

Do not reuse hose.

Figure 14.43 - 4 Instructions for Disassemble of Swagelok Push on Hose Fittings



Caution !

Follow the guidelines given in this section of the manual to reduce the possibility of liquid leakage.

14.44 Pneumatic Pipe work

The pneumatic pipe work will generally be formed from nylon flexible pipe of either 8mm OD or 12mm OD.

Adapters are placed on each component where the air supply is required. These adapters are of a push in type. After the pipe has been cut to length, it can be pushed into the adapter.

The pipe should be cut cleanly and squarely to ensure that no leaks are present.

Once the pipe has been placed in the adapter, it can be removed by pressing on the outer collar and pulling the pipe back.

The pipe work network will be illustrated on the pneumatic schematic drawing provided.

Care should be taken when carrying out hot work above the machine, as hot objects falling on to the tube will damage it.

	Warning !
	Potential Hazard to Pressurised system. Ensure system pressure has been removed at main isolator and that isolation valve has been locked out.

14.45 Taper Lock Bushes

General Vacuum Equipment use taper lock bushes to secure drive pulleys to their shafts. The manufacturers give precise instructions on how to install and remove taper locks.

	Warning !
	Ensure that the main machine isolator is switched off and isolated by a lock before attempting to carry out work on this section of the machine.

Installing

1. Ensure that the mating tapered surfaces are completely clean and free from oil and dirt.
2. Insert the taper lock bush in the pulley hub so that the screw holes line up.
3. Lightly oil hole threads and grub screws.
4. Place screws loosely in the holes, which are threaded in the pulley hub.
5. Clean the shaft onto which the pulley is to be fitted.
6. Fit the pulley and taper lock onto the shaft as one unit.
7. Check the alignment of the pulleys.
8. Use an Allen Key (Hexagonal wrench) to tighten screws gradually and alternately to the required torque.
9. After operating the pulley under normally operating conditions, check the screws for tightness.

Removal

1. Slacken all the screws by several turns.
2. Remove one or two screws and place in the required "jacking" holes
3. Lightly oil the jacking screws.
4. Tighten the screw(s) alternately until the bush is loosened in the pulley and is free on the shaft.
5. Remove the pulley and taper lock bush from the shaft.

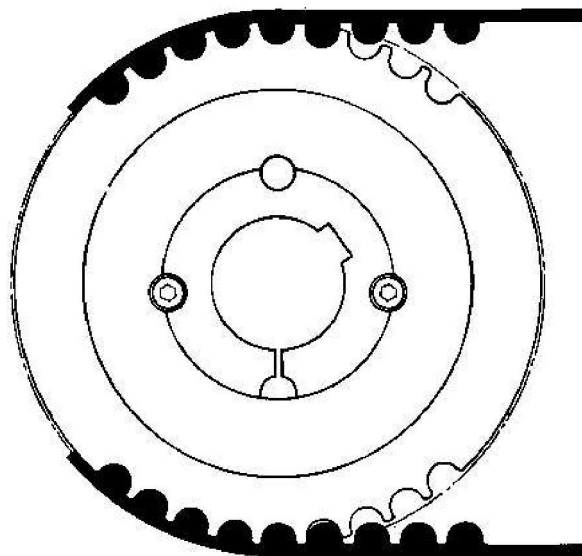


Figure 14.45 - 1 Taper Lock Bush

14.46 Rotary Vacuum Rated Shaft Seals

To allow the rotation of internal components within the chamber and pumps, rotary shaft seals are used. This section of the manual will discuss the operation of such seals.

Operational Theory

Operation of a shaft seal depends upon a balance between oil surface tension and environmental conditions. The shaft seal inner diameter, when inserted over a shaft sleeve, serves as a barrier to molecule flow from the high-pressure side (atmosphere) to the low-pressure side (vacuum).

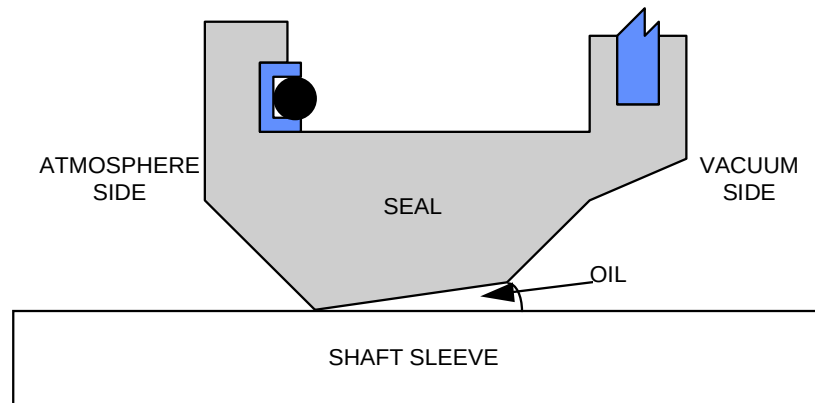


Figure 14.46 - 1 Seal Against Wear Sleeve Cross Section

The sealing edge is lubricated by a film of oil between the edge and the sleeve surface by a capillary action. This oil film is limited on the atmospheric side of the seal by a meniscus whose surface tension prevents free oil flow from the cavity. Variables such as the shaft sleeve condition and installation technique may act alone or together to affect this delicate balance.

Shaft Seal Failure

Shaft conditions such as those listed below will lead to a restricted life cycle of the seal;

- Lack of lubrication
- Eccentricity of rotation
- Surface damage
- Incorrect seal material

The seal will start to damage the shaft sleeve, once it is damaged. The oil from the cavity will then leak into the chamber. Indication of this failure is that the gravity oil bottles will empty rapidly over a few days. The seal and maybe also the sleeve will have to be replaced.

14.47 Unwind/Rewind Chucks and Reel Shaft System

Reel Shaft System

Checking for Wear

The metal segments that grip the core should be checked on a regular basis for wear and be replaced when necessary.

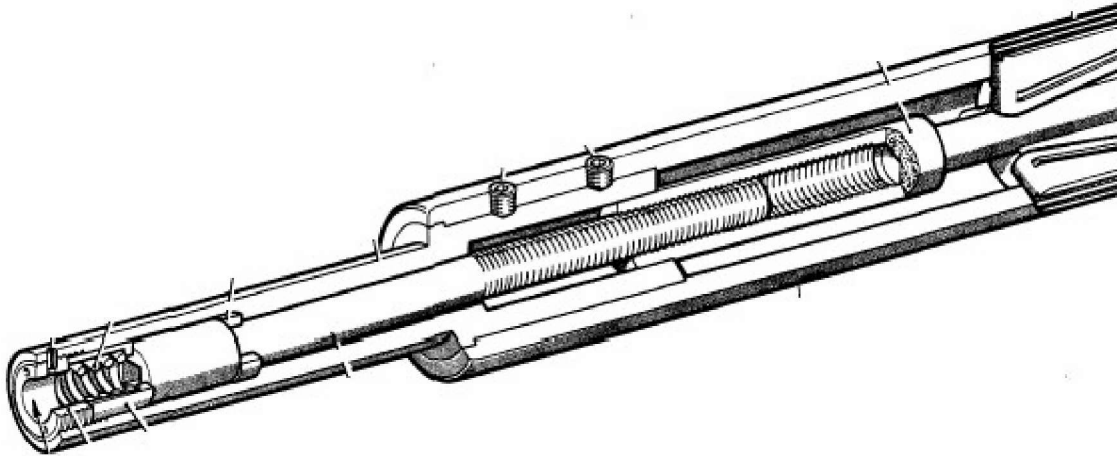


Figure 14.47 - 1 Exploded View of Mechanical Reel Shaft

Replacement of Parts

Follow the instructions given in the proprietary manual supplied

Parts are available from General Vacuum Equipment.

Unwind / Rewind Chucks

The chucks used to locate the reel shafts (arbors) in the machine are manufactured by General Vacuum Equipment. Both chucks have areas that locate the bearings, which are located on the reel shaft.

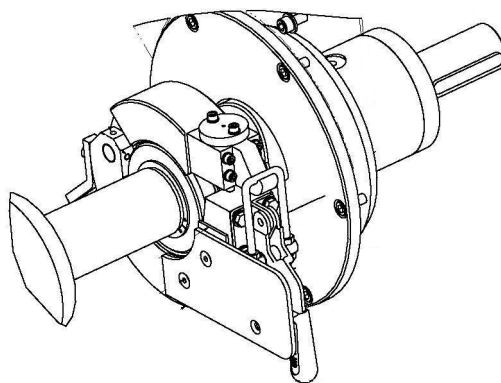


Figure 14.47 - 2 Drawing of Chuck Assembly

The chuck on the faceplate side of the mechanism has a section that passes through the faceplate, to allow it to be driven from outside the chamber. There are matching female and male triangular sections on the chucks and shafts.

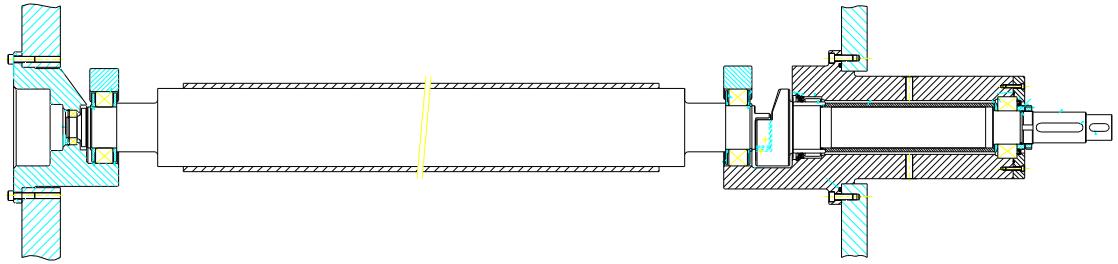


Figure 14.47 - 3 Cross Section of Reel Shaft in Chucks

In each driven chuck there are two replaceable wear sections, which are supplied.

These have a hardened surface, which is slightly less value than that of the reel shafts. Over a period of time these will wear and the drive system instability will change. This can be seen in tension feedback displayed measurements changing.

Checking the tolerances between the reel shafts and the wear plates can be accomplished by rotating the clamped shaft by hand and monitoring for large changes in the smoothness of the movement.

Replacement plates can be purchased from General Vacuum Equipment.

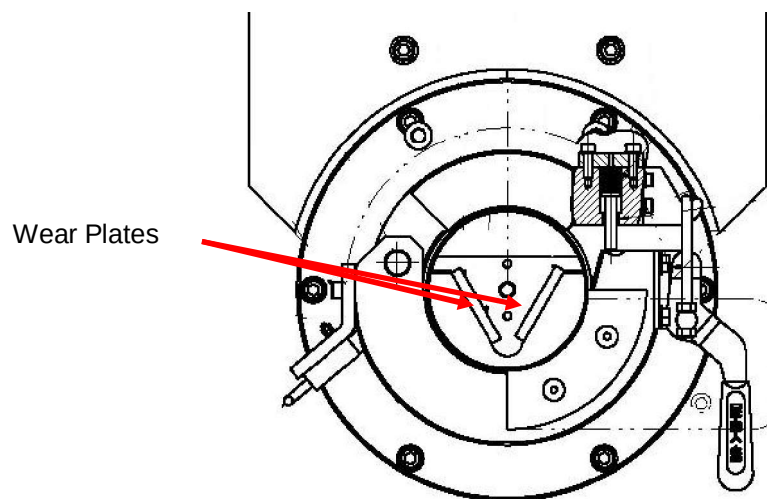


Figure 14.47 - 4 Chuck Wear Section

The chucks are self-locating as the reel shaft is lowered into them. A manual clamping device is provided to lock the chuck closed.

Changing Shaft Seals and Bearings

The seals and bearings selected for this application should provide a long operating life of the assembly. However, there will be a time when replacement of components will be necessary.

The following procedure should be used to remove the driven chuck assembly and replace the bearings and seals.

Removing the Chuck from the Machine

The following procedure should be used;

1. Lock out the main machine isolator to remove all power from the machine.
2. Remove any reel shafts that maybe in the machine.
3. Remove the guard from over the timing belt and pulleys.

4. Loosen the motor location jacking and fixing bolts to allow the belt to be removed.
5. Remove the pulley off the chuck assembly shaft.
6. Drain the oil from the chuck housing into a clean container to allow it to be used again.
7. Support the housing of the chuck with a sling and lifting device from above.
8. Remove the bolts (A1) on the drawing and withdraw the assembly through the hole in the faceplate.
9. Remove the chuck to a clean area to allow work to be carried out on it.

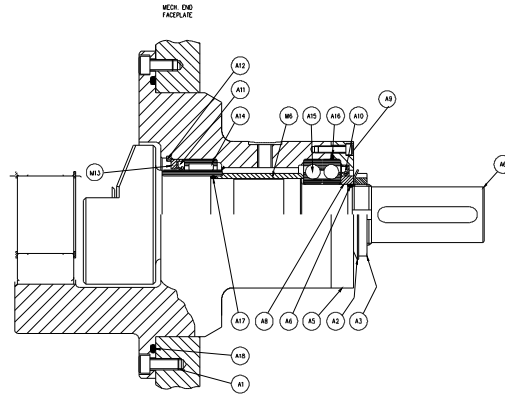


Figure 14.47 - 5 Cross Section Drawing Of Driven Chuck

Removing the Shaft from the Chuck Housing

1. Remove the clamping arm (B1) from the chuck by loosening the screws (B2) and withdrawing the locating pin (B3). Place in the storage area.

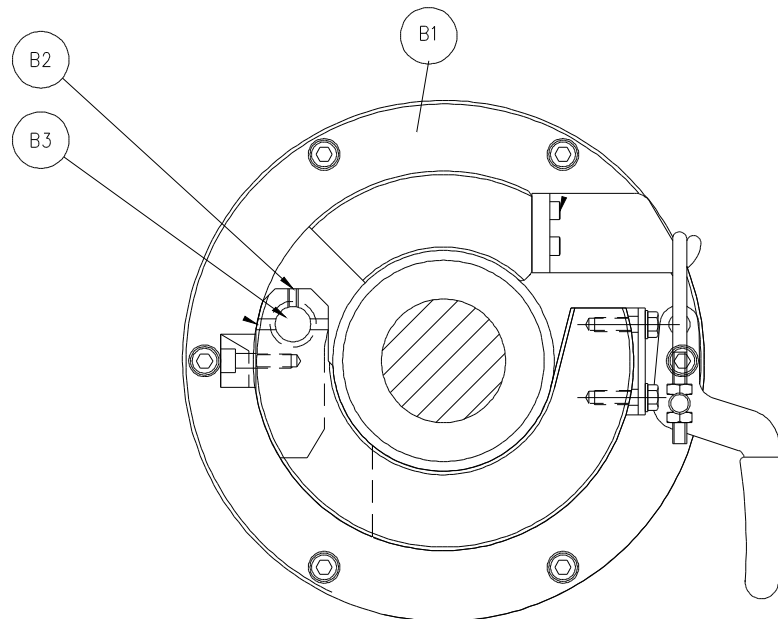


Figure 14.47 - 6 End View of Driven Chuck

2. Press back the engaged tab from below the locknut (A2)
3. Remove the locknut (A3) and tab washer (A4).
4. Remove the screws (A4) and remove the end cover (A5) from the assembly. The o-ring (A6) will be loose at this time and should be removed and placed in the storage area.
5. Ensure that the clamping arm (B1) has been removed.

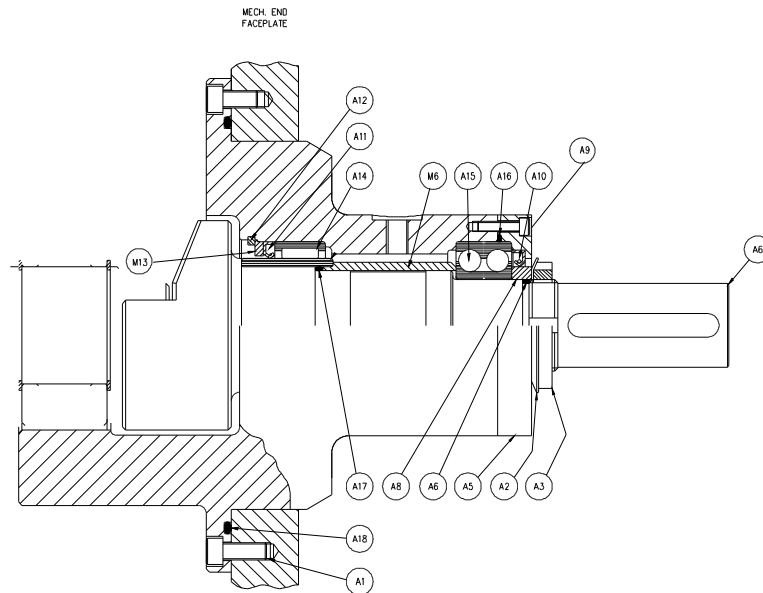


Figure 14.47 - 7 Cross Section Drawing Of Driven Chuck

6. Drive the shaft (A6) out of the chuck housing using a rubber mallet. DO NOT USE A STEEL HAMMER, AS DAMAGE WILL OCCUR. The spacer (A7) can be removed and placed in the storage area.
7. The sleeves (A8) (A9) can be checked for wear marks.

Replacing the Seal (A10) In the Cover (A4)

1. Tap the old seal out of the housing.
2. Check for damage to the sealing surfaces in the cover
3. Wipe the sealing side of the seal with a light coating.
4. Use a hydraulic press and an adapter to place the seal in the cover cap without damaging it.

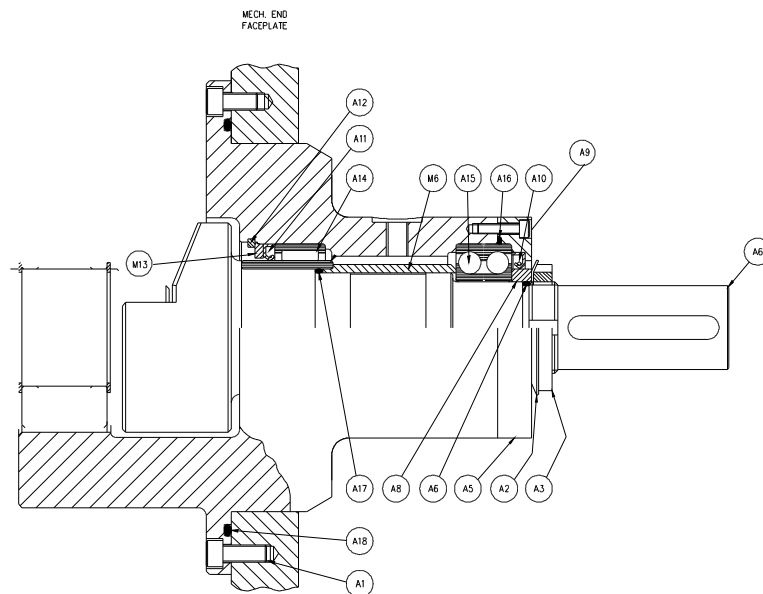


Figure 14.47 - 8 Cross Section Drawing Of Driven Chuck